

The Lean formalization of Gwynne–Miller’s LQG metric theorems: deviations from the sources and errata

Generated from DEVIATIONS.md and ERRATA.md at commit 28310f87

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Deviations from the sources

Source: DEVIATIONS.md.

Every place where the Lean formalization departs from the stated text of a source is recorded here, in plain English, in the same commit as the change. This covers: readings of the paper's text, conventions, parameter ranges, definition choices (§1–§2), the statements of cited literature results (§3), and places where a proof does not follow the published proof it cites (§3). The user reviews all entries at the end; approvals are tentative until then.

Entry format: **ID** · source wording (quote + location) · Lean choice (name, file) · kind · reason · approval status · could this change the content of a main theorem? (yes/no, why).

Entries written before 2026-10-03 may cite Lean files or declarations that were removed in the dead-code cleanup (D133); they are in git history at commit 481f8aa0.

1. Statement of Theorems 1.1 and 1.2

Entries S1–S7 are proposed by P1-STMT (2026-10-01); Lean code in LQGMetric/Statement/Thm11.lean and Thm12.lean, details in STATEMENT_SPEC.md.

S1 · Thm 1.1: "converge ... to a random metric on $\mathbb{C}$ " (GM l. 230) · theorem11Shape: the limit Y is a random continuous function on $\mathbb{C} \times \mathbb{C}$ ($\mathbb{C}(\mathbb{C} \times \mathbb{C}, \mathbb{R})$) which is a.s. a finite metric on $\mathbb{C}$ (IsMetricFn); it is not required to be a continuous metric (inducing the Euclidean topology) · reading · continuity of the limit is automatic (a local uniform limit in probability of the a.s. continuous LFPP metrics is a.s. continuous), so asking for it costs nothing; "continuous metric" would be a strengthening that GM do not state in Thm 1.1 (it follows from DDDF) · approved by the user 2026-10-01 (D38) · no.

S2 · Thm 1.1: "a random metric on $\mathbb{C}$ which is a.s. determined by h " (GM l. 230), with h a whole-plane GFF plus a bounded continuous function (GM l. 216, the class for which D_h^{ϵ} is defined) · for each probability space and each such h there is a limit Y with $Y = F(h)$ a.s. for a measurable F (AEDeterminedBy, F9) · reading · this is GM's sentence; that one F works for all h is Thm 1.2's content · approved by the user 2026-10-01 (D38) · no.

S3 · "converge in probability ... " as $\epsilon \rightarrow 0$ (GM l. 230, the limit variable is not named) · along the filter $\mathcal{N}[>] \ 0$ of all real $\epsilon \rightarrow 0^+$ (not a sequence), in the sense of TendstoInProbLU (F8) · reading · the natural meaning; it implies convergence along every sequence · approved by the user 2026-10-01 (D38) · no.

S4 · Thm 1.2: "There is a γ -LQG metric D such that the limiting metric of Theorem 1.1 is a.s. equal to D_h whenever h is a whole-plane GFF plus a bounded continuous function" (GM l. 320) · theorem12Shape, existence part: there is a strong γ -LQG metric D such that for every such h , $\alpha_{\epsilon}^{-1} D_h^{\epsilon} \rightarrow D_h$ in probability (local uniform topology) · reading · equivalent given Thm 1.1 (limits in probability are a.s. unique), and stated without referring to Thm 1.1's Y · approved by the user 2026-10-01 (D38) · no.

S5 · Thm 1.2 uniqueness: "there is a deterministic constant $C > 0$ such that if h is a whole-plane GFF plus a continuous function, then a.s. $D_h = C \tilde{D}_h$ " (GM l. 321) · one $C > 0$ chosen before the probability space and h ; " $D_h = C \tilde{D}_h$ " as equality of the functions on $\mathbb{C} \times \mathbb{C}$ (all u, v , one null set) · reading (literal) · approved by the user 2026-10-01 (D38) · no.

S6 · "Fix $\gamma \in (0,2)$ " (GM l. 320; Thm 1.1 inherits it from l. 162) · Theorem11 := $\forall \gamma$, theorem11Shape γ with $0 < \gamma \rightarrow \gamma < 2 \rightarrow$ inside each shape · convention · approved by the user 2026-10-01 (D38) · no.

S7 · α_{ϵ} is defined with the normalized whole-plane GFF (GM l. 223) while Thms 1.1–1.2 use any GFF plus a bounded continuous function · the same in Lean (aEps uses normGFFLaw) · literal, listed so that nobody "fixes" it · approved by the user 2026-10-01 (D38) · no.

2. Definitions

Entries F1–F13 are proposed by P1-FOUND (2026-10-01); see FOUNDATIONS.md for the Lean code.

F1 · "Let $\mathcal{D}'(\mathbb{C})$ be the space of distributions ... equipped with the usual weak topology ... a measurable function $h \mapsto D_h$ from $\mathcal{D}'(\mathbb{C})$ " (GM l. 293–294; σ -algebra not stated) · DistC (mathlib Distribution) with the cylinder σ -algebra of the pairings (DistC.measurableSpace) · reading · it makes every GFF a random variable; for duals of nuclear spaces it is known to agree with the Borel σ -algebra of the weak topology (Fernique 1967; not checked for $\mathcal{D}'$ here) · approved by the user 2026-10-01 (D38) · could change content only if the paper meant a strictly larger σ -algebra (then our uniqueness quantifies over fewer metrics); the user should confirm.

Update 2026-10-01: no longer a deviation. LQGMetric.borel_weakStar_eq (LQGMetric/Field/RandomDistBorel.lean) proves that the cylinder σ -algebra equals the Borel σ -algebra of the weak-*

topology on $\mathcal{D}'(U)$, which is exactly Berestycki–Powell’s definition of a random distribution (arXiv:2404.16642, definitionGFF.tex l. 902–905). **F2** · “there exists a coupling of h with a random continuous function f such that the law of $h - f$ is that of a whole-plane GFF” (GM l. 211) · IsGFFPlusCont: h measurable and $\exists f : \Omega \rightarrow C(\mathbb{C}, \mathbb{R})$ measurable on the same probability space with $h - f$ a whole-plane GFF · reading · how GM’s proofs use f · approved by the user 2026-10-01 (D38) · no: the two classes give the same notion of LQG metric and the same theorems. Sketch: given a coupling elsewhere, disintegrate it along h (condDistrib, $C(\mathbb{C}, \mathbb{R})$ is Polish) to get $\Omega' = \Omega \times C(\mathbb{C}, \mathbb{R})$ carrying f ; a set $B = \{\omega : \neg \Phi(h \omega)\}$ whose preimage in Ω' is contained in a null set is contained in the P -null measurable set $\{\omega : \kappa_\omega(N'_\omega) = 1\}$, so $\forall^m$ statements, outer probabilities and “ $\exists F$ measurable, $Y = F(h)$ a.s.” transfer back to Ω . (Lemma to prove only if someone needs the coupling form.) In IsGFFPlusBddCont, “ f bounded” means each sample is bounded with a possibly random bound ($\forall \omega, \exists M$), as in D3: the literal reading of “random ... f bounded” (audit 2026-10-01, B1).

F3 · whole-plane GFF “defined modulo a global additive constant” (GM l. 214), DS/IG4 normalization (IG4 l. 1082: covariance $-\log|x-y|$) · IsWholePlaneGFF: pairings with mean-zero test functions are a centered Gaussian process with covariance $\int \varphi(x)(-\log|x-y|)\psi(y)$ · definition (equivalent characterization of the law, instead of the series $\sum \alpha_n f_n$) · standard, no completions needed · approved by the user 2026-10-01 (D38) · no.

F4 · circle average $h_r(z)$ (GM l. 223 footnote, DS §3.1: the a.s. defined L^2 -limit) · circleAvg $h \ r \ z := \lim_n$ circle average of $x \mapsto \langle h, \psi_{\{n, x\}} \rangle$ (mollifier at scale 2^{-n}), junk 0 · definition · pathwise and measurable; equals the classical average for continuous functions and (proof obligation) a.s. the DS circle average for GFF + continuous · approved by the user 2026-10-01 (D38) · no.

F5 · “ $h_\varepsilon(z) := (h \ p_{\{\varepsilon^2/2\}})(z)$... interpreted in the sense of distributional pairing” (GM l. 175–177) · heatMollify $\varepsilon \ h \ z := \lim_n \langle h, p_{\{\varepsilon^2/2\}}(z, \cdot) \chi_n \rangle$ with a smooth cutoff χ_n (1 on $B_{\{n+1\}}$), junk 0 · definition (the kernel is not a test function) · agrees with the pairing for tempered h , in particular a.s. for GFF + bounded continuous · approved by the user 2026-10-01 (D38) · no.

F6 · “ a_ε be the median of the D^ε_h -distance between the left and right boundaries of the unit square” (GM l. 223) · aEps: lower median $\inf\{m : P(X \leq m) \geq 1/2\}$ under normGFFLaw · reading · “the median” presupposes uniqueness; the lower median is always a median · approved by the user 2026-10-01 (D38) · no if the median is unique (expected: the law should be atomless with connected support); if not unique, Theorem 1.1 is stated for this choice — the user may want to approve. Proof obligation (audit 2026-10-01, B5): GM l. 592–594 (“the multiplicative constant factor is 1”) silently use that the median of the limit’s left–right distance is unique; the Lean proof supplies it (decision D13 / D-A2; task P2-M1D’s no-gap lemma).

F7 · “the infimum is over all piecewise continuously differentiable paths from z to w ” (GM (1.4), l. 217) · IsPiecewiseC1Path (paths $[0,1] \rightarrow \mathbb{C}$, finitely many C^1 pieces on closed subintervals, as LQGDimension’s IsAdmissiblePath), length as a lintegral in $[0, \infty]$, lfpDist = toReal · convention · avoids junk values of the Bochner integral · approved by the user 2026-10-01 (D38) · no.

F8 · “converge in probability w.r.t. the local uniform topology on $\mathbb{C} \times \mathbb{C}$ ” (GM l. 230) · TendstoInProbLU: for all $R, \delta > 0$, (outer) $P(\sup_{B_R(0)^2} |X_\varepsilon - Y| \geq \delta) \rightarrow 0$ · reading · equivalent to convergence in probability for the standard metric of that topology · approved by the user 2026-10-01 (D38) · no.

F9 · “a.s. determined by h ”, “determined by $h|_U$ ” (GM l. 230, 298) · AEDeterminedBy: $Y = F \circ h$ a.s. with F measurable; locality as $\exists F$ measurable on $\mathcal{D}'(U)$ (cylinder σ -algebra, $h|_U = \text{restrictTo } U \ h$) with $D_h(z, w; U) = F(h|_U)(z, w)$ for all $z, w \in U$ on one null set · reading · standard; equivalent to “measurable w.r.t. the P -completion of $\sigma(h)$ ” (GM_B M2) for standard Borel targets, and the locality form (one null set for all z, w) is the strong reading · approved by the user 2026-10-01 (D38) · no.

F10 · “a metric d on U which induces the Euclidean topology” (GM l. 283) · IsContinuousMetric d for $d : C(\mathbb{C} \times \mathbb{C}, \mathbb{R})$: metric axioms + euclidean_of_small · equivalent formulation · a metric inducing the Euclidean topology is continuous on $\mathbb{C} \times \mathbb{C}$, and conversely · approved by the user 2026-10-01 (D38) · no.

F11 · “We set $d_\gamma := 2/\chi$ ” (DG l. 214), χ from DZZ Thm 1.1 (bounds $C^{-1}\delta^\chi\{-\chi + \iota\} \leq D_{\{\gamma, \delta\}}(u, v) \leq C\delta^\chi\{-\chi - \iota\}$); GM cite d_γ via DG (GM l. 202–206) · chiDZZ $\gamma := \text{the } \chi > 0 \text{ with } \log D_{\{\gamma, \delta\}}(u, v) / \log \delta^{-1} \rightarrow \chi \text{ a.s. for every zero-boundary GFF on the open unit square (QZ IsZeroBoundaryGFF0n, measure QZ qAreaMeasure0n), } D_{\{\gamma, \delta\}} = \text{DZZ’s LGD (open balls, rational centres, mass } \leq \delta^2, \text{ paths unrestricted); junk } 1/2 \text{ if none; } d\text{Gamma} = 2/\text{chiDZZ} \cdot \text{definition} \cdot \text{DZZ’s convention because DZZ’s proof is the one formalized (P1-INV-DIM §7); the limit form is equivalent to DZZ’s bounds} \cdot \text{approved by the user 2026-10-01 (D38)} \cdot \text{no once DZZ Thm 1.1 is proved; junk } 1/2 \text{ keeps the targets non-degenerate before that. Conventions (audit 2026-10-01, B2): DZZ later use balls with rational centre and radius (DZZ l. 2276) and DG restrict paths to the closed square (DG l. 222); neither changes } \chi \text{ (radii shrink to rationals by a Lebesgue-number argument; a path from } u \text{ to } v \text{ must cross a fixed annulus around } u \text{ inside the square). The Lean definition follows DZZ l. 123. Uniqueness (B3): QZ’s K3.exists_zeroGFF0n openSquare gives IsZeroBoundaryGFF0n openSquare on an } \Omega : \text{Type, so IsLGDExponent } \gamma \ \chi \text{ holds for at most one } \chi \text{ (LQGMetric.isLGDExponent_unique, LQGMetric/Dimension/ChiUnique.lean, proved).}$

F12 · Axiom V: "there is a deterministic constant c_r ... the closure of this set of laws w.r.t. the Prokhorov topology is contained in the set of laws on continuous metrics" (GM l. 440–443) · `tightnessAcrossScales`: one function `c` chosen before the probability space and the GFF; closure in `mathlib`'s `ProbabilityMeasure` topology on $C(C \times C, \mathbb{R})$; "supported on continuous metrics" as $\forall^m d \partial \mu, \text{IsContinuousMetric } d \cdot \text{reading} \cdot$ the law of $h \bmod$ constants is unique · approved by the user 2026-10-01 (D38) · no.

F13 · probability spaces unrestricted (GM) · $\forall \{ \Omega : \text{Type} \}$ (universe 0) · convention · universe-polymorphic quantification inside a `Prop` structure is awkward; every law on a standard Borel space is realized on a `Type` · approved by the user 2026-10-01 (D38) · no.

Entries below (2026-10-01, P2-MERGE) collect definition-level proposals from `decisions/DEC-*.md`, the blueprint inventories and finished proof tasks.

DIM-3 · d_γ in DG's convention ($\text{mass} \leq \varepsilon$, paths in the closed square) (GM/DG) · `dGamma` via DZZ's Liouville graph distance ($\text{mass} \leq \delta^2$, rational centers, paths unrestricted, a.s. limit form; D9) · convention · DZZ's form is the one proved · proposed · no: equivalence of exponents ($\varepsilon = \delta^2$) is proved as a lemma.

D-DDDF-1 · ψ_δ printed without $\sqrt{\pi}$ (DDDF T:355) · ψ_δ carries $\sqrt{\pi}$ like φ_δ · reading · (2.23) needs it · proposed · no.

D-DDDF-2 · (2.20) upper limit $2^{\{-2K+2\}}$ · $\psi_{\{K,n,P\}}$ integrates $t \in [2^{\{-2n\}}, 2^{\{-2K\}}]$ · reading · printed range double counts · proposed · no.

D-DDDF-3 · $\sigma_t := r_0 \sqrt{t} |\log t|^{\{\varepsilon_0\}}$ · $\sigma_t := r_0 \sqrt{t} (1 + |\log t|)^{\{\varepsilon_0\}}$ · definition of an auxiliary object · printed form vanishes at $t = 1$, making " $\sup_t \sqrt{t}/\sigma_t < \infty$ " false · proposed · no.

D-DDDF-4 · " ℓ with $P(L \leq \ell) = p$ " (DDDF DD:67) · generalized quantiles `IsQuantile` (`Prob/Quantile.lean`; D22) · convention · avoids an unproved atomlessness claim · proposed · no: downstream estimates hold for any quantile.

D-DDDF-10 · §6.1 η_t, φ_t printed without $\sqrt{\pi}$ · $\sqrt{\pi}$ inserted in both · reading · consistency with §2 · proposed · no.

P2-MEDIAN · medians / p-quantiles (GM α_ε , DDDF ℓ, ℓ) · `IsQuantile`, lower median (`Prob/{Quantile,QuantileMap,QuantileLimit,Median}.lean`) · convention + own elementary proof (right-continuity of the CDF; Portmanteau on closed half-lines, Billingsley Thm 2.1 via `mathlib`) · realizes D-DDDF-4 and the S8/S1.17 median-limit step · proposed · no.

P2-MG2-b · CONF: geodesic = "path of minimal length" · `IsGeodesicPath`: length = distance of endpoints (`Metric/Geodesic.lean`) · convention · equivalent in length spaces (`isGeodesicPath_iff_forall_le`) · proposed · no.

P2-GCONC-d · "centered Gaussian vector" (DZZ L2.1, DFGPS L4.2) · $X_i = \langle v_i, Z \rangle$ with Z standard Gaussian (`Gaussian/Concentration*.lean`) · convention (DZZ's own $X = AZ$ step) · proposed · no.

P2-PERC-b · finite-range dependence of box states (DG L3.11, DDLGD P4.2, DDDF P18) · product inequality for finite families of boxes pairwise at ℓ_∞ -distance $> r$ (`Perc/*.lean`) · definition · weaker than independence, so the theorem is stronger · proposed · no.

P2-FCM-a · Cameron–Martin for the whole-plane GFF, shift by $(h, \varphi)_\nabla$ · stated for the law of the mean-zero pairings on $\text{TestC0} \rightarrow \mathbb{R}$; φ a test function, $(h, \varphi)_\nabla := \langle h, -\Delta \varphi / (2\pi) \rangle$ (`Field/{Green,CameronMartin}.lean`) · convention · that is all `IsWholePlaneGFF` determines · proposed · no.

3. Cited literature results and proof routes

Entries below (2026-10-01, P2-MERGE). IDs are the proposers'; GA-n = blueprint/GM_A.md §8 item n, CF-n = blueprint/Confluence.md §7 item n, DIM-n = blueprint/Dimension.md §7.7 item n.

3.1 From decisions/DEC-A.md and DEC-B.md **DA1** · "Ang Thm 1.9 ... $1 - \xi_Q \geq 0$, which implies $\xi_Q - 1 - \xi^2/2 < 0$ " (GM U:1056; DFGPS T:2453) · GM.S2.5 from DG eqn-exponent-mono, DG Thm 1.5 at small γ' and `LQGDimension` `Blueprint.LambdaAsymptotics`; 6-line combination own (D12) · proof route of a cited result (Ang Thm 1.9 not formalized) · Ang needs a DGFF layer, Ding–Li, Lawler, Makarov · proposed · no: only the proof of an intermediate inequality changes. (= DIM-1.)

DA2 · DZZ Thm 1.1's bound $\chi \leq 4[(1+\gamma^2/4) - \sqrt{(1+\gamma^4/16)}]/\gamma^2$ (DZZ:142–146) · not formalized; only $\chi \leq 2$ by a first-moment covering argument (DIM.S-chi-le-2) · statement of a cited result (weaker part used) · only " χ bounded" is needed · proposed · no: d_γ uses only the existence part (F11).

DA3 · GM apply DFGPS Thm 1.2 (internal-crossing median, T:285) as if normalized by GM's α_ε (U:586–590) · keep α_ε (D7) and add GM.S1.16' (ratio bounded, limits are multiples of weak metrics) (D13) · own argument filling a silent identification · keeps Thm 1.1 as stated · proposed · no. (= GA-1.)

DA4 · "Therefore, the multiplicative constant factor is 1" (U:590) needs a unique median · own lemma GM.S1.18' (no

gap in the law of $D_h(L,R)$ by Cameron–Martin + Weyl + IVT, adapting Gwynne 1909.08588 L2.5 · own argument filling a gap (same gap in DFGPS L2.14) · no source · proposed · no. (= GA-2.)

DA5 · ”This axiom implies the tightness of various functionals of D_h ...” (U:449–452; T:370–374; U:1434, 3605) · own proofs S2.4a–e (D14); U connected and bounded in S2.4c; S2.4e also uses DFGPS T1.5 and LM L3.1 · own argument + corrected attribution · no source; S2.4e does not follow from Axiom V alone · proposed · no: intermediate lemmas, every use has U a bounded domain. (= GA-3.)

DA6 · ”This tail σ -algebra is trivial ...” (U:1205) · replaced by GM L2.7 on n translates $E(z)$ (D15) · proof route · tail triviality has only a sketched source (HS L2.2) · proposed · no.

DA6' · ”since $\sigma(\bigcap_{r>0} h|_{\mathbb{C} \setminus B_r(0)})$ is trivial ...” (DFGPS T:1734) · replaced by DFGPS L3.3 (= LM L3.1) along radii 8^j · proof route · as DA6 · proposed · no.

DV-B11 · GM L4.13–L4.14, (4.38) (tex:2050–2105): prime ends, prime-end closure, L4.14 for every compact connected $\mathcal{K}$ · L4.14 assumes $\partial\mathcal{K}$ is a Jordan curve; $Cl'(X)$:= closure; arcs := images of circle arcs (D17) · reading / hypothesis added · GM uses L4.14 only for $\mathcal{B}^\bullet_{t_k}$ (Jordan boundary, GPS L2.4; Carathéodory) · proposed · no: same content in every use. (Supersedes GM_B’s DV-B11.)

DV-B12 · GM (4.6) (tex:1645) s_k, t_k with unit $c_{re}^{\{h_r(z)\}}$, used as stopping times · unit $\tau_{\ell r}$; $K := [(a/c_2)\varepsilon^{\{-\beta\}} - 1]$; $c' := c/c_1$ in F_k (4.40); Thm 2.16 with $\tau := s_k \wedge \tau_{2\ell r}$; $a \rightarrow \min\{a, \ell\}$ (D16) · fix of a gap · GM’s s_k, t_k are not stopping times · proposed · no: internal to the proof of GM Thm 4.2.

DV-B13 · MS Prop 2.1, injectivity paragraph (mapmaking_final.tex:610), a loop-separation argument · injectivity via QZ is $ClosedEmbedding_bdryMap$, hypothesis proved from QZ janiszewski · own argument · Jordan curve theorem not in mathlib/QZ · proposed · no.

DV-CONF-J · CONF uses without proof that $\partial\mathcal{B}^\bullet_s$ is a Jordan curve (confluence-final.tex:533, 560, 572, 627, 1741) · proved as GM.S-Jordan following GPS L2.4 / MS Prop 2.1 · implicit fact supplied from a later paper · proposed · no. (= CF-4.)

DEV-LM-g · LM Thm 1.7 Step 3 (local-metrics-final.tex:1075–1084): fixed compact range of P · variance argument for $D(z,w;\bar{V})$ with ε^2 -near-minimal paths; (5.18) by LM.S5.e (Arzelà–Ascoli + loop removal) (D18) · fix of a gap · $P = P_\varepsilon$ depends on ε · proposed · no: Thm 1.7, Cor 1.8 as stated.

D-DDDF-14 · DDDF L12 (tightness.tex:744–757), DF Thm 3.1 Step 1: conformal $F: E_p \rightarrow E'$ mapping marked arcs · Lemma 12' with explicit F_{ij} = affine $\circ G \circ M_{ij} \circ G^{-1} \circ$ affine, only inclusions required (D19) · own argument (auxiliary lemma changed) · no modulus monotonicity / reflection in mathlib/QZ · proposed · no: Props 10, 14 unchanged.

D-DDDF-11 · DDDF P29 (tex:1501) for any bounded D ; (6.96) via Brownian bridges · P29 only for $D = (-1,2)^2$, (6.96) from the explicit image-series Dirichlet heat kernel (D19) · statement of a cited result restricted + own route · DFGPS uses only this D (tex:883) · proposed · no.

D-DDDF-13 · DDDF Thm 1(2)/P29 cover $t \in (0,1/2)$, DFGPS needs $\varepsilon \in (0,1)$ · $\varepsilon \in [1/\sqrt{2},1)$ handled directly (uniformly bi-Lipschitz metrics, Arzelà–Ascoli) (D19) · own argument for an omitted parameter range · proposed · no.

3.2 From the blueprint inventories **GA-4** · GM footnotes U:1219, U:1263 (S3.2, S3.3): ” $|z-w| = \beta r$ ”, ” $v \in \partial B_r \subset B_r$ ” · corrected; $\beta = (1-\alpha)/2$ instead of $1-\alpha$ · corrected detail · proposed · no: results keep the form $\exists\beta$.

GA-5 · P3.6 $\Rightarrow$ P3.4 counting (S3.5): $\#\mathcal{K}^\varepsilon \neq \lfloor \nu \log_\varepsilon \varepsilon^{-1} \rfloor$ · P3.6 proved under ” $(\nu-\mu)/2 \cdot \log_\varepsilon \varepsilon^{-1}$ radii” · strengthened intermediate statement · proposed · no.

GA-6 · GM L3.8 interval $[\varepsilon^{\{1+\mu\}r}, \varepsilon r]$ · read as $[\varepsilon^{\{1+\nu\}r}, \varepsilon r]$, plus proof details (i)–(iii) · typo reading + corrected detail · proposed · no.

GA-7 · GM L2.10 used with L a whole circle (U:1071) · cite DFGPS P4.1, which includes circles · statement of a cited result · proposed · no.

GA-8 · GM L2.7’s RN estimate · derived from MQ L4.1’s proof (MQ compare to a zero-boundary GFF, GM to the marginal law) · statement of a cited result (to confirm, D23) · proposed · no.

DV-B1 · GM Thm 4.2 proved only for $\lambda_3 = 1$ (”general case identical”) · proved for general λ_3 (§5 uses $\lambda_3 = 3$) · own completion · proposed · no.

DV-B2 · GM L4.11 applies GM L2.6 (= LM L3.1) whose ratio hypothesis is not implied · use every m -th radius · fix of a gap · proposed · no.

DV-B3 · GM P4.3 radii with ratio $8 < \lambda_4/\lambda_1 = 16$ · keep every other radius, apply Thm 4.2 with $\mu/2$ · fix of a gap · proposed · no: §6 unaffected.

DV-B4 · GM Remark 4.10 ($\mathcal{E}_r \subset \mathcal{E}^{\{ \ell r \}}(a)$) not literally true · add $\bigcap_{z \in rU} \mathcal{E}^{\{ \ell r \}}(a)$ to $\mathcal{E}_r$ and prove its probability $\rightarrow 1$ (as GM announce, tex:1161) · fix of a gap · proposed · no.

DV-B5 · GM's unproved claim $1 - \xi Q \leq 2\xi$ (tex:2550) · choose $\chi' > \max\{\xi(Q+2), 1\}$ · proof route avoiding an unproved claim · proposed · no.

DV-B6 · E_r condition 10 and Λ (GM §5) · read with $|(h, \varphi)_\nabla|$, $\Lambda := \exp(\Lambda_0 + 2 \max_{\{\varphi \in \mathcal{G}_r\}} (\varphi, \varphi)_\nabla)$, so that (5.9) holds · reading / fix · GM's (B) $\Rightarrow$ (5.5) for $h - \varphi$ is not literal · proposed · no.

DV-B7 · GM L4.18, L4.15 Step 4: binomial domination + Hoeffding · Azuma–Hoeffding (mathlib) · proof route · proposed · no.

DV-B8 · GM's "Proposition 3.4" (tex:2669, 2851), "Proposition 3.2" (2681) · read as Proposition 3.5 · reading · proposed · no.

DV-B9 · GM L2.13 cited as "[CONF, Lemma 3.4]" · CONF Lemma 3.5 (arXiv-final), translated by Axiom IV · statement of a cited result (numbering) · proposed · no. (= CF-1 part.)

DV-B10 · GM P4.3 conclusion $P([s, t]) \subset B_{\{\lambda_{2r}\}}(z)$ · stated as proved: $P(s), P(t) \in B_{\{3r/2\}}(z)$, $P([s, t]) \subset \bar{B}_{\{3r\}}(z)$ · statement of an intermediate result changed · §5 does not prove GM's form; §6 does not use it · proposed · no.

DEV-DFGPS-1 · DFGPS L2.2 constant $(2+\zeta)$ · unspecified deterministic A · parameter change of an internal lemma · only T:733 uses it · proposed · no.

DEV-DFGPS-2 · DFGPS L2.3 via Borell–TIS (AT Thm 2.1.1) · Maurey–Pisier (LQGDimension) · proof route · only a sub-Gaussian tail is used · proposed · no.

DEV-DFGPS-3 · misprints (T:1448–1450, 2371, 2375, 697, 874, 1031, 1686, 1566, 1803, 1818) · read as intended (list in blueprint/DFGPS.md §7) · reading · proposed · no.

DEV-DFGPS-4 · DFGPS L3.2 applies LM L3.1 to annuli $\mathbb{A}_{\{r/2, 2r\}}$ normalized by $h_{\{3r\}}(z)$ · $r_k := 3 \cdot$ (every third dyadic radius), $s_1 = 1/6$, s_2 slightly above $2/3$ · own bookkeeping · proposed · no.

DEV-DFGPS-5 · DFGPS L2.8 cites only DDDF Thm 1 and §6.1 · uses DDDF Thm 1(2) + Prop 28 + P29 + BBI Ex 2.4.19 · statement of a cited result (to confirm, D23) · proposed · no.

DEV-DFGPS-6 · DF Lemma 7.1 "the same proof" · used in adapted form (local uniform convergence on $\mathbb{C}$, GM's $e^{\{\xi\}} \cdot D$) · statement of a cited result · proposed · no.

DEV-DFGPS-7 · DFGPS P4.1 (:2531) uses $\xi < 2/d_{\gamma} < 1$ via DG Prop 1.7 · $\xi = \gamma/d_{\gamma} < \gamma/2 < 1$ from $d_{\gamma} > 2$ · proof route · proposed · no. (= DIM-4.)

DEV-DFGPS-8 · DFGPS S6, S8, S9, S11 and the L3.20 inclusion are not proved in the paper · own arguments (S6 = D14) · own argument · proposed · no.

D-DDDF-5 · "the uppermost geodesic" · measurable $(1+\eta)$ -near-geodesics, $\eta \rightarrow 0$ at the end · proof route · proposed · no.

D-DDDF-6 · DDDF Prop 10's l' · includes the factor $C l$ (as in DF Props 4.5–4.6) · reading · proposed · no.

D-DDDF-7 · positive association in the limit · own small argument without positive densities · own argument · proposed · no.

D-DDDF-8 · (3.37) $3\epsilon^C \cdot 3\epsilon^{1/C}$ (as in the proof and DF Thm 3.1) · reading · proposed · no.

D-DDDF-9 · (4.49), (4.50) · read $C e^{-c s^2/\log s}$ · reading · proposed · no.

D-DDDF-12 · DFGPS:1097's bound on $c_{\{\delta r\}}/c_r$ · uses DDDF (6.99) + a polynomial bound on λ_{δ} , not (1.3) alone · statement of a cited result (to confirm, D23) · proposed · no.

DEV-LM-a · LM L1.4 proof (tex:262) $h|_{\{U \setminus \bar{V}\}} \cdot h|_{\{U \setminus V\}}$ · reading (typo) · proposed · no.

DEV-LM-b · LM L2.2(1) via QLE Lemma 6.8 (GFF on $\mathbb{D}$, Hausdorff limits) · monotone-limit argument S2.2a, needs compact closed balls · proof route · proposed · no. (= CF-5.)

DEV-LM-c · LM L3.3 "the same proof works" (MQ L4.1) · passage to the marginal law (S3.3a) by an own short argument · own argument (to confirm, D23) · proposed · no.

DEV-LM-d · LM L3.1 binomial domination by coupling · iterated conditioning + Chernoff · proof route · proposed · no.

DEV-LM-e · LM L4.1, L4.2 · re-indexing into three residue classes (S4.1a); enlarge K (S4.2a) · own bookkeeping · proposed · no.

DEV-LM-f · LM Thm 1.6 proof · selection-free route S4.6a ($K = \bar{B}_R(0)$, event G_R) · proof route · proposed · no.

DEV-LM-h · CONF:1187 "LM Lemma 2.2"; GMSH L2.1 typos · read as LM Lemma 2.1 (= GMSH L2.2) · reading · proposed · no. (= CF-1 part.)

DEV-LM-i · bibliographies give math/0605337 for [SS13] · cited paper is arXiv:1008.2447 (D26) · reading · proposed · no. (= CF-3.)

CF-1 · GM's "[CONF, Lemma 3.4]" (GM:1097), "[CONF, Eq. (2.19)]" (GM:2052) · arXiv-final CONF Lemma 3.5 and (2.18) · statement of a cited result (numbering) · proposed · no.

CF-2 · CONF C:1488 $e^{\{-\xi_{h_r}(0)\}}$; C:1128 $\text{dist}(z, \partial U)$; C:381 "from 0"; C:472 local-set definition · read as $e^{\{\xi_{h_r}(0)\}}$, $\text{dist}(u, \partial U)$, "from z ", SS13 form (SS:2041) · reading · proposed · no.

DIM-2 · DZZ Thm 1.1 "amalgamation", no proof written · assembled from DZZ P5.1 + P3.17 by Borel–Cantelli along dyadic δ and monotonicity · own glue argument · proposed · no.

DIM-5 · DG Thm 1.1 $\neq$ DZZ Thm 1.1 literally (conventions) · conversion by an own elementary lemma · reading · proposed · no.

3.3 From finished proof tasks P1-MG · GM's partition length; path concatenation/reversal; shortest paths (BBI) · equality with eVariationOn and the length identities are own elementary proofs (< 100 lines each); shortest paths via BBI Prop 2.5.19, Thm 2.5.14 (Arzelà–Ascoli), not Petrunin's midpoint proof (Metric/{CurveLength, LengthSpace, CurveNatural, LengthGeodesic}.lean) · own elementary proof / proof route · proposed · no.

P2-MG2-a · Petrunin, Menger's lemma: "extend from dyadics by completeness" · $P_t := \lim g_n \lfloor t \cdot 2^n \rfloor$ (Metric/Midpoint.lean) · proof route (same mathematics) · proposed · no.

P2-MG2-c · sub-path, additivity, concatenation, set-distance facts used without comment · own elementary proofs (Metric/{LengthLimit, Geodesic, SetDist}.lean) · own elementary proof · proposed · no.

P2-EFRON · Efron–Stein (BLM), applied to a conditional law in LM tex:1017 · abstract probability measure (conditional application left to the consumer); countable version assumes summability (else trivial); resampling forms assume (X_i, X'_i) jointly independent (Prob/EfronStein*.lean) · statement of a cited result · BLM's standard setting · proposed · no.

P2-GCONC-a · Adler–Taylor Lemma 2.1.4 (characters + unwritten approximation) · integration along a rotation ($\alpha = \sin \theta$) with Gaussian integration by parts · proof route · proposed · no.

P2-GCONC-b · Adler–Taylor Lemma 2.1.6 smoothing · two stages (mollification, then truncation + Fatou; C^1 suffices) · proof route · proposed · no.

P2-GCONC-c · DZZ L2.1 via Borell–Sudakov–Tsirelson isoperimetry · derived from Lipschitz concentration (both tails; $E d \leq \sigma \sqrt{(2 \log(1/c))}$, $C = \max 1 \sqrt{(2 \log(1/c))}$); also gives DFGPS L4.2's prefactor-free form (Gaussian/Concentration*.lean) · proof route · sharp constant kept (D20) · proposed · no.

P2-SKOROKHOD · Billingsley Thm 6.7 · variants: null-boundary balls around a dense sequence, radii in $(\varepsilon/4, \varepsilon/2)$; $P_n(\cdot|B) := P_n$ when $P_n(B) = 0$; regime index (largest m with $n_m \leq n$); extra a.s. event in the "same cell" step; tightness criterion with Hölder constants (own ~ 100 -line argument from mathlib's Arzelà–Ascoli + union bound) (Prob/{PolishContinuousMap, TightHolder, Skorokhod*}.lean) · proof route / own elementary proof · proposed · no.

P2-PERC-a · planar duality (cited without proof by DG L3.11, DDLGD P4.2, DDDF P18) · $K \times L$ box rectangle, edge-connected good / diagonally-connected bad, own write-up of the exploration-path proof of the Hex theorem (Gale 1979; Kesten 1982 §2.4) on the square lattice (Perc/*lean) · own elementary proof · proposed · no.

P2-FCM-b · GFF Cameron–Martin via eigenfunction expansion (Berestycki–Powell Prop lem:CMGFF) · QZ's abstract Gaussian-process Cameron–Martin theorem + a Green identity (Field/{Green, CameronMartin}.lean) · proof route · proposed · no.

P2-FINV · change of variables in $\log \text{Cov}$; "normalized field = measurable function of the field mod constants" (used implicitly) · own elementary arguments (recentering by a mean-one ρ , $\hat{N} g = \langle g, \rho \rangle + N(g - \langle g, \rho \rangle)$) (Field/{GFFInvariance, GFFLaw}.lean) · own elementary proof · proposed · no.

P2-PITT · Pitt (Ann. Probab. 10 (1982), main theorem) and Esary–Proschan–Walkup 1967 · $\text{LQGMetric.Pitt.integral_mul_le_of_monotone}$ and the approximation chain in $\text{LQGMetric/Gaussian/{PittSmooth, PittLipschitz, AssociationSets, AssociationEvents, AssociationFun}.lean}$ · own elementary arrangement of the standard approximation steps: smooth from Lipschitz by bump convolution; closed upper sets from Lipschitz functions via $\max(0, 1 - k \cdot \text{dist}(\cdot, U))$; measurable upper sets via compact inner regularity and the closed upper set $K + [0, \infty)^n$; bounded measurable monotone functions via the layers $\lfloor n f \rfloor / n$ (EPW's reduction); the core inequality by the standard rotation interpolation $\varphi(\theta) = E[F(x) G(\cos \theta x + \sin \theta y)]$ (Pitt's paper itself could not be downloaded) · kind: proof route · reason: Pitt's paper unavailable; the published approximation is "standard" and unwritten · proposed · no (the proved statement is Pitt's theorem in full generality: any mean, degenerate covariance, bounded measurable monotone functions).

P2-FMEAS · FOUNDATIONS §9 items 3–4 · $\text{LQGMetric/Field/{Measurable, MeasurableAvg, StandardBorelMetric, StandardBorelMetric}.lean}$ · measurability of the field operations, Borel-ness of the set of continuous metrics (characterised via uniform separation on balls $\bar{B}_R(0)$ and a compactness argument), second countability of $\mathcal{D}_K$, countable generation of the

cylinder σ -algebra: standard facts proved by own elementary arguments, no specific published proof followed · kind: own elementary proof · proposed · no.

P2-FHEAT · GM footnote (uniqueness-final.tex l. 212: boundedness of f makes h^*_ε finite; no proof given) · LQGMetric/Field/HeatMollify{Cont,Var,Point}.lean: IsWholePlaneGFF.ae_tendsto_heatMollify, IsGFFPlusBddCont.ae_tendsto_heatMollify · own elementary argument where no published proof exists: test functions split into a mean-zero part plus a multiple of a fixed bump ρ ; truncation differences $\leq K e^{-n}$, supported in B_{n+3} ; variance bounded with $|\log t| \leq t + 1_{\{t \leq 1\}} |\log t|$; summable first moments · kind: own elementary proof · proposed · no.

DV-BINOM-1 · LM S3.1a (local-metrics-final.tex l. 744–749), GM L4.18, GM S4.10 (“stochastically dominates a binomial”) · LQGMetric/Prob/BinomialDomination.lean (measureReal_binomDom_*) · no coupling is formalized; the tail bounds the papers deduce from the domination are proved directly by iterated conditioning (exponential supermartingale) · kind: proof route (own elementary argument, as the LM inventory suggested) · proposed · no.

DV-BINOM-2 · GM Lemma 4.18 (uniqueness-final.tex l. 2284–2318, eqn-uncond-to-cond) is **false as printed** (off-by-one: $K = 1$, $m = 1$, $\alpha = 0.99$, $\delta = 0.98$, independent events of probability 0.98 give probability $0.9604 > e^{-1.92}$) · two corrected forms proved in LQGMetric/Prob/BinomialGM.lean: gm_uncond_to_cond (bound $e^{-2\delta^2 m}$, thresholds with $n = K+1$ indices) and gm_uncond_to_cond_K (the paper’s thresholds, bound $\exp(-2(\delta m - \alpha)^2/(m+1))$ for $\alpha \leq \delta m$) · kind: statement of an internal lemma (correction) · reason: the printed statement is false · proposed (decision D29) · no: the only use, GM L4.21 ($\alpha = 1/2$, $\delta = 1/4$, $m = \lfloor 4\varepsilon^{\theta K} \rfloor$), loses only a constant factor in the exponent with either form.

DV-BINOM-3 · GM_B inventory DV-B7 (use mathlib’s Azuma–Hoeffding for GM L4.18) · superseded: mathlib’s version needs one variance parameter per index and gives $\exp(-2\delta^2 m^2/(K+1))$ instead of $e^{-2\delta^2 m}$; the exponential-supermartingale argument above is used instead (mathlib’s Hoeffding lemma enters through bernoulli_hoeffding) · kind: proof route · proposed · no.

DV-C1 · GM, CONF, MQ, DFGPS throughout (e.g. U:1340–1360 “ $E_r(z) \in \sigma(\dots)$ ”, GM_B M2–M6, C:1546, MQ:482–486) never discuss measurability of events quantified over continua or over all geodesics, and write “ $\in \sigma(X)$ ” / “determined by X ” · Lean: such events are universally measurable sets of fields (Lusin), “ $\in \sigma(X)$ ” is read as “a.s. equal to an event of $\sigma(X)$ ” (AEEventDeterminedBy), σ -algebra identities hold modulo null sets (AESigmaLE), and “ $P[E] \geq 1-\varepsilon$ ” is stated as “ $P[E^c] \leq \varepsilon$ ” · convention (measurability reading of the text) · the papers’ events are in general analytic or coanalytic, not Borel; every probability computed in the papers is unchanged mod null sets · proposed · could it change a main theorem? no: Theorems 1.1/1.2 are stated with $\forall^m$, AEDeterminedBy and outer measures (FOUNDATIONS §4, §7), which this convention does not touch.

DV-C2 · GM U:648 (“the D_h -geodesic”), GM_B M1/M4, CONF C:505–566 and DDDF T:950 use “the geodesic” and other choices as random objects without discussing measurability · Lean: events about the a.s.-unique geodesic are $\exists/\forall$ -events over all geodesics; when a random variable is needed, the geodesic is a measurable Lusin–Souslin selector (valid a.s. for fields whose law is absolutely continuous to the GFF’s modulo constants); other choices are first elements of fixed countable families · convention (measurability reading) · the papers’ objects are only defined a.s. and their measurability is implicit · proposed · could it change a main theorem? no: these objects appear only inside proofs; the selector equals the paper’s geodesic a.s.

DV-C3a · GM U:852, CONF C:471–474, LM tex:164/499 define local sets for arbitrary couplings (conditional independence), SS L3.9 for a GFF on a simply connected domain · Lean: local sets only for sets that are measurable functions of the field, defined by “ $\{A \subseteq U\}$ is a.s. determined by $h|_U$ for every open U ” on the field space · definition change (restricted generality) · every local set used on the path is a function of h ; GM’s own second sentence states this equivalent form · proposed · could it change a main theorem? no: only lemmas about h -determined sets are needed.

DV-C3b · SS l. 1999–2003 / CONF C:473: $\sigma(A, h|_A) := \bigcap \delta \sigma(A, h|_{A_\delta})$ with δ -neighbourhoods · Lean: $\bigcap_n$ over WP’s dyadic hulls (squares meeting the closed 2^{-n} -neighbourhood), Borel sets modulo null sets · definition change (equivalent by the sandwich $B_{2^{-n}}(A) \subseteq \text{int } A^{(n)} \subseteq B_{3 \cdot 2^{-n}}(A)$) · avoids restricting a distribution to a random open set · proposed · no.

DV-C3c · SS L3.9(3) / WP Prop 4.11 (conditional law given $\mathcal{F}_A$ = harmonic part + zero-boundary GFF on $\mathbb{C} \setminus A$), used by CONF L3.6, L3.10, T3.9 and GM L4.15 · Lean: the strong Markov property is proved and used in WP’s dyadic form (independence on each $\{A^{(n)} = a\}$ from functionals independent of $h|_{\{\text{int } a\}}$) together with the backward martingale limit to $\mathcal{F}_A$ · statement of a cited result (weaker form; the full form needs Green functions of random domains, absent everywhere) · consumers argue through countably many deterministic domains, as CONF’s own proof does (C:1424–1447) · proposed · no (if a consumer needs the full form, it is escalated, not assumed). (QLE L6.8 $\rightarrow$ S2.2a is DEV-LM-b, confirmed.)

DV-C4a · MQ:502–506: “We will further condition on the sets $\bar{B}_h(x,r)$ and $\bar{B}_h(y,u)$ (which are local) ... the conditional law of $h + A\varphi$ in U_i given the values of h outside of U_i is mutually absolutely continuous ...” · Lean: φ is

taken from a fixed countable family of bumps (MQ already choose φ as a deterministic function of the balls, MQ:504); for each fixed φ the unconditional Cameron–Martin theorem and Fubini over A replace the conditioning on local sets; admissibility of φ is part of the event $\cdot$ place where we do not follow the published proof $\cdot$ the conditional step needs the conditional law given random local sets and SS L3.10, which none of our consumers need otherwise; the repair keeps MQ’s argument and constants $\cdot$ proposed $\cdot$ could it change a main theorem? no: MQ Thm 1.2 is proved as stated (for weak metrics).

DV-C4b $\cdot$ MQ:486 (“since D_h induces the Euclidean topology, the probability of (i) tends to 1”) and MQ:504 (geodesics between points of $B(x_\ell, \delta)$ “contained in $B(x_\ell, \xi)$ ”) $\cdot$ Lean: (i) is proved from bounded compactness of (C, D_h) (CONF.S-bcpt = DFGPS.L3.8; a continuous metric alone does not give it), and the containment radius is $\xi + \delta$ (support of φ still in $B(x_i, 2\xi)$, $\delta < \xi$) $\cdot$ reading / fix of an imprecision $\cdot$ proposed $\cdot$ no.

P2-PERC-c $\cdot$ exclusivity of rectangle crossings (used implicitly by DDDF, DDLGD and DG when gluing crossings) $\cdot$ LQGMetric.perc_not_goodLR_and_badTB, percGoodLR_meets_percBadTB (LQGMetric/Perc/Exclusive*.lean) $\cdot$ own lattice crossing-parity argument (discrete Jordan-curve parity, after Courant–Robbins ch. V) $\cdot$ kind: own elementary proof $\cdot$ proposed $\cdot$ no.

P2-PERC-d $\cdot$ DZZ Lemma 3.7 (LBM_LGDarXiv.tex l. 1009–1016) derives a good enclosure of an annulus from an annulus duality $\cdot$ LQGMetric.perc_annulus_enclosure, perc_annulus_peierls (LQGMetric/Perc/Annulus*.lean) $\cdot$ the enclosure is built from long-way crossings of the four side rectangles (as in Ding–Dunlap Prop 4.2 / DDDF Prop 18 step 1), and the failure probability is bounded by four rectangle Peierls bounds: $4(2N+1)(8\theta)^{N-n+1}$ instead of $9K(8p^{1/(2\kappa+1)})^K$ (same form) $\cdot$ kind: proof route $\cdot$ proposed $\cdot$ no.

P2-CI-a $\cdot$ GM Lemma 4.9 (uniqueness-final.tex l. 1839–1880, ratio formula for $P[H \cap E \mid \mathcal{G} \vee \sigma(E)]$) $\cdot$ LQGMetric.condExp_inter_le_of_sup_singleton (LQGMetric/Prob/CondIndepDetermined.lean) $\cdot$ tested directly on $F' \in \mathcal{F}$ using $F' \cap E = g \cap E$ a.s. with $g \in \mathcal{G}$; the hypothesis is used up to a null set (blueprint/GM_B.md l. 129) $\cdot$ kind: proof route (own elementary argument) $\cdot$ proposed $\cdot$ no.

P2-CI-b $\cdot$ conditional independence (SS13 L3.5, LM l. 259–273) $\cdot$ LQGMetric.CondIndepEv (LQGMetric/Prob/CondIndepEv.lean), event form $P[a \cap b \mid G] = P[a \mid G]P[b \mid G]$ a.s., so that the results hold without StandardBorelSpace Ω ; wrappers to mathlib’s CondIndep (standard Borel Ω) and QZ’s CondIndepCE are proved $\cdot$ kind: convention $\cdot$ proposed $\cdot$ no.

P2-CI-c $\cdot$ Markov form of conditional independence and its converse, the π – λ extension SS leave implicit, the bridge to AEDeterminedBy, the $p = p^2$ argument in LM S5.d $\cdot$ LQGMetric/Prob/CondIndep*.lean $\cdot$ own elementary proofs of standard facts $\cdot$ proposed $\cdot$ no.

P2-CI-d $\cdot$ LM S5.d (two-copies criterion, local-metrics-final.tex l. 956–986) assumes the two copies have the same conditional law $\cdot$ aeDeterminedSigma_of_condIndepEv_of_ae_eq drops that hypothesis (it follows from $X = Y$ a.s.) $\cdot$ kind: statement of an internal step (more general) $\cdot$ proposed $\cdot$ no.

P2-TOPO1-a $\cdot$ facts used without comment by LM L4.3 / DFGPS L3.5 (a circle ∂B_ρ disconnects; separating two circles separates the inner disc from the outside) $\cdot$ LQGMetric/Topo/Disconnect.lean $\cdot$ own elementary arguments (intermediate value theorem) $\cdot$ proposed $\cdot$ no.

P2-TOPO1-b $\cdot$ LM L4.3 (local-metrics-final.tex l. 858–875) / DFGPS L3.5 (l. 1573–1596) $\cdot$ LQGMetric/Topo/CircleUnion.lean $\cdot$ “the circle of a $\mathcal{B}_2$ -disc meets a $\mathcal{B}_1$ -circle” by an explicit convexity argument (sphere_inter_sphere_nonempty); “connected union of circles $\Rightarrow$ path” via path components $\cdot$ kind: proof route (small) $\cdot$ proposed $\cdot$ no.

P2-TOPO1-c $\cdot$ LM L4.3 states a circle point within 2ε of z_2 ; the paper’s argument only gives about 4ε $\cdot$ exists_joinedIn_biUnion_sphere_of_cover takes $b \in A$ and gives $< 2\varepsilon$; the consumer applies it with $b = P(t)$, $t < t_J$ close to $t_J (2\varepsilon + o(1))$ $\cdot$ kind: statement of an internal step (constant) $\cdot$ proposed $\cdot$ no ($\varepsilon \rightarrow 0$).

P2-TOPO1-d $\cdot$ DFGPS Def 3.7 (“paths which disconnect the inner and outer boundaries”) $\cdot$ LQGMetric.Disconnects, ContMetric.aroundDist $\cdot$ read as: every path in C from E to F meets the set, with the inner and outer boundaries passed explicitly $\cdot$ kind: reading (decision D34) $\cdot$ proposed $\cdot$ no.

P2-FMEAS-b $\cdot$ FOUNDATIONS §9 item 4 (StandardBorelSpace (DistOn U); published background: L. Schwartz, *Radon measures on arbitrary topological spaces*, Part II — $\mathcal{D}'$ is a Lusin space; Trèves ch. 13–14) $\cdot$ LQGMetric.standardBorelSpace_distOn (LQGMetric/Field/StandardBorel{Ext,DistOn,Range}.lean) $\cdot$ own coordinate argument instead of formalizing Schwartz’s theorem: coordinates are integer combinations of dense sequences in $\mathcal{D}_{[K_n]}$; the range is characterised by additivity, consistency and finite-order seminorm bounds on each K_n (Borel), with a density-extension lemma $\cdot$ kind: proof route $\cdot$ proposed $\cdot$ no.

P2-ELEM-a $\cdot$ DDDF Lemma 22 (proof via an independent copy of Z) $\cdot$ LQGMetric.Analysis.quantile_gap_sq_le_variance $\cdot$ Chebyshev at distance $(\ell - \ell)/2$ from the mean instead (no product measure); gives the stronger $(\ell - \ell)^2 \leq 4p^{-1} \text{Var } Z$, which implies the printed bound $\cdot$ kind: own elementary proof, same statement $\cdot$ proposed $\cdot$ no.

P2-ELEM-b $\cdot$ DDDF Lemma 25 (tightness.tex l. 1221–1258; the lower bound “is similar”; condition $(C_1 + C_2)^2 +$

$(C_1+C_2)C_3 \leq C_3^2) \cdot \text{LQGMetric.Analysis.exists_exponent_of_weakMul} \cdot$ one induction on the absolute value covers both bounds; the printed condition drops a factor 2 — the correct one is $(C_1+C_2)^2 + 2(C_1+C_2)C_3 \leq C_3^2$, we take $C_3 = 3(C_1+C_2) \cdot$ kind: typo in the source (statement unchanged) and own completion of an unwritten half · proposed · no.

P2-ELEM-c · GM proof of Lemma 1.10 (“ $\mathbb{f}_b D_h \rightarrow D_h$ in law $\Rightarrow \mathbb{f}_b \rightarrow 1$ ”, no proof given) · $\text{LQGMetric.Analysis.eq_one_of_map_const_mul_eq_map}(X > 0, kX \sim X \Rightarrow k = 1) \cdot$ own elementary proof (iterate $\mu(-\infty, t] = \mu(-\infty, t/k]$) of an implicit step; the GM S1.11 measurable variant is proved for Borel-measurable $\mathbb{f}$ (GM say Lebesgue measurable suffices; only the continuous version is used) · kind: own elementary proof · proposed · no.

P2-ELEM-d · DG eq. (1.7b) and the route-B inequality algebra (decision D12) · $\text{LQGMetric/Analysis/DimAlgebra.lean} \cdot$ elementary algebra the sources state without proof · kind: own elementary proof · proposed · no.

P2-INT-a · DFGPS Lemma 2.11 (lqg-metric-estimates-final.tex l. 971–988; used at l. 1013 for a D_h not yet known to be a length metric) · $\text{LQGMetric.ContMetric.internal_eq_of_tendsto_internal_closure} \cdot$ weaker hypotheses ($D, \tilde{D}$ need not be length metrics; pointwise convergence of $D^n(\cdot, \cdot; \tilde{V})$ instead of uniform), $U = \mathbb{C}$; own proof structure ($D = \tilde{D}$ near every point of $V \Rightarrow$ local isometry $\Rightarrow$ equal internal metrics; locality of length = BBI Prop 2.3.4) instead of “lengths of paths at positive distance from ∂V agree” · kind: proof route, statement more general · proposed · no.

P2-INT-b · GM S4.2 (“the first time it enters $B_r(z)$ ”, uniqueness-final.tex $\approx$ l. 1697) · $\text{IsGeodesicCurve.firstHit_outside} \cdot$ stopped at the first hitting time of closure $B_r(z)$; agrees unless the geodesic touches $\partial B_r(z)$ before entering, and only this version gives GM’s “does not hit $\partial B_r(z)$ except at x ” · kind: reading · proposed · no.

P2-INT-c · GM S3.1(a) (“determined by $D(\cdot, \cdot; V)$ ”) · $\text{ContMetric.infEDist_frontier_eq_of_internal_eq, ball_eq_of_internal_eq} \cdot$ formalized as the functional statement (same internal metric on $V \Rightarrow$ same $d(u, \partial V)$, same balls); measurability is WP-10 · kind: reading + own elementary proof · proposed · no.

P2-INT-d · LM Lemma 1.1 (“length metric”) · $\text{internalEDist_univ_internalSpace} \cdot$ read in the $[0, \infty]$ -valued infimum form; IsLengthSpace only when all internal distances are finite · kind: reading · proposed · no.

P2-INT-e · LM S-int (d) (scaling of internal metrics) · $\text{internalEDist_image_of_edist_eq} \cdot$ stated for a bijection scaling distances by c (for D and cD on $\mathbb{C}$ take the identity) · kind: convention · proposed · no.

P2-GTAIL-a · DZZ Lemma 2.3 (LBM_LGDarXiv.tex l. 364–373; Fernique 1975, Adler 1990 Thm 4.1) · $\text{LQGMetric.SupTail.dzz_lemma23, dzz_lemma23_exists} \cdot$ E max bound by Dudley chaining ($\text{LQGDimension.ChainBox}$) with explicit $C_F = 20\sqrt{5}\sqrt{(3\sqrt{2})}$, instead of Adler’s entropy theorem; continuous version from Kolmogorov–Čentsov (sixth moments) after extending the field to $\mathbb{R}^2$ by the nearest-point map onto the box (continuous on all of $\mathbb{C}$ — slightly stronger) · kind: proof route · proposed · no.

P2-GTAIL-b · Borell–TIS for continuous fields (Adler–Taylor Thm 2.1.1, p. 50; first part derives $E \sup < \infty$ from a.s. boundedness) · $\text{LQGMetric.SupTail.borellTIS_iSup_continuous, tail_iSup_le} \cdot$ $E \sup < \infty$ enters as a hypothesis or via uniform finite-maximum bounds ($\text{integrable_iSup_of_continuous}$); own elementary steps for the degenerate case and the Fatou-type limit · kind: statement of a cited result (integrability as a hypothesis; consumers supply it) · proposed · no.

P2-FHEAT-b · GM footnote l. 212 (no proof that h^*_ε is continuous / the truncations converge locally uniformly) · $\text{IsWholePlaneGFF.ae_tendstoLocallyUniformly_heatMollify, IsGFFPlusBddCont.ae_tendstoLocallyUniformly_heatMollify, memLp_heatMollify_sub, tendsto_eLpNorm_heatIncApprox} \cdot$ ($\text{LQGMetric/Field/HeatMollify}\{\text{Lip, Kolm, Unif, L2}\}.\text{lean}$) · own argument along FOUNDATIONS §3: Lipschitz kernel bound, Gaussian fourth moments, dyadic Kolmogorov sup-tail bound ($\text{QZ kolm_sup_tail_N}$), Borel–Cantelli, Weierstrass M-test, Minkowski + Fatou in L^2 · kind: own proof where no published proof exists · proposed · no.

P2-FCIRC-a · commutation of distributions with integrals of test-function curves (standard: Hörmander Thm 2.1.3) · $\text{TestIntegral.exists_testIntegral} \cdot$ ($\text{LQGMetric/Field/CircleAvgIntegral.lean}$) · own short argument via a finite-jet Hahn–Banach factorization instead of the textbook Riemann-sum proof · kind: own elementary proof · proposed · no.

P2-FCIRC-b · Duplantier–Sheffield §3.1 / Prop 1.1 proof (0808.1560 l. 1166–1172: exponential decay + Borel–Cantelli) · $\text{logCov_circDiff_succ_le, ae_tendsto_mollAvg} \cdot$ ($\text{LQGMetric/Field/CircleAvg}\{\text{Var, Rate, Limit}\}.\text{lean}$) · the variance rate is an own elementary bound (L^1 norm $\times$ sup of the $1/r$ -Lipschitz log potential) instead of DS’s exact covariance; a.s. existence proved for the standard mollifiers bumpTest (independence of the mollifier is the next leaf) · kind: proof route · proposed · no.

DV-CONF-214a · CONF Lemma 2.14 proof, sector property at C:876 (stated without proof) · $\text{div_norm_mem_circArcOpen_of_mem_ball, disjoint_ball_arcPt} \cdot$ ($\text{LQGMetric/Complex/CircleArcBall.lean}$) · own elementary proof (Jordan’s inequality on $\arg(z \cdot \bar{u})$) · proposed · no.

DV-CONF-214b · CONF Lemma 2.14 (confluence-final.tex l. 832–894): “at most 8 arcs with $r_I \geq \pi/4$ ” (C:855) and the statement without an additive constant · $\text{confL214_long_arcs} \cdot$ ($\leq (400/\pi)^2$, own disjoint-balls area argument;

only universality is used) and the corrected statement $\#good \geq (1 - A C^{-2} \text{area}(U)) n - N_0$ (decision D36) · kind: statement correction of a cited lemma · reason: the printed statement is false for small n · proposed · no (CONF L3.10 uses it only for large n).

DV-CONF-214c · CONF L2.14 proof (Koebe $1/4$ for $\text{dist}(\varphi(w), \partial U)$; Brownian motion / conformal invariance at C:864) · `confL214_koebe_dist` via the Schwarz-lemma bound (`QZ infDist_compl_le_schwarzPick`), harmonic functions and `harmonicOnNhd_comp_holo` instead of Brownian motion · kind: proof route (equivalent standard facts) · proposed · no.

M1A-1 · blueprint/M1.md §5 TIP_smul (an auxiliary lemma of the M1 plan, not a paper statement) · `LQGMetric.GM.gm_tip_smul_of_aemeasurable` · adds the hypothesis $\forall p, \text{AEMeasurable}(Y \cdot p) P$ (the plan's statement is false with outer measures) · kind: definition/statement of an auxiliary lemma · proposed (D37) · no.

M1A-2 · steps GM leave implicit in §1.4 (uniqueness-final.tex l. 452–465, 507–509, 519, 588, 591–593) · `LQGMetric/Papers/GM/S1/{Subseq,StrongWeak}` · lean · own elementary proofs: length, internal metrics and Weyl scaling of C·D scale by C; Axiom IV' from IV at $r = 1$; $\Lambda = |\xi Q| + 2$ in (1.15); closure of a singleton set of laws; the convergence-in-probability lemmas · proposed · no.

P2-WEYL-a · Weyl-scaling identities GM states without proof (GM.S1.6, GM.S5.W; uniqueness-final.tex l. 262–302) · `LQGMetric/Metric/Weyl{Basic,Scaling,Concat,Length,PathLength,Local,Cont,Metric,LQG}` · lean · own elementary proofs: constants, monotonicity, scaling by C, affine coordinate change, internal metrics and path lengths of $e^{\xi f} \cdot D$, locality, $e^{\xi f} \cdot D$ is a continuous length metric · kind: own proof (no published proof exists) · proposed · no.

P2-WEYL-b · GM (1.5)–(1.6) · `LQGMetric.weylScaleOn` (Weyl scaling with paths constrained to U), proved equal to the internal metric of $e^{\xi f} \cdot D$ on open U · kind: convention (helper definition) · proposed · no.

P2-FHEAT-c · covariance of $h^* \varepsilon$ (GM states it without proof) · `IsWholePlaneGFF.covariance_heatMollify_sub` (`LQGMetric/Field/HeatMollify{Cov,LogCov}` · lean) · own standard argument: L^2 approximation by the Gaussian pairings of mean-zero truncations, continuity of covariance in L^2 , dominated convergence for $\log \text{Cov}(|\log t| \leq t + 1_{\{t \leq 1\}} |\log t|)$, Gaussian first moments); the white-noise/heat-kernel form is not yet derived · kind: own proof · proposed · no.

P2-TOPO2-a · GM S4.6 (a path from the interior to the exterior meets the frontier; uniqueness-final.tex l. 2232–2236) and DDDF L11 = DF Lemma 4.8 (sub-crossing extraction; DF l. 607–618) · `disconnects_of_frontier_thickening`, `RectCross.crossing_thin_subrect` (`LQGMetric/Topo/{Shortcut,RectCross}` · lean) · own elementary arguments for the steps the papers leave implicit (last visit to one level before the first visit to the next); L11's rectangle family stated explicitly, DF's squares replaced by the rotated rectangles containing them ("isometric to $[0, a/2] \times [0, b/2]$ ") · kind: own elementary proof · proposed · no.

P2-TOPO2-b · CONF Lemma 3.7, topological step (confluence-final.tex l. 1467–1468: "we can choose"; no proof) · `exists_frontier_closedBall_disconnects`, `exists_frontier_ball_disconnects` (`LQGMetric/Topo/ArcDisconnect*` · lean) · own argument ($W = \bigcap_{y \in Y} \text{closedBall } y \text{ d}$; rerouting along gauge level sets); "first hits K at a point of I " formalized as $\text{range } \gamma \cap K \subseteq \{x\}$; closed ball for $\text{diam } Y \leq \varepsilon r$, the paper's open ball only for $\text{diam } Y < \varepsilon r$ · kind: own proof + reading · proposed · no (consumers can use the closed ball or a slightly larger radius).

DEV-CI-1 · LM S3.3b (local-metrics-final.tex l. 636–637, commented lemma l. 651–672) · `LQGMetric.densityKernel`, `hasCondDistrib_pair_densityKernel` (`LQGMetric/Prob/CondLawRN` · lean) · conditional densities are $\mathbb{R}_{\geq 0}$ -valued (an a.e.-finite RN derivative can be chosen finite everywhere; makes the kernel s-finite); the statement is given as `HasCondDistrib` (mathlib's uniqueness lemma needs `IsFiniteKernel`) · kind: convention · proposed · no.

DEV-CI-2 · conditionally i.i.d. copies (LM l. 306–332; Kallenberg Thm 6.3, Prop 6.13 cited for statements only) · `LQGMetric.condCopyMeasure` (`LQGMetric/Prob/CondCopies` · lean) on $\Omega \times \beta$, conditional independence in event form · own bookkeeping construction · kind: own elementary proof · proposed · no.

BP-M2 (group) · Blueprint Props for GM §§2–6's cited results (`LQGMetric/Blueprint/{M2Defs,CONFDefs,LMResults,MQGeodesic,DFGPSestimates,CONFResults}` · lean; full texts in blueprint/M2.md §1) · proposed readings: BP-M2-1 LM only for $U = \mathbb{C}$, $n = 2$, $N = 0$ (D21), internal metric of $e^{\xi h_r(z)} \cdot D$ read as $e^{\xi h_r(z)}$ times D 's internal metric, LM Lemma 2.1 only for V disjoint from ∂D ; BP-M2-2 MQ Lemma 4.1 with the harmonic part fixed (GM L2.7 derives the marginal comparison, GA-8); BP-M2-3 DFGPS L3.22 typo corrected, DFGPS P4.3 in its source form (centre 0) with GM's centre- z filled-ball version as node GM.S4.8; BP-M2-4 S2.4 extra hypotheses (U bounded/connected, $U = B_R(0)$ for S2.4e, explicit disconnecting path for S2.4d); BP-M2-5 $\sigma(A)$ of a random closed set = Effros σ -algebra, $\sigma(A, h|_A)$ via D32's dyadic hulls; BP-M2-6 c_-, C_- deterministic (constant chosen before the probability space, via uniqueness of the GFF law mod constants); BP-M2-7 T4.2 for general λ_3 (DV-B1), geodesic as the D31 selector, conditional probabilities via `condExp`, P4.3 with $\mu/2$ (DV-B3) and the proved containment (DV-B10); BP-M2-CONF-1, 3–7 CONF results at a general centre z , an arc of $\partial \mathcal{B}^\bullet$ is a connected subset of the boundary, harmonic part via conditional expectations, two typos corrected

($\mathcal{E}_r(a)$ condition 2 with $e^{\{\xi h\}}$), infinite radii in $[0, \infty]$, shared L3.5–T3.9 parameters; **BP-M2-CONF-2** (needs a decision before CONF L2.4 is proved): “leftmost geodesic” (never defined in CONF) read as a uniform limit on $[0, s]$ of geodesics to points $\varphi(t_n)$ of $\partial \mathcal{B}^\bullet$, $t_n \downarrow t_0$, for a counterclockwise Jordan parametrization φ (CONF l. 548–556) · kind: statements of cited results / readings · proposed · no (CONF-2 pending).

P2-RANDDIST · Berestycki–Powell, arXiv:2404.16642, Lemma L:measurability (definitionGFF.tex l. 907–912; proofSLE.tex l. 470–540 via $\mathcal{S}^*$, cutoffs, polar sets, Alaoglu and Banach–Steinhaus) and the identification of random distributions with processes of pairings · LQGMetric/Field/RandomDist{,Borel,Complete,Conv,Version}.lean: borel_weakStar_eq, measurable_weakStar_iff, map_eq_of_finiteDim, measurableSet_convSet, measurableSet_conv_event, exists_version · the convergence lemma is proved by localising to $\mathcal{D}_{\{K_n\}}$ directly (no $\mathcal{S}^*$), with an $\varepsilon/3$ equicontinuity argument in place of Alaoglu; completeness and barrelledness of $\mathcal{D}_K$ are proved (absent from mathlib); the σ -algebra equality is an own polar-set argument (BP state the definition, not the equality) · kind: proof route · proposed · no.

DV-CONF-214d · CONF L2.14 proof (C:863, Brownian exit probability from the half-disc H_I) · LQGMetric.CONF.l214_minorant (LQGMetric/Papers/CONF/L214Harm.lean) · a harmonic minorant built by Poisson reflection of a tent on a cap inside J^- , with an own elementary Poisson-kernel lower bound at w_I ; arc lengths $r_I \leq 1/10$ instead of $\pi/4$ (only the universal constant N_0 of D36 changes; long arcs counted by confL214_long_arcs_of_le) · kind: proof route · proposed · no.

DV-CONF-214e · CONF L2.14 proof (C:864–867, Beurling’s estimate for Brownian motion) · LQGMetric.CONF.l214_beurling_reach · the analytic contrapositive statement via QZ beurlingHarmStmt_holds; Brownian motion not used · kind: proof route · proposed · no.

P2-DDDFLEN · DDDF length observables (tightness.tex l. 449–498, 908, 950, 964) · LQGMetric/Papers/DDDF/Len{Basic,Countable,Obs}.lean · conventions: $\ell(p)$ = lower quantile of the law of L , $\ell(p) = \ell(1-p)$, λ_δ , λ_n lower medians (D-DDDF-4, D22; no atomlessness claim); geodesics replaced by measurably indexed near-minimizers from one countable family of paths valid for every continuous weight (D-DDDF-5; own soft argument from separability of $C(\mathbb{C}, \mathbb{R})$ and the weight comparison); paths are piecewise C^1 (as in GM/DFGPS) instead of DDDF’s “smooth curves” (l. 462) — the equality of the two infima is not formalized and all later uses go through this definition; the S2.c sandwich and median lemmas are own elementary proofs · kind: conventions + proof route · proposed · no.

DV-MEAS-1 · LM S-int (a) (measurability of $D \mapsto D(\cdot, \cdot; V)$; blueprint/LocalMetrics.md row LM.S-int) · ContMetric.internal_eq_chainInf, measurable_internal (LQGMetric/Meas/Internal.lean) · stated as “agrees on length metrics with a Borel function of (D, z, w) ” via a countable chain formula (own argument, as the blueprint suggests) · kind: convention · proposed · no.

DV-MEAS-2 · decisions/DEC-C.md signature of AESigmaLE (P typed as a measure for the last local instance m_2 — a typo) · LQGMetric.AESigmaLE takes P as a measure for the ambient σ -algebra · kind: correction of a signature typo · proposed · no.

DV-MEAS-3 · countable-reduction lemmas CR1, CR2, CR5, CR6 of decision D30 · LQGMetric/Meas/CR.lean · own elementary proofs (each < 30 lines); Lusin’s theorem (Kechris Thm 21.10) via QZ analyticSet_nullMeasurableSet; Lusin–Souslin selector via a credited copy of QZ exists_measurable_of_partialGraph · kind: own elementary proof / reuse · proposed · no.

DV-MEAS-4 · “choose in a deterministic manner” (D31 rule 4; GM §5, CONF) · LQGMetric.firstIdx, firstChoice (LQGMetric/Meas/Choice.lean) · the first index of a fixed enumeration satisfying a measurable condition, junk index 0 if none · kind: convention · proposed · no.

P2-FCIRC-c · Duplantier–Sheffield §3.1, Prop 3.3; Hu–Miller–Peres Prop 2.1 · ae_circleAvg_affineComp, exists_gaussianReal_of_ae_tendsto, exists_continuous_version_circleAvg (LQGMetric/Field/CircleAvg{Affine,GaussLim,Cov,Proc,GP,Kolm,Bridge}.lean) · own elementary arguments: leaf (b) by comparing two mollifications through log-potential bounds (DS use the circle measure directly); Gaussianity of a.s. limits via characteristic functions; joint continuity via $q = (\log r, \operatorname{Re} z, \operatorname{Im} z)$, eighth moments and $\operatorname{Var} \leq 2(|r-s| + |z-w|)/\min(r,s)$, with QZ’s Kolmogorov–Čentsov · kind: proof route · proposed · no.

P2-M1D-1 · decision D13 / D-A2 (iii) (annulus $\mathcal{A} = \{1/8 \leq \operatorname{dist}(\cdot, L) \leq 1/4\}$) · LQGMetric.GM.gm_S1_18 (LQGMetric/Papers/GM/S1/NoGap.lean) · the closed $1/4$ -neighbourhood of L with a first-exit bound $\theta(t) \geq e^{\{\xi t\}} \cdot D_h(L, \{\operatorname{dist}(\cdot, L) = 1/4\})$; test function $f = \varphi - (\varphi_1(0)/\psi_1(0))\psi$ with explicit bumps · kind: proof route (own simplification) · proposed · no.

P2-M1D-2 · steps GM leave implicit around l. 582–593 · LQGMetric/Papers/GM/S1/{FieldAux,Median}.lean · own elementary proofs: Lipschitz bound and compact-open continuity of crossFn, eventual positivity of α_ε , positivity of the circle average of a nonnegative bump, finiteness of lfppDistE along segments; Cameron–Martin node GFF.CM-wp proved in the null-set form S1.18’ needs (test functions with zero average over ∂D , normalized h) · kind: own elementary proof · proposed · no.

M2A-1 · GM Prop 2.2 (uniqueness-final.tex l. 885–936; GM applies LM Thm 1.6 to h directly) · LQGMetric.GM.Bilip.gm_P2_2 (LQGMetric/Papers/GM/S2/Bilip.lean) · LM Thm 1.6 is applied to the normalized field $h - h_1(0)$ on a reference probability space and transferred to every GFF by uniqueness of the normalized law plus Weyl scaling by a constant (gives the deterministic constant C of reading BP-M2-6); GM l. 914 “jointly local” read as “ ξ -additive” (already noted in GM_A) · kind: convention / proof route · proposed · no.

M2A-2 · GM S2.3, S1.23, S3.4, S6.1 (no proofs in GM) · LQGMetric/Papers/GM/S2/BilipLocal.lean, S3/DefsLemmas.lean · own elementary proofs (“local and determined by $h \Rightarrow$ conditionally independent”, the bounds $C^{-1} \leq c_* \leq C$, the reciprocal identities under $D \leftrightarrow \bar{D}$, the 8-adic counting of S6.1) · kind: own elementary proof · proposed · no.

ZB-1 · zero-boundary GFF via its Green covariance (Berestycki–Powell §1.2; Sheffield §2) · LQGMetric.IsZBGFFProcess, IsZeroBoundaryGFF (LQGMetric/Field/ZeroBoundary*.lean) · defined by the dual Dirichlet form $\langle \varphi, \psi \rangle_{H^{-1}(U)}$ (QZ zeroGFFTestCov, expanded bilinearly on $\varphi^+ - \varphi^-$), a law statement on pairings; the Green identity $G_U = -\log|x-y| + \text{harmonic}$ is proved per domain (bounded simply connected via the Riemann map; explicit disc and half-plane formulas) · kind: definition (equivalent formulation) · proposed (D39) · no.

ZB-2 · steps used without proof · LQGMetric/Field/{ZeroBoundary*, GreenFn*}.lean · own short arguments: polarization of QZ’s diagonal kernel form with signed densities; complementary components of convex sets are unbounded (ray argument); translating bounded domains into $\mathbb{H}$ (QZ states results only for $D \subseteq \mathbb{H}$); the harmonic part via dslope · kind: own elementary proof · proposed · no.

DV-CONF-214f · CONF L2.14 proof, “this path necessarily divides U ... must disconnect 0 from I ” (C:866–869) · LQGMetric.CONF.l214_sep, DisconnectsIn (LQGMetric/Papers/CONF/L214Sep*.lean) · own argument (QZ’s winding-number crossing lemma lwfSide_crossing after a Möbius map to $\mathbb{H}$), stated for a connected open set with two boundary points rather than a path; “disconnected by a path” becomes “disconnected by a set”, as L3.10 uses it (C:1586) · kind: proof route + reading · proposed · no.

P2-CONF29 · CONF Lemma 2.9 (confluence-final.tex l. 676–709: “locally compact”; “a.s. continuous at f ”; (2.11) with $\leq$ and probability $\geq 1 - 2^{-n}$) · LQGMetric.CONF.fkg_continuous_gaussian (LQGMetric/Papers/CONF/L29{Approx,FKG}.lean) · adds σ -compactness (the proof uses a compact exhaustion, CONF:686); “ Φ a.s. continuous at f ” read as $\forall^m \omega$, ContinuousAt $\Phi(f \omega)$ in the local uniform (compact-open) topology; “Gaussian random continuous function” = measurable map into $C(X, \mathbb{R})$ with Gaussian finite-dimensional marginals; strict inequalities and $P[\text{bad}] < 2^{-n}$ in (2.11) · kind: implicit hypothesis + readings · proposed · no.

DA7 · GM (3.1) and “determined by $h|_{B_R(0)}$ ” in the proof of GM Lemma 3.1 (uniqueness-final.tex l. 1186–1206) · LQGMetric.GM.locEv, gmEv_subset_locEv, aeEventIn_locEv_addConst (LQGMetric/Papers/GM/S3/Deterministic*.lean) · (3.1) used in strict form with rational u, v ; locality routed through an internal-metric event on $B_R(z)$ for points of $B_{R/2}(z)$, between (3.1) and $\{\bar{D} > C \cdot D\}$, making measurability and locality countable; tail triviality replaced by GM L2.7 on rescaled translates (DA6, D15) · kind: own argument (GM gives none) · proposed · no.

M1B-1 · GM Lemma 1.10 proof, Axiom V for $D^{\{b\}}$ (uniqueness-final.tex l. 497–505: “therefore implies”) and S1.10 (l. 515–517: “it is easy to see”) · LQGMetric.GM.gm_sl_8, tight_smul_family, gm_sl_10 (LQGMetric/Papers/GM/S1/{Dilate,WeakStrong}.lean) · $(h(\cdot/b))_{\text{br}}(0) = h_r(0)$ a.s. so only the deterministic ratio c_{br}/c_r remains (BP-M1-4); “limit laws supported on metrics” by continuity of $(a, \mu) \mapsto \text{law of } a \cdot X \text{ plus Prokhorov}$; S1.10 via one coordinate $D_h(0, 1/b)$ and the continuous version of circle averages (iterating the distribution function); vacuous case $\beta = 0$ when no GFF exists; BP-M1-3 sign convention $\sharp_b = b^\beta$ · kind: own elementary proofs of implicit steps · proposed · no.

EX-1 · existence of the whole-plane GFF as a random distribution (Sheffield math/0312099 §2, Berestycki–Powell ch. 1: Hilbert-space / Sobolev / Bochner–Minlos constructions) · LQGMetric.GFFExist.exists_wholePlaneGFF, exists_normalizedWPGFF, LQGMetric.GM.existsNormalizedGFF (LQGMetric/Field/Exist*.lean) · constructed instead as $\partial_{\text{red_im}}$ of a continuous antiderivative of white-noise rectangle pairings (decision D40); same law (centred Gaussian pairings with covariance logCov on mean-zero tests) · kind: proof route (not following the published construction) · proposed · no (the theorem concerns existence only; the law is the GFF’s).

EX-2 · standard facts used in EX-1 · LQGMetric/Field/Exist{Kernel, KernelL2, Cov, Rect, Field, FTC, KerId, SFub, SFub2}.lean · own elementary proofs: weighted Cauchy–Schwarz, L^2 and increment bounds for the kernels, integration by parts against rectangles, a stochastic Fubini identity, the signed (mean-zero) Frullani identity with $t = 2\tau^2$ · kind: own elementary proof · proposed · no.

DA5-addendum (P2-TIGHT) · GM S2.4 (uniqueness-final.tex l. 449–452; DFGPS T:370–374; no published proof) · LQGMetric.GM.Tight.gm_S2_4a, gm_S2_4a_sep, gm_S2_4b, gm_S2_4b', gm_S2_4c, gm_S2_4d (LQGMetric/Papers/GM/S2/Tight*.lean) · own proofs per decision D14 using only Axioms I–III, IV', V: S2.4a/b by compactness (Cantor’s intersection theorem on the compact closure of the laws) instead of extracting subsequences; uniformity in the centre via uniqueness of the normalized GFF law; U bounded, open, connected in S2.4c; S2.4d with $\forall x, y$ in place of

the set distance; results stated as $P[\text{bad event}] < \epsilon \cdot \text{kind} \cdot \text{own proofs} \cdot \text{proposed} \cdot \text{no}$.

DV-M2C-1 · GM Lemma 2.11 (uniqueness-final.tex l. 1062–1072; “by translation invariance”) and DFGPS Prop 4.1 (its ϵ_0 chosen after the field) · `LQGMetric.GM.gm_L2_11` (`LQGMetric/Papers/GM/S2/ThinAnnulus.lean`) · reduced to the normalized translated field $h(\cdot + z) - h_1(z)$ via Axioms IV', III and scaling of internal metrics; DFGPS Prop 4.1 applied to a reference normalized GFF, ϵ_0 taken from that law and transferred (`Measure.le_map_apply`) · kind: own elementary transfer · proposed · no.

DV-M2C-2 · GM S1.2 (l. 645–648; MQ assume $x \neq y$) · `LQGMetric.GM.gm_S1_2` · the case $z = w$ proved directly (the constant curve is the only geodesic) · kind: own one-line argument · proposed · no.

DV-M2C-3 · GM S1.2 for $h - \varphi$ (D33 transfer corollary) · `gm_S1_2_addFun`, `ae_notMem_addFun_of_ae` (`LQGMetric/Papers/GM/S2/GeodesicsCM.lean`) · stated for deterministic test functions φ , via Cameron–Martin on mean-zero pairings plus recentring invariance (Axiom III) · kind: own argument; scope (random φ not covered) · proposed · no.

DA8 · GM Lemma 3.1 (“the statement for c_* is proven in an identical manner”) · `LQGMetric.GM.lowerRatio_eq_inv`, `gm_L3_1` (`LQGMetric/Papers/GM/S3/Deterministic*.lean`) · c_* obtained from the C_* statement for the swapped pair $(\tilde{D}, D)$ via $c_* = (\sup D/\tilde{D})^{-1}$ · kind: own elementary step · proposed · no.

DV-CONF-LM1 · CONF L2.4 (C:533–537): “each D_h -geodesic from 0 to y lies (weakly) to the right of P_y^- ... if we stand at y and look outward from $\mathcal{B}^{\bullet}_s$ ”; CONF defines no “right of” · Lean: `WeaklyRightOf` $D \preceq z \preceq Q \preceq P$ = for all $t \in (0, s)$, every positive Jordan lift (φ, θ) of $\partial \mathcal{B}^{\bullet}_t$ and all angle lifts α of $P \rightarrow z$, β of $Q \rightarrow z$ on $[t, s]$ with $\alpha(s) = \beta(s)$, the positions u, v ($\varphi u = P t$, $\theta u = \alpha t$; $\varphi v = Q t$, $\theta v = \beta t$) satisfy $v \leq u$; `leftmost` := geodesic to y such that every geodesic to y is `WeaklyRightOf` it (rightmost symmetric), in `Blueprint/CONFDefs.lean` · reading of undefined text (order formalized through the universal cover, as MS Prop 2.2, MS:619–622) · the statement of L2.4 becomes literal; the earlier Blueprint reading omitted the clause · proposed · no: GM uses `leftmost` geodesics only through T2.15/T2.16 counts and the arcs $\mathcal{J}_k$, whose meaning is unchanged (a.s. the two readings define the same geodesic, by CONF’s proof).

DV-CONF-AN1 · CONF C:572–575, 627: “Let $\pi : \mathbb{R} \times [0, 1] \rightarrow \mathcal{A}$ be a universal covering map normalized so that $\pi(\mathbb{R} \times \{0\}) = \partial_{\text{in}} \mathcal{A}$...”, `wind(P)` := difference of first coordinates (C:577) · Lean: covering $w \mapsto z + e^w$ restricted to the preimage of $\mathcal{B}^{\bullet}_s \setminus \text{int } \mathcal{B}^{\bullet}_s$; `wind(P)` := $(v - u)/(2\pi)$ where u (resp. v) is the position of the lifted start (resp. end) on the lift of $\partial \mathcal{B}^{\bullet}_s$ (resp. $\partial \mathcal{B}^{\bullet}_s$) given by positive Jordan lifts (`IsWindNum`) · model change (another covering of the same annulus; CONF’s covering exists only via the annulus theorem) · winding numbers depend on the choice of covering in CONF too; L2.5 and L2.6 are proved for every choice · proposed · no.

DV-CONF-AN2 · CONF L2.5 proof (C:600): “Since also $\hat{x}' < \hat{x}$, it follows that P and P' must cross”, and L2.7 proof (C:633): “the path $\hat{P}_q$ divides $\mathbb{R} \times [0, 1]$ into two connected components” (both “obvious” in the strip model) · Lean: deterministic lemma TOPO-ORD (below), own proof (no published proof in the log-cover model; suggested route: J2 exterior map $\Phi_{\{K_1\}}$ plus radial rays of $\Phi_{\{K_2\}}$ to reduce to two disjoint proper rays in a closed half-plane, then a square-crossing lemma from QZ janiszewski / the Eilenberg criterion, or a `loopDeg` computation) · own argument for an implicit topological fact · proposed · no.

DV-CONF-ARC1 · CONF C:620, 1513, 1537–1547 (“arcs ... open, half-open, or closed”; $\mathcal{J}_k$ a finite collection of arcs “chosen in a manner depending only on $(\mathcal{B}^{\bullet}_s, h)$ ”) · Lean: `arc` = nonempty connected subset of the Jordan curve $\partial \mathcal{B}^{\bullet}_s$ (includes points and the whole curve); $\mathcal{J}_k$ represented by the $\mathcal{J}_0$ -indexed family $I_j^{\wedge\{k\}}$ (CONF’s own equivalent definition, C:1547); “chosen depending only on $(\mathcal{B}^{\bullet}_\tau, h)$ ” = the graph of each I_j is universally measurable for `localSigma` $\otimes$ Borel (D30) · encoding · the only change is that degenerate arcs (points, the whole curve) are allowed. CONF needs them (singletons in GM U:1652; the whole curve at C:1513) · proposed · no.

DV-CONF-212 · CONF L2.12 (C:785–802): events $E_{\{r_k\}} \in \sigma((h - h_{\{r_k\}}(0))|_{\mathcal{A}_{\{S_{1r_k}, S_{2r_k}\}}(0)})$, proof “same argument as Lemma 2.11” or “Lemma 2.11 and Lemma 2.13” · Lean: events required to lie in $\sigma((h - h_{\rho(0)})|_{\mathcal{A}})$ for every $\rho > 0$ (i.e. determined modulo additive constants), proof by applying LM L3.1 to the reversed finite list of radii; CONF L2.13 is not formalized · parameter range (the lemma is stated for a smaller class of events) and own proof route (a third, shorter argument) · the only consumer, CONF L3.4, uses constant-invariant events (C:1154) · proposed · no.

GA-8b / P2-M2B · GM Lemma 2.7 (uniqueness-final.tex l. 964–1004) and MQ Lemma 4.1 (radii $7r/8, 15r/16$) · `Blueprint.MQLem4_1Gen` (D41; general radii $\rho_1 r < \rho_2 r < r$ and an arbitrary centring constant a with $|g - a| \leq M$ on $B(z, \rho_2 r)$ — MQ’s proof l. 566–589 gives this form: the cutoff times $(h - a)$ lies in H_0^1 with energy bounded in terms of M); the configuration is translated away from the unit circle and the field renormalized before applying `LM Lem2_1` (forced by the `LM Lem2_1` reading BP-M2-1; GM do not do it); GM L2.7’s conditional lower bound passes through the zero-boundary law by Cauchy–Schwarz (MQ Remark 4.2) rather than GM’s conditional-vs-marginal RN derivative; typo GM l. 1002 ($1 - \tilde{p}^{\{\#\}} \rightarrow 1 - (1 - \tilde{p})^{\{\#\}}$), l. 983 ($H_r \rightarrow H_z$); own configuration-free L^2 bound instead of translation invariance; independence of the zero-boundary GFF on components assembled from QZ’s splitting lemma · kinds: statement of a cited result (generalized radii), proof route, readings, own arguments · proposed · no.

GA-6a-e / P2-M2E · GM S3.6, Lemmas 3.7–3.9 (l. 1325–1404) · LQGMetric/Papers/GM/S3/GoodAnnulus*.lean · L3.8–L3.9 stated for every C' (GM restricts to $C' \in (0, C_-^*)$, unused); (B) with the $(\nu-\mu)/2$ count (GA-5); in L3.9 U bounded and w within $\varepsilon^{1+\nu}r/100$ of x (GM’s statement fails for unbounded U / points near $\partial(rU)$); L3.8’s crossing bound on the closed annulus $3r/4 \leq |w| \leq r$; closed-annulus L2.11’ (D42) · kinds: generalization, reading, correction · proposed · no.

DV-M2I-1..4 · GPS arXiv:2010.07889 Lemma 2.4, Miller–Sheffield arXiv:1506.03806 Prop 2.1 (mapmaking_final.tex l. 555–612), GM (4.38) and S4.5 (l. 2052–2058) · LQGMetric/Complex/JordanMap*.lean, LQGMetric/Papers/GM/S4/{JordanBasic, JordanPunct, Jordan, JordanBdy, JordanLC, DcSet}.lean · (1) the Jordan-curve theorem for $\partial\mathcal{B}^\bullet_s$ and J2 are proved conditional on node J1b (FilledBallBdyLC, MS l. 570–601 verbatim, open); the Jordan curve is the injective continuous image of the unit circle; J2 in punctured-disc form $\Psi(b) = z + 1/\varphi(b)$; (2) the generic disc map takes HasHoloSqrt U (or “every component of $\mathbb{C} \setminus U$ unbounded”) instead of simple connectivity (QZ lacks hasHoloSqrt_of_isSimplyConnected); (3) own elementary arguments for J1d, the inversion transport of ULC and cut points, boundary connectivity, S4.5’s metric axioms; (4) S4.5’s upper bound against the internal Euclidean metric of U and curves with interior in U · kinds: conditional proof, definition reading, own arguments, reading · proposed · no.

DV-CONF-212-impl · decision D47 · LQGMetric.CONF.confLem2_12a_of_LM, confLem2_12b_of_LM (LQGMetric/Papers/CONF/L212.lean) · the $j = 0$ slot uses $E'_0 = E_{\{K+1\}}$ at radius $S_1S_2r_{\{K+1\}}$ (one closed formula; that index is never counted) · kind: implementation detail of the decided route · proposed · no.

P2-GMC-1.3 · DZZ L2.10 and the LGD definition (LBM_LGDarXiv.tex l. 121–124, 673–694) · LQGMetric/Dimension/{GMCircle, GMCTest, GCMoment, GCMass, GMCBall, LGDBasic, LGDScale}.lean · (1) the first-moment constant includes the junk value of $\limUnder$ from QZ’s avgReg (only the constant; a.s. existence of the limit on the square is the open node GMC.EXIST-SQ); (2) own elementary proofs of the graph-distance facts (monotonicity in δ , $D \geq 1$, finiteness, scaling, rational vs free centres, similarity invariance) and cut-off/exhaustion lemmas; (3) DZZ L2.10 so far only for $0 < p \leq 1$ (any set) and $p = 1$ at ball scale · kinds: convention, own proofs, partial result · proposed · no.

GA-6e/f (P2-M2E round 2) · GM Lemma 2.11 (l. 1062–1072) and Lemma 3.8 (l. 1384–1389, “the same is true with $\tilde{D}_h$ ”) · LQGMetric.GM.gm_L2_11_closed (LQGMetric/Papers/GM/S2/ThinAnnulusClosed.lean), gm_gaLong_prob (S3/GoodAnnulusLong.lean) · Lemma 2.11 for the closed annulus (D42), stated in complement form $P(E^c) \leq 1 - p$ (stronger for events not known to be measurable); the $\tilde{D}$ case of condition 2 reduced to the D case through Prop 2.2 · kinds: variant of an internal lemma, proof route · proposed · no.

DV-CONF-214g · CONF L2.14 proof (C:866–870) · LQGMetric.CONF.l214_pull, l214_signs (LQGMetric/Papers/CONF/L214Pull.lean) · transporting the separation through the conformal map Φ (closed embedding of the closed disc) and the sign computation for the cap points are own elementary steps · proposed · no.

DV-DZZ-S2-1..3 · DZZ §2 (LBM_LGDarXiv.tex l. 380–645) · LQGMetric/Papers/DZZ/S2*.lean · (1) dzz_sum_sup_tail: sub-boxes of side $\leq (\rho/2)^i$ and geometric levels $\lambda(1-\theta)\theta^i$ ($\theta = \rho^{1/4}$) instead of DZZ’s i^{4-2^i} grids and $\lambda(i+1)^{-2}$; hypothesis weakened from $e^{-\Omega(i^2)}$ to $K\rho^i$; (2) dzz_lemma26_core applies Fernique and Borell–TIS to $x \mapsto X_x - X_u$ on the $(2k+2)^2$ boxes of side δ instead of the max over DZZ’s grid first; (3) the box version of L2.6 reduced to the $\mathbb{C}$ version via a 1-Lipschitz projection · kind: bookkeeping / own elementary steps · proposed · no.

P2-LFPP-a..f · DFGPS l. 666–684, 810–813, 909–913; GM l. 216–220, 300–302 · LQGMetric/LFPP/*.lean · (a) own elementary proofs (path calculus, chain approximation of the infimum over paths, local Weyl comparison, IVT exit-time lower bound); (b) measurability of lfppCross etc. stated as AEMeasurable (a.e. equal to a measurable countable infimum); (c) S5 bounds with $|\xi|$, assuming the truncations of h converge; (d) S10 tightness criterion one direction only (compact connected X ; DFGPS do not use the converse); (e) S4 localized mollifier $\psi_\varepsilon = \text{ContDiffBump}$ with radii $\sqrt{\varepsilon}/2, \sqrt{\varepsilon}$ (an exact pairing, continuous for every h ; locality deterministic) and $\tilde{D}(\cdot, \cdot; U)$ read as the infimum over piecewise C^1 paths inside U (equality with the internal metric of $\tilde{D}$ not proved); (f) D2.4 allows dyadic levels $k \in \mathbb{Z}$ (M2Defs.dyadicSq has $n \in \mathbb{N}$) · kinds: own proofs, conventions, readings · proposed · no.

GA-6g · GM L3.7 condition 1 (“unique geodesic”, l. 1327) and decision D48’s route · LQGMetric.GM.Unique, measurableSet_midUnique, uMeasurableSet_gaCompareB (LQGMetric/Papers/GM/S3/GoodAnnulus{Unique, Compare}.lean) · “unique geodesic” replaced by the midpoint criterion (each M_t , t rational, has at most one point), Borel for every continuous metric and equivalent to uniqueness when geodesics exist between all pairs (a.s. for weak metrics) — own short argument reaching D48’s conclusion without the upper-semicontinuity route · kind: own argument replacing a decided route · proposed · no.

DGo-1..4 · Ding–Goswami arXiv:1610.09998 §3 (DGo:483–704) · LQGMetric/Papers/DGo/{Smoothness, GaussianTail, CouplingTail, CircleKernel}.lean · (1) Lemma 3.4 by per-box tails plus a union bound (the paper’s route needs the Y_R jointly Gaussian; ours does not; same constant), squares instead of rectangles, side length b instead of diam R in (I), Dudley replaced by our DZZ L2.3 chaining bound; (2) $\text{Var}(G_{\{v;4\}}) \leq 1$ by convexity of the norm and Lemma 3.2 instead of the $I_1 - 2I_2 + I_3$ computation; (3) Prop 3.3: V a closed

square cut into $\lceil 2b/\delta \rceil^2$ sub-squares, explicit K for $\delta \leq \delta_1 < 1$, $\Omega(x^2) = x^2/(2\sigma^2)$; Props 3.2/3.3 proved conditionally on the variance bounds (3.9), (3.10) (open: R0–R4, the Dirichlet-kernel identities, shared with DZZ L2.5–2.9 and DDDF P29); (4) numbering: DG’s ”DGo Lemma 3.1” = arXiv Lemma 3.2, DG’s ”DGo Prop 3.2” = two-sided arXiv Prop 3.3 · kinds: proof route, readings, conditional result · proposed · no.

DDDF-F1..F3 · DDDF tightness.tex l. 280–352 (smoothness stated without proof at l. 292; Props 2, 3) and DF arXiv:1809.02607 l. 1823–1880 (Lemmas 10.1, 10.2) · LQGMetric/Field/WhiteNoiseC1*.lean, LQGMetric/Papers/DDDF/Field{Grad, Osc, Max, ExpTail, OscExp, MaxExp}.lean · (1) the C^1 version via Kolmogorov–Čentsov on the difference-quotient field in three parameters plus a deterministic ”jointly continuous difference quotients $\Rightarrow C^1$ ” lemma (own); (2) (2.16) by Fernique (DZZ L2.3) + Borell–TIS on each of the 4^n squares with scale-correct covariance bounds and a union bound, instead of DF’s scale-by-scale summation (same ingredients, different constants); (3) (2.10), (2.11), (2.17) follow DF L10.1/10.2 and DDDF’s proof of (2.17) with $\text{Var } \varphi_{\{0,n\}} = n \log 2$ (DF: $(n+1) \log 2$), corner grid, and a tangent-line bound replacing the final Gaussian-integral bound · kinds: own proof, proof route · proposed · no.

DEV-MARKOV-1..4 · IG4 Prop 2.8 (1302.4738 l. 1111–1127), BP Thm 1.52, Sheffield Thm 2.17, GMSH Lemma 2.2 (1807.07511 l. 603–628) · LQGMetric/Field/Markov*.lean · (1) the decomposition as the projection onto $H_0^1(V)$ through the Cameron–Martin isometry (BP’s route) instead of IG4’s series expansion; (2) limits under independence via characteristic functions (textbook argument, own write-up); (3) own write-up of how $h_1(0) = 0$ and $V \cap \partial D = \emptyset$ give independence from $\sigma(h|_{\mathbb{C} \setminus V})$ (GMSH treat it in one sentence); (4) weak harmonicity read off the isometry · kinds: proof route, own write-ups · proposed · no.

DV-CONF-POS1, DV-CONF-L25a · CONF l. 533–600; MS Prop 2.2 · LQGMetric/Papers/CONF/{Lift, LiftGeod, LiftPos, L25}.lean · S-PosLift own elementary proof ($\varphi(t) = z + 1/f(e^{-it})$), orientation via a homotopy-lifted log of $f(w)/w$; CONF’s ”segments before w do not coincide” formalized as ”lifted starts differ: $\neg(P a = Q a \wedge \alpha a = \beta a)$ ”; CONF L2.5’ proved conditionally on TOPO-ORD (open) · kinds: own proof, reading, conditional result · proposed · no.

D-DDDF-PSI-1..6 · DDDF tightness.tex l. 322, 350–490; DZZ Lemma 2.7 (l. 548–575) · LQGMetric/Field/WhiteNoisePsi*.lean, LQGMetric/Papers/DDDF/Psi*.lean · (1) Lemma 4 for ψ by a pointwise split instead of the q_t route; (2) Borell–TIS on $|D_k|$ per box instead of DZZ’s oscillation/grid split (own summation over k); (3) constants: (2.23) with $(\pi t)^{-1}$, (2.24) with a constant C instead of 1; (4) $X_{[a,b]}$ bounded directly on $[0, \max(a,b,1)]^2$; (5) block lemmas for arbitrary measurable disjoint blocks (half-open dyadic blocks); (6) (2.27) with the factor $e^{\{|\xi| \sqrt{(\max(1, \log(C/\epsilon))/c)}\}}$ · kinds: proof routes, constants, conventions · proposed · no.

DV-M2G-1..3 · GM uniqueness-final.tex l. 1609–1716, 1782–1938 · LQGMetric/Papers/GM/S4/{Setup*, Conditional*}.lean · (1) GM.S4.2 stops P at the first hitting time of the closed ball $cl B_r(z)$ instead of GM’s ”first time it enters $B_r(z)$ ” (this is what makes ”does not hit $\partial B_r(z)$ except at x ” true); (2) stabCond reads (4.11)’s ”hits ∂B ” in the same arc” as: all hit points of all such geodesics lie in one arc $\text{arcOf } x_0$; (3) the (4.19) count is done per radius with disjoint grid cells (GM’s per-ball area bound $\epsilon^{\{2+2\nu\}}(\log)^{\{-1\}}$, l. 1815, is loose; the per-radius count gives GM’s stated $\epsilon^{\{-2\nu-\zeta\}}$) · kinds: readings, own route · proposed · no.

DG-B1..B4 · DG arXiv:1807.01072 (Lemma 2.5 l. 736–770, Thm 1.5 l. 336–356, (eqn-exponent-mono) l. 770–779) · LQGMetric/Papers/DG/*.lean · (1) DG’s field h_δ modelled as an `IsGFFCircleAverage` process (a version continuous in z , DS Prop 3.1; exists `isGFFCircleAverage`), paths reparametrized to $[0,1]$; (2) ” $\delta^{\{\lambda+o(1)\}}$ w.p. $\rightarrow 1$ ” read as: for each $\eta > 0$ the outer probability that $\delta^{\{\lambda+\eta\}} \leq X \leq \delta^{\{\lambda-\eta\}}$ fails tends to 0; (3) (1.5b) restricted to bounded connected U and K with two distinct points (as literally stated it fails for one-point K and for K meeting two components of U), its $D(K, \partial U)$ half omitted (unused); (4) DG Lemma 2.5 proved only for $U = (-2,2)^2$, $z = 0$, $w = 1$, $\delta \in (0,1)$; the conditional expectation given h becomes Tonelli on $\Omega \times \Omega$ and the infimum over paths is taken after integrating (no measurable path selection); measurability of the LFPP distance (upper semicontinuity on $C(\mathbb{C}, \mathbb{R})$) own argument; DG’s display (eqn-lfpp-mono) writes D_h where the proof uses $D_{\{h\}}$ — the $\tilde{h}$ version is formalized · kinds: conventions, readings, statement of a cited result (restricted), own arguments · proposed · no.

DV-CONF-214c-b, 214d-b · CONF (arXiv:1905.00381) Lemma 2.14 proof, confluence-final.tex l. 855–894 · LQGMetric/Papers/CONF/L214Final*.lean · (c) the Beurling estimate is applied with $K = \varphi(\partial D) \cup \{p_0 + t : t \geq 0\}$ (p_0 a point of $cl U$ of maximal real part, so the ray lies outside U) instead of $K = \partial U$, so that K leaves $B_{\{Rd\}}$ in every case (Brownian motion from U hits ∂U before the ray: same probabilistic content); J^+ ”symmetrically” realised as $g^+ = g^- \circ (z \mapsto -\bar{z})$; (d) the disconnecting set $\varphi(cl(W^- \cup W^+))$ has diameter $\leq 2Rd$ (CONF: $4Rd$), arc threshold $\ell \leq 1/10$ instead of $\pi/4$, hence $N_0 = 10^6$ in place of ”at most 8” (only universality of the constants matters, D36) · kinds: own adaptation, constants · proposed · no.

BP-DF-1..10 · DFGPS (lqg-metric-estimates-final.tex) and the results it cites: DDDF tightness.tex DD:155–166, 1385–1393, 1498–1506, 1615–1621; LM LM:317–329, 523–531, 284; DF DF:1280–1285, 1344; DG DG:343–346, 1096–1102, 1425–1437 · LQGMetric/Blueprint/DFGPSInputs*.lean, blueprint/DF.md · (1) ”tight”: the random function is continuous on $K \times K$ and its laws on $C(K \times K, \mathbb{R})$ are tight; subsequential limits along $\delta_n \rightarrow 0$; ”bi-Hölder”: some constants and exponents exist; (2) λ_δ computed from a reference white noise; (3) DDDF (1.3) read as a bound

for $\delta \rightarrow 0$; (4) DDDF Prop 29 for bounded D with $\text{cl } U \subset D$ compact, $p_{\{t/2\}^*h}$ the pairing with $p_{\{t/2\}(x-\cdot)}1_D$ (heatBdd); (5) LM results for $U = \mathbb{C}$, the conditionally i.i.d. pair $(D, \bar{D})$ the canonical copy `condCopyMeasure`; (6) LM Def 1.5 with $n = 1$ is `IsXiAdditive2` $\xi \in P \cap D \cap \bar{D}$; (7) DF Lemma 7.1 adapted: metrics on $\mathbb{C}$, GM's `weylScale`, local uniform convergence, DFGPS's localization (T:1040–1048) as a hypothesis, factor $e^{\{2\|\xi\|M\}}$ correcting DFGPS's " $C > \sup\|f^n\|$ " (T:1046), balls in place of squares; (8) the zero-boundary GFF on $\mathbb{S}(1)$ a zero-extended distribution with a circle-average version continuous on $\mathbb{S}$; (9) (super)polynomially high probability as in `DFGPSEstimates`; (10) DG (1.5b) second half for bounded U and nonempty K · kinds: readings, conventions, statement of cited results (adapted) · proposed · no.

DV-J1b-a, DV-J1b-b · Miller–Sheffield arXiv:1506.03806 Prop 2.1 (`mapmaking_final.tex` l. 570–601); GPS arXiv:2010.07889 Lemma 2.4 · `LQGMetric/Papers/GM/S4/JordanJ1b*.lean` · (a) in MS l. 585–601 the exit point of a near-geodesic through two auxiliary circles is used, and the low- D part of those circles is covered by finitely many components, instead of the crossing point of two geodesics γ and η (MS assume without proof that the geodesic must cross a given short geodesic η ; the variant does not need that claim); (b) MS's "it is not hard to see" (l. 582) formalized as boundary bumping + crossing continua (proved) + the sector lemma `GMSector` (open) · kinds: own variant of the published proof, implicit fact made explicit · proposed · no.

P2-GMC-4, P2-GMC-5 · Duplantier–Sheffield arXiv:0808.1560 Prop 1.1, §3.1; `QuantumZipper`'s free-field-on- $\mathbb{H}$ existence argument · `LQGMetric/Dimension/GMCSq*.lean` · (4) the a.s. identification of `avgReg` with the circle average is proved one point at a time (summable L^1 increments + Fatou) instead of QZ's continuous modification; (5) own elementary proofs of the lower Lipschitz bound of the conformal map of the square (integrated difference quotient) and of the Lipschitz bound for the harmonic part h_S of the Green function · kinds: own adaptation, own elementary proofs · proposed · no.

DV-E1 · CONF l. 477 / GM l. 854 "stopping time for the filtration generated by $(\mathcal{B}^\bullet_s, h|_{\{\mathcal{B}^\bullet_s\}})$ " · `Blueprint.filledBallSigma` t becomes the generated filtration $\bigsqcup_{s \leq t} \sigma(\mathcal{B}^\bullet_s, h|_{\{\mathcal{B}^\bullet_s\}})$ (was $\sigma(\mathcal{B}^\bullet_t, h|_{\{\mathcal{B}^\bullet_t\}})$ alone, a filtration only a.s.); `IsFilledBallStoppingTime/IsClosedBallStoppingTime` ask $\{\tau < t\} \in \mathcal{F}_t$ (optional time, i.e. stopping time of the right-continuous filtration $\mathcal{F}_{\{t+\}}$, the usual convention) instead of $\{\tau \leq t\} \in \sigma(\mathcal{B}^\bullet_t, h|_{\{\mathcal{B}^\bullet_t\}})$ · kinds: definition change (convention) · proposed (D49) · yes: `Blueprint` defs in `M2Defs` and the hypotheses of `CONFLem2_1`, `CONFLem3_6At`, `CONFThm3_9At` (texts unchanged).

DV-RECTX · Maehara, Amer. Math. Monthly 91 (1984) 641–643 (lemma); Kulpa, Amer. Math. Monthly 104 (1997) 545–550 (2-D Poincaré–Miranda) · `LQGMetric/Topo/RectMeet.lean` (`RectMeet.rect_crossings_meet`) · the published proofs use the Brouwer fixed point theorem, which is not in `mathlib` at our pin; we use the standard winding-number proof with `QuantumZipper`'s `loopDeg` (homotopy invariance, degree-1 circle): if $\gamma_s - \delta t$ never vanished, the boundary loop of the square would have degree 0 (contract the square) and degree 1 (edge-wise half-plane homotopy to $e^{2\pi i u}$) · kind: does not follow the published proof (own write-up of the standard argument, ≈ 200 lines) · proposed · no.

DV-BMT · DZZ Lemma 2.10 first moment, D-A1 (DIM.S-chi-le-2) · `LQGMetric/Papers/DG/BallMass.lean` · the a.e.-measurability of $\omega \mapsto \mu_{\{h(\omega)\}}(B(x, \rho))$, used implicitly in Markov's inequality, is proved by an own elementary argument: monotone convergence over the cutoffs $\min(1, n \cdot \text{dist}(\cdot, B^c))$ plus `QuantumZipper`'s measurable limit formula `Prop16Area.Meas.Psi` on the a.s. vague-limit event · kind: implicit fact, own argument (cost rule) · proposed · no.

D-DDDF-15, D-DDDF-16, D-DDDF-17 · DDDF arXiv:1904.08021 `tightness.tex` 1494–1601 (Prop 29, (6.96)); Feller II §X.5 (method of images) · `LQGMetric/Field/HeatKernelSquare*.lean`, `LQGMetric/Papers/DDDF/P29*.lean` · (15) the Brownian-bridge estimate behind (6.96) is replaced by the explicit image-series Dirichlet heat kernel of the square (D-DDDF-11); the fold identity, the image-kernel Chapman–Kolmogorov, and the decay, Lipschitz and L^1 bounds are own elementary proofs; (16) DDDF use (6.96) for y near ∂D (l. 1568), outside its stated range $x, y \in U$ (l. 1561), where it fails; fixed by a near/far split with the L^1 bound (`abs_firstKer_le`); (17) the second-term time split is at $|x-x'|$ instead of $\sqrt{|x-x'|}$ (which gives only $O(\sqrt{|x-x'|})$), giving the stated $O(|x-x'|)$; the third term's increment bound ("Similarly") uses the Lipschitz property of $1_D \cdot p^\Delta$ from the zero boundary values · kinds: own route, gap fixed, own argument · proposed · no.

DV-E0 (D49 applied) · CONF l. 477, GM l. 854 · `LQGMetric/Blueprint/M2Defs.lean` (`filledBallSigma`, `IsFilledBallStoppingTime`, `IsClosedBallStoppingTime`), `LQGMetric/Papers/GM/S4/Setup.lean` (`gm_stop_mul`, `gm_filledBallSigma_mono`, `gm_S4_12` without the monotonicity hypothesis) · applied as decided in DV-E1/D49 · proposed · no.

D-DDDF-L6-1 · DDDF L6 coupling (`tightness.tex` 541–543, DF 385–395: "the rest of the white noises are chosen to be independent") · `LQGMetric/Field/WhiteNoisePushCoupling.lean` (`coupledNoise`, `isWhiteNoise_coupledNoise`), `LQGMetric/Papers/DDDF/L6Decomp.lean` · $\tilde{W}(g) := W(Tg) + W'(g \mathbb{1}_{\{(0, \infty) \times V^c\}})$ with a second white noise W' on the same space, independent of W (hypothesis `IndepFun W W'`); a space carrying only W is enlarged by a product to carry W' . The paper's joint independence of $W|_{U^c}$, $W|_U$, $\tilde{W}|_{V^c}$ follows (disjoint supports + $W \perp W'$). Also on times $t \leq 0$ (WN-1) $\tilde{W}$ takes its noise from W' , which is never seen by $\tilde{\varphi}$ · reading · pro-

posed (P2-DDDFL6). **D-DDDF-L6-2** · DDDF L6 sign/labels · $\varphi_H := -\varphi_3$ and $\varphi_L := -(\varphi_1 + \varphi_2)$ (DDDF write $\varphi_\delta - \tilde{\varphi}_\delta \circ F = \varphi_1 + \varphi_2 + \varphi_3$ but state (eq:Decompo) for $\tilde{\varphi}_\delta \circ F - \varphi_\delta$); the split of $T\tilde{k}$ at $t = \delta^2$ is done by restriction of the kernel (equal to DDDF's $\int_{\{\delta^2|F'|^{-2}\}^{\wedge}\{\delta^2\}}$ since $T\tilde{k}$ vanishes for $t < \delta^2|F'(y)|^{-2}$) · reading · proposed (P2-DDDFL6).

DV-E1-b · GM l. 854, CONF l. 477 (stopping-time claims without proof) · LQGMetric/Papers/GM/S4/SetupStop.lean · that τ_R is a filled-ball stopping time is proved surely (`gm_tauR_isStop`) under D49's generated filtration and optional times; closedness of $\mathcal{B}^\bullet$'s and the rational-time description of $\{\tau_R < t\}$ are own elementary proofs; the hit-of-a-closed-set event follows the standard Effros/Fell argument (Molchanov, Theory of Random Sets, §1.1.1) · kind: paper gives no proof (own elementary argument) · proposed · no.

DEV-MKA-1, DEV-MKA-2 · Sheffield math/0312099 §2.2; Hörmander ALPDO I §4.1 · LQGMetric/Field/MarkovAdm*.lean, MarkovWeyl*.lean · (1) admissibility on unbounded V with $V \cap \partial\mathbb{D} = \emptyset$: split V at the unit circle, map the exterior piece into the disc by $z \mapsto z^{-1}$ and push the density forward (own elementary proof; no published proof of this exact statement found); (2) distributions commute with integrals of test functions by factoring through finite jets with Hahn–Banach, not Schwartz's Riemann-sum proof (as in `CircleAvgIntegral`) · kinds: own proofs · proposed · no.

GA-6h, GA-6i, GA-6j · GM uniqueness-final.tex l. 1320–1404 (Lemmas 3.7–3.8) · LQGMetric/Papers/GM/S3/GoodAnnuLus{L38Trans,L38,Meas,Meas2,Meas3}.lean · (h) L3.7 and L3.8 take `DFGPSLem3_8` (geodesic existence, used implicitly by GM) as an extra hypothesis (D50); (i) in L3.8 the scales are enumerated with `Nat.nth`, index 0 not counted, indices past the last selected scale padded with trivial events, $K_0 = \max(c, 1) \cdot e^a$, case $N = 1$ separately; (j) own elementary arguments for GM's "obvious" locality steps l. 1348–1358 (sphere-to-sphere distance and distance to ∂U through the internal metric, geodesics as internal geodesics, midpoint sets) · kinds: additional cited hypothesis, bookkeeping, own proofs · proposed · no.

MKD-i, MKD-ii · Sheffield math/0312099 §2.6 Thm 2.17 (tex l. 690–725); IG4 Prop 2.8 (1302.4738 l. 1110–1127); BP Thm 1.52, Lemma 1.53 (2404.16642) · LQGMetric/Field/MarkovGermVer*.lean · (i) germ measurability of the harmonic part ($LM\ L2.1 = GMSH\ L2.2$) through the Hilbert-space reduction $\text{gaussSpace} \ominus S(B_\varepsilon(V^c)) \subseteq H_0^1(V)$; the cited sources assert the measurability without proof, the argument is our own following Sheffield §2.6; (ii) `gradEnergy_cut` from the weak identity via the covariance symmetry instead of the pointwise $\Delta u_\psi = -2\pi\psi$ · kinds: own argument, proof route · proposed · no.

DFA0-1..3 · DFGPS T:1446–1508, 1783–1841 · LQGMetric/Papers/DFGPS/{L3_4Tail,L3_4,L3_11,L3_2}.lean · (1) L3.4 takes the supremum over r in an arbitrary countable $S \subseteq [\varepsilon^{\wedge}\{1+\nu\}r, \varepsilon r]$ (`circleAvg` is the a.s.-defined limit, not a continuous version; all uses involve countably many radii); (2) L3.4 with an explicit constant for all $\varepsilon \in (0, 1)$, allowing $\nu = 0$ (used at T:1803, 1818) and only $q > 0$ (stronger statement); (3) L3.11's optimization over a partition $\{\alpha_k\}$ done per n by completing the square; (3.4) printed with $h_r(0)$ is read as $h_r(z)$ (its use at T:1542 requires it) · kinds: reading, parameter range, proof route, misprint · proposed · no.

DEV-DFA4-1..4 · DFGPS T:1659–1723 (Thm 1.5 proof, Steps 1–3) · LQGMetric/Papers/DFGPS/T1_5{Disc,Chain,Trans}.lean · (1) Step 1's loops around square annuli (`eqn-square-around`, node S9) replaced by one horizontal and one vertical rectangle crossing per vertex, which meet by `RECT-X` (the blueprint's S9 idea with two rectangles instead of four); (2) Step 3's exit neighbourhood a sup-norm box of half-side $3s/2$, next vertex = exit point rounded toward the current vertex, annulus crossed Euclidean $3s/4 \rightarrow 3s/2$ inside $B_{\{2s\}}$ (the paper's $B_{\{\delta r\}}(S_z)$ makes the rounding ambiguous on its boundary); (3) the sum $\sum_{j=0}^J$ (T:1712–1714) has one more term than crossings: the path runs on to $\text{Re} \geq r+2s$ so each vertex is followed by a full crossing; misprints T:1686 (exponent $\pi(j)$) and T:1721 (`(eqn-lfpp-upper)`) read as in `DFGPS.md`; (4) own elementary arguments: rounding to grid neighbours, sub-crossing extraction, time step from uniform continuity · kinds: proof route, reading, misprints, own arguments · proposed · no.

DEV-DFGPS-B2a..B2c · DFGPS T:688–799 (Lemmas 2.1–2.3); Hu–Miller–Peres Prop 2.1; Adler–Taylor Thm 2.1.1 · LQGMetric/Papers/DFGPS/{L2_3Tail,L2_3,L2_2Small,L2_2,L2_1Ratio}.lean · (a) L2.3: the bound uniform in k from the scale-invariant covariance bound `incCov_self_le` instead of scale invariance of the law; blocks centred at $H(2^k, 0)$, with $h_{\{2^k\}}(0) - h_1(0)$ handled by Gaussian marginals + Borel–Cantelli instead of Brownian-motion growth; supremum over B_R instead of $B_{\{R \cdot 2^k\}}$; Borell–TIS (Adler–Taylor) replaces Maurey–Pisier, so DEV-DFGPS-2 no longer applies; (b) L2.1–L2.3 stated for a jointly continuous version H of the circle-average process (`IsCircleAvgVersion`, existence `exists_isCircleAvgVersion`); (c) " $\hat{D}^\varepsilon/D^\varepsilon \rightarrow 1$ uniformly" in L2.1 stated as $c^{-1}\hat{D}^\varepsilon \leq \hat{D}^\varepsilon \leq cD^\varepsilon$ for every $c > 1$ and all small ε (meaningful also where distances are 0 or ∞) · kinds: proof route, readings · proposed · no.

DEV-L36-1..4 · DFGPS T:1628–1650 (Lemma 3.6, S11); DG Prop 3.16 (DG:1447–1452) · LQGMetric/Papers/DFGPS/L36*.lean · (1) whole-plane circle averages compared directly, without the white-noise field (only a continuity estimate is needed from DG Lemma 3.7; `LQGDimension` proves it for the whole-plane field); (2) the S11 lower bound with $K = [0, 1/4] \times [0, 1]$, $U = (-1, 1/2) \times (-1, 2)$, cutting the polygon at its first crossing of $\text{Re } z = 1/2$ (own elementary argument); (3) the "equal in law" step done pathwise (a.s. identity on the countable grid + measurability of the graph distance); (4) the upper half is open: DG Thm 1.5 (1.5b) has no rate, so paths from a leftmost vertex (within

δ of the side) cannot be joined to the sides as T:1645–1647 claim; decision pending (D52) · kinds: proof route, own argument, gap in the source · proposed · no.

DV-GMSEC · Miller–Sheffield mapmaking_final.tex l. 582 (“it is not hard to see”; no written proof); Newman Ch. V not available · LQGMetric/Topo/Sector*.lean (GM.gmSector) · own argument from sourced tools: each crossing continuum carries a continuous log of $y - x$ (Eilenberg, Burckel Ex. 4.37), the three continua lift under $x + e^w$ to disjoint crossers of $[\log r_1, \log r_2] \times [-R, R]$, straight segments in log coordinates and lifted paths in the component containing W give connected sets avoiding one lift and containing the other two, contradicting the ordering of three disjoint crossers, which follows from the continuum crossing lemma (from RECT-X by δ -thickenings and clamping) · kind: own argument (no source at hand) · proposed · no.

E3a-1, E3a-2 · GM uniqueness-final.tex l. 1682–1716 (Lemma 4.6); Kechris Thm 14.7 (Lusin separation) · LQGMetric/Papers/GM/S4/L46Meas{Det,*}.lean · (1) “ $\text{dist}(z, \partial \mathcal{B}^\bullet) \geq \lambda_4 \varepsilon \Rightarrow B_\rho(z) \cap \mathcal{B}^\bullet = \emptyset$ ” own elementary proof (a segment from z into the hull crosses its boundary); (2) GM’s “determined by $h|_{\{\mathbb{C} \setminus B_r(z)\}}$ ” made precise through Lusin separation on a Borel set carrying the law instead of an explicit measurable function (own argument; needs the measurability inputs L, hB); the ball is $B_{\{\lambda_3 r\}}(z)$ with general $\lambda_3 \leq \lambda_4$ ($\lambda_3 = 1$ would need $\lambda_4 \geq 1$) · kinds: own arguments, reading · proposed · no.

DEV-DFGPS-L35 · DFGPS Lemma 3.5 (T:1573–1596) and Prop 3.1 (T:1414–1420) · LQGMetric/Papers/DFGPS/L3_5.lean, LQGMetric/Papers/DFGPS/Nodes.lean (Prop3_1) · (a) U is assumed connected in L3.5 and in the node statement DFGPS.Prop3_1: Prop 3.1 says only “open”, but its proof uses connectedness (“since U is connected”, T:1580) and the statement is false without it (K_1, K_2 in different components give infinite distance); (b) L3.5 drops the paper’s assumptions that U is bounded and $K_1 \cap K_2 = \emptyset$ (not needed); ε_0 depends on M (implicit in the paper); (c) the conclusion is strengthened: every circle used has $\text{closedBall } w(2r) \subseteq rU$ (Prop 3.1’s upper bound needs $A_{\{r/2, 2r\}}(w) \subseteq rU$; the paper’s choice of balls does not ensure it literally; here it holds because the balls meet the fixed path $r\gamma_0$ and $3\varepsilon < \text{dist}(\gamma_0, \partial U)$); (d) the event F^{ε_r} is abstracted as `GoodCover good 1 M  $\varepsilon$  r` · kinds: reading (hypothesis the proof uses), dropped assumptions, strengthening · proposed · no.

DV-E2 · GM uniqueness-final.tex l. 1654, 1660–1676 (Lemma 4.5) · LQGMetric/Papers/GM/S4/L45Meas.lean · GM Lemma 4.5 in σ -algebra form is proved modulo three explicit “determined by $(\mathcal{B}^\bullet|_{\{t_k\}}, h|_{\{\mathcal{B}^\bullet|_{\{t_k\}}\}})$ ” hypotheses (GMarcFamDet, GMGeodSelDet, GMConfPtSel) that GM use without proof (l. 1654, 1665–1675; pending P2-LOCMEAS); the arc is encoded by its hit events on open sets; $\mathcal{F}_k$ uses the product σ -algebra on paths $[0, 1] \rightarrow \mathbb{C}$ (= Borel on continuous paths); GMConfPtSel uses a deterministic countable family of opens (a sufficient form of “the arc determines the point”) · kinds: implicit facts made explicit (open), reading · proposed · no.

KHK-1.3 · DZZ arXiv:1807.00422 LBM_LGDaXiv.tex l. 400–411, 437–441, 467–469, 491–499; Berestycki–Powell §1.2; Revuz–Yor Ch. I §3; Karatzas–Shreve §5.6.B · LQGMetric/Field/KilledHeat*.lean · (1) p_A defined by DZZ’s bridge formula with a canonical planar BM; DZZ’s density definition (eq:heat_kernel) proved as a theorem (D53); (2) the reflection principle for the bridge maximum proved by Doob’s time change plus QZ’s drift-hitting law instead of the strong-Markov reflection DZZ cite; (3) DZZ’s constant C in the bridge-maximum bound taken as 3 · kinds: definition (pointwise version), proof route, constant · proposed · no.

D-DDDF-L6-3..5 · DDDF arXiv:1904.08021 tightness.tex l. 539–646 (Lemma 6, “Third term” l. 641–645); DF Lemma 4.4 · LQGMetric/Papers/DDDF/L6Var{H,L,G}.lean · (3) in the φ_H variance bound DDDF’s $|F x - F y| \geq |x - y|/C$ is replaced by the change of variables $z = F(y)$ ($dz = |F'|^2 dy$) and integrating the Gaussian over $\mathbb{C}$: only $1 \leq |F'| \leq M$ is needed and $\text{Var } \varphi_H \leq M^4/2$ for all x and $\delta > 0$; (4) pointwise variance bound for the W' part $\varphi_{\{2,2\}}, \|g_{\text{Ker } x}\| \leq 1/(\pi d(F(K), V^c)^2)$, via $e^{\wedge\{-v\}} \leq 1/v$ (DDDF only treat its increments); (5) φ_L ’s increment and variance bounded by $2\pi(\|W\text{-kernel}\|^2 + \|W'\text{-kernel}\|^2)$ via $(a+b)^2 \leq 2a^2 + 2b^2$ instead of independence · kinds: own variant, own elementary argument, route · proposed · no.

DEV-DFGPS-B1a..c · DFGPS T:837–866 (L2.6, L2.7), T:971–988 (L2.11) · LQGMetric/Papers/DFGPS/{L2_6Path,L2_6,L2_6Law,L2_7,L2_11}.lean · (a) L2.6 formalized for a whole-plane GFF as an a.s. identity with $h^r = \text{affineComp } r \circ h$ (as DFGPS state it); the plan’s node (every $h, \text{affineComp } r^{-1}$) was false and is removed (D54); (b) the “standard change of variables” $h^{\wedge\{r,\}_\varepsilon/r\}}(z) = h_\varepsilon(rz)$ needs, with the truncated definition of h^*_ε , an a.s. cutoff-independence step: own elementary proof (Gaussian decay of the truncation difference via `ae_summable_abs_of_decay` applied to $h(a\cdot)$, extended from a countable dense set by continuity); (c) L2.11 stated for $U = \mathbb{C}$ (the only case DFGPS use, T:1013, 1018), L2.7 with “metric on X ” read as continuous; both re-export Metric/LengthLimit and Metric/InternalLimitC with weaker hypotheses · kinds: statement correction (to the paper’s form), own argument, readings · proposed · no.

DV-CONF-TO1 · CONF arXiv:1905.00381 C:598–600 (“paths with reversed boundary order must intersect”, in the universal cover $\mathbb{R} \times [0, 1]$ of the annulus); MS Prop 2.2 · LQGMetric/Papers/CONF/TopoOrd*.lean (topoOrd, conf_L25) · own formalization of the strip argument: the inner boundary is straightened by a log chart built from the Carathéodory map of the inverted exterior of K_1 (generic to_disc_map; Pommerenke Thm 2.6 via JordanMap.jm_jordan_closedDisc_extension'), the outward rays come from K_2 ’s chart, two rectangle crossings meet by RECT-X; supporting facts (K and $K \setminus \{z\}$ connected, ∂K a Jordan curve via $\varphi \circ \arg$, chart properties, boundary-order mono-

tonicity) are own elementary proofs; CONF's covering exists only via the annulus theorem, the chart replaces it (no Schoenflies or Jordan curve theorem needed) · kind: published proof not followed (own argument) · proposed · no.

DV-MKH-1 · Weyl's lemma (leaf H of LM L2.1 / GMSH L2.2; cited, not proved, in the sources); Hörmander ALPDO I Thm 4.4.1; Evans PDE §2.2.3 · LQGMetric/Field/MarkovWeyl2*.lean · formalized by Hörmander's mollification argument in mean-value form; the radial step (radial tests of equal mass differ by a Laplacian) from Newton's theorem via the mean-value property of $\log\|w - \cdot\|$; Δ of the log potential by differentiating the convolution twice; own radial mollifiers rbump (mathlib's ContDiffBump is not provably radial) · kind: standard textbook argument for a cited fact · proposed · no.

DEV-DFGPS-L21a..e · DFGPS T:666, T:688–738 (Lemma 2.1) · LQGMetric/Papers/DFGPS/L2_1{Cont,Bdd,Tail,}.lean · (a) bounded continuous part via the explicit bound $|f^* - \hat{f}^*| \leq 2M e^{-1/(8\varepsilon)}$ ($p_{\{\varepsilon^2/2\}} \leq 2e^{-1/(8\varepsilon)}$ $p_{\{\varepsilon^2\}}$ off $B_{\{\sqrt{\varepsilon/2}\}}(z)$; the paper says "similarly"); (b) tail estimate with $\max\{A \log(1/r), \log r, 1\} \leq A/r + r + 1$ and weight $\leq e^{-1/(8\varepsilon)} e^{-r^2/2}$ for $\varepsilon \leq 1$ (paper: "exponentially fast"); (c) joint continuity in (ε, z) proved for every distribution (no continuous modification needed); (d) DEFINITION ISSUE (being fixed by P2-DFB2c): LFPP. locBump uses mathlib's ContDiffBump, whose profile someContDiffBumpBase C is a Classical. choice and not provably radial, while DFGPS T:666 require a radial ψ ; fix: build ψ_ε from ContDiffBumpBase. ofInnerProductSpace C; (e) the open node Lem2_1GffApprox includes existence of the limit defining $h^*_\varepsilon(z)$, which the paper leaves implicit · kinds: own proofs, reading, definition fix, implicit fact · proposed · no.

DEV-DF71-a..c · Dubédat–Falconet arXiv:1809.02607 Lemma 7.1 and proof (LiouvilleMetricStarScale.tex DF:1288–1344); DFGPS L2.12 localization T:1036–1048 · LQGMetric/Papers/DFGPS/DFLem71*.lean (dfLem7_1 : Blueprint.DFLem7_1) · (a) DF cut geodesics by repeated midpoints, which needs exact geodesics ("strictly intrinsic", DF:1297); we cut a near-optimal path of GM (1.6) at $P(iL/N)$, no exact geodesics needed; (b) DF's lower bound passes to a subsequence and a limit of chains; here path lengths are uniformly bounded, so N is uniform over $\bar{B}_R(0)^2$ and the same two estimates give local uniform convergence directly (one lemma dfL71_sym for both directions); (c) DF:1305 ($r^{-1}(R(\cdot))$ keeps short paths in a compact set) replaced by the positive D-distance from $\bar{B}_{\{r'\}}(0)$ to $\partial B_{\{r'+1\}}(0)$ plus the crossing lemma (own elementary argument, ≈ 30 lines) · kinds: proof routes, own argument · proposed · no.

DEV-DFGPS-T15-a..c · DFGPS T:1659–1723 (Thm 1.5 proof), T:1414–1420 (Prop 3.1) · LQGMetric/Papers/DFGPS/T1_5*.lean (dfgpsScaling_of_nodes) · (a) "by Axiom V" in Steps 2–3 read as Prop 3.1 at centre 0 for two boxes with segments as K_1, K_2 (Step 2, lower bound: sides $\{0\} \times [-r, 2r], \{r\} \times [-r, 2r]$ in $(-2r, 3r)^2$; Step 3, upper bound: $\{0\} \times [3r/8, 5r/8], \{5r/4\} \times [3r/8, 5r/8]$ in $(-r/8, 11r/8) \times (r/4, 3r/4)$); (b) parameters $\zeta = \min(\zeta, 1)/4$, $A = \delta^{\{-\zeta\}}$, $p = 3/\zeta$, $\eta = 1/4$ in L3.6, union bound over $\leq \delta^{\{-2\}}$ grid points; "sending $\zeta \rightarrow 0$ " replaced by $\delta^{\{\zeta/4\}} \leq 1/8$; (c) uniformity of Prop 3.1 in the centre from translation invariance of the law proved on the canonical space via transField and a.s. identities · kinds: reading, parameter choice, own argument · proposed · no.

KAHANE-1..3 · Kahane, Ann. Sci. Math. Québec 9 (1985); Rhodes–Vargas arXiv:1305.6221 Thm 2.1; Berestycki–Powell arXiv:2404.16642 Thm 3.18 · LQGMetric/Gaussian/Kahane*.lean · (1) interpolation path $\cos \theta \cdot X + \sin \theta \cdot Y$ ($= \sqrt{t} Y + \sqrt{1-t} X$ with $t = \sin^2 \theta$); (2) the function class: F of class C^2 on $(0, \infty)$ with $|F|, |F'|, |F''| \leq C(y+y^{-1})^k$ (covers y^q , $q \notin (0, 1)$) instead of general convex F of polynomial growth; centred vectors, strictly positive weights; (3) own elementary bounds $y^a \leq (y+y^{-1})^k$ for $|a| \leq k$ and the integrand bounds · kinds: parametrisation, statement of a cited result (restricted class), own proofs · proposed · no.

D-DDDF-14 (supplement), D-DDDF-14b · DDDF tightness.tex l. 744–760 (Lemma 12), l. 545 (Lemma 6 hypotheses); DF LiouvilleMetricStarScale.tex l. 640–660 · LQGMetric/Papers/DDDF/L12*.lean (L12.lemma12') · (14 suppl.) $K = \alpha(G(\bar{D}_\rho))$ with $\alpha(w) = (w_0/2 + iH/2) + i(w_0/\pi)w$; marked pieces $\alpha(G(\rho e^{\pm i\theta}))$ for θ in m equal-ANGLE subintervals of $[\theta_1, \pi - \theta_1]$ (DF: "subarcs of, say, equal length"); $M_{\{ij\}} = \rho \cdot \text{mob}(\cdot/\rho)$ with an explicit Möbius family; $U = \alpha(G(\bar{D}_{\{r_1\}}) \cap \text{strip})$ with derivative bounds by compactness instead of Schwarz reflection; the hypothesis $a/b < 1 < a'/b'$ is not needed (stronger statement); own elementary arguments throughout, as D-B4 prescribes; (14b) L12' does not provide convexity of U or K , which DDDF Lemma 6 assumes (l. 545); the final L6 statement is to be stated without convexity where possible (P2-DDDFL6C informed) · kinds: own construction, open reconciliation · proposed · no.

GA-S3.3, GA-5 (applied), GA-S3.7, GA-3.17 · GM uniqueness-final.tex l. 1222 (footnote), 1263–1309, 1440–1499 · LQGMetric/Papers/GM/S3/Attained*.lean · (S3.3) "the event (A) is contained in $\bar{G}_r(C', 1-\alpha)$ " is false as written ($v \in \partial B_r(0)$ is not in the open ball): P3.4 is applied with $C_1 = (C' + C_-)/2$ and v moved radially inward by continuity, $\beta = (1-\alpha_-)/2$; (GA-5) in S3.5, $\#\mathcal{K} \geq v \log_\varepsilon^{-1} - 1$ replaces $\lfloor v \log_\varepsilon^{-1} \rfloor$ and (B) counts $(v-\mu)/2 \cdot \log_\varepsilon^{-1}$, with $\varepsilon \leq 8^{\{-2/(v-\mu)\}}$; (S3.7) GM's re-routing argument in midpoint-set form; (3.17) GM's sequence t_j and indices $J, \bar{J}$ replaced by an induction on the number of steps with explicit end errors ω (restructuring of GM's argument); the footnote statement S3_2fn needs β uniform over all GFFs (law uniqueness mod constants, prob_eq_of_ae_addConst_iff, not mentioned by GM) · kinds: misprint/false step fixed, bookkeeping, restructuring · proposed · no.

MKD2-1..3 · Sheffield math/0312099 §2.6 Thm 2.17; IG4 Prop 2.8; BP Thm 1.52, Lemma 1.53, §1.8; Folland Real

Analysis Thm 6.18 · LQGMetric/Field/MarkovGermVer2*.lean · (1) decay $|u_\psi(x)| \leq C/|x|$ for mean-zero ψ and the $O(R^{-2})$ cutoff limit: own elementary proof; (2) Schur bound $\log \text{Cov } p \leq 2L_R \int p^2$ for the log kernel on $\bar{B}_R$ (Schur test via Tonelli): own elementary proof; (3) the germ step (b) for bounded V follows the cutoff argument of Sheffield Thm 2.17 / BP Lemma 1.53, but instead of L^2 limits of the CM functions the weighted mean $c = \int \zeta^2 f / \int \zeta^2$ makes $\zeta^2(f-c)$ mean-zero, and the covariance-Poincaré bound $\int \zeta^2(f-c)^2 \leq e^2 L$ and a Caccioppoli-type bound are derived directly from orthogonality (own arrangement of the published argument) · kinds: own proofs, own arrangement · proposed · no.

DV-M2H-1..4 · GM uniqueness-final.tex l. 1948–1990 (Def of $\mathcal{E}_r$, Remark 4.10, Lemma 4.11), l. 2005–2010 ((4.36)) · LQGMetric/Papers/GM/S4/Regularity*.lean · (1) the upper bound in condition 3 only for $z \neq w$ (the internal metric over the empty ball is ∞); (2) condition 5 with $\epsilon \in (0, a] \cap \{2^{-n}\}$ (GM prints “ $(0, ar] \cap \{2^{-n}\}r$ ”, typo); (3) condition 6 with the grid $(\epsilon r/4)\mathbb{Z}^2$ of GM L2.13 (CONF L3.5) instead of the printed $(\epsilon^{1+\nu}r/4)\mathbb{Z}^2$ (condition 6 is used only through Remark 4.10, and L2.13 controls only that grid); (4) condition 1 via GMS2_4e with V a ball (GM cite “Axiom V tightness”); DV-B4 implemented as condition 7, so the L4.11 union bound uses $(1-p)/7$ instead of $(1-p)/6$ · kinds: readings, typo, proof route · proposed · no.

KHSQ-1..3 · Feller II §X.5; Karatzas–Shreve Problem 2.8.8; Berestycki–Powell §1.2 · LQGMetric/Field/HeatKernelSquareGreen*.lean · (1) square-summability of the sine coefficients from the heat kernel’s L^1 bound and $s \rightarrow 0$ instead of Bessel’s inequality (own elementary argument); (2) 1D trigonometric orthogonality directly from the fundamental theorem of calculus (own elementary proof); (3) the Green identity on the square (P29 R1) via the Dirichlet eigenfunction expansion instead of the potential $u_\sigma = \pi \int P_\sigma ds$ (equivalent; avoids the regularity of u_σ) · kinds: own proofs, proof route · proposed · no.

MKH2-1, MKH2-2 · Hörmander ALPDO I Thms 4.1.4, 4.4.1; Berestycki–Powell definitionGFF.tex l. 842–862 · LQGMetric/Field/MarkovWeyl3.lean, MarkovVer*.lean · (1) $T(\varphi * \rho_\epsilon) \rightarrow T\varphi$ via $T(\varphi * \rho_\epsilon) = \int \rho_\epsilon(t) T(\varphi(\cdot - t)) dt$ and continuity of $t \mapsto T(\varphi(\cdot - t))$ (translation clamped by $\text{proj } B$) instead of convergence $\varphi * \rho_\epsilon \rightarrow \varphi$ in $\mathcal{D}$ (equivalent standard argument); (2) null-set modification making the version vanish off $\text{cl } V$, and continuity in probability via the L^2 projection bound plus Gaussian variance convergence: own elementary arguments · kinds: own steps · proposed · no.

LOCMEAS-a..c · GM uniqueness-final.tex l. 1345–1360 (L3.7 “hence determined by $h|_U$ ”); Kechris Thm 14.7; BBI Thm 2.4.16(2); LM Lemma 1.1 · LQGMetric/Meas/LocalEvent*.lean, LQGMetric/Papers/GM/S3/GoodAnnulusMeas4*.lean · (a) the “determined by $h|_U$ ” step of GM L3.7 is our own descriptive-set-theory argument (D51): Lusin separation on a Borel set ($\text{LocalEvent.lensSet}$ of boundedly compact length metrics, a.s. via DFGPS Lem3_8) carrying the law; GM give no proof; (b) paths $[a, b] \rightarrow \mathbb{C}$ replaced by $C([0, 1], \mathbb{C})$ through affine reparametrization; path length lower semicontinuous; “range η disconnects” a closed condition (own proofs); (c) condition 2 used in its Borel form gaLongB with chainInf in place of the internal metric (equal for length metrics) · kinds: own argument for an unproved step, own proofs · proposed · no.

DEV-DFB34-1..3 · DFGPS T:872–898 (Lemma 2.8); LM Lemma 2.1; Berestycki–Powell §1.2 · LQGMetric/Papers/DFGPS/{MarkovNorm, L2_8Couple, L2_8}.lean · (1) the Markov property at $h_r(z) = 0$ derived from LM L2.1 by scaling and translation (σ -algebra and thickening bookkeeping own elementary proof); (2) L2.8 read as in BP-DF-1: limits along $\epsilon_n \rightarrow 0$, “induces the Euclidean topology” encoded as a metric continuous on the compact $S \times S$ (equivalent); (3) the zero-extended $\hat{h}$ is the extension zbBdd of $\hat{h}|_V$ to bounded densities (BP-DF-8); the coupling recentres h by $h_r(z)$, a random constant the paper absorbs in “plus a continuous function” · kinds: own bookkeeping, readings · proposed · no.

KAHANE2-a, KAHANE2-b · Berestycki–Powell 2404.16642 Thms 3.21, 3.23 (GMCproperties.tex l. 1200–1350); Rhodes–Vargas Thm 2.11; Ahlfors §4.3.4 · LQGMetric/Gaussian/Kahane2Const.lean, LQGMetric/Dimension/GMCMomentPos*.lean · (a) the Green comparison $G_\vee \leq G_\mathbb{H}$ ($h_S \leq \log 3$) by Schwarz–Pick for the inverse conformal map instead of the maximum principle (own route; the sources take it as standard); (b) Kahane with an additive covariance constant realized on the product space $P' \otimes N(0, c)$ (modelling choice) · kinds: own route, modelling · proposed · no.

DEV-DFGPS-P39a..c, P310a..b · DFGPS T:1766–1775 (P3.9), T:1871–1916 (P3.10) · LQGMetric/Papers/DFGPS/P3_{9,10}*.lean · (P39a) for $p < 0$, Prop 3.1’s lower bound with $U = \mathbb{C}$ and K_1, K_2 closed balls around two points of K (P3.9 does not assume U connected, P3.1 needs it): reading of “follows from the lower bound of Proposition 3.1”; (P39b) chaining the covering squares through a connected K ($\text{internalDiam_le_two_sum}$): own elementary argument the paper leaves implicit; (P39c) $c_{\{r\}}$ vs c_r at fixed ρ via Axiom V’s $c(\delta r)/c(r) \leq \Lambda \delta^{\{-\Lambda\}}$ instead of Thm 1.5 (both constant-order); (P310a) the layer-cake integration of T:1914 in dyadic form with measurable hulls (no measurability of internalDiam), uniformity over realizations via the canonical space and law uniqueness; (P310b) Step 1’s rectangle geometry must change: the paper applies Prop 3.1 with K_1, K_2 the short sides of an open rectangle R , which are not inside R ($K \subseteq U$ fails) — to be decided with diamTailRS_of · kinds: reading, own arguments, route, open geometry fix · proposed · no.

DEV-DFGPS-P31a..d · DFGPS Prop 3.1 proof T:1522–1568; Lemma 3.2 T:1452–1456; (1.2) T:336 · LQGMetric/

Papers/DFGPS/P3_1*.lean, L3_2Node.lean · (a) "assume U bounded" (T:1525, 1566) made concrete: U_0 a thickening of $K_1 \cup \gamma_0 \cup K_2$ (γ_0 a path in U), bounded, connected, open, $U_0 \subseteq U$; lower bound directly in rU , upper bound in rU_0 then monotonicity (own construction); (b) "by the triangle inequality" over the circles (T:1550) as a chaining lemma: a connected set covered by finitely many closed sets of diameter $\leq L$ has diameter $\leq (\#sets+1)L$ (own proof); (c) the circle count bounded by $(2(8R+1)^2+1)\varepsilon^{\{-6\}}$ instead of $\varepsilon^{\{-4-o(1)\}}$ (any polynomial bound suffices); (d) explicit parameters $s = 32p\xi + 2p(2\Lambda+6) + 16$, $q = s$, $M = s^2/8 - 4$, $\varepsilon = A^{\{-1/(2(2\Lambda+\xi s+6))\}}$ for the paper's $\varepsilon = A^{\{-b/\sqrt{M}\}}$ with "M arbitrarily large"; "O_ $\varepsilon(\varepsilon^M)$ uniformly in r " in the node Lem3_2 read as $\exists K \varepsilon_0 > 0$, $P[\text{not } F] \leq K\varepsilon^M$ for $\varepsilon \in (0, \varepsilon_0)$, all r · kinds: own constructions, parameter choices, reading · proposed · no.

GA-P36a..e · GM uniqueness-final.tex l. 1418–1499 (Prop 3.6 proof, (3.14)–(3.18)) · LQGMetric/Papers/GM/S3/AttainedP36*.lean · (a) the end segments bounded via the last time $|P - z| \leq 2\varepsilon r$ and the first later time $|P - w| \leq 2\varepsilon r$, three end pieces by C_ω each, instead of GM's $J, \bar{J}$ and (3.15) (same estimate); (b) (3.18), stated by GM without proof ("Axiom V ... and the triangle inequality"), proved from GM.S2.4a (separated points, $b = \beta/2$) and S2.4b (modulus of continuity) for D only, with $3C_s' = (C_2 - K)s$, plus $\tilde{D} \leq C_* D$ (from RatiosAre + P2.2): own argument; (c) "by continuity" made precise: bound for rational pairs with $|z-w| > \beta r/2$, then density of $\mathbb{Q}^2$ and closedness for $|z-w| \geq \beta r$; (d) the closed confinement set confSet (implied by GM's event), confinement into $\bar{B}_{\{R\}}$, $U = B_{\{R+1\}}(0)$ in L3.9; (e) $C'' = (K + C_-)/2$ with $K = C' + A/(A+1)(C_- - C')$ inside GM's interval (3.11) · kinds: proof details, own argument for an unproved step, reading, conventions · proposed · no.

D-DDDF-L9-1, D-DDDF-L9-2, D-DDDF-L13-1 (applied with D-DDDF-5, D-DDDF-7) · DDDF tightness.tex l. 689–722 (L9), l. 771–780 (L13); DF Lemma 4.7 (DF:576–596), DF §2.3 (DF:226–252); Pitt 1982; Tassion Prop 4.1 · LQGMetric/Papers/DDDF/L9*.lean, L13*.lean · (L9-1) in L9's proof $\alpha = s/\sigma^2$ instead of DDDF's $s/(2\sigma^2)$ (with $s/(2\sigma^2)$ the displayed identity $e^{\{\alpha^2\sigma^2/2\}}e^{\{-\alpha s\}} = \varepsilon$ is false); "ε small" made explicit as $\varepsilon < e^{\{-\xi^2\sigma^2/2\}}$; (L9-2) the weight $e^{\{\xi\}}$ is kept (σ becomes $\xi\sigma$), any marked rectangle; the left-tail step argues directly with generalized quantiles instead of DDDF's contrapositive (which needs an atom-free law); only one-point laws of Ψ are used, independence = IndepFun of the function-valued fields; (L13-1) DF's approximation uses the field's gradient, here its modulus of continuity, with dyadic rounding to the lower-left corner instead of DF's centre; D-DDDF-5 applied (measurably selected near-geodesics, slack $\rightarrow 0$), D-DDDF-7 applied (assoc_of_tendsto, own argument) · kinds: corrected constant, conventions, own arguments · proposed · no.

DF-L28-a..d · DFGPS T:872–898 (Lemma 2.8), T:909–913 (tightness criterion) · LQGMetric/Papers/DFGPS/L2_8Proof*.lean · (a) the converse Arzelà–Ascoli step (a compact subset of $C(Y, \mathbb{R})$ has a uniform modulus) and "tight + random Lipschitz constant $\Rightarrow$ tight" are own elementary arguments (the paper: "by combining", T:887–888); (b) continuity of $D_h \varepsilon(\cdot, \cdot; S)$ on $S \times S$ from the segment bound and the triangle inequality (implicit in the paper); (c) a.e.-measurability of $\omega \mapsto \text{lfppSqC}$ via chains in $\mathbb{Q}[i] \cap S$ and ContinuousMap.measurable_iff_eval (own argument); (d) the bound on $\|f\|_\infty$ taken as the measurable $\sup \text{Abs } f$ (supremum over a dense sequence) · kinds: own arguments · proposed · no.

DEV-L32-a, DEV-L32-b · DFGPS Lemma 3.2 (T:1452–1456, proof T:1474–1481) · LQGMetric/Papers/DFGPS/L3_2{Meas, Main}.lean · (a) the event $E_r(z; C)$ is shown to be determined on the larger annulus $\mathbb{A}_{\{r/2, 5r/2\}}(z)$ instead of $\mathbb{A}_{\{r/2, 2r\}}(z)$, because the circle $\partial B_{\{2r\}}(z)$ in its second condition is not inside the open annulus $\mathbb{A}_{\{r/2, 2r\}}(z)$; LM L3.1 allows any $s_2 < 1$ and is applied with $s_2 = 5/6$, conclusion unchanged; (b) own elementary arguments for implicit details: grid points by rounding coordinates toward 0 (settles DFA0's boundary concern), scales $2^{\{-(k_0+3j)\}}r$, $j = 1..K$, $k_0 = \lfloor \log_2 \varepsilon^{-1} \rfloor$, $K = \lfloor (\nu \log_2 \varepsilon^{-1} - 1)/3 \rfloor$, LM L3.1 with $b = 1/2$, $a = 3 \log 2 \cdot \tilde{M}/\nu$, $\tilde{M} = M + 2(M+1+\nu)$, constants $K = 81c e^{\{4a/3\}}$, $\varepsilon_0 = 2^{\{-4/\nu\}}$; internal diameter and crossing distance as countable sup/inf over dense pairs (continuity of internal metrics) · kinds: reading (necessary enlargement), own bookkeeping · proposed · no.

DV-AUDM2-1 · DFGPS Lemma 3.20 (T:2317–2330, first display; T:2321) · LQGMetric/Blueprint/DFGPSEstimates.lean (DFGPSLem3_20) · the first display is read for $u \neq v$ only: for $u = v$ the ball $B_{\{2|u-v|\}}(u)$ is empty, the internal metric $D_h(u, u; \emptyset)$ is ∞ , and the printed bound $|(u-v)/r|^{\chi} = 0$ fails; the lemma is meant for distinct points (D55) · kind: reading of the paper's text · applied (P2-APPLY-M2) · no change of content.

DV-AUDM2-2 · GM §4.3, Theorem 4.2 (1) and U:1556 · LQGMetric/Papers/GM/S4/Regularity.lean (RegPar.rr, regC5) · the radii $r^{\varepsilon}_k \in [\varepsilon^{\{1+\nu\}}r, \varepsilon r]$ depend on the scale r : $\text{RegPar.rr} : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{N} \rightarrow \mathbb{R}$ ($r \mapsto \varepsilon \mapsto k \mapsto r^{\varepsilon}_k$), condition 5 uses $rr \ r$; GM: "Suppose $r > 0$ and we are given ... $\mathcal{R}$ " (D59) · kind: GM-internal definition, no change of content · applied (P2-APPLY-M2).

DV-AUDM2-3 · GM U:223 footnote (circle-average process, citing Duplantier–Sheffield, *Liouville quantum gravity and KPZ*, §3.1); GM §4.3 conditions 4 and 7 / Remark 4.10 · LQGMetric/Papers/GM/S4/Regularity.lean (regC4, confRegH, regC7, regEvent), RegularityDet.lean · events with a supremum or intersection over a continuum of centres use a fixed jointly continuous version H of $(r, z) \mapsto h_r(z)$ (DFGPS.IsCircleAvgVersion), passed as a parameter; condition 7 is confReg with $c_R e^{\{\xi H_R(z)\}}$ (confRegH), a.s. equal to confReg for each fixed z (confRegH_ae_eq); gm_S4_11_ii, gm_S4_3, gm_S4_3_K assume $H_r(0) = h_r(0)$ and $H_r(z) = h_r(z)$ at ω (a.s. true for each fixed z). Blueprint Props keep the literal circleAvg (D60). Condition 2 (centres $z \in B_{\{4\ell r\}}(rV)$ in scaleFac ... z) still uses the literal circleAvg (not listed in D60) · kind: convention · applied (P2-APPLY-M2).

DV-DFC2-1..3 · DFGPS T:1628–1650 (Lemma 3.6), T:1638; DG arXiv:1807.01072 Prop 3.21 (DG:1603–1610, rate announced DG:356), DG:309, DG:584 · LQGMetric/Papers/DFGPS/L36Upper*.lean, LQGMetric/Blueprint/DFGPSInputsDG2.lean (D52) · (1) the left–right crossing of $\mathbb{S}$ that DFGPS cite to DG Thm 1.5 (whose (1.5b) has no rate) is built as a chain of dyadic boxes $2^{-k-1}B(1+i, 3/4)$ towards the corners, each handled by DG Prop 3.21 on the rescaled field, plus straight lattice walks for the last δ^{1-a} near the corners and fractional-moment ($\theta \in (0, 1]$) Markov bounds ($\xi_Q > \xi^2/2$ cannot be used: no lower bound on d_γ is proved) — own argument; (2) continuum path $\rightarrow$ graph path: arclength pieces of length $[\delta, 2\delta]$, cut points rounded to $\delta\mathbb{Z}^2$, joined by 8-neighbour walks of ≤ 3 steps, oscillation $\text{osc}_{\{8\delta\}} h_\delta$ of the whole-plane field used directly (instead of “the same argument as DG Prop 3.16” and DG Lemma 3.7) — own argument; (3) NEW CITED RESULT Blueprint.DGProp3_21 (DG Prop 3.21) with constants p, C, δ_0 depending only on γ, U, K, ζ (before the probability space; needed to apply it to $\approx \log \delta^{-1}$ rescaled fields), U connected (DG’s “domain”) · kinds: own arguments, statement of a cited result (added, D52) · proposed · no.

KHSQ2-1, KHSQ2-2 · Evans PDE §5.5 Thm 2; QZ K3.exists_radialApprox_tendsto · LQGMetric/Field/{HeatKernelSquareGreen6, HeatKernelSquareGreen7, GreenSquareCut, GreenSquareMem, GreenSquare, GreenSquare2}.lean · (1) the Green identity on the square via the Dirichlet sine eigenfunction expansion: own assembly of standard facts (mode orthogonality, Parseval from the heat kernel tending to δ , Riesz identification; DDDF only cite the identity); (2) the sine modes in H^1_0 of the square via explicit smooth cutoffs and Vitali’s theorem instead of the general Lipschitz-vanishing-on- ∂U fact (own elementary proof) · kinds: own arguments for standard facts · proposed · no.

D-DDDF-L6-convex, D-DDDF-L6-split, D-DDDF-L6-Taylor · DDDF tightness.tex l. 539–646 (Step 3(A), (B), $\varphi_{\{2,1..3\}}$); DF Lemmas 4.1–4.2 · LQGMetric/Papers/DDDF/L6Ker*.lean, L6Tail.lean · (convex) DDDF’s convexity of U, V, K dropped: instead of $|F x - F y| \geq |x-y|/C$ (from $\|(F^{-1})'\|$ and convex V) we prove $\kappa|x-y| \leq |F x - F y|$ for $x \in K, y \in U$ by Taylor in balls $B(x, \varepsilon) \subseteq U$ and, off the diagonal, injectivity + open mapping + compactness of K + boundedness of U (own argument; matches what L12’ supplies); (split) time split at $t = |x-x'|$ (DDDF’s $t \leq \sqrt{|x-x'|}$ after $t \leftrightarrow 2t^2$), Step 3(A) for $t > |x-x'|$ via $(e^{-p} - e^{-q})^2 \leq (p-q)^2(e^{-2p} + e^{-2q})$, $|x-x'| \geq \varepsilon$ from the uniform bounds; (Taylor) constant $\|F''\|$ instead of $\frac{1}{2}\|F''\|$ · kinds: hypothesis change (weaker), own arrangement, constant · proposed · no.

DFA5-1, DFA5-2 · DFGPS Lemma 3.8 proof T:1732–1739; LM Lemma 3.1 · LQGMetric/Papers/DFGPS/L3_8.lean (dfgpsLem3_8_of (hT15 : DFGPSScaling) (hL31 : LMLem3_1a) : DFGPSLem3_8) · (1) DFGPS get infinitely many good scales from triviality of $\bigcap_r \sigma(h|_{\mathbb{C} \setminus B_r(0)})$ (no source); we use LM Lemma 3.1 on decreasing radii $8^{m+K}/8^k$ (chance that none of $8^m..8^{m+K-1}$ is good $\leq c e^{-K}$), no inversion needed; (2) instead of $t \mapsto h_{\{e^t\}}(0)$ being a Brownian motion, the Gaussian lower tail of $h_{\{8^n\}}(0) - h_1(0)$ (variance $n \log 8$) with Borel–Cantelli, giving $c_r e^{\xi h_r(0)} \rightarrow \infty$ along $r = 8^n$ (all the proof needs) · kinds: published step replaced by a cited result, own argument · proposed · no.

P2-MKV-1, P2-MKV-2 · Sheffield math/0312099 §2; Berestycki–Powell §1 (definitionGFF.tex l. 842–862) · LQGMetric/Field/MarkovVer2*.lean · (1) Poincaré inequality along rays for C^1 functions vanishing on $\partial\mathbb{D}$, $\int_{\{B(0,A)\}} f^2 \leq A^3 \int \|Df\|^2$ (classical Friedrichs argument, own write-up); (2) the zero-boundary GFF extended by zero is a random distribution via Kolmogorov–Čentsov on its rectangle field plus a stochastic Fubini step (route of P2-EXIST) instead of the published Sobolev-series proofs · kinds: own arguments · proposed · no.

GA-S3fn · GM footnote l. 1222 (footnote-G-prob, cited l. 1265) · LQGMetric/Papers/GM/S3/AttainedFn.lean (gm_S3_2fn) · (a) GM pick a close pair “along a D-geodesic” with $|z-w| = \beta r$; we use a path whose D-length is within ε of $D(u, w)$ (Axiom I), cut it into a fine partition and take a consecutive pair with ratio $> C$ (Euclidean distance $< 1/2$, lower bound $1/(m+1)$ chosen later) — no geodesic existence needed (no DFGPSLem3_8), avoids the last-piece caveat; (b) R, β and the N covering balls obtained in one step by continuity from below over $m \in \mathbb{N}$ on Borel events built from a dense sequence, $\beta = 1/(2(m+1))$, $P[\bar{G}_1(C'', \beta)] \geq \beta$; β uniform over all GFFs via a reference GFF and prob_fnEv_eq (translation invariance mod constants + Axiom IV’) · kinds: own bookkeeping (literal steps adapted) · proposed · no.

P2-M2L-1..5 · GM uniqueness-final.tex (5.1) l. 2662–2668, L5.5 l. 2890–2912, (5.14)–(5.15) l. 2936–2940, L5.6 l. 2940–2994, L5.7 l. 2997–3020, L5.8 l. 3044–3110 · LQGMetric/Papers/GM/S5/*.lean · (1) $\mathcal{R}_0$ uniform over all whole-plane GFFs; the deterministic sets H, V, U chosen per probability space (as GM, who fix one field; Prop 4.3 then needs a canonical space + transfer); (2) the constants p_1, ε_1 (and ρ, ε_0) chosen after c'_1, c'_2, η (GM’s own proof makes p_1 depend on ε_1); (3) GM’s tightness event “points not in a single quarter-annulus are far apart” replaced by “points at Euclidean distance $\geq r/4$ are far apart”, so the geodesic lies in $B_{\{r/4\}}(u)$ inside one of four fixed half-planes — same $p_0/8$ (own elementary argument); (4) exists_fiber_ge (counting by finite subadditivity, own proof); (5) L5.7 stated for every open $V \subseteq B_{\{3r\}}(z)$ (all GM’s proof uses); L5.5 with $0 < cs, 0 < Cs$ instead of RatiosAre (more general) · kinds: conventions, parameter order, own arguments · proposed · no.

DF-L28-LIM, DF-L28-POS, DF-L28-MID · DFGPS T:860–866 (L2.7 = BBI Ex. 2.4.19), T:872–898; BBI 2.4.10, 2.4.16(2), 2.4.17 · LQGMetric/Papers/DFGPS/L2_8Lim*.lean, L2_8GffRed.lean · (LIM) “combining with Lemma

2.7” (T:888) via the closed set of pseudo-metrics with midpoints + portmanteau instead of a Skorokhod coupling followed by lem-bbi (same content: BBI’s midpoint proof); (POS) how positivity of limits survives bi-Lipschitz comparisons with random ε -independent constants (implicit, T:887–888, 897–898): own standard argument (ae_posOffDiag_of_dominated: portmanteau on open/closed sets along a convergent subsequence + compactness of $\{|x-y| \geq \delta\}$); (MID) midpoints of the internal LFPP metric: own elementary proof (cut a near-optimal path at half its length, IVT); Lem2_8Gff uses ”positive off the diagonal” for ”induces the Euclidean topology” (equivalent for a continuous metric on a compact square) · kinds: route, own arguments, reading · proposed · no.

D-DDDF-18..20 · DDDF tightness.tex l. 541–543, 653–662, 725–736, 783–810, l. 795; DF Thm 3.1 proof (DF:624–678), Props 4.5–4.6 · LQGMetric/Papers/DDDF/{P10Node,RSW*}.lean · (18) Prop10 stated in law form with L' computed for the same field $\varphi_{\{0,n\}}$ (equivalent since $\tilde{\varphi}$ has the law of φ), C uniform in n , probabilities are outer measures of $[0,\infty]$ -valued lengths; (19) instead of rectangles isometric to $[0,a/2^p] \times [0,b/2^p]$ plus law invariance of φ under rigid motions, Prop 10 is applied directly to the moved domains $g^{-1}(K)$ with $F \circ g$ (no law invariance needed); (20) the constant absorption at l. 795 via $|\log(\varepsilon/c)| \leq (1+2|\log c|)|\log(\varepsilon/C)|$ (own elementary step); the uniform-over- $[A,B]$ Prop 14 is not given by DDDF’s proof as written (DF L4.8’s $j(b/a)$ blows up as $a/b \rightarrow 1$): own monotonicity reduction to (A,B) and (B,A) (pending) · kinds: statement form, own route, own steps · proposed · no.

P2-DFA6b-1..4 · DFGPS Props 3.9/3.10, T:1766–1775, T:1871–1912 · LQGMetric/Papers/DFGPS/P3_10Tail{Geom,Chain,Prob,Good,Main}.lean, P3_9Final.lean · (1) *Modified crossing geometry*: the paper applies Prop 3.1 with K_1, K_2 the short sides of an open $s \times s/2$ rectangle R (not inside R , so $K \subseteq U$ fails). In relative coordinates of a dyadic square $w + s[0,1]^2$ we use instead horizontal crossings from $\{re = 1/16\}$ to $\{re = 15/16\}$ inside the strips $im \in (1/8+e, 3/8+e)$, $e \in \{0, 1/2\}$ (Prop 3.1 with $U = (0,1) \times (1/8+e, 3/8+e)$, $K_1, K_2 = \{1/16\} \times [3/16+e, 5/16+e]$, $\{15/16\} \times [3/16+e, 5/16+e]$) and one vertical crossing from $\{im = 1/16\}$ to $\{im = 15/16\}$ inside $re \in (1/8, 3/8)$; these meet inside the square and the parent’s horizontal crossing at the child’s height meets the child’s vertical crossing (crossing lemma RectMeet.rect_crossings_meet); three crossings per square instead of the four of the ”#” (status: place where we do not follow the published proof literally; same argument). (2) *Levels and Step 3*: the paper uses $A = 2^{\lfloor \zeta n \rfloor}$ for $n \geq N_C$ and a triangle inequality over the level- N_C squares (Step 3); we use $A_n = C^{\lfloor \zeta \rfloor} 2^{\lfloor \zeta n \rfloor}$ ($\zeta = \xi(Q-q)/4$) at every level $n \geq 0$, whose union bound still decays like $C^{\lfloor -\zeta p \rfloor}$ for any p , so the root square $\mathbb{R}$ itself is in the dyadic tree and Step 3 is not needed; the resulting exponent $\tilde{C} = K_1 C^{\lfloor \xi + \zeta \rfloor}$ is the paper’s. (3) The parameters $q \rightarrow Q, \zeta \rightarrow 0$ are coupled (lem3_11 slack $Q-q$, A -exponent and scaling slack $\xi(Q-q)/4$) and chosen by continuity (exists_q_exponent). (4) Prop 3.9: the paper combines (eqn-diam-moment0), the Gaussian tail and Theorem 1.5 at the level of moments; we combine at the level of tails (union bound with $s = t^\varepsilon$) and integrate the tail, and the chaining over the cover (internal diameter of $K \leq 2 \Sigma_k \text{diam } S_k$) is our own elementary argument (internalDiam_le_two_sum, P2-DFA6).

DV-M2H2-1, DV-M2H2-2 · GM uniqueness-final.tex U:1985–1987 (conditions 3–5 of $\mathcal{E}_r$); DFGPS L2.3 (T:757–768); Adler-Taylor Thm 2.1.1; LM Lemma 3.1 · LQGMetric/Papers/GM/S4/RegularityCond{3,4,5}.lean · (1) GM give no detail for condition 4: proved with DFGPS L2.3’s chaining + Borell-TIS at scale r , uniformity in r checked on the covariance bounds (as DEV-DFGPS-2) rather than through the law of h ; (2) in condition 5, LM L3.1 is applied to every m -th T4.2 radius $\lambda_5 r_{\{mj\}}$ with $s_1 = \lambda_1/\lambda_5, s_2 = \lambda_4/\lambda_5$ and $(\lambda_1/\lambda_4)^m \leq \lambda_1/\lambda_5$, because T4.2 (1) gives only $r_{\{k+1\}}/r_k \leq \lambda_1/\lambda_4$ while LM L3.1 needs $\leq \lambda_1/\lambda_5$ (GM silent) · kinds: own arguments for unstated steps · proposed · no.

DV-LM21-AE, DV-LMASM-b · LM Lemma 2.1 (lem-whole-plane-markov, l. 425–429), GMSH Lemma 2.2, Sheffield math/0312099 §2.6 · LQGMetric/Blueprint/* (LMLem2_1), LQGMetric/Field/MarkovAsm*.lean · (AE) the harmonic clause of the cited-result Prop LMLem2_1 holds for a.e. ω (D61; the every- ω form was a Lean modelling artefact); (b) pairing-level statements moved to a distribution-valued version via the countable pairJ coordinates (own bookkeeping) · kinds: statement of a cited result (to its a.s. form), own bookkeeping · proposed · no.

DZZ-PRE-1..4 · DZZ arXiv:1807.00422 LBM_LGDarXiv.tex l. 424–609 (Lemmas 2.5–2.8) · LQGMetric/Papers/DZZ/S2L{5,6,7,8}*.lean · (1) own elementary steps: for $a \leq 0 < c$ the reflection band uses $1 - e^{\lfloor -2c^2/t \rfloor} \leq \sqrt{2c}/\sqrt{t}$; in the variance bounds $0 \leq k' \leq k \Rightarrow (k-k')^2 \leq k^2 - k'^2$, cross-term lower bound via monotonicity of p in the domain; (2) the variance-truncation bound proved as $K\rho^i$ with explicit $\kappa = \min((\log 4)^2/16, 1/100)$ instead of DZZ’s $e^{\lfloor -\Omega(i^2) \rfloor}$ (all the sum argument uses); bridge exit via $|X| \geq r \Rightarrow$ one coordinate exceeds $r/3$; (3) GAP in the paper: the η half of L2.5 (”similar argument”, l. 460–461) needs BridgeShellBound (bounded density of the radial maximum of the planar bridge) — D62; (4) L2.8 uses $\Delta_0 = -\eta_1$ where DZZ write η_1 (sign irrelevant); statements over all $u, v \in \mathbb{C}$ (includes $u, v \in \mathbb{V}$) · kinds: own steps, route, gap in the source, convention · proposed · no.

MKD3 · Sheffield Thm 2.17; BP Lemma 1.53, §1.8 · LQGMetric/Field/MarkovGermVer3*.lean · leaf (D) of the Markov property for unbounded V : instead of the cited cutoff argument, reduction to the bounded case by Hilbert-space truncation with logarithmic cutoffs at ∞ (QZ K3.logCutoff), a uniform truncation bound from a 2D logarithmic Hardy inequality (FTC along rays + Cauchy-Schwarz in polar coordinates), and a Poincaré inequality on a ring around $\partial\mathbb{D}$ from the Schur bound — own elementary proofs of standard facts (zero capacity of ∞ in 2D) · kind: published proof not followed verbatim, same mathematics · proposed · no.

DF-L29-CONN, DF-L210-R, DF-L210-PROK, DF-L25-LAW · DFGPS T:817–830, 903–1019 (L2.5, L2.9, L2.10), T:953–957 · LQGMetric/Papers/DFGPS/L2_{5,9,10}*.lean · (CONN) L2.9 and L2.5 B for dyadic domains with connected closure (the paper’s own “WLOG $\bar{W}$ connected”; real-valued metrics would turn the infinite distance between components into 0); (R) in L2.10, R depends on (Ω, P, h) rather than only on the law of h (consumers fix h); $\liminf$ along $\mathcal{N}[>0]$, distances in $[0, \infty]$; (PROK) own argument for “by Lemma 2.8 there exists R” (T:953–957): Prokhorov + portmanteau on increasing open events (exists_eventually_measure_ge); (LAW) L2.5 B’s “coupling” read as the limit law μ on the countable product DyProd (Borel σ -algebra), internal-metric equality in lem2_11’s form · kinds: reading, dependence of constants, own argument · proposed · no.

E3b-1..3 · GM uniqueness-final.tex l. 1682–1716 (Lemma 4.6), l. 1348 · LQGMetric/Papers/GM/S4/L46Meas{B1,B2,B3,B4,C1}.lean · (1) D30 measurability (own elementary argument): the complement of the filled ball characterized by paths avoiding the dense-sequence points of the metric ball, distance to the frontier replaced by distance to the metric ball, τ read off the dense sequence on lenSet ; (2) the events whose trace on E is a.s. in M form a σ -algebra (traced σ -algebra glue for L4.6 (c), own argument); (3) (b) needs $\rho \leq r$ (geodesics avoiding $B_r(z)$ stay in U ; automatic for GM’s $\lambda_3 = 1$) · kinds: own arguments, parameter restriction · proposed · no.

DEV-P41-1..5 · DFGPS Prop 4.1 proof T:2455–2582; DZZ L2.1; DS §3.1 · LQGMetric/Papers/DFGPS/P4_1*.lean · (1) $\xi < 1$ from $\text{GMxiQBound} + \chi \leq 2$ instead of DFGPS’s “ $\xi < 2/d_2 < 1$ ” (own elementary argument); (2) Paley–Zygmund via an AM–GM bound instead of Cauchy–Schwarz; (3) the power-sum bound via Bernoulli’s inequality (own proofs); (4) only one-sided bounds of (4.6)–(4.7) used ($\text{Var} \geq \log \varepsilon^{-1}$, $\text{Cov} \leq \log \varepsilon^{-1} - \log(|j-k|+1) + A_0$), $\lambda = (p \log \varepsilon^{-1} + \min(\log a, 0))/\xi$ absorbing the Paley–Zygmund constant A (T:2578); (5) planned: annulus radii $2\varepsilon r - 4\varepsilon r$ with centre spacing $8\varepsilon r$ (instead of $2\varepsilon r - 3\varepsilon r$, $6\varepsilon r$) to reuse $\text{prop3_1_centre_ann}$, and cyclic index intervals for the whole-circle case (DFGPS’s Step 2 assumes a path must pass every intermediate centre, which fails on a circle; Step 4 applied only to short blocks, since (4.7) fails near antipodal wrap-around) · kinds: own proofs, route, gap fixes · proposed · no.

KAHANE3-a..e · Berestycki–Powell arXiv:2404.16642 GMCproperties.tex l. 1205–1250, 1270–1430 (T:finiteposmoments), 1622–1745 (T:negmom); Rhodes–Vargas Thms 2.11–2.12 (no proofs) · LQGMetric/Dimension/GMCMomentPos2*.lean, GMCMomentNeg.lean, GMCSqCont.lean · (a) comparison field: circle averages of the same GFF around the middle third (two-sidedly log-correlated) instead of BP’s exactly scale-invariant field; BP’s scaling relation replaced by Kahane with an additive constant (same estimate); (b) off-diagonal step: each checkerboard group decoupled by Kahane against independent copies on a product space, the i.i.d. sum bounded by a Rosenthal-type inequality (own proof) instead of BP’s multinomial expansion + induction on $[q]$; (c) discretization by an exact identity (averaging Riemann sums over a random offset in each cell gives the cell integral) + Jensen in the offset (own device); (d) covariance lower bound circleCov_ge_log and $hS \geq c$ on compacts: own elementary proofs from the Green-function formulas; (e) negative-moment bootstrap with the N squares of one checkerboard group instead of BP’s two separated sets D_1, D_2 · kinds: own routes and devices · proposed · no.

D-DDDF-21..23 · DDDF tightness.tex l. 653–676 (Prop 7, Cor 8), 771–780 (L13), 783–810 (P14), 482–490 ((2.27)); DF §2.3, Thm 3.1 proof · LQGMetric/Papers/DDDF/{L13Cross,RSWHigh,RSWUnif,RSWUnifQ,C8,C8High}.lean · (21) the uniform-over- $[A,B]$ Prop 7 via monotonicity in the shape ($\text{rectLen_rectAB_mono}$: cut the crossing at its first hit of $\text{Re} = x$), reduced to shapes (A,B) and (B,A) — DDDF give no argument and DF L4.8’s $j(b/a)$ is not uniform as $a/b \rightarrow 1$; holds on all of $[A,B]^2$ without $a/b < 1 < a'/b'$; (22) (3.37), (3.39) stated for $3\varepsilon^{1/C} < 1$ (where ℓ is a real quantile), high-quantile results take $\xi \geq 0$ (square-root trick), crossings with no admissible path dropped before the square-root trick, trivial case $\varepsilon^{1/N} \geq 1/3$ separated; derived $C \geq 1/2$ for (3.36) and $C \geq 1$ for (3.37); (23) Cor 8 bookkeeping (“follows from Props 5 and 7”): max_log_le , tailExp_le , level_le , XAB_mono , abs_log_div_mono (own elementary steps) · kinds: own route, readings/ranges, own steps · proposed · no.

DZZ-S3-1..6, DZZ-S3-centres · DZZ arXiv:1807.00422 §3.1 (l. 765–798), Lemma 3.1 (l. 799–803, 833–848), notation l. 287–316, L3.4 · LQGMetric/Papers/DZZ/S3*.lean · (1) box membership by the index $[2^n x]$ capped at $2^n - 1$ (half-open boxes except the right/top edges of $\mathbb{V}$; each level partitions $\mathbb{V}$) where DZZ leave boundaries open (l. 778); (2) D' counted in cells (graph distance + 1, as in the proof of L3.5, l. 1086), T when a point lies in no cell; (3) $\text{Var } \eta_\varepsilon(v)$ as the deterministic etaVar (= the variance under a white noise, hasLaw_etaInf); (4) L3.1 via exponential-moment (Chernoff) bounds instead of DZZ’s Gaussian tail; the maximal-cell half (omitted by DZZ, l. 847) with exponent $(p-1)(2 - p\gamma^2/2) = (4-\gamma^2)^2/(8\gamma^2)$, $p = (\gamma^2+4)/(2\gamma^2)$; (5) the L3.1 event also asserts every $v \in \mathbb{V}$ lies in a cell (implicit in DZZ); (6) ξ -admissible pairs indexed by $\delta \in (0,1)$, HighProb as $P(E_\delta)^c \leq \delta^c$ for all small δ ; (centres) L3.4’s event on dyadic box centres (D64) · kinds: conventions, readings, own steps, additions · proposed · no.

DEV-DFB2c-1..3, NOTE-radial · DFGPS T:666 (radial ψ_ε), T:711–734 (polar formulas, tail), T:722–723 (continuity) · LQGMetric/LFPP/Localized*.lean, LQGMetric/Papers/DFGPS/L2_1*.lean · (1) DEFINITION CHANGE (internal, not the frozen layer): ψ_ε (LFPP. locBump , now $\mathbb{C} \rightarrow \mathbb{R}$) built from mathlib ’s $\text{radial ContDiffBumpBase.ofInnerProductSpace}$ instead of $\text{ContDiffBump/someContDiffBumpBase}$ (a Classical.choice , not provably radial), same radii, so that DFGPS’s radial symmetry holds (locBump_radial); (2) joint existence and continuity of $h^*_\varepsilon(z)$ in (ε, z) (lem2_1HeatJoint , implicit in the paper): three-parameter Kolmogorov in $(\log s, z)$ + own s -

Lipschitz bound for the heat kernel; (3) swapping the cutoff centred at 0 for a radial cutoff centred at z (ae_tendsto_heatTrunc_sub_radTrunc); NOTE: the frozen statement layer's cutoff n and bumpTest (Statement/Field.lean) also use someContDiffBumpBase and are not provably radial — harmless for the frozen statements (limit definitions), but proofs needing radial symmetry must use their own radial cutoffs (e.g. radCut) · kinds: internal definition fix, own arguments, caveat · proposed · no.

E3c-1, E3c-2 · GM uniqueness-final.tex l. 1648–1654 (S4.1), 1692 ((4.11)), 1701–1716 (L4.6) · LQGMetric/Papers/GM/S4/L46MeasD*.lean · (1) on the a.s. event of GM.S4.1, Stab rewritten as $\{(z,r) \in \mathcal{Z}_k\} \setminus \text{Split}(gm_stabCond_iff_not_split, \text{own elementary argument})$; GM's "hits $\partial \mathcal{B}^\bullet$ in the same arc" read as $\neg \text{Split}$; its universal measurability reduced to analyticity on lenSet of the conf/arc relation and of the avoiding-geodesic relation (GMArcRelAn, GMAvoidRelAn); GM do not discuss measurability (own argument, D30 convention); (2) gm_L4_6_hArc, gm_L4_6_c assume a.s. $w \notin \mathcal{B}^\bullet_{\{t_k\}}$ and $r < \lambda_4 \epsilon r$ (implicit in GM's setup) · kinds: own argument, reading · proposed · no.

D-DZZ-BR1, D-DZZ-L26a, D-DZZ-L26b · DZZ l. 459–461 ("similar argument"), l. 516–538, l. 530 · LQGMetric/Field/KilledHeatScale.lean, LQGMetric/Papers/DZZ/{S2Bridge, S2BridgeLemmas, S2L6Log, S2L6Max}.lean · (BR1) the bounded density of the planar bridge's radial maximum (the gap D62) proved by an own analytic argument: shell probability $= G(t/r^2) - G(t/(r-d)^2)$ by Brownian scaling, $G(\tau) = 2\pi\tau K(\tau)$ with K the killed kernel of the unit disc at its centre, K log-convex (Chapman–Kolmogorov + Cauchy–Schwarz), a discrete log-convexity lemma $K(\tau) - K(\tau+s) \leq (4s/\tau) \cdot \sup_{x \geq \tau/2} K(x)$, $K(x) \leq \min((2\pi x)^{-1}, 1/(\pi x^2))$; constant 256; (L26a) the log-variance bound for η (l. 530, stated without proof): own elementary proof (split the time integral at $|u-v|^2$, L2.5 at scale $|u-v|$ above, $p \leq (2\pi s)^{-1}$ below); (L26b) the union bound in L2.6(2) with $2\lceil 1/\delta \rceil^2$ boxes of side 3δ via DGo L3.4 instead of DZZ's grid (bookkeeping) · kinds: own proofs, bookkeeping · proposed · no.

DFA7-1..4 · DFGPS Prop 3.18 (T:2247–2254, 2383–2385), L3.20 (T:2317–2330), L3.22 (T:2367–2377) · LQGMetric/Papers/DFGPS/{L3_19, L3_20*, L3_22*, P3_18}.lean · (1) the paper's scale $2^{\{-k\}}\epsilon$ with grid mesh $2^{\{-k-2\}}\epsilon r$ cannot fit both $u, v \in B_{\{tr\}}(z)$ and $B_{\{2tr\}}(z) \subseteq B_{\{2|u-v|\}}(u)$: scales $\epsilon(3/4)^k$, mesh $tr/40$, z the grid point nearest the midpoint (L3.20) or nearest u (L3.22), L3.19/L3.21 on $\bar{B}_R(0) \supseteq B_{\{\epsilon r\}}(K)/r$ (same union bound); (2) the closed squares of L3.20's second display via nested open dyadic sub-squares and an infinite concatenation of paths (internalEDist_le_tsum_of_chain), the factor $2/(1-2^{\{-s\}})$ absorbed by applying L3.19 with $s = (\chi + \xi(Q-2))/2$ (own argument); (3) the proof of L3.22 cites "(eqn-ep-diam)" (T:2375) but the lower bound needs L3.21 (eqn-ep-cross) — used L3.21; (4) Lem3_19U, Lem3_19SqU, Lem3_21U: constants before the field ("uniformly", D57); they imply the nodes Lem3_19, Lem3_19Sq, Lem3_21 · kinds: proof details, own argument, misprint, statement form · proposed · no.

P2-M2L2-1..3 · GM uniqueness-final.tex l. 1200–1203 (law mod constants), l. 2997–3018 (L5.7 proof) · LQGMetric/Papers/GM/S5/{GoodRadiiLaw, Tubes57*}.lean · (1) measurability of $F_r(z)$ through a Borel form (geodesic uniqueness via MidUnique, condition (3) on a dense sequence); needs SepFromMeas, which GM do not state — restricted to square tubes (D66); (2) own elementary arguments for facts GM use implicitly: a geodesic from inside $B_\rho(z)$ to outside crosses $\partial B_\rho(z)$; $\bar{D}(u, \partial B_{\{2r\}}(z))$ is attained, hence determined by the internal metric of $B_{\{3r\}}(z)$; $\{u \mid w \in O_u\}$ is open; $c_* = cs$, $C_* = Cs$ give $cs \cdot D \leq \bar{D} \leq Cs \cdot D$ pointwise; (3) law invariance of $P[\text{attainedLow}]$ (GM l. 1200–1203) applied to the universally measurable Borel form · kinds: measurability convention, own arguments · proposed · no.

DV-AUDM2-3 (extension), M2H3-c2, M2H3-c7 · GM uniqueness-final.tex l. 1940–1990 (Remark 4.10, L4.11 and proof) · LQGMetric/Papers/GM/S4/Regularity*.lean · (ext.) condition 2 of $\mathcal{E}_r$ uses the continuous circle-average version H instead of the literal circleAvg (continuum of centres), as D60 did for conditions 4 and 7 (definition change, internal); (c2) own argument for condition 2 (GM give none): grid $(\ell r/8)\mathbb{Z}^2$ covering $B_{\{4\ell r\}}(rV)$, Axiom V via GMS2_4a (annulus crossings at scales $r, \ell r$) and GMS2_4b (diameter of $B_{\{\ell r/2\}}(w)$), condition 4 at both scales comparing $h_r(z)$ with $h_r(w)$, number of grid points independent of r ; (c7) own argument for condition 7 (DV-B4; Remark 4.10 has no proof): parts 1–2 from condition 2 at scale r ($\ell \leq 1$), part 3 from condition 3 at scale ℓr with $\chi_2 = (\chi + \xi(Q-2))/2$ plus condition 4 (factor $e^{\chi}|\xi|M$ absorbed by $a^{\chi_2 - \chi}$), part 4 from condition 6; CONF L3.8 no longer needed; a chosen after (Ω, P, h) (same law for all whole-plane GFFs) · kinds: internal definition change, own arguments, quantifier order · proposed · no.

E2R-a..d · GM uniqueness-final.tex l. 1654, 1660–1676 (L4.5) · LQGMetric/Meas/LocalEventRandom*.lean, LQGMetric/Papers/GM/S4/L45Det*.lean · (a) GM's "determined by $\mathcal{B}^\bullet_{\{t_k\}}$ and h " proved through dyadic hull pieces with Axiom II and Lusin separation (aeEventIn_localSigma_of_saturated, random-set version; own argument); (b) deterministic locality of $\mathcal{B}^\bullet_{\{t_k\}}$, its geodesics, leftmost geodesics, arcs and Conf_k under equal internal metrics (GMLocData), $\tau_R > 0$ surely: own elementary proofs; (c) disjoint arcs of a Jordan curve have distinct closures and hit patterns (L45DetJ, own proof); (d) E2a and GMConfPtSel via finite random sets and decoders instead of GM's implicit joint measurability; GMArcFamDet ($\forall x \in \mathbb{C}$, stronger than GM need) no longer used · kinds: own arguments, route · proposed · no.

DFB2d-1, DFB2d-2 · DFGPS T:715–717 (polar coordinates via the circle-average process); Hörmander ALPDO I Thm 4.1.1 · LQGMetric/Papers/DFGPS/L2_1Radial*.lean · (1) the radial pairing formula (only sketched in the pa-

per): for every ω mollify by convolution and use polar coordinates (`mathlib Complex.integral_comp_polarCoord_symm`), then $L^1(P)$ convergence of the polar integrals of the mollified circle averages with rate $(2/\sqrt{r}) \cdot 2^{-n/2}$ ($E|X| \leq (EX^2/B + B)/2 + \text{telescoping}$), summable so a.s. convergence without subsequence — own elementary argument; (2) joint measurability of the circle-average version $H(\text{aemeasurable_circleAvgVersion})$, needed for Tonelli; $\text{IsCircleAvgVersion}$ gives only a.s. equality for each r) by dyadic approximation from above in r — own argument forced by the Lean modelling · kinds: own arguments · proposed · no.

DZZPRE3-a, DZZPRE3-b · DZZ l. 521, 527–529 (L2.6), 594–645 (L2.8, L2.9) · `LQGMetric/Papers/DZZ/{S2L6Exp,S2L9Push,S2L9Band}.lean` · (a) L2.6 second inequality from the tail bound by the layer-cake formula and the Gaussian integral (implicit “this yields”), stated for continuous versions of $\tilde{h}_\delta, \eta_\delta$; (b) L2.9’s coupling realised as the affine case of DDDF’s conformal white-noise coupling (`phi_coupledNoise_affine`) instead of DZZ’s abstract gluing; for non-dyadic a , (`eq-coupling-hat-h-eta-2`) needs L2.8 along scales $a \cdot 2^{-\{j\}}$, which DZZ do not prove (their derivation works as written only when a is a power of 2): supplied by a sub-band bound (own gap-filling argument, `dzz_lemma28_scaled` pending) · kinds: own steps, gap fix · proposed · no.

DFA7b-1, DFA7b-2 · DFGPS T:2264–2300, T:2337–2357; Duplantier–Sheffield §3.1; Janson Thm 1.7 · `LQGMetric/Field/CircleAvgIndep*.lean`, `LQGMetric/Papers/DFGPS/L3_19Proof*.lean` · (1) independence of $h_\rho(z) - h_r(z)$ and $(h - h_\rho(z))|_U$ (T:2275) from joint Gaussianity + zero covariance, the covariance limit via Newton’s theorem (log potential of $\sigma_{\{z,\rho\}} - \sigma_{\{z,r\}}$ constant on $\tilde{B}_\rho(z)$) instead of the Markov property the paper implicitly uses (own write-up of a standard fact); (2) anticipated: in the square part of L3.19 the normalization circle (radius $2\epsilon r$ about the corner) differs from Prop 3.10’s (radius ϵr) — a Hölder step bridges them (left out by “proven similarly”); in L3.21 the ball is enlarged to contain $\tilde{B}_{\{2\epsilon r\}}(z)$ before locality applies · kinds: own write-up, implicit steps · proposed · no.

NEGMOM-1..3 · Robert–Vargas arXiv:0807.1030 Prop 3.6 (`main.tex` l. 1031–1037), adapting Molchan (CMP 179, 1996); Berestycki–Powell `GMproperties.tex` l. 1626–1745 · `LQGMetric/Dimension/GMCMomentNeg2*.lean` · (1) base case of negative moments via Robert–Vargas’s Laplace-transform recursion with Kahane (D63) instead of BP’s Girsanov + FKG; constant bookkeeping (window $[T, T']$, $m = 2k_0 + 3$, $\delta = 1/D$) and the Paley–Zygmund step are own arrangement; (2) Laplace $\rightarrow$ moment via a dyadic sum $y^{\wedge\{p\}} \leq T^{\wedge p} + \sum_k T^{\wedge p} e^{2^{\wedge\{p(k+1)\}}}$ instead of BP’s Gamma identity (own elementary proof); (3) bootstrap following BP T: negmom with $m = 2$, base interval from $y^{\wedge q} \leq 1 + y^{\wedge\{q_0\}}$ instead of Jensen (own step) · kinds: sourced alternative route (D63), own steps · proposed · no.

DFB34d-a..e · DFGPS T:872–898; DDDF Thm 1, §6.1, Prop 29 (DD:1498–1510); Revuz–Yor Ch. I Thm 2.1 · `LQGMetric/Papers/DFGPS/L2_8Gff*.lean` · (a) the continuous version of $p_{\{\delta/2\}} * \tilde{h}$ (assumed by DDDF) by an own Kolmogorov argument, square domains only; (b) the range $\epsilon \in [\epsilon_1, 1)$ by an own direct argument (joint continuity, segment bound, a_ϵ lower bound; the paper is silent); (c) positivity of zero-boundary limits transferred from the Prop 29 coupling at the level of laws (DDDF §6.1, cited at T:881); (d) tightness for near-range ϵ via per- ϵ events (DDDF Thm 1(2) versions are chosen per δ); (e) λ_δ ’s independence of the white noise proved (`lambdaDelta_eq_of_isWhiteNoise`), discharging reading BP-DF-2 · kinds: own arguments, route · proposed · no.

DZZ3B-1..3 · DZZ arXiv:1807.00422 l. 861–903 (L3.4 and proof; fine extension l. 885–891) · `LQGMetric/Papers/DZZ/S3L4*.lean` · (1) DEFINITION (D64, D64a): the η -part of $\mathcal{E}_{\{\delta,\alpha\}}$ at dyadic centres, second point the centre of any box of level $\geq m+j$ (`nbrFineEvent`); (2) on centres the first and third of DZZ’s four triangle-inequality terms vanish; the space term uses L2.5’s variance bound and a Gaussian (Chernoff) tail instead of L2.6, the finer-centre term uses L2.6 with $k = 1$ (not the published proof verbatim); any $\alpha > 0$ works (DZZ’s large α is only needed for continuum maxima); (3) own elementary steps: asymptotics in $L = \log \delta^{-1}$ (`l34_eventually`), `HighProb.inter`, Chernoff with a variance upper bound · kinds: definition (decided), route, own steps · proposed · no.

E2S-1, E2S-2 · GM uniqueness-final.tex l. 1665–1675 (L4.5 proof), 1890–1912 (L4.7 proof) · `LQGMetric/Papers/GM/S4/{L45Sel,L45Sel2,L47Asm}.lean` · (1) “ $P|[0,s_k]$ is determined by $(\mathcal{B}^{\wedge\{t_k\}}, h|)$ ” formalized by decoding each $P(s_k u)$ from countably many local field events (rational half-planes, finite arc-hit prefixes) instead of a measurable geodesic selection (own argument; null-measurability `hnullGW`, `hnullGC` open — descriptive set theory, not in GM); (2) locality of $d(z,x)$ for $x \in \mathcal{B}^{\wedge\{t\}}_T(\text{gm_dist_eq_of_locData})$, implicit in GM; own proof); geodesics for both metrics on `lenSet` via `exists_isGeod01_of_bcpt` · kinds: own arguments · proposed · no.

DF-L210-REF, DF-L210-MEAS, DF-L29-GEOM, DF-L29-MEAS, DF-L29-POS, DF-L25-TIGHT · DFGPS T:908–936 (L2.9), T:951–966 (L2.10), T:973–981 (L2.5); Billingsley §7 · `LQGMetric/Papers/DFGPS/L2_{10,9,5}Proof*.lean` · (L210-REF) reference normalized GFF `fieldScale 1` ($h - f$), h compared with $h - f$ directly (L2.6 holds for any whole-plane GFF, the “ $1-2(1-p)$ ” step with $p' = (1+p)/2$); (L210-MEAS) law transfer of event (2.10) via a measurable version on countable dense sets (`lfppDistChain`); (L29-GEOM) “geometric considerations” (T:921–922) by a coordinatewise clamp; (L29-MEAS) measurability of $\omega \mapsto D_{h^\epsilon(\cdot, \cdot; \tilde{W})}$ from continuity of $\psi \mapsto D_\psi(\cdot, \cdot; \tilde{W})$ + an a.e.-measurable field; (L29-POS) the square-annulus positivity argument (T:927–934) fails at reentrant corners of $\tilde{W}$: replaced by one square $S \supseteq \tilde{W}$, $D(\cdot, \cdot; S) \leq D(\cdot, \cdot; \tilde{W})$ and `ae_posOffDiag_of_le_mul` (own argument); (L25-TIGHT) tightness in L2.5 A via $D_{h^\epsilon} \leq D_{h^\epsilon(\cdot, \cdot; S)}$ on squares + the ball-restriction criterion instead of L2.8 on $S_{\{Rr\}}$ with the agreement event (own simplification) · kinds: own arguments, gap fix · proposed · no.

DG3A-1..6 · DG arXiv:1807.01072 l. 916–925, 994–1036, 1044–1094, 2160–2164 (L3.2–L3.6, (3.6)–(3.7), App. A); DZZ L2.3, L2.6; DDDF L4 · LQGMetric/Papers/DG/S3L*.lean · (1) (3.7) read with $\bar{T} = \delta^{\{-2-\gamma^2/2\}} e^{\{-\gamma \max \hat{h}_\delta\}}$, $\underline{T} = \delta^{\{-2-\gamma^2/2\}} e^{\{-\gamma \min \hat{h}_\delta\}}$ (DG write $\exp(-\max \hat{h}_\delta)$, missing the γ their (3.6) produces: typo); (2) L3.4: U placed in a square, increment bound $\text{Var} \leq \sqrt{2}|z-w|/\delta$ for all z, w (DDDF L4) instead of $|z-w|^2/\delta^2$ for $|z-w| \leq \delta$, union bound over DZZ’s boxes of side 3δ ; (3) L3.5: grid in a square containing U with mesh $\leq \delta/2$, explicit exponent $p = \zeta$; (4) L3.6: grid mesh $\delta/(2A)$, L3.4 at scale δ/A with $\zeta/6$ and at scale δ with distance $(C+1)\delta$ and $\zeta/3$ — own repair (DG apply L3.4 at scale δ/A to distances of order δ); (5) dgLGD with open balls of positive radius, L3.2’s tail over the closure of K ; (6) App. A’s last step: $E[\max]$ by chaining (DZZ L2.3) under the increment bound; DG’s “uniformly over w ” in App. A holds only in integrated form near $\partial B_{\{1/10\}}$ · kinds: typo, routes, own repair, conventions · proposed · no.

DEV-DF-A10-1..5 · DFGPS T:2593–2744 (L4.4, L4.5, P4.3) · LQGMetric/Papers/DFGPS/{L4_4, L4_5*, P4_3*}.lean · (1) L4.5: both sides normalized at scale $r' = \varepsilon^{\{-M\}} r$ (the comparison of $h_{\{r'\}}(0)$ with $h_r(0)$ cancels); grid mesh $\varepsilon' \rho/4$, L3.4 with $v = 1$, $q = 8$, P3.1 in the whole plane ($U = \mathbb{C}$) for $\bar{B}_{\{1/4\}}(0)$ and $\partial B_{\{1/2\}}(0)$ (own choices); (2) L4.5 uses the polynomial upper tail $\text{diam}_K \text{upper_tail}$ instead of “Prop 3.9 + Markov” (Markov would need measurability of the internal diameter); (3) P4.3 (eqn-tau-upper): the point $z \in \partial B_k$ with $D(0, z) < s$ from a near-geodesic + IVT; sharpened covering lem4_4_half (“ B_k meets ∂B_s ” is not implied by radius $\geq 2\varepsilon r$); (4) P4.3 Step 2: inductive first-hitting times τ_k (not attained for an open neighbourhood) replaced by a maximal chain; (5) GAP: times before s for metric balls with holes — D68 (filled-ball version DFGPS Prop4_3F); the case $0 \in B_k$ (not treated in the paper) to be handled in the filled-ball proof · kinds: own choices, route, gap (decided) · proposed · no.

D-DDDF-L6F-1, D-DDDF-P10-1 · DDDF tightness.tex l. 545–646 (L6), 725–736 (P10), 701–722 (L9); DF Props 4.5–4.6, L4.7 (DF:534–608), $F \circ \pi$ length inequality DF:552–566 · LQGMetric/Papers/DDDF/{L6Final*, P10*}.lean · (L6F-1) Lemma 6 Step 2 without convexity of K : Kolmogorov/Fernique on finitely many boxes inside a closed thickening of K , versions glued a.s.; DDDF’s “smooth” weakened to “continuous on K ” (all Prop 10 uses); the continuous version of φ_H is $YH := \tilde{\varphi} \circ F - \varphi - YL$; (P10-1) DF Prop 4.5’s conditional Markov (first-moment) bound replaced by the Lemma 4.7 tail (tail_cross) for both parts (same form of I'); near-geodesics selected with additive slack $1/n$ (a crossing of a general K can have length 0); statement needs $\xi > 0$ (DDDF’s standing assumption); law transfer to the enlarged space $\Omega \times \Omega$ with two independent white noises (own routine glue) · kinds: weakened regularity (sufficient), route, own glue · proposed · no.

E3d-1..3 · GM uniqueness-final.tex l. 1648–1654 (GM.S4.1), l. 1690–1712 (L4.6), l. 1655–1688 (L4.5); BBI Prop. 2.5.19 / Thm 2.5.14 (via MetricGeometry.exists_pathLength_eq_edist) · LQGMetric/Papers/GM/S4/L46MeasE1..E10.lean · (1) measurability of the arc relation, avoiding geodesics and the L4.5 hit events (GMArcRelAn , GMAvoidRelAn , hnullE/W/GW), which GM do not discuss: own descriptive-set-theory argument (D65) — the filled ball is Borel in (d, s, x) through a dense sequence of $\mathbb{C}$ and a dense sequence of paths, Jordan parametrizations as $\mathbb{C}(\mathbb{R}/2\pi\mathbb{Z}, \mathbb{C})$ with angle lift $t + \psi(t \bmod 2\pi)$, geodesics as $\mathbb{C}([0, 1], \mathbb{C})$, approximating sequences as $\mathbb{N} \rightarrow \mathbb{R} \times \mathbb{C}([0, 1], \mathbb{C})$; (2) the minimality clause of IsAvoidGeod rewritten on lenSet as $\text{len} \leq \sup_m \text{chainInf}(\mathbb{C} \setminus \bar{B}_{\{r-1/(m+1)\}}(z))(z, x)$ by an own Arzelà–Ascoli argument copied from the BBI route of $\text{exists_pathLength_eq_edist}$; (3) null-measurability for the law of h from “analytic on lenSet ” + $D_h \in \text{lenSet}$ a.s. (Lusin) · kinds: own arguments (measurability not in GM) · proposed · no.

NEGMOM2-a..c · Berestycki–Powell GMCproperties.tex l. 1211–1218, 1719–1745 (T:negmom); DZZ L2.10 (l. 673–682) · LQGMetric/Dimension/GMCMomentNeg3*.lean · (a) Jensen $(\int g \, d\nu)^p \leq \int g^p \, d\nu$ for $p < 0$ on a probability space in $\mathbb{R}_{\geq 0}^\infty$ via Hölder (exponents $1+q, (1+q)/q, q = -p$): own proof of a standard fact; (b) lower covering of a square by offset-averaged dyadic Riemann sums ($\text{lintegral_w_le_sqQ}$, reverse of $\text{setLIntegral_sqIn_le}$): own device; (c) limit passage for negative moments via a continuous plateau test function $1_Q \leq f \leq 1_{\{B(x, \rho/2)\}}$ + Fatou instead of portmanteau for compact sets · kinds: own proofs/devices · proposed · no.

DEV-P41-5 (final), DEV-P41-6, DEV-P41-7 · DFGPS Prop 4.1 (T:2444–2582: Step 1 T:2464–2470, Step 2 T:2486–2492, Step 3 T:2494–2511) · LQGMetric/Papers/DFGPS/P4_1Step*.lean, P4_1Main.lean · (5) the block of centres chosen per path (for a whole circle a path may go round either way, so “one block for all paths” is false), centres for arcs on the full circle, the angle lifted along the path ($\text{IsCoveringMap.exists_path_lifts}$); the tube geometry (left to the reader) is an own write-up; (6) annuli $2\varepsilon r \rightarrow 4\varepsilon r$ inside $B_{\{(16/3)\varepsilon r\}}$ with centres spaced 9ε (DFGPS: $2\varepsilon r \rightarrow 3\varepsilon r$, spacing 6ε), because the available annulus estimate is Prop 3.1 for $\bar{B}_{\{3/4\}} \rightarrow \partial B_{\{3/2\}}$ in B_2 at scale $(8/3)\varepsilon r$; (7) “send $\zeta \rightarrow 0$ ” made explicit: loss $\eta = \min(p/4, \zeta^2/(8p))$, Step 4 with $p - 2\eta$, annulus constant $A = \varepsilon^{\{-\eta/2\}}$, Thm 1.5 with $\zeta = \eta/2$ · kinds: correction (per-path blocks), own geometry, parameters · proposed · no.

DZZ3D-a..d · DZZ arXiv:1807.00422 l. 905–1080 (Def 3.6, L3.7, (eq-berlin1), (eq-LQG-tilde-B-Phi)) · LQGMetric/Papers/DZZ/{S3D6, S3VarBdry, S3L7Mass, S3L70pen, S3L7Indep}.lean · (a) the unstated variance fact (l. 975) by an own elementary argument (Var monotone in the scale, $\|a\|^2 - \|b\|^2 \leq \|a-b\|(\|a\| + \|b\|)$, DZZ L2.5 kernel form, etaVar_le ; error $O(\sqrt{L})$ within DZZ’s $O(\alpha\sqrt{L} \log L)$) — no killed-kernel lower bound needed; (b) (eq-berlin1)/(eq-LQG-tilde-B-Phi): the normalized exponent $\gamma\eta - \gamma^2/2 \cdot \text{Var}$ bounded by $1.2 \log t^{-1}$ with tail $\leq t^{\{1.2\}}$ (Markov on the exponential martingale) instead of DZZ’s $\eta \leq 1.5 \log t^{-1}$ (which near $\partial\mathbb{W}$, where the killed η has smaller band variance, gives only $t^{\{2-1.5\gamma\}}$, not $\leq t^{\{0.8\}}$ for $\gamma > 0.8$) — same $t^{\{0.8\}}$ conclusion; (c) Def 3.6 reading: “neighbouring” =

DyBox.Neighbour, $B'_i \subseteq B_{\text{large}} \setminus B$ with closed boxes (interior of B'_i disjoint from B), Φ takes the LQG measure as a parameter, $\Psi \mathbb{N}_{\infty}$ -valued (" $\Psi \leq \lambda$ " means finite and $\leq \lambda$); (d) etaRad_le_of_le elementary bound on the truncation radius instead of the monotonicity of $\sqrt{u} \log u^{-1}$ DZZ implicitly use (l. 999) · kinds: own arguments, gap fix near the boundary, readings · proposed · no.

****P2-M2L3-1** (obsolete by D69: gm_L5_7 holds for every open V)** · GM l. 2997–3018 (L5.7), 3110–3126 (use in L5.8) · LQGMetric/Papers/GM/S5/Tubes57SqMain.lean, Tubes57Sq.lean, Tubes58.lean · L5.7 and its use in L5.8 for square tubes only (D66), under the open node SepFromMeasSq (Borel measurability of $\{u \mid \text{SepFrom } V \text{ (nearComp } V \delta u) \text{ a d}\}$ for square tubes; strictly weaker than SepFromMeas) · measurability convention · proposed · no.

P2-M2L3-2 · GM (5.17), l. 2970–2978, condition (3) of L5.6 · LQGMetric/Papers/GM/S5/Tubes56Diam.lean, Tubes56Geom.lean · (5.17) is taken at every dyadic level (DFGPS L3.20 second display, gm_L56_sqDiam), not only for the squares of $\mathcal{S}_{\{\varepsilon_1 r\}}$: GM's closed squares of $\mathcal{K}$ can touch ∂V , so their internal paths may leave V and condition (3) does not follow directly; the chain argument through dyadic squares inside V is left to L56Geom · reading of the paper (gap fix) · proposed · no.

P2-M2L3-3 · GM L5.6 (l. 2959–2995), L5.8 (l. 3063–3126) · LQGMetric/Papers/GM/S5/Tubes56.lean, Tubes58*.lean · L5.6: $b_1 \leq 1 - \alpha$ instead of $b_1 = 1 - \alpha$ (the statement assumes only $\alpha \geq 3/4$), $p_1 = p_0/(9 \cdot 2^{\wedge\{L_2\}})$; L5.8: arcs encoded as finite index sets with $\#A \cdot \delta \leq 100$; own elementary arguments for the grid-translate transfer of DFGPS L3.20 to the centre z , the box count, and the $20\rho r$ separation of $\mathcal{Z}$ (linkPt_sep) · parameter conventions, own arguments · proposed · no.

DFA7b-2, DFA7c-1, DFA7c-2 · DFGPS arXiv:1905.00380 Lemma 3.19 and proof (T:2264–2300) · LQGMetric/Papers/DFGPS/L3_19Fin*.lean · (DFA7b-2) Prop 3.10 is normalized at the square's corner while (3.31) needs the centre: the Gaussian $h_{\rho}(w) - h_{\rho}(z)$ (variance ≤ 2) is absorbed by Hölder with $q = (p + 4dy/\gamma^2)/2$ — own argument for the paper's "proven similarly" (T:2300); (DFA7c-1) the paper's $o_{\varepsilon}(1)$ made explicit by $\delta = \zeta/(4p)$ (in Thm 1.5 and in the threshold $\varepsilon^{\wedge\{s+\delta\}}$); the switch from $h_r(z)$ to $h_r(0)$ (T:2297–2298) costs $\varepsilon^{\wedge\delta}$ plus the Gaussian tail L34.tail_far_part; the square part uses radius εr at the centre (the square lies in $\tilde{B}_{\{\varepsilon r\}}(z)$) instead of $2\varepsilon r$; (DFA7c-2) own elementary limits gauss_small, exp_small · own arguments, explicit constants · proposed · no.

P2-L47-a..d · GM l. 1701–1716 (Lemma 4.6 (c)), l. 1890–1912 (proof of Lemma 4.7), (4.59) · LQGMetric/Papers/GM/S4/L47Inputs{A,W,G,Z}.lean, L47Final.lean · (a) in L4.6 (c) " $w \notin \mathcal{B}^{\bullet}_{\{t_k\}}$ " is not assumed a.s. (it fails for large k) but derived on $\text{Stab} \cap \text{Hit}$ by an own elementary argument (gm_geod_mem_filledBall_of_end; _nw versions of P2-E3c's lemmas); (b) " $G \in \sigma(h|_{\{C \setminus B_r(z)\}})$ " in the proof of L4.7 is formalized a.s. (AEEEventIn, as GM's "by locality" gives it); Lemma 4.9 is applied with $\mathcal{G}' = m\Omega \cap \{\text{a.s. } \mathcal{G}\text{-events}\}$ (own standard argument, gm_L4_7_pair_ae, gm_condExp_ae_eq_aeSigma); (c) hr uses $r \leq \varepsilon r < \lambda_4 \varepsilon r$, i.e. $\lambda_4 > \lambda_3 = 1$, true in GM's application ($\lambda_4 = 4$) although GM Thm 4.2 also allows $\lambda_4 = \lambda_3$; (d) the trace of the hull σ -algebra uses one dyadic level n with $2 \cdot 2^{-n} < \lambda_4 \varepsilon r - \rho$ (own step) · a.s. readings, own arguments · proposed · no.

P2-E2T-1 · GM arXiv:1905.00383v3 Lemma 4.5, l. 1655–1688 · LQGMetric/Papers/GM/S4/L45Pos1.lean, L45Pos2.lean · null-measurability of the decoding events gmGeodCEv: own argument (GM do not discuss measurability; extends D65). Counting events with positive arc-hit clauses are analytic (injective tuple witnesses, gmP_leEncardAn); events with negative clauses are recovered by inclusion–exclusion $N_G(F) = N_{\{GU\{n\}\}}(F) + N_G(FU\{n\})$ (gmP_encard_split, gmP_nullMeas_count), valid because Conf_k is a.s. finite (GM S4.1). A positive-only decoding would fail (two disjoint arcs of a Jordan curve can have one closure inside the other) · own argument · proposed · no.

DZZPRE4-a..d · DZZ arXiv:1807.00422 l. 594–645 (L2.8, L2.9), l. 740–759 (L2.12) · LQGMetric/Papers/DZZ/{S2L8Scaled, S2L9Asm, S2L12Chain, S2L12Lower}.lean · (a) L2.8 along scales $a \cdot 2^{\wedge\{j\}}$ ("the same derivation"): each scale split into the dyadic telescope plus one sub-band inside a dyadic band, then dzz_sum_sup_tail — the derivation DZZ do not write out; (b) L2.9: part (1) stated as " $\tilde{W}$ is a white noise"; C chosen per probability space (law universal); V_1 any subset of $\mathbb{V}^{\wedge\{\xi\}}$ (boxes are a special case) — convention/parameter range; (c) L2.12 deterministic step (l. 756–758): own elementary proof (intermediate value on $t \mapsto |P(t) - u|$ and a length count; grid of spacing $s = \delta^{\wedge\{c_j\}}$ instead of $2^{\wedge\{-k_{\delta}\}}$); (d) L2.12 lower half: small-ball lower tail taken as the explicit hypothesis BallMassLowerTail (replacing DZZ's (eq-max-white-noise-process), (eq-LQG-negative-moment) for $\tilde{M}$) — still open, to be discharged · own arguments, conventions, open hypothesis · proposed · no.

DZZ3C-a..g · DZZ arXiv:1807.00422 l. 1218–1270 (L3.8, Cor 3.9, L3.10), l. 849–853 (eq-very-crude), l. 556–559 (eq-variance-truncation) · LQGMetric/Papers/DZZ/{S3L8, S3L9, S3L10, S3L10Asym, S3L10Var, S3L10Eta, S3L10Exp}.lean · (a) Cor 3.9 states the factor the proof gives, $(\delta/\delta')^3 e^{\wedge\{(\log \delta^{-1})^{0.9} + (\log \delta^{-1})^{0.8} + (\log \delta'^{-1})^{0.9}\}}$ (D_η version $(\delta/\delta')^3 e^{\wedge\{10(\log \delta^{-1})^{0.9} + (\log \delta'^{-1})^{0.9}\}}$) — DZZ's single factor $e^{\wedge\{(\log \delta^{-1})^{0.9}\}}(\delta/\delta')^3$ does not follow from P3.2 and L3.5 as stated; (b) P3.2 and L3.5 enter with a separate diameter exponent $\xi' > \xi$ (uniform forms DZZProp32U, DZZLemma35U), constant of L3.5 uniform over sequences (D71); (c) in L3.8 $M^{\wedge\{\xi^2\}}$ is the limit along $a \cdot 2^{\wedge\{-n\}}$ (used implicitly by DZZ), normalisations V arbitrary, chaos limits tested on rational balls (IsChaosLimit); (d) l310_asym absorbs $h = O(\sqrt{\log \delta^{-1}})$ into the $(\log \delta^{-1})^{0.9}$ terms (own step); (e) dzz_var_compare ((eq-var-compare) for $(\tilde{h}, \eta)$) not proved in DZZ, own proof by

their band argument; (f) in (eq-oferberlin1) Cauchy–Schwarz replaced by the pointwise AM–GM bound $|d|_S \leq (\epsilon d^2 + 1_S/\epsilon)/2$; (g) `logMinLGD` uses the junk value $\log 0 = 0$ when the minimum is infinite · statement as proved, parameter separation, own steps, convention · proposed · no.

DV-M2J-1..5 · GM l. 2004–2277 (§4.4), l. 2527–2579 (L4.22, L4.16), l. 1679–1726 · `LQGMetric/Papers/GM/S4/ManyGood*.lean` · (1) P4.12, L4.22, L4.16 assume $\lambda_4 > 1$ (GM l. 2555 “since $\lambda_4 \geq 1$ ”; T4.2’s only application P4.3 has $\lambda_4 = 4$) — parameter range; (2) P4.12 stated for dyadic ϵ only (condition 5 of $\mathcal{E}_r$ is imposed for $\epsilon \in \{2^{-n}\}$) — reading; (3) S4.6 covers the level set at d_0 with $[d_0 \pm 2\sigma] \subseteq [\lambda_4 \epsilon r, 2\lambda_4 \epsilon r]$ instead of at $2\lambda_4 \epsilon r$ (with GM’s level the grid point can fall outside $\mathcal{Z}_k$) — gap fix; (4) L4.22 chains the Hölder bound over consecutive circle points at spacing $\leq \epsilon r/4$ instead of GM’s balls $B_{\{\epsilon r/2\}}(w)$ ((4.40) does not follow literally from condition 3, whose balls $B_{\{2|u-v|\}}(u)$ can leave $B_{\{\epsilon r\}}(w)$) — gap fix; (5) L4.22/L4.16 use the Hölder unit $c_{re}^{\{h_r(0)\}}$ of condition 3, allow $\text{dist}(z, \mathcal{B}^{\{t_k\}}) = 2\lambda_4 \epsilon r$ (closed range of $\mathcal{Z}_k$); L4.16 for an arbitrary parametrization (“points reached after D-length t_k ”); its Hölder lower bound needs $|u-v| \leq ar$ (intermediate value, implicit in GM) · parameter range, readings, gap fixes · proposed · no.

DEV-DF-A10b-1, -2 · DFGPS arXiv:1905.00380 Prop 4.3 (T:2593–2600), proof T:2683–2744, for filled balls (D68) · `LQGMetric/Blueprint/DFGPSEstimatesF.lean`, `LQGMetric/Papers/DFGPS/P4_3F{Top,Data,Det,Main}.lean` · (1) route: for filled balls the excursion argument also runs backwards in time for geodesic times $\leq s$ — every ball around a point of $\partial \mathcal{B}^{\bullet}_s$ has a boundary point q' with $D(0, q') \geq s$ (exists_sphere_ge), backward bound $s - \tau \leq C(\tau - \sigma)$ (step_back, chain_count_back), time set split at s with one maximal chain per half; (2) gap fill: the case $0 \in B_k$ (not covered in the paper) by a dichotomy (exists_sphere_lt_or_subset), excursion end time = next chain time τ_{k+1} , count $nB = O(\log \epsilon^{-1})$. The a.s. length property is handled as a null set (outer-measure bound). DEV-DF-A10-5 resolved by D68 and this proof · own arguments, gap fill · proposed · no.

DZZ3E-i..iv · DZZ arXiv:1807.00422 l. 970–1015 (proof of L3.7) · `LQGMetric/Papers/DZZ/S3L7{Count,CountPsi,MainBox,Perc,PercIndep,PercChain}.lean` · (i) box count $8(2^{\{k\}}+2)$ instead of DZZ’s $8\epsilon/t$ (l. 980; +2 counts corner boxes); (ii) “ $\mathcal{E}_{\text{open}} \subseteq \{\Psi \leq \lambda\}$ ” (l. 994–996, not spelled out) by own proof `psiLe_of_bdry` (cells touching ∂B are no finer than the boxes of $\mathcal{B}_\partial$; cells form an antichain); (iii) independence for index distance $> r$ under $2R + 2ts \leq r \cdot \epsilon$ (DZZ: $\kappa = 2$ with $p^{\{1/(2\kappa+1)\}}$); the Perc library uses $(r+1)^2$; (iv) enclosures near $\partial \mathbb{V}$ via a clipped-annulus Peierls bound (D72; DZZ do not address boxes near $\partial \mathbb{V}$); own elementary exists_desc, anc_anc, exists_isChain_cover · own arguments, gap fix · proposed · no.

P2-SEPDEC-1 · GM L5.6 condition (2) (l. 2948–2950), L5.8 condition (2) (l. 3052–3054), proofs l. 2982–2989, 3111–3127 · `LQGMetric/Papers/GM/S5/Defs.lean` (SepNear, tubeEvent, linkEvent), `Tubes57Main.lean`, `Tubes58Geom.lean`, `Tubes58.lean`, `SepMeas56.lean` · condition (2) (“the component of $V \setminus O_u$ containing $z - 2r$ is at distance $\geq \epsilon r$ from the other components”) is required at every point u' of a neighbourhood of u , not only at u (D69). GM’s proof gives this verbatim, and the one later use (l. 3478) needs it only at u . The robust form makes condition (2) an open condition on u , which GM’s Lemma 5.7 needs for measurability (GM do not discuss measurability) · reading of the paper (measurability) · proposed · no.

P2-SEPDEC-2 · GM L5.6 statement (l. 2945: $V_r(z)$ made of squares of $\mathcal{S}_{\{\epsilon_1 r\}}(B_{\{2r\}}(z))$, $z \pm 2r \in V_r(z)$), L5.8 statement (l. 3049: squares of $\mathcal{S}_{\{\epsilon_0 r\}}(\mathbb{A}_{\{r/2, 2r\}}(0))$, $x, y \in U$) · `LQGMetric/Papers/GM/S5/Defs.lean` (L5_6, L5_8), `SepMeas56.lean` (L56GeomN), `Tubes58Geom.lean` (TubeProps, L58Geom), `Tubes56.lean` · the squares are those meeting the closed ball $\text{cl } B_{\{2r\}}(z)$ and the closed annulus $\{r/2 \leq |w| \leq 2r\}$. With GM’s open sets the statements are false: for $z = 2r$, the point $z - 2r = 0$ is a grid corner and the squares to its left miss $B_{\{2r\}}(z)$, so $z - 2r \notin V$. The square count in L5.6’s pigeonhole uses the box of $B_{\{3r\}}(z)$, $L = \lceil 6/\epsilon_1 \rceil + 1$ · correction of a paper error (boundary case) · proposed · no.

DV-D70 · GM arXiv:1905.00383 §4.2, (4.8) l. 1640–1650 and proof of Lemma 4.7 l. 1890–1912 (D70, decisions/DEC-47.md) · `LQGMetric/Papers/GM/S4/L47Meas{A,F,G,C,Final}.lean` · convention: GM §4’s conditional arguments (conditioning on $\mathcal{F}_k$) are run on a complete probability space (`[P.IsComplete]`); T4.2, stated for every probability space, is recovered by passing to P .completion (`gm_isWholePlaneGFF_completion`; probabilities are unchanged). GM never mention measurability; the a.s. unique geodesic P is not assumed measurable, its measurability on the completed space is proved (`gm_measurable_geod_of_complete`). Own argument (no source discusses it); status: convention.

DF-L25-LEN, DF-L25-POS, DF-L25-B, DF-L25-CUT · DFGPS arXiv:1905.00380 Lemma 2.5 (T:993–1019) · `LQGMetric/Papers/DFGPS/L2_5Proof*.lean`, `L2_5Final.lean` · (LEN) length property of limits in A (T:1005–1012): instead of joint convergence with $D_{h^{\epsilon}}(\cdot, \cdot; S_{\{R\}})$ plus Lemma 2.11, a local midpoint condition ($z \text{SetC}$, closed) is passed to the limit and geodesics are built by Menger’s construction inside $S_s(0)$ (Petrinin, Pure metric geometry, arXiv:2007.09846, l. 596–613); no coupling; the agreement event is used as in the paper; (POS) positivity on $S_r(0)$ via a domination lemma whose comparison laws depend on p through $R = R(p)$; (B) $D_{\{h,W\}}(\cdot, \cdot; W) = D_h(\cdot, \cdot; W)$ by splitting the proof of Lemma 2.11 into two closed conditions passing to limits in law ($D \leq D_W$ on $\bar{W}$; $D_W(u,v) \leq D(u,v)$ when $D(u,v) < D(u,w) \forall w \in \partial \bar{W}$), then the local-isometry step as in the paper, instead of a Skorokhod coupling; (CUT) `iInf_frontier_lfpD0n_le` · own elementary first-hit cut lemma. Note: global instances

(Countable/CompactSpace/PolishSpace) on DyProd, DyFam, closure W declared in L2_5ProofBSp · route changes, own lemmas · proposed · no.

DFA7d-1, DFA7d-2 · DFGPS Lemma 3.21 and proof (T:2337–2357), via the steps of L3.19 (T:2275–2298) · LQGMetric/Papers/DFGPS/L3_21Proof*.lean, DFGPSM2Asm2.lean · (1) the crossing distance $D_h(B_{\{\varepsilon\}}(z), \partial B_{\{2\varepsilon\}}(z))$ is made local in $B_{\{3\varepsilon\}}(z)$ and normalized by $c_{\{3\varepsilon\}}e^{\{\xi_h\}_{\{3\varepsilon\}}(z)}$ (Prop 3.1 with $K_1 = \bar{B}_{\{1/3\}}(0)$, $K_2 = \partial B_{\{2/3\}}(0)$, $U = B_1(0)$ at scale 3ε ; Thm 1.5 with 3ε) — constants only, own detail for “the proof is similar”; (2) for a length metric the distance from $A \subseteq B_\rho(z)$ to $\partial B_\rho(z)$ equals the internal distance in any $V \supseteq \bar{B}_\rho(z)$ (first-hit argument), used implicitly by the paper (Axiom II) · own details · proposed · no.

DEV-DFGPS-L28-SUP, -CMP, -MED, -RATIO · DFGPS arXiv:1905.00380 Lemma 2.8, T:872–898 · LQGMetric/Papers/DFGPS/L2_8Fin*.lean · (SUP) the zero-boundary version of Lemma 2.1 (“same proof”, T:883–885) is proved with a Gaussian sup bound (DZZ Lemma 2.3 + Borell-TIS, SupTail.tail_iSup_abs_box) on the tail densities instead of the polar formula + Lemma 2.2; explicit ε -bounds are own elementary estimates; (CMP) the a.s. comparisons (T:886–888) are restated per ε with probability bounds uniform in ε , because DDDF’s versions Y_δ are chosen separately for each δ ; measurability bookkeeping (dense countable sets, rational ε , joint continuity) own; (MED) “this implies $\lambda_\varepsilon \asymp \text{median}$ ” (T:888–891) via quantiles of the left–right crossing functional crossPhi (Prokhorov + portmanteau), own standard argument; (RATIO) the ratio $a\text{EpsDF}(\varepsilon/s)/a\text{EpsDF}(\varepsilon)$ for general squares (not justified in the paper, T:893–894) from DDDF (6.99) (Blueprint.DDDFEq6_99, D73) · own arguments, gap fill · proposed · no.

GMCID-a..d · Berestycki, An elementary approach to GMC (arXiv:1506.09113) §4, main.tex l. 680–700; DZZ l. 430–433, 648–658, 677–682 · LQGMetric/Dimension/GMCIdent*.lean · (a) approximations normalized by $CR(z)^{\{\gamma^2/2\}}e^{\{\gamma\bar{h} - \gamma^2/2 \cdot \text{Var } \bar{h}\}}$ (equal to the Duplantier–Sheffield circle-average measure) instead of DZZ’s $e^{\{\gamma\bar{h}\}2^{\{\gamma^2/2\}}}$ (deterministic factors with finite limit; irrelevant for exponents) — convention; (b) own elementary proofs where Berestycki gives none (conditional Fubini, L^1 continuity of $e^{\{aW(g)\}}$, first-moment Borel–Cantelli, Bochner-mean identity for the kernels); (c) the white-noise time filtration $\mathcal{G}_n$ replaces Berestycki’s Karhunen–Loève filtration (DZZ l. 653 points to it); (d) $ae_tendsto_wnGMC$ is stated under the coupling hypotheses $WNCircleCoupling$ and $\mathcal{G}_\infty$ -measurability until II (exists_wnCoupled) is proved · convention, own arguments, pending hypothesis · proposed · no.

P2-M2O-1..3 · GM arXiv:1905.00383v3 §6, l. 3570–3676 · LQGMetric/Papers/GM/S6/{Thm19, GLowLaw, Prop61Det}.lean, LQGMetric/Assembly/M2.lean · (1) GM (6.4): $\bar{D}(z, w)$ bounded by the triangle inequality and $\bar{D} \leq C_* \cdot D$ pointwise ($le_upperRatio_mul$, $away_sep$) instead of $\bar{D}$ -lengths of segments of the geodesic (same bound) — proof route; (2) GM treat $P[G_r(c', \beta)]$ as a deterministic number; proved ($prob_GLow_eq$) by universal measurability and invariance under adding constants (as $prob_attainedLow_eq$) — implicit fact; (3) the M2 assembly uses MQLem4_1Gen (D41) instead of the fixed-radii MQLem4_1 — statement of a cited result · proposed · no.

DV-D74 · GM arXiv:1905.00383v3 Thm 4.2 (l. 1554–1571), Prop 4.3 (l. 1582–1593), Prop 6.1 Step 1 (l. 3588–3599) · LQGMetric/Papers/GM/S3/Defs.lean (T4_2, P4_3, AsmM2), LQGMetric/Papers/GM/S6/Prop61Step1.lean · (1) statement change (stronger conclusion than GM states): Theorem 4.2 also says the start and target points lie outside the ball of radius $\lambda_4 r$ around the point z it produces. GM’s statement omits this, but their proof gives it (the balls used are at distance at least $\lambda_4 \varepsilon r$ from the filled metric ball containing the start point, and are only used while the target is farther away), and GM use it when they apply Prop 4.3 in Prop 6.1. (2) statement change (explicit hypothesis): Prop 4.3 assumes the path selector is almost surely a D_h -geodesic, which GM leave implicit by writing “the D_h -geodesic”. (3) the M2 assembly uses MQ Lemma 4.1 with general radii (D41) instead of the fixed radii 7/8, 15/16. (4) In Prop 6.1 Step 1 the geodesic is a chosen (a.s. existing) geodesic; c_1', c_2' are taken at the thirds of (c_-, C_-) , $\mu = \nu_-/2$, $\nu = \nu_-$. Status: decided (D74).

DF-B7 (P2-DFB7, 2026-10-01): DFGPS Lemma 2.12 (lem2_12, Papers/DFGPS/L2_12*.lean)

- **DF-B7-MOLL** (own elementary proof): DFGPS T:1036–1037 state without proof that $f^n * p_{\{\varepsilon_n^2/2\}} \rightarrow f$ locally uniformly. Proved directly ($tendstoUniformlyOn_heatMollify_seq$, L2_12Moll.lean): split the heat-kernel integral at $|u| = \delta$, use uniform continuity of f near 0 and the Gaussian tail bound $p_s \leq 2e^{\{-\delta^2/(4s)\}} p_{\{2s\}}$ far away. Status: own argument for an approximation-of-identity fact.
- **DF-B7-LOC** (reading): the localization (eqn-square-bdy-weyl) is formalized with the distance to the complement of the open square $S_{\{r'\}}(0)$ instead of to $\partial S_{\{r'\}}(0)$ (equivalent for LFPP by the first-hit cut $lfppC_cut$; the complement form is a closed condition, so it passes to the limit law without a path argument), then converted to Euclidean balls/spheres for DFLem7_1 (loc_of_agree). Status: equivalent reformulation.
- **DF-B7-LEN** (implicit facts made explicit): the node Lem2_12 does not assume that D_h is a length metric; the paper uses it implicitly (D_h is a subsequential limit, a length metric by Lemma 2.5). We prove it a.s. from

lem2_5_lim applied to the law of D_h (D_h is a.e.-measurable as an a.s. limit; a.s. convergence gives convergence in law, `mathlib tendstoInDistribution_of_ae_tendsto`). Also $\alpha_{\{\varepsilon_n\}} > 0$ for large n is derived from $D_h(0, 1) > 0$ (no positivity lemma for `aEpsDF` exists). Status: proofs of implicit facts, no statement change.

P2-M2M-1..5 · GM arXiv:1905.00383v3 §5.4–5.5, l. 3180–3430 · LQGMetric/Papers/GM/S5/{EventDefs, EventStmts, ShortcutHit, ShortcutDist}.lean · (1) statement form: Lemma 5.11 (L5_11) is stated deterministically at a field $g \in E_r$; the a.s. inputs it uses (D and $\tilde{D}$ are length metrics at g , Weyl scaling $D_{\{g-\varphi\}} = e^{\{-\xi\varphi\}} \cdot D_g$ and $\tilde{D}_{\{g-\varphi\}} = e^{\{-\xi\varphi\}} \cdot \tilde{D}_g$ for the bump φ , and the bi-Lipschitz bounds $c_* \cdot D \leq \tilde{D} \leq C_* \cdot D$ at g and $g - \varphi$) are hypotheses; these hold a.s. because $\mathcal{G}_r$ is finite. (2) P5.2 (C) quantifies over every pair of first hitting points instead of a measurable choice (GM footnote l. 3344: any choice works); stronger than GM. (3) parameter ranges (EData.Ranges): $a \in (0, 1)$ (as chosen at l. 3324; l. 3186 says $(0, \varepsilon_0)$), $\zeta \in (0, \varepsilon_0/100)$ and $\theta \in (0, \zeta/100)$ as chosen in the proof of Lemma 5.10 (l. 3322, 3326); Lemma 5.10 uses $p = 1 - (1 - p)/100$ (l. 3309, not the $/2$ typo at l. 3185). (4) the bump choices f_r, g_r enter L5_10/P5_2 existentially for each r (GM: chosen deterministically from U resp. W); only the constants, including Λ_0 , must be uniform in r , as in GM. (5) own elementary proof: $X \subseteq \text{int } \bigcup \mathcal{S}_s(X)$ (subset_interior_squares_m2m), used for $[x, (3/2 - \theta)x] \subseteq W_r^{\wedge} x$, which GM use implicitly in the proof of Lemma 5.13 (l. 3410, 3427) · statement forms, reading, own argument · proposed · no.

DV-M2K-1..2 (P2-M2K, 2026-10-01) · GM arXiv:1905.00383v3 §4.5, Lemma 4.21 (l. 2361–2398), Prop 4.17 (l. 2404–2433), proof of Thm 4.2 (l. 2437–2446) · LQGMetric/Papers/GM/S4/{IterateCount, IterateRate, IterateP417, IterateUnion, IterateT42, IterateWitness}.lean · (1) parameters of Lemma 4.18 (corrected form D29, $n = K+1$ indices): in the first application $m = \lceil 4\varepsilon^{\wedge} \theta K \rceil + 4$ instead of $\lceil 4\varepsilon^{\wedge} \theta K \rceil$ (so that the count threshold $(1 - \varepsilon^{\wedge} \theta)K$ of Prop 4.12 dominates $n - m/4$); in the second application $\alpha = 1 - p/2$, $\delta = p/4$ with $p = \kappa/2 - e \geq \varepsilon^{\wedge} \{2\nu + \zeta/2\}$ (GM: $\alpha = 1 - \varepsilon^{\wedge} \{2\nu + \zeta/2\}$, $\delta = \varepsilon^{\wedge} \{2\nu + \zeta/2\}/2$), so that $P[E'_k | \mathcal{F}_k] \leq 1 - p < \alpha$ is strict as Lemma 4.18 requires, and $m = n - m_1$ instead of $\lceil (1 - 4\varepsilon^{\wedge} \theta)K \rceil$; parameter ranges $\theta < \beta$ and $4\nu + \zeta < \beta$ (GM: $4\nu < \beta \wedge \theta, \zeta$ small) — parameter choice inside the proof, same conclusion. (2) own elementary proofs of facts implicit in the paper: the number of grid points of $\varepsilon^{\wedge} q r \mathbb{Z}^2$ in rU is $\leq 4(\rho+2)^2 \varepsilon^{\wedge} \{2q\}$ (disjoint grid cells), and the witness $(z, r) \in \mathcal{Z}^{\wedge} \mathcal{G}_k$ satisfies $z, w \notin B_{\{\lambda_4 r\}}(z)$ (from $\text{dist}(z, \partial \mathcal{B}^{\wedge} \{t_k\}) \in [\lambda_4 \varepsilon r, 2\lambda_4 \varepsilon r]$, $z \in \mathcal{B}^{\wedge} \{t_k\}$ and $\partial \mathcal{B}^{\wedge} \{t_k\} \subseteq \tilde{B}_L(z)$ with $L + 3\lambda_4 \varepsilon r < |z - w|$; the conjunct of D74) · proof route / own argument · proposed · no.

DV-M2J2-1..3 (P2-M2J2, 2026-10-01) · GM arXiv:1905.00383v3, proof of L4.15 (l. 2125–2170) and of P4.12 (l. 2257–2276) · LQGMetric/Papers/GM/S4/{P412Step12, P412Sigma, P412Final}.lean · (1) reading: the scale $\varepsilon^{\wedge} \kappa$ of L4.15 Steps 2–3 ($R_k, \sigma_k, B_{\{16\varepsilon^{\wedge} \kappa r\}}$) is any dyadic $\delta = 2^{\{-m\}} \leq \varepsilon^{\wedge} \kappa$, because condition 6 of $\mathcal{E}_r$ (regC6) is imposed only at dyadic scales; (4.39) is proved for every such δ (p412_eq439); with the D16 time unit $\tau_{\{\ell r\}}$ the replacement of $h_r(\theta)$ by $h_r(z)$ (condition 4) is not needed. (2) own elementary argument (GM implicit): GM writes $P'(t_k)$ for “the” point where a $D(\cdot, \cdot; C \setminus \text{cl } B_r(z))$ -geodesic hits $\partial \mathcal{B}^{\wedge} \{t_k\}$; we prove that all its hitting points coincide (p412_avoid_hit_eq: a D -geodesic to a point of $\partial \mathcal{B}^{\wedge} t$ stays in $\mathcal{B}^{\wedge} t$, which avoids $B_r(z)$, so by minimality the length between two hits is 0) and run GM (4.42) with the connected set $Q((u_1, T]) \cup \text{cl } B_r(z)$ after the last exit time u_1 (p412_hit_mem_of_dU). (3) L4.15 Step 1 monotonicity $X_{\{s, t'\}} \subseteq X_{\{s, t\}}$ is proved for the D44 leftmost geodesics (hitSetDD, p412_hitSetDD_mono); Blueprint.hitSet/CONFThm3_9At still use the superseded one-sided-limit IsSideGeod, for which restriction is not available (flag for the orchestrator: align hitSet with D44) · proposed.

DV-NEGMOMU-1 (P2-NEGMOMU, 2026-10-02) · DZZ Lemma 2.12 lower half (LBM_LGDDarXiv.tex l. 750–755) · LQGMetric/Papers/DZZ/S2L12Neg{Cmp, Ball, Main, Cor}.lean · proof route: the uniform scaled negative moments $E_{\mu_h(B(w, s))^{\wedge} p} \leq C s^{\{-A\}}$ (negU_ball_neg_moment, any $p < 0$, $A = p(p-1)/2 - 2p$) are proved by Kahane comparison of the level- m dyadic square with a fixed reference square plus an independent constant-covariance Gaussian (Berestycki–Powell arXiv:2404.16642, GMCproperties.tex l. 1211–1218), rather than by the GMC scaling identity DZZ quote; BallMassLowerTail ($p = 1$ in the tail, Markov with moment -1) is the form consumed as hT by dzz_lemma212_lower_exp. No change of statement.

DV-DFB34f-1..3 (P2-DFB34f, 2026-10-01) · DFGPS arXiv:1905.00380, proof of Lemma 2.8, T:893–894 (“By Lemma 2.6 and the scale and translation invariance of the law of h , modulo additive constant, it suffices to prove the lemma for $S = [0, 1]^2$ ”) · LQGMetric/Papers/DFGPS/{L2_8GenHeat, L2_8GenRatio, L2_8GenTrans, L2_8GenMain}.lean · (1) own argument for the “standard change of variables” with translation, $(h(\cdot + a))^{\wedge} \varepsilon(z) = h^{\wedge} \varepsilon(z + a)$ a.s. (ae_heatMollify_translate): the two cutoff truncations differ only where the heat kernel is $O(e^{\{-n\}})$, summable by IsWholePlaneGFF.ae_summable_abs_of_decay, exactly as the scaling case ae_heatMollify_affineComp (L2_6.lean); (2) the reduction uses $\tilde{g} = g(s + a)$ without recentring (the unit-square case lem2_8Gff_unitSq holds for any additive normalization), so no random factor $e^{\{\xi\}}$ appears; (3) the normalization change $\alpha_{\varepsilon} \rightarrow \alpha_{\{\varepsilon/s\}}$, not mentioned in the paper, is bounded via DDDF (6.99) per decision D73 (aEpsDF_ratio_bdd, Blueprint hypothesis DDDFEq6_99). Status: proof detail / cited-input use (D73).

DV-DFB8-1..3 (P2-DFB8, 2026-10-02) · DFGPS arXiv:1905.00380, proof of Lemma 2.14 (lem_lfpp-constant, T:1073–1098) · LQGMetric/Papers/DFGPS/{L2_14Ratio, L2_14}.lean · (1) cited result: the Λ -bounds (T:1096–1097) are said to be “immediate from DDDF Thm 1, Eq. (1.3)” and $\alpha_{\varepsilon} \asymp \lambda_{\varepsilon}$; (1.3) alone only gives $\lambda_{\{\delta\eta\}}/\lambda_{\eta}$

up to $e^{\{O(\sqrt{|\log \eta|})\}}$, unbounded as $\eta \rightarrow 0$, so the proof also uses DDDF (6.99) (DDDFEq6_99, already an input of blueprint/DF.md row 17) · kind: statement of a cited result (extra input) · (2) own elementary bookkeeping: $e^{\{K\sqrt{L}\}} \leq e^{\{K^2/4\}}e^{\Lambda}$ turns the $e^{\{O(\sqrt{|\log \delta|})\}}$ error into a power of δ ; $\delta \in [\delta_0, 1)$ is handled by writing $a_{\{\delta\eta\}}/a_{\eta}$ as a quotient of two small- δ ratios; $c_r > 0$ follows from the same bounds; the final exponent is $\Lambda+1$ (L214.ratio_bounds, L214.scaling_const_bounds) · (3) statement gap (question for the orchestrator, open): the existence argument for c_r (T:1080–1095) uses (eqn-scaled-conv), i.e. convergence in law of $r^{-1}e^{\{-\xi h_r(0)\}}a_{\varepsilon^{-1}D_h^{\varepsilon}}(r, \cdot)$, which needs joint convergence of $(h, a_{\varepsilon^{-1}D_h^{\varepsilon}})$ (as in Lemma 2.13, where the pair is taken), not only of the metric as in the node Lem2_14; it also takes the median of the internal left–right crossing of the limit metric, a functional that is not continuous for the local uniform topology on $C(\mathbb{C} \times \mathbb{C})$ (needs the internal-metric tightness of L2.8 and L2.11), and needs that median to be unique (S8, not argued in the paper) · status: (1)(2) proposed; (3) decided by D78 (DV-D78).

DV-M2L56-1..2 (P2-M2L56, 2026-10-01) · GM arXiv:1905.00383v3, proof of Lemma 5.6, condition 3 (l. 2991–2994) · LQGMetric/Papers/GM/S5/{Geom56Step, Geom56Whit, Geom56Cond3, Geom56Main}.lean · (1) own elementary argument (GM leave it implicit, cf. P2-M2L3-2): a point of a closed square S of the tube is joined to the centre of S inside V by a Whitney chain through dyadic squares of side $2^{-k-4}s$ (16 steps per level), cost $\leq (2 + 32/(1-2^{-\chi}))t$ (geom56_whitney); the centres of two squares sharing a point of O_u are joined through that point. (2) GM’s graph-distance bound 40 is replaced by 43^2 (the number of squares meeting $B_{\{20\varepsilon_1 r\}}(u)$); $\kappa = (2 + 2 \cdot 43^2)(2 + 32/(1-2^{-\chi}))$ is still independent of ε_1 · own argument / constant · proposed · no.

DV-D77 · GM arXiv:1905.00383 L5.6 condition (2) (l. 2949–2951) and proof (l. 2982–2989); L5.8 condition (2) (l. 3053–3055) and proof (l. 3120–3126); §5.4 l. 3476–3486 · LQGMetric/Papers/GM/S5/Defs.lean (SepDiscNear, tubeEvent), Tubes57Main.lean, SepMeas56.lean (L56GeomN), Geom56Main.lean (L56ConstrN), Tubes58.lean · condition (2) now says, at every point u' near u , both that the component of $V \setminus O_{\{u'\}}$ containing $z - 2r$ is at distance $\geq \varepsilon r$ from the other components, and that $z + 2r$ is not in that component (the other way round for v). The second part is GM’s own wording (“removing O_u disconnects $V_r(z)$ ”, title and proof l. 2988). The earlier Lean reading left it out and was then true for any tube whose complement of O_u stays connected. linkEvent (GM L5.8) now uses the same reading (SepDiscNear $(U \times y) (20\varepsilon r) u \times y (\varepsilon r)$ and $\dots v \times y \dots$, LQGMetric/Papers/GM/S5/Defs.lean), as written out in decisions/DEC-77.md §4; the interim gap noted earlier is closed (GMLEM-REC, 2026-10-03, AUDIT-GMLEM-F issue 4) · reading of the paper (restores the paper’s condition) · decided (D77) · no.

DV-D78 (P2-DEC78, 2026-10-02; supersedes DV-DFB8-3) · DFGPS arXiv:1905.00380, Lemma 2.14 (T:1073–1079, proof T:1080–1098) and Lemma 2.13 (T:1060–1069, proof T:1104–1119) · LQGMetric/Papers/DFGPS/Nodes.lean (Lem2_14, Lem2_13), LQGMetric/Papers/DFGPS/L2_14.lean (lem2_14, L214.liminf_scaling, L214.ratio_rel) · (1) statement change of a node: the paper says the limit $c_r = \lim r a_{\{\varepsilon/r\}}/a_{\varepsilon}$ exists along the subsequence; we only state (and prove) that $c_r := \liminf$ is a positive cluster point of the ratios with the Λ -bounds, for every sequence $\varepsilon_n \rightarrow 0$. Reason: the paper’s limit argument uses the median of the crossing of $[0, 1]^2$ inside the square for the limit metric, which is not determined by the limit (T:999–1000) and needs a unique median (S8, not argued), and nothing downstream uses the limit. (2) Lemma 2.13 is stated for a coupling with joint convergence of $(h, a^{-1}D^{\varepsilon}_h)$ (as T:1058, T:1062 say), and its constants depend only on the sequence ε_n (paper: on the law of D_h). (3) own elementary argument: the liminf bookkeeping (L214.liminf_scaling) · kind: statement of a node / own argument · status: decided (D78).

DG3B-1, DG3B-2, DG3C-1 (P2-DG3B / P2-DG3C, 2026-10-01/02) · DG arXiv:1807.01072, Lemmas 3.1, A.1, A.2 (DG:966–974, 2129–2244) · LQGMetric/Papers/DG/AppA*.lean · (1) an own integrated shell estimate (integral_killedHeat_ball_diag_sub_le) replaces DG’s pointwise “uniformly over w ” bridge estimate, which is not literally true near $\partial B_{\{1/10\}}$; (2) an own semigroup / Schur-test route replaces DG’s bridge coupling at $\pm\varepsilon$ (DG:2183–2232), and proves DG’s “easily checked” continuity of $h^{\wedge}U_{\{1, \infty\}}$ (Chapman–Kolmogorov at time $1/4$, $\int p_U(\sigma; u, \cdot) \leq 2R^2/\sigma$); (3) A.1 and the $(h^{\wedge}U, \hat{h})$ case of (3.5) are stated for a box $K \subseteq \{z : B(z, 1/10) \subseteq U\}$ (DG’s K for a square U is a square), with $h^{\wedge}U$ the white-noise field $\lim h^{\wedge}U_{\{t, \infty\}}$ (DG:2142–2146) until its identification with the zero-boundary GFF (RV Lemma 5.4; needs killedHeat_sqOpen); valid for all $A \geq 0$ (DG: $A > 1$) · own arguments, form pending identification · proposed · no.

D-CROSSCUT-1, -2 (P2-CROSSCUT, 2026-10-02) · GM arXiv:1905.00383, proofs of L4.14 (l. 2478–2479) and L4.13 (l. 2081) · LQGMetric/Topo/Crosscut{Radial, Sep, Main, Theta}.lean · (1) “an arc of $\partial B \setminus K$ divides $C \setminus K$ into one bounded and one unbounded component” proved as Topo.Crosscut.crosscut_circle by an own elementary route (radial collars give at most two components; Eilenberg’s criterion with logarithms glued over the connected K , Janiszewski-style, Burckel Ex. 4.37) instead of Pommerenke Prop. 2.12 (Riemann map + Jordan curve theorem); no Jordan hypothesis on ∂K ; (2) the separation core of L4.13 (“one of the components V of $C \setminus (\mathcal{B} \cup X)$ is bounded”) as theta_sep / theta_bounded (two applications of sep_glue), own argument without prime ends or the Jordan curve theorem · own arguments for cited standard facts · proposed · no.

DV-M2N-1..3 (P2-M2N, 2026-10-02) · GM arXiv:1905.00383v3, Def- $\mathcal{G}$ and Lemma 5.4, l. 2724–2862 · LQGMetric/Papers/GM/S5/Prop43{CM, Frk, Out, Dir}.lean · (1) Lemma 5.4 without conditional laws or the Markov decomposition: integrate over events of the conditioning σ -algebra (invariant under $h \mapsto h - \varphi$) with the unconditional

Cameron–Martin formula and a finite partition by the value of φ (`condExp_le_cm`, `condExp_le_cm_out`) — own route, same mathematics; (2) $\Lambda := e^{\{3\Lambda_0\}}$ instead of GM’s $e^{\{\Lambda_0\}}$: the DV-B6 constant $\exp(\Lambda_0 + 2 \max(\varphi, \varphi) \cdot \nabla)$ depends on r while P4_3 needs Λ before R ; cross term via $|\langle \nabla \varphi, \nabla \psi \rangle| \leq \frac{1}{2}(|\nabla \varphi|^2 + |\nabla \psi|^2)$ and Green’s identity (`exp_dir_subTest_le`) — own elementary proof; (3) $\mathcal{G}_r(z)$ (`frkE`) defined at centre z for h instead of at 0 for $h(\cdot+z) - h_1(z)$ (l. 2846), the G parameter taking the translated $\mathcal{G}_r$ — equivalent a.s. by translation invariance. DV-M2N-4 (condition (4) of `GeoIterateHyp` modulo constants) pending decision D79 · own route, constants, convention · proposed · no.

DV-D75 (P2-DEC7576, 2026-10-02) · GM `arXiv:1905.00383v3` Thm 4.2 (l. 1554–1571), Lemma 4.7 (l. 1782–1788), Lemma 4.11 (l. 1979–1990), Prop 4.12 (l. 2022–2024), Lemma 4.22 (l. 2555) · `LQGMetric/Papers/GM/S4/{Regularity,RegularityProb,RegularityCond2Prob,RegularityCond3,RegularityCond4,RegularityCond5b,Regularity` · lean · (1) no deviation, made explicit: the constants of Lemma 4.11 (a, V), Prop 4.12 (C, ε_0) and (4.19) are chosen before the probability space, the field and the events $E_r(z)$, radii $r \in \varepsilon_k$, as GM’s “depending only on ...” says; (2) own argument (planned, node `gm_T4_2_of_gt1`): GM’s proof of Lemma 4.22 uses $\lambda_4 \geq 1$, which Theorem 4.2 does not assume; the case $\lambda_4 \leq 1$ is reduced to $\lambda_4 > 1$ by multiplying all λ_i by an integer s and dividing the base scale by s (radii, events and the grid are rescaled accordingly; the conclusion is unchanged). Status: decided (D75).

DV-D76 (P2-DEC7576, 2026-10-02) · CONF `arXiv:1905.00381` Thm 3.9 (C:1506–1509), Lemma 2.4 (C:533–537) · `LQGMetric/Blueprint/CONFResults.lean` (`hitSetLM`, `CONFThm3_9At`) · statement of a cited result made literal: Theorem 3.9 counts the boundary points hit by leftmost geodesics in the sense of CONF Lemma 2.4 (“every geodesic to the endpoint lies weakly to its right”, D44), no longer by the one-sided-limit reading of the earlier Blueprint (which is not CONF’s notion). The other Blueprint CONF statements and GM §4’s confluence sets still use the earlier reading (flagged in DEC-76). Status: decided (D76).

DV-D79 (P2-DEC7576, 2026-10-02) · GM `arXiv:1905.00383v3` Thm 4.2 (l. 1554–1571), Prop 4.3 (l. 1582–1593), Lemma 5.4 (l. 2801), l. 214, 1200–1205, 3344 · `LQGMetric/Papers/GM/S3/{Defs,SigmaMod}.lean`, `LQGMetric/Papers/GM/S6/{GeodSelN,Prop61Step1}.lean`, `LQGMetric/Papers/GM/S5/Prop430Out.lean` · (1) reading: in Theorem 4.2 (4) the conditioning on the field outside $B_{\{r\}}(z)$ is modulo additive constants (GM’s field is defined modulo constants; GM say so at l. 1200–1205 and 2801); (2) convention: the radii set is $\mathcal{R} \subset (0, \varepsilon_0 r]$ (GM write $(0, \varepsilon_0]$, which with base scale r contradicts hypothesis (1)); (3) explicit hypotheses GM leave implicit: the events $E_r(z)$, $\mathcal{G}_r(z)$ and the geodesic selector are unchanged by adding a constant to the field (GM l. 2801; $D_{\{h+c\}} = e^{\{\xi c\}} D_h$), and in Prop 4.3 the selector is a measurable function of the field (GM’s “the geodesic” is the a.s. unique one, MQ Thm 1.2); (4) Prop 6.1 Step 1 now uses a measurable, constant-invariant selector built from MQ Theorem 1.2 and the Lusin–Souslin theorem (own construction around cited inputs) and takes `MQThm1_2Weak` instead of `DFGPSLem3_8`. Status: decided (D79).

DFB9-1.4 (P2-DFB9, 2026-10-02) · DFGPS `arXiv:1905.00380`, Lemmas 2.17, 2.19, Def 2.15 (T:1132–1282) · `LQGMetric/Papers/DFGPS/{L2_19CI,L2_19,L2_17,L2_17Ind,L2_17Exh,L2_17Step4}.lean` · (1) L2.19’s change of conditioning as weak union up to null events (`aeClosure`; the circle-average identities hold only a.s.) — own proof; (2) in L2.19 $D_{\{h\}} = e^{\{-\xi h_r(z)\}} D_h$ (the paper’s identification at T:1191, consistent with `IsXiAdditive2`); Def 2.15 stated one V at a time; empty V handled trivially — reading; (3) exhaustion of V by dyadic domains with connected closure (the class of `Lem2_5B`) — own proof; (4) L2.17: ξ -additivity base case for h from the $(z, r) = (0, 1)$ case via $h = h - h_1(0)$ a.s.; length property from L2.5 A; the Step 4 gap (T:1268–1278: conditioning on $h|_V \rightarrow h|_{\{cl V\}}$ does not follow from (eqn-limit-metric-ind) as written; counterexample $X = \xi$, $Z' = \xi \text{ XOR } Y$) is open, decision D80 · reading, gap · proposed · no.

DV-M2K2-1.3 (P2-M2K2, 2026-10-02) · GM `arXiv:1905.00383v3`, (4.9) l. 1650, Lemmas 4.19/4.20 (l. 2318–2360, 2593–2611), Prop 4.17 · `LQGMetric/Papers/GM/S4/Iterate2*.lean` · (1) GM’s $\mathcal{F}_k$ used through `gmFilt k = \bigsqcup_{j \leq k} \mathcal{F}_j ($\mathcal{F}_j \subseteq \mathcal{F}_k$ holds only up to null sets in the model; conditional expectations agree a.s., gm_condExp gmFilt); (2) the measurability statements of L4.19/L4.20 used a.s.; Prop 4.17 applied as gm_P4_17_ae; (3) F_k built from the countably many candidate pairs (gmF0) instead of (4.39) at every z ; GM’s $s_{\{k+1\}} \leq \tau_{\{2\ell r\}}$ left out of F_k (it holds on $\mathcal{E}_r$, gm_S4_3, gm_filledBall_s_subset_ball); the geodesic part of F_k and the λ_5 question go to decision D81 · a.s. readings, construction · proposed · no.`

DZZ312-1..6 (P2-DZZ312, 2026-10-02) · DZZ `arXiv:1807.00422`, §3.4 (Def 3.11 – Lemma 3.16, l. 1272–1500) and Cor 3.3 (l. 859–862) · `LQGMetric/Papers/DZZ/{S3L12*,S3L13Meas,S3L16*}.lean` · (1) Cor 3.3 proved with rate $(\log \delta^{-1})^{-0.1}$ instead of DZZ’s $e^{\{-(\log \delta^{-1})^{0.9}\}}$, which does not follow from Prop 3.2’s bound $|\log D - \log D'| \leq (\log \delta^{-1})^{0.9}$ (D82); (2) Def 3.11: w ranges over the open box C_{large}° (DZZ’s “ $w \in C_{\text{large}}$ ” fails at the right/top edge with half-open cells); (3) Def 3.14 uses $M < \delta^2$, the bad event of (eq-B-good) $M \geq \delta^2$, matching the partition rule “split iff $M \geq \delta^2$ ” (null difference); (4) (eq-B-good): band $\eta^{\{s\}}_{\{\varepsilon s\}}$ instead of $\eta^{\{\varepsilon s/\log s^{-1}\}}_{\{\varepsilon s\}}$; Chernoff bound with $\lambda = 2/\gamma^2 + 1/2$ on the normalized exponent instead of DZZ’s field threshold; requires $\alpha^* \geq (\lambda\gamma\alpha+1)/q$; own `etaBandVar_le`; (5) L3.12 stated uniformly in u, v ; α_0 replaced by “ $\mathcal{E}_{\{\delta, \alpha^*\}}$ holds with high probability”; $\mathcal{E}_{\{\delta, \alpha\}} = \text{eventEDeltaAlpha}$ (D64) in $\mathcal{E}_{\{\delta, \alpha^*, u, v\}}$, `eventEFine` in L3.16; own counting lemmas and asymptotics; (6) L3.13’s conditional law stated via `condExp` of indicators indexed by $\delta' < s_{\{B_i\}}$, $x \in (B_i)_{\text{large}}$ (DZZ’s display is inconsistent; follows their sentence and φ_F); $\mathcal{E}_{\{\delta, \alpha^*, u, v\}}$ is not literally $\mathcal{V}_\delta$ -measurable (η -increments), so L3.13 is an a.s.

equality of conditional expectations on that event · statement as proved, readings, own steps · proposed · no.

D-P18-a.c (P2-DDDF16c, 2026-10-02) · DDDF arXiv:1904.08021, Prop 18 (tightness.tex l. 900–938) · LQGMetric/Papers/DDDF/P18{Zero,Comb,Law,Geom,Strip,Blocks,Step3}.lean · (a) Step 3 counting: instead of “each of the $\geq 2^k$ visited blocks forces a crossing of one of the four $2^{\lfloor k \rfloor}(1,3)$ rectangles around it”, the strips $[j\delta, (j+1)\delta]$ with even j give $\lfloor 2^k/2 \rfloor$ sub-arcs on disjoint time intervals, each crossing the 1×3 block rectangle left–right or a 3×1 block rectangle vertically (same rectangles; the factor $1/2$ goes into $e^{-C\sqrt{k}}$); Prop 2’s bound on $|\varphi_{\{0,k\}}|$ instead of the infimum; bad probabilities $< p/3$ instead of $p/2$; union bound over $3 \cdot 2^k(2^{k+1})$ block rectangles; P18Step3 stated for $p \leq p_0$ (Cor 17 holds for small p ; the paper says “depending on p ”) — own argument; (b) combination: m with $2^m \leq s^2/(B \log s) < 2^{m+1}$ so all exponents are $\leq D\sqrt{(2^m m)} \leq s$ (DDDF: $s^2 = 2^m$, $e^{\{cs\sqrt{(\log s)}\}}$); small s into C ; Step 4 covers $2^n \leq s^2$ — own bookkeeping; (c) $n = 0$ (not treated by DDDF): $\varphi_{\{0,0\}} = 0$ a.s., so $L_{\{3,1\}} \leq 3 < e^s \cdot \Lambda_0 \ell_0$ · own arguments · proposed · no.

P2-M2M2-1.5 (P2-M2M2, 2026-10-02) · GM arXiv:1905.00383v3, §5.3–5.4 (l. 3217–3222, 3374–3550) · LQGMetric/Papers/GM/S5/{Event2Fin,Shortcut2*}.lean, EventStmts.lean · (1) EtaChoice should contain $2cs^{-1}C\eta < 1$ (GM’s “small η ”, l. 2929), for now the hypothesis $hpos$ (D83); (2) GM l. 3505 compares distances from u and from p as if they were the same point; repair by the triangle inequality needs $1 + 3\eta \leq C/c$ ($h\eta 3$, own argument, Shortcut2Eq541; D83); (3) $L5_11$ needs $0 < S.c.r$ (used by Lemma 5.12; D83); (4) own elementary proofs: exists_cross_m2m2 (one sentence in GM, l. 3455), the $\varphi < \xi^{-1} \log 2$ neighbourhood trick in 5.14 (factor $1/2$ absorbed by slack), the limit $\varepsilon \rightarrow 0$ in gm_L5_14_enter; (5) the entry step (GM l. 3477–3489: P^φ visits O_u and later O_v) is a possible gap in GM — the first point of $\tilde{P}^\varphi$ may lie near y' — open, decision D83 (Fable). D79 (3) applied in EventStmts.lean ($\exists U \text{ fb gb before } \forall \Omega$ in $L5_10$, $P5_2$) with the invariance clause added to $P5_2$ · gap fixes, own arguments · proposed · no.

DZZ3FG-1.7 (P2-DZZ3F / P2-DZZ3G, 2026-10-02) · DZZ arXiv:1807.00422, Lemma 3.7 (l. 958–1035), (eq-280318b), Lemma 3.5 (l. 1037–1083) · LQGMetric/Papers/DZZ/{S3L7Fin*,S3L5*}.lean · (1) L3.7 percolation with the clipped-annulus Peierls bound (D72), $r = 3$, $\alpha > 4 \cdot C_{mc}$ (slightly stronger than DZZ); (2) continuous paths discretised into $\ast$ -chains of boxAt boxes and the $\mathbb{Z}^2$ -chain encoded into DyBox enclosures (exists_path_sites, hasEnclosure_of_sites) — own steps; (3) (eq-280318b) by approxDist_mono etc.; (4) (eq-B-good-Phi): band $\eta^{\{s\}}_{\{ts\}}$ instead of DZZ’s $\eta^{\{\varepsilon s\}}_{\{ts\}}$, threshold $1.2 \cdot \log t^{-1}$ on the normalized exponent, row-to- D' step own elementary proof, stated for every $\alpha > 0$ uniformly in B and x ; (5) start pieces: eq-B-good-Phi applied to the ≤ 9 boxes per level with $u \in B_{\text{large}}$ (what S_u needs) instead of “all dyadic boxes containing u ”; (6) (Eq.lowerboundforDprime) own elementary proof (DZZ give none); (7) Lemma 3.5 with $\iota = C_{Mc}/2$ (DZZ $C_{Mc}/3$), $\alpha = 4C_{mc} + 1$, for $\xi d < C_{Mc}$ (DZZ $0 < \xi < C_{Mc}/3$, $\xi d = \xi$; D71), Lemma 3.1 also used at δ' (implicit in DZZ), the crossing claim in the deterministic form L35Crossing with own constants (DZZ: λ/ε^2 and $2\delta^{\wedge\{-i\}}(\delta/\delta')^3$); its proof (nested enclosed regions) pending D84 (since decided and applied: DV-D84) · parameters, own steps · proposed · no.

DV-D80 (P2-DEC80, 2026-10-02) · DFGPS arXiv:1905.00380, Lemma 2.17 proof, Steps 2–4 (T:1224–1282) · LQGMetric/Papers/DFGPS/L2_17CoreFM.lean, decisions/DEC-80.md · a place where we do not follow the published proof: Step 4’s passage from conditioning on $h|_{\mathcal{V}}$ to conditioning on $h|_{\{cl \mathcal{V}\}}$ (T:1276–1277) does not follow from (eqn-limit-metric-ind) (counterexample in DEC-80 §1). We instead keep $\sigma(h|_{\{cl \mathcal{V}\}})$ on the left side of the independence from the LFPP stage on (where it holds by the Markov property and the locality of $\hat{D}^\varepsilon$) and pass it to the limit with the law of h held fixed (L217.tendsto_integral_mul_of_fixed_marginal, a standard fact proved by L^1 -approximation of bounded measurable functions by bounded continuous ones; own elementary proof, ≈ 100 lines); Step 4 is then plain weak union. Lean-specific: the field coordinate of every joint law is taken as the countable coordinates $\text{pairJ } T \circ h$ (mathlib’s compact-convergence topology on DistC has no Prokhorov/regularity theory and is not known to be Borel for the cylinder σ -algebra); the auxiliary coupling of Step 3 is built on a new space and only $\text{Indep of comap } \sigma$ -algebras of $(\text{pairJ} \circ h, D_h)$ is transported back to the given space, the final conditional independence being produced there · decided · D80.

KHSQ3-1, KHSQ4-1 (P2-KHSQ3 / P2-KHSQ4, 2026-10-02) · killed heat kernel of the square = image (sine) series, used by GMC identification (DZZ (eq:Green_fxn) l. 400–403) and DG Appendix A · LQGMetric/Field/KilledHeatSq{Markov,Stop,Lim,Open,Ker}.lean · (1) proved by optional stopping of the space-time martingale $e^{\{\lambda_p s/2\}} \varphi_p(z + B_s)$ at the first exit on a dyadic grid, $\text{mesh} \rightarrow 0$, equal sine coefficients + Parseval, then Chapman–Kolmogorov for the pointwise statement (Karatzas–Shreve §4.3, Problem 2.8.8), instead of Feller II §X.5’s iterated reflection; (2) the grid-vs-open staying step (N) by Brownian scaling ($z + \mu^{\{-1/2\}} B_s, \mu \downarrow 1$; B_t has no atom on the boundary lines) instead of the two-sided bridge-maximum law · own assembly of standard arguments · proposed · no.

DV-D81 (P2-DEC81, 2026-10-02) · GM arXiv:1905.00383v3 Lemma 4.22 (l. 2527–2540, (4.39)), Lemma 4.19 (l. 2322–2327, 2584–2601), Lemma 4.20 (l. 2349), Thm 4.2 (2) (l. 1559) · LQGMetric/Papers/GM/S4/Iterate3Geo.lean, Iterate2L419F.lean · (1) definition change: the geodesic part of F_k is GM’s (4.39) with the internal distances taken in the fixed open set $\mathbb{C} \setminus cl B_{\{\varepsilon r\}}(z)$ (GM: $\mathcal{B}^{\wedge\{-s_{\{k+1\}}\}} \setminus cl B_{\{\varepsilon r\}}(z)$), the circle replaced by a countable dense subset (LM Lemma 1.1, continuity of the internal metric on an open set) and the explicit right

side $t_k + c\epsilon^{\chi} \epsilon_r^{\xi h_r(z)}$ (not a function of $h|_{\mathcal{B}^{\bullet}_{s_{k+1}}}$) by a rational threshold $\theta < s_{k+1}$ (GM l. 2540: the right side is $< s_{k+1}$); locality follows from GM's own exit argument (l. 2597–2601) instead of being read off from the definition; (2) reading/gap: GM l. 2349 says $E_r(z)$ is determined by $h|_{\mathcal{B}_{\lambda_4 r}(z)}$, but condition (2) of Thm 4.2 normalizes by $h|_{\mathcal{B}_{\lambda_5 r}(z)}$ ($\lambda_5 > \lambda_4$), so $E_r(z)$ is determined by $h|_W$ for open $W \supseteq A_{\lambda_1 r, \lambda_4 r}(z) \cup \partial B_{\lambda_5 r}(z)$; the ball part of F_k uses the radius $(2\lambda_4 + \lambda_5)\epsilon_r$ (GM: $2\lambda_4\epsilon_r$), proved on $\mathcal{E}_r$ by one step of the Hölder bound (own elementary proof, packet B1 of DEC-81) rather than GM's circle covering. Status: decided (D81).

DV-L56P, DV-L56C (P2-M2L56b, 2026-10-02) · GM arXiv:1905.00383v3, proof of Lemma 5.6, l. 2963–2968 (construction), 2976–2977 (connectivity), 2982–2989 (condition (2)) · LQGMetric/Papers/GM/S5/Geom56C*.lean · (P) L56Paths is GM l. 2963–2966 with $\pi_{\pm}$ as preconnected sets in the closed ball $cl B_{2r}(z)$ (matching D69) and a constant $b > 0$ depending on α (GM: $b = 1 - \alpha$; only $b > 0$ is used) — statement reading; (C) near u and v the tube uses axis-parallel grid corridors (D69): L_- replaced by the corridor at u plus $[z, c_u]$, L_+ by the corridor at v plus $[c_v, v']$; condition (2) (D77 SepDiscNear, incl. the disconnection clause) by an own argument (sepDisc_of_corridor: outside $B_{20\epsilon_r}(u')$ the corridor-and- π_- side is connected and at distance $\geq \epsilon_r$ from the rest of the tube); own elementary corridor connectivity/depth/grid arguments (GM leave them implicit) · not following the published construction near u, v · proposed · no.

DV-DFB8b-1, -2 (P2-DFB8b, 2026-10-02) · DFGPS arXiv:1905.00380, Lemma 2.13 (T:1060–1071), proof T:1101–1119 · LQGMetric/Papers/DFGPS/L2_13{Conv, Scale,}.lean · (1) $\rightarrow e^{\{-\xi h_r(0)\}c_r^{-1}D_h(r, \cdot)}$ in law along $\mathcal{E}$ (T:1114–1116) by an own argument: $h_r(0)$ is the a.s. limit of the continuous pairings $(h, \text{circBump } m \ 0 \ r)$, then a three-term estimate against bounded Lipschitz test functions, uniform in n (DEC-78 §4.3); (2) "the closure of this set of laws is supported on continuous metrics" (T:1111–1112, 1118) used only for the $\delta \rightarrow 0$ limit points $T = \bigcap_m \text{closure}\{\text{law}(a_{\delta^{-1}}D^{\delta}_h) : \delta < 1/(m+1)\}$, which is all the argument needs · own argument, reading · proposed · no.

DV-L58paths (P2-M2L58c, 2026-10-02) · GM arXiv:1905.00383v3, proof of Lemma 5.8, Steps 1–2 (l. 3068–3091) · LQGMetric/Papers/GM/S5/Geom58P{1..5}.lean · the points $\mathcal{Z}$ are an own explicit choice — two windows of n points on the upper half of $\partial B_r(0)$ at real parts $-r/4 + 10R_j$ and $r/8 + 10R_j$ ($R = \delta r/(500n)$), so $\#A = 2$ (GM only ask $\#A \cdot \delta \leq 100$) — and the paths are an own construction (staircases inside a window; for $\hat{L}_x / \hat{L}_y$ a stub, a vertical, a radial, an arc at radius $1.25r$ or $1.75r$ and a final radial to x or y ; the window chosen so that x lies outside its angular band). GM give neither the points nor the paths (l. 3069–3086), so this is elementary own work filling in the paper, not a departure from its argument · own construction · proposed · no.

DV-D83 (P2-DEC83, 2026-10-02) · GM arXiv:1905.00383v3, §5.3–5.5: (5.15) l. 2925–2930, Lemma 5.6 (1)–(2) l. 2944–2958, Lemma 5.8 (1)–(2) l. 3047–3060 and construction l. 3086–3088, Lemma 5.15 proof l. 3477–3489, l. 3505 · LQGMetric/Papers/GM/S5/{Defs, EventDefs, EventStmts, ShortcutDist}.lean, decisions/DEC-83.md · gap fixes and a reading of the proof: (1) GM's entry step assumes $\tilde{P}^{\wedge}\phi$ starts near $\mathfrak{x}'$ and that $\tilde{P}^{\wedge}\phi$ enters 0_u before 0_v ; neither follows from E_r as defined (the exit point of $\mathcal{B}_{\sigma_r}(z; D_h)$ may lie near y' ; condition (2) is symmetric in $(u, x) \leftrightarrow (v, y)$). We use the first and last hitting times of $cl B_{3r}(0)$ by $\tilde{P}^{\wedge}\phi$ (which lie on the sphere, where condition (4) applies) together with Lemmas 5.12–5.13, and we add to condition (1) of Lemmas 5.6 and 5.8 the bound $\tilde{D}_h(u, v) \leq (c_*/C_*)^2 \tilde{D}_h(v, \cdot)$ at v (GM's proof via Lemma 5.5 gives it verbatim), so that the shortcut times can be taken in either order; the parameter range gains $\zeta < \delta/100$ (GM: $\zeta < \epsilon_0/100$ with $\epsilon_0 < \delta/500$ in their construction). (2) GM's "the component of $(U \cup W) \setminus 0_u$ containing $\mathfrak{x}'$ lies at distance $\geq \epsilon_0 r$ from the others" (l. 3480–3483) uses that U near y is attached to y (true for GM's U , built from squares along a smooth path $\hat{L}_y$ ending at y); we record this as the clause (T5) of IsTubeFam (every point of $U \cap B_{4\epsilon_0 r}(y)$ is joined to y in $U \cap B_{5\epsilon_0 r}(y)$, likewise at x), to be supplied by Lemma 5.8 with radial stubs at x, y (as D69's horizontal stubs at the z_j). GM's inclusion $B_{2\zeta r}(0_u) \subset 0_u \cup (B_{2\zeta r}(\partial U) \setminus W)$ (l. 3484) fails at the outer rim of 0_u ; the argument uses it only off the two sides, where it holds. (3) EtaChoice records GM's "small η ": the denominator of (5.15) positive, and $1 + 3\eta \leq C_*/c_*$ for the repaired l. 3505 (P2-M2M2-2); L5_11 takes $\theta < \epsilon_r$ · decided · D83.

D-P18-S1a..c (P2-DDDF18S1, 2026-10-02) · DDDF arXiv:1904.08021, Prop 5 (tightness.tex l. 423–451), Prop 18 Step 1 (l. 900–910) · LQGMetric/Papers/DDDF/{P5ULarge, P18S1Conv, P18S1Geom, P18S1Chain}.lean · (a) DDDF Prop 5 says the constants depend on a and b ; for Step 1 (l. 906–907) the same proof with the size dependence made explicit (prop5_XAB_large), giving a tail bound for $X_{\{3k, k\}}$ uniform in k (prop5_X3k_tail); (b) the gluing of annulus crossings (DDDF: "via the figure" only, l. 902–905) by an own elementary argument (hub of each circuit, simple path in the graph of good sites, triangle inequality for ℓ_{fppD0n} , constant $6k^2 + 2$ in place of DDDF's Ck^2); (c) planned: a site is open when each of its four crossings is $\leq \ell_{\{3, 1\}}$ (stronger than DDDF's "sum $\leq 4\ell_{\{3, 1\}}$ "; closed probability still $\leq 4p$) · own arguments, explicit constants · proposed · no.

DV-D84 (P2-DEC84, 2026-10-02) · DZZ arXiv:1807.00422, Def 3.6 and remark l. 920–939, proof of Lemma 3.5 l. 1071–1079 · LQGMetric/Papers/DZZ/{S3L5X*, S3L5Cross, S3L5Main}.lean, decisions/DEC-84.md · (1) Def 3.6 enclosures stay literal (topological separation by closed boxes); the crossing argument is run on the dyadic square grid of a level N finer than all cells involved, via the bridge sqSep_of_enclosesBox (own elementary argument: polygonal path through square centres, integer arithmetic; DZZ use the separation without comment); (2) the deterministic

crossing claim takes the hypothesis that all δ' -cells have level $\leq N_0$ (Lemma 3.1 at δ' , implicit in DZZ), needed for corner contacts of cells; (3) DZZ's nesting of enclosed regions (" C_i encloses u ", the i_r recursion) is made rigorous on the grid with the facts F1–F4 of DEC-84 §4 (F4, a Jordan-type step, by 4-connectedness of $\mathbb{V} \setminus C_{\text{large}}$ and a free corner of $\mathbb{V}$) — own argument; DZZ's $|C_i| \leq \lambda/\varepsilon^2$ is replaced by the crude $\#B(C, \varepsilon) \leq 4^{k+2}$ times λ (DZZ: $\varepsilon^{\{-2\}}\lambda$) · reading, own steps, hypothesis made explicit · decided · D84.

DV-L414-compact, DV-L415-scale, DV-P412-ball (P2-M2J2b, 2026-10-02) · GM arXiv:1905.00383v3, §4.4 (l. 2071–2276), §4.7 (l. 2457–2520) · LQGMetric/Papers/GM/S4/P412b*.lean · (L414-compact) the draft GML4_14' asked for a compact $Y \subseteq C \setminus K$, which is false (K the closed unit disc, $y = 1$: $C^{\wedge} \varepsilon_y$ accumulates on ∂K); restated with GM's own open arc $Y^{\wedge} \varepsilon_y$ (l. 2516), connectedComponentIn (sphere $c \cap r \setminus K$) y_0 with $r \leq 8\varepsilon$ — reading; (L415-scale) $\tau = s_k \wedge \tau_{\{2\ell r\}}$ and comparison of the normalization with $\tau_{\{\ell r\}}$ via conditions 2–3, needing $2\beta < \omega\beta_C$ — own argument; (P412-ball) (4.41) transferred from $B_{\{\lambda_2 r\}}(z)$ to its centre z (p412b_ext_transfer; GM prove (4.42) for $P \cap B_r(z) \neq \emptyset$ while $Z^{\wedge} E_k$ gives $P \cap B_{\{\lambda_2 r\}}(z) \neq \emptyset$) — gap fix. Open questions → D86 · proposed · no.

GMCID2-1 (P2-GMCID2, 2026-10-02) · DZZ arXiv:1807.00422, (eq:Green_fxn) l. 400–403, (eq-cov-tildeh) · LQGMetric/Dimension/GMCID2{Green, Circ}.lean · killedGreen_openSquare ($G_{\mathbb{V}} = \pi f p_{\mathbb{V}}$, stated without proof in DZZ) by an own elementary proof: R1 and the kernel form of the covariance applied to shrinking normalized bumps, limit by continuity of $G_{\mathbb{H}}(\varphi, \varphi)$ off the diagonal and dominated convergence in s ; G2 (pi_inner_measKerL2_circle) by Tonelli + the semigroup identity + this identity · own argument · proposed · no.

DV-L56paths (P2-M2L56c, 2026-10-02) · GM arXiv:1905.00383v3, Lemma 5.6, l. 2963–2966 (paths π_-, π_+ ; GM: "We can choose", no construction) · LQGMetric/Papers/GM/S5/Geom56P{1..4}.lean · own elementary construction: π_- runs straight out to radius $7/4$ in a direction g with $\text{Re}(g \bar{e}) \leq -1/2$ and $|g - \bar{e}| \geq 1/2$, along the circle of radius $7/4$ avoiding the direction of $z + 2r$, then straight out to $z - 2r$; π_+ runs along the circle of radius $3/2$ from v' avoiding g , then straight out to $z + 2r$; constant $b = \min(1/8, 1 - \alpha)$ instead of GM's $b = 1 - \alpha$ (only $b > 0$ is used) · own construction (none published) · proposed · no.

DV-M2N2-a..d (P2-M2N2, 2026-10-02) · GM arXiv:1905.00383v3, Lemmas 5.3–5.4, Prop 4.3 (l. 1578–1596, 2724–2862) · LQGMetric/Papers/GM/S5/Prop43b*.lean · (a) Lemma 5.4 by a union bound over the finitely many bump functions instead of GM's measurable random φ (footnote l. 3344); Λ becomes $(N+1)e^{\{3\Lambda_0\}}$ (needs the uniform bound N of D87 (3)); (b) $\mathfrak{E}$ taken as its constant-invariant core constCore (GM's "unaffected by adding a constant", l. 2801, holds only a.s.); (c) own arguments for the measurability of $\mathfrak{E}$, existence of hitting points, the crossing lemma, the every-other-scale extraction (exists_sparse_radii); (d) Lemma 5.3 uses the internal metrics of D and $\tilde{D}$ in $B_{\{4r\}}$ ($\lambda_4 = 4$ as GeoIterateHyp requires) via the bi-Lipschitz bounds (GM's " $D(P(s), P(t)) = t - s$ " is not available with the $[0, 1]$ parametrization) · own route, a.s. reading · proposed · no.

DV-D86a (P2-DEC86, 2026-10-02) · GM arXiv:1905.00383v3, proof of Lemma 4.14, l. 2482–2518 · GML4_14' (S4/P412bL414.lean; proof route decisions/DEC-86.md) · GM's step "every unbounded connected subset of $C \setminus \mathcal{K}$ whose closure contains y meets X " (l. 2485, 2501) is false when $y \in \text{cl } X$, and "there is one such arc Y_B " (l. 2482) fails when $y \in \partial B$; replaced by $(\dagger)$ (bounded components of $C \setminus (X \cup K)$ with $X \subseteq \text{int } B$ lie in $U(\alpha)$ for an arc α of $\partial B \setminus K$ on the boundary of the unbounded component of $C \setminus (B \cup K)$), radii $\rho \in (3\varepsilon/2, 2\varepsilon]$, $\tilde{\rho} \in (6\varepsilon, 8\varepsilon]$ chosen with $y \notin \partial B$, and the shadow lemma (S) from local connectivity of $C \setminus K$ at ∂K (Jordan boundary; Pommerenke 1992 Thm 2.1/2.6); constant 16ε kept · place where we do not follow the published proof · proposed · no.

DV-D86b (P2-DEC86, 2026-10-02) · GM L4.15, definition of C_y (l. 2107) vs L4.14 (l. 2099); Step 3 (*) l. 2162–2165, l. 2242–2244 · LQGMetric/Papers/GM/S4/P412cStep3.lean (p412c_step3) · only the L4.14 version dcSetC is used; the boundary point $P(t_k) \in \partial B^{\bullet}_{\{t_k\}}$ of the L4.15 version is handled by the exit lemma p412c_exit (shadow lemma), proving GM's (*) for such points instead of assuming it · reading of the paper's text · proposed · no.

DV-D86c (P2-DEC86, 2026-10-02) · GM (4.11) and the definition of $D_h(\cdot, \cdot; C \setminus \text{cl } B_r(z))$ -geodesics, l. 1692–1697 · IsAvoidGeod (S4/SetupStab.lean), hypothesis hfin of p412b_stab_of_ext/p412b_core · GM tacitly take these geodesics to have finite length; we prove finiteness on $\mathcal{E}_r$ from condition 3 (internal Hölder upper bound, l. 1962) and the length property (finite-length path to $x \in \partial B_r(z)$ along the outward normal, step ratio $2/3$) — own argument for an implicit fact (planned, DEC-86 (3)) · own argument · proposed · no.

DV-D86d (P2-DEC86, 2026-10-02) · GM L4.13 (l. 2071–2093; $B^{\bullet}_s$ a closed Jordan domain, used implicitly) · LQGMetric/Papers/GM/S4/P412cInt.lean, P412cLC.lean · IsPreconnected (interior $B^{\bullet}_s$) by an own elementary argument (exterior map Ψ of gm_filledBall_conformal' + Eilenberg's criterion, QuantumZipper CA. Topo.not_separates_of_hasLogOn, Burckel Ex. 4.37) instead of the Jordan curve theorem; local connectivity of $C \setminus B^{\bullet}_s$ at its boundary read off Ψ (Pommerenke 1992 Thm 2.1/2.6 for the general fact) · own argument · proposed · no.

DV-D88 · GM L4.14 (l. 2097–2100, proof l. 2455–2520) · GML4_14' (Papers/GM/S4/P412bL414.lean) · hypothesis " $C \setminus K$ is locally connected at every boundary point" (LocConnAt) added to the lemma; GM's lemma is applied only to filled metric balls, where it is proved (p412c_filledBall_locConnAt, from the exterior conformal map) · added hypothesis, discharged in the application (D88) · proposed · no.

GMCID3-1 (P2-GMCID3, 2026-10-02) · DZZ arXiv:1807.00422 l. 648–658 (identification in law); Duplantier–Sheffield arXiv:0808.1560 Prop 1.1 · LQGMetric/Dimension/GMCIdent3{Law,Vague,Wn}.lean · white-noise GMC transferred to an arbitrary zero-boundary GFF by law transfer through the circle family (map_circVec_eq, qAreaMeasureOn_circExt, ae_iff_wn; D85) instead of DZZ’s implicit GFF = $\sqrt{\pi} W(K)$ — own argument via uniqueness of centred Gaussian laws; DS Prop 1.1’s existence argument split into the measurable event VagueGood plus a deterministic step so that it transfers · own argument · proposed · no.

DFB10-1..4 (P2-DFB10, 2026-10-02) · DFGPS arXiv:1905.00380, proof of Lemma 2.17 (T:682, 1150–1155, 1218–1282; D80) · LQGMetric/Papers/DFGPS/L2_17Core*.lean · (1) “is a.s. determined by $h|_V$ ” (T:1233) via Lusin separation through the pairJ coordinates — own standard argument; (2) $\varphi\mathfrak{h}$ realized as harmFn (φ times the pairing of $\mathfrak{h}$ with a radial bump), equal to $\varphi\mathfrak{h}$ a.s. — representation choice; (3) stable convergence, Slutsky for metric spaces, and closeness in probability from the multiplicative bounds of Lemma 2.1 (Häusler–Luschgy 2015 Thm 3.2; Billingsley Thm 3.1) — own standard arguments the paper uses implicitly at T:1240–1260; (4) the stage- n independence uses $\hat{D}^\varepsilon$ while the limits come from D^ε via Lemma 2.5, matched through Lemma 2.1 (the paper’s Step 1) in law rather than pathwise · own arguments, representation · proposed · no.

DV-M2K3-1..3, DV-D81-2, DV-D75, DV-D79-1 (P2-M2K3, 2026-10-02) · GM arXiv:1905.00383v3, Lemmas 4.19 (l. 2584–2601), 4.20 (l. 2333–2356), 4.22 (l. 2527–2565), Thm 4.2 (l. 1559–1561) · LQGMetric/Papers/GM/S4/Iterate3*.lean · (M2K3-1) the circle bound of (4.39) from its dense subset, without $|z - z| > \rho$ (gmGeoSet already forces $z \notin \text{cl } B_e(z)$, gm_geoSet_far, gm_exists_last_hit’); (M2K3-2) the exit argument for transferring avoid-geodesics, incl. the infinite-length case (gm_avoidGeod_transfer_geo); (D81-2) own elementary proof of gm_regEvent_ball_subset; (M2K3-3) the E-piece uses $W = B_{\{\lambda_5 r + \lambda_4 \varepsilon r\}}(z)$; (D75) the reduction $\lambda_4 \leq 1 \rightarrow \lambda_4 > 1$ by the integer scaling $s = \lceil 1/\lambda_4 \rceil + 1$ (gm_T4_2_of_gt1) — own argument; (D79-1) the far normalization fieldSigmaClosed_eq_fieldSigmaClosed0 — own elementary argument · own arguments · proposed · no.

DFB12-1..5 (P2-DFB12, 2026-10-02) · DFGPS arXiv:1905.00380, Lemma 1.3 (T:381–385, citing Schramm–Sheffield Lemma 4.5), Lemma 2.20 (T:1297–1336), Thm 1.2 proof (T:1339–1386) · LQGMetric/Papers/DFGPS/{L1_3,L2_20*,ExistenceAsmV}.lean · (1) L1.3 by an own argument (fixed-marginal lemma + countable dense partition) instead of the paper’s commented-out tightness/subsequence proof (T:388–398); completeness of β not used; (2) L2.20’s last sentence (T:1335, footnote T:383): Lemma 1.3 applied for each R to $C(\bar{B}_R \times \bar{B}_R, \mathbb{R})$ with the coordinates pairJ $T \circ h$ (D80) — reading; (3) T:1325–1330: uniform bounds by Prokhorov compactness; the internal-diameter bound by GM S2.4c chaining since (eq:tight) at T:370–374 is unproved in the paper; the paper’s “ $(1-p)/2$ ” read as “ $1 - (1-p)/2$ ” (typo); the copy $\bar{D}$ shown a.s. a length metric at the level of laws; (4) “translation invariance of the law of h modulo additive constant” (T:1323) proved as lem2_20Transl (LFPP translation identity + uniqueness of limits in law); (5) “By Lemma 2.13, also Axiom V holds” (T:1356) as tightAcrossScales_of_ref via Weyl constants and map_normalize_eq · own arguments, readings, typo fix · proposed · no.

DZZ316-a, -b (P2-DZZ316, 2026-10-02) · DZZ arXiv:1807.00422, Lemma 3.16 (l. 1365–1404), proof of Lemma 3.12 (l. 1436–1499) · LQGMetric/Papers/DZZ/{S3L316P1–P4,S3L12S1–S6}.lean · (a) L3.16 follows our L3.7: band $\eta^{\{\varepsilon^2 s\}}_{\{ts\}}$, threshold $1.2 \cdot \log t^{-1}$, $\theta = t^{\{1/100\}}$, $t = \varepsilon$; $\alpha > 4C_{mc}$, $\alpha \geq 2\gamma\alpha + 1$; the “boundary boxes enclose” step DZZ leave implicit is the own lemma hasEnclosure_ring; (b) L3.12: the surgery claim stated for sequences without repeated cells with at most $32(\varepsilon)^{-2}$ extra cells (DZZ: $4(\varepsilon)^{-2}$, which undercounts since cells of side $\geq \varepsilon^2 s$ meeting C_{large} stick out of it; with repeated cells cutting a segment need not shrink ψ); counting by a potential function instead of DZZ’s informal argument; own proofs that $\mathcal{C}_0$ can be taken without repeated cells and that the cell graph is connected · parameters, gap fixes, own arguments · proposed · no.

DV-M2N3-a..c (P2-M2N3, 2026-10-02) · GM arXiv:1905.00383v3, Lemma 5.3 (l. 2775–2789), Prop 4.3 (l. 1582–1593), §5.1 (l. 2722–2836) · LQGMetric/Papers/GM/S5/Prop43c{L53,Asm,Main,Final}.lean · (a) GM’s “by Axiom II” in Lemma 5.3 made precise: internal metrics on $B_{\{4r\}}(z)$ read on a dense sequence of pairs (LM Lemma 1.1), the stopped path read at dense times (own argument: the last-exit time is measurable), Lusin separation (D30/D51), null-measurability from measurableSet_frK on the law space with the length-metric set replaced by the complement of a null Borel hull; (b) GM’s “ η small enough” (l. 3174) replaced by the explicit $\eta = \min(1/2, (c_2 - c_1)/(4(c_1 + kc_2)), (k-1)/4)$, $k = C_-/c_-$ — own elementary argument; (c) P4_3 needs $\nu s < 1$, matching GM l. 1586 (D91) · own arguments, explicit constants · proposed · no.

DV-L414-net, DV-L414-dagger (P2-M2J2c, 2026-10-02) · GM arXiv:1905.00383v3, Lemma 4.14 (l. 2097–2102), proof l. 2457–2520, repaired per D86 · LQGMetric/Papers/GM/S4/P412d{Arc,Dag,Uniq,L414}.lean · (net) GM’s grid $(\varepsilon/4)\mathbb{Z}^2$ (l. 2477) replaced by a finite $\varepsilon/4$ -net of the 3ε -neighbourhood of K ; ball radius $\rho \in [3\varepsilon/2, 2\varepsilon]$ chosen so that $y \notin \partial B$ for every net centre; p412d_far rules out witnesses X more than 3ε from K ; final ball radius $\tilde{\rho} \in \{7\varepsilon, 8\varepsilon\}$ with $\tilde{\rho} \neq |y - c_1|$; (dagger) (†) of D86 proved without assuming X connected; only local connectivity at y is used (the IsJordanCurve hypothesis of GML4_14' is no longer needed by the proof) · own construction · proposed · no.

DV-D90 (P2-DEC90, 2026-10-02) · DFGPS arXiv:1905.00380, Lemmas 2.13, 2.17, 2.20 (hypothesis “(h, $D^{\{\varepsilon n\}}_h$) $\rightarrow$ (h, D_h) in law”, T:1062, T:1150, T:1297) and proof of Theorem 1.2 (T:1342–1384) · LQGMetric/Papers/DFGPS/{Nodes,L2_17,L2_13,L2_20Transl,CoordApprox,T12Nodes}.lean, decisions/DEC-90.md · (1) definition change

of nodes: the joint convergence in law is taken with the field read through its countable coordinates $\text{pairJ } T \circ h$ (the Polish topology in which Prokhorov applies), not mathlib 's compact-convergence topology on distributions; the hypothesis is weaker, so the lemmas as stated are stronger than before; the circle average $h_r(z)$ is approximated a.s. by single coordinates (own elementary argument: density of the generators, convergence in measure, Borel–Cantelli; CoordApprox) · (2) a place where we do not follow the published proof (planned, packets of DEC-90): the map D is glued from limits of the localized LFPP internal metrics on dyadic domains (measurable for every distribution), instead of the a.s. limit of the global LFPP (T:1343–1349) extended to unbounded f by patching locality functions “in a manner which does not depend on φ ” (T:1378–1381); locality then holds by construction (the paper’s Step 2 argument (eqn-close-pts-msrble), T:1362–1374, is not used); Weyl scaling for all continuous f uses DF Lemma 7.1 in its literal form on a compact set; D is the Euclidean metric off the good set (the paper’s T:1385 comment) · decided · D90.

DV-D89a (P2-DEC89, 2026-10-02) · GM arXiv:1905.00383v3 Thm 4.2 condition (2), l. 1559–1561 (“the (a.s. unique) D_h -geodesic from z to w stopped at the last time it exits $B_{\{\lambda_4 r\}}(z)$ ”) · `stopLastExit`, `lastExitTime` in `LQGMetric/Papers/GM/S3/Defs.lean`; `LQGMetric/Papers/GM/S5/Prop43bL53b.lean` · definition change / reading: GM’s geodesic is parametrized by D_h -length; our selector’s geodesic is parametrized on $[0,1]$ proportionally to its total length L , and $\eta(\min(t, T))$ would make L a function of the stopped path and $h_{\{B_{\{\lambda_4 r\}}\}}$ (not $\mathcal{F}_{\{k+1\}}$ -measurable, so GM l. 2356 would fail). The stopped path is now $u \mapsto \eta(uT)$, $T = \sup\{t : \eta t \in B_{\{\lambda_4 r\}}(z)\}$ ($\emptyset$ if the geodesic never meets the ball): the stopped curve reparametrized in proportion to its own length, which carries the same information as GM’s unit-speed stopped path given $h_{\{B_{\{\lambda_4 r\}}\}}$ (its length is local) · decided (D89) · no (the hypothesis (2) of T4.2 is the one GM intend; P4.3’s events are unchanged, `frkDistIn_stop_iff`).

DV-D89b (P2-DEC89, 2026-10-02) · GM l. 1611 (“we will also impose the assumption that $\lambda_3 = 1$ (the proof when $\lambda_3 \neq 1$ is identical, just with extra factors of λ_3 in various subscripts)”), l. 1891–1914, (4.10)–(4.11), Lemma 4.20 · planned nodes `T4_2PairOne`, `T4_2Pair`, `gm_T4_2Pair_of_one`, `gm_T4_2Gt1S_of_pair` (decisions/DEC-89.md, `LQGMetric/Papers/GM/S4/Iterate3Pair*.lean`) · own argument making GM’s remark precise: the chain (stab events at the candidate radius r , conditioning radius $\leq r$, Prop 4.12 at dyadic ε) is proved for one pair (z, w) with $\lambda_3 \leq 1$ and dyadic ε ; general $\lambda_3 < \lambda_4$ and general ε follow by reparametrizing the radius variable ($r' = \lambda_3 r$, the primed data lam/λ_3 , $\lambda_3 \cdot \text{Rad}$, $E(\cdot/\lambda_3)$, the chain run at the dyadic scale $\varepsilon' \in [2\lambda_3\varepsilon, 4\lambda_3\varepsilon)$ with $v' = 2v$), before the union bound over the grid $\varepsilon^{\text{qr}}\mathbb{Z}^2$ (which forbids a scaling at the level of Theorem 4.2) · decided (D89) · no (T4_2 is unchanged).

DV-D89c (P2-DEC89, 2026-10-02) · GM Thm 4.2 hypotheses allow $\lambda_3 = \lambda_4$ (l. 1555); the proof uses only $\lambda_4 \geq 1$ (l. 2555), while the Lean chain needs $\lambda_4 > \lambda_3$ (`gm_L4_7_final` hr, P2-L47-c/d; `GMP4_12At 1 < R.lam 3`) · planned node `gm_T4_2Gt1_of_strict` : `T4_2Gt1S` $\rightarrow$ `T4_2Gt1` (DEC-89 packet F) · own argument: the case $\lambda_3 = \lambda_4$ is reduced to $\lambda_3 < \lambda_4$ by enlarging λ_4 to $\lambda_4' \in (\lambda_4, \min(\lambda_5, \lambda_4^2/\lambda_1))$ and using every other radius of (1) $(\mu/2)$; (2) transfers since the geodesic stopped at the last exit of the larger ball determines it stopped at the last exit of the smaller one; (4) has fewer pairs to check; the conclusion is stronger. Fallback if this packet is cut: add `lam 2 < lam 3` to `T4_2` (P4.3 uses $\lambda = (1/4, 2, 3, 4, 5)$) · decided (D89) · no.

DV-D84 addition (P2-DZZ35X, 2026-10-02) · DZZ Lemma 3.5, ends of the crossing (l. 1058–1077), (`Eq.boundDprime`) · `LQGMetric/Papers/DZZ/S3L5Y{Sq,End,Main,Lemma}.lean` · own elementary arguments on the level- N grid ($N = N_0 + k + 1$): point case — the δ' -geodesic from u to $\partial C_{\text{large}} \cap \mathbb{V}$ as a path of squares plus at most 2 squares to leave C_{large} (edges of C_{large} never lie on $\partial \mathbb{V}$, a parity argument) meets the ring; connected case ($C_{\text{start}} = \emptyset$) — the squares whose closures meet A are 4-connected (closed-cover argument), one lies in the ring and is joined to the square of a point of A by ≤ 2 steps (≤ 3 cells); DZZ’s $2\delta^{\wedge\{-i\}}(\delta/\delta')^3$ for the two ends becomes $2(R + 4)$ · own arguments, constants · proposed · no.

GMCID4-1..3 (P2-GMCID4, 2026-10-02) · DZZ arXiv:1807.00422 l. 135–141, 648–683; Berestycki arXiv:1506.09113 §4 · `LQGMetric/Dimension/GMCIdent4*.lean` · (1) DZZ l. 656–665 proved for the white-noise field with the GFF entering only through the law of its circle family (D85); the Lévy step uses an own measurable-version argument (`exists_iSup_version`, `wnFieldVer`); (2) `tendsto_lgdRatio_iff_dyadic`: own elementary reduction of the real δ -limit to dyadic δ by monotonicity of D_δ (DZZ use dyadic δ implicitly, l. 135–141), needed for measurability of the LGD event; (3) $\tilde{M}_{\{\gamma, \delta\}}$ defined as $c\text{Dens}^{-1} \cdot M_\gamma$ and then shown to be DZZ’s limit (eq-def-tilde-M, l. 676–683) instead of being defined as the martingale limit; the identification with DZZ’s band field holds for each z a.s. · own arguments, definition route · proposed · no.

P2-M2M5-1..3 (P2-M2M5, 2026-10-02) · GM arXiv:1905.00383v3, proof of Lemma 5.10 (l. 3307–3335), E_r (l. 3240–3266) · `LQGMetric/Papers/GM/S5/L510{A,B}.lean` · (1) failure probability $q = (1-p)/9$ per condition instead of GM’s $(1-p)/100$ — convention; (2) condition (4)’s internal-distance bound and condition (8) (GM: “tightness across scales”) by an own short argument: a short D_h -path cannot reach $\partial \mathbb{A}_{\{r, 4r\}}$ (GM.S2.4a, S2.4b, length-space axiom); $\zeta := \min(\zeta_1, \varepsilon_0/200, \delta/200)$ (condition (6) only weakens as ζ shrinks), $\theta := \min(\beta/2, \zeta/200)$; (3) statement issue: `L5_6/L5_8` must choose V/U before $\forall \Omega$ (D92) · convention, own argument · proposed · no.

P2-M2M3-1..3 (P2-M2M3, 2026-10-02) · GM arXiv:1905.00383v3, Lemma 5.12 (l. 3391–3401), entry step l. 3477–3489 repaired per D83 (a)–(c) · `LQGMetric/Papers/GM/S5/Shortcut3*.lean` (+ D83 P1 edits in `Defs`, `Tubes56`,

Tubes57Det/Main, Tubes58, Shortcut2L511) · (1) `dist_le_weyl_of_geod_m2m3`: own elementary argument for a fact GM use tacitly (if a $D_{\{h-\varphi\}}$ -geodesic stays where $\varphi = 0$ before its last point, then $D_h \leq D_{\{h-\varphi\}}$ between its ends), from `internal_eq_of_eq_const`; (2) own elementary geometry of the entry step (tubes W^x , W^y stay apart, $|x-y| \leq |\lambda x - \mu y|$; tube tips near $\partial B_{\{3r\}}(0)$; sides separated via (T5); the crossing window; GM's inclusion at l. 3484 in the corrected form of D83 step 5); (3) `gm_L5_12_sum` duplicates the inequality inside `gm_L5_12`'s proof (ShortcutHit.lean) · own arguments · proposed · no.

GMCID5-i, -ii (P2-GMCID5, 2026-10-02) · DZZ arXiv:1807.00422, $\tilde{M}$ (l. 676–683), moment bounds (l. 684–688) · `LQGMetric/Dimension/GMCID5{Ind,Mom}.lean` · (i) independence of $\tilde{M}_{\{\gamma,\delta\}}$ from $\tilde{h}_\delta$ formalised as "has a version measurable for the fine-scale white noise" (`exists_fine_version_integral_tildeM', indep_integral_tildeM`), own measurable-version argument; (ii) (eq-LQG-positive-moment) so far only for $0 < p \leq 1$ ($C = \pi^p$, open U ; `lintegral_tildeM_rpow_le`); the range $1 < p < 4/\gamma^2$ and the negative moments open; DZZ's "square or ball" read as open sets (closed sets would need $\tilde{M}(\partial B) = 0$). Existence of a zero-boundary GFF on $\mathbb{V}$ from QuantumZipper K3.`exists_zeroGFF0n` (re-exported as `exists_zeroGFF_openSquare`) · formalisation choice, partial · proposed · no.

DV-T12a, DV-T12b (P2-DFT12a, 2026-10-02) · DFGPS arXiv:1905.00380, Thm 1.2 proof Step 0 (T:1342), gluing (T:1343–1349, 1376–1385; Euclidean junk T:1385) · `LQGMetric/Papers/DFGPS/T12P{0,1A,1B}.lean` · (a) `patchD_eq` takes as hypothesis the agreement of `locLim` with $D(\cdot, \cdot; \text{sqW } n)$ on $\text{sqW } n \times \text{sqW } n$ instead of DEC-90's equality with `intExt` (false when no continuous extension of the internal metric to the closed square exists; equivalent otherwise, `intExt_eq`); (b) own elementary arguments: $D(\cdot, \cdot; \text{sqW } n)$ is finite on $\text{sqW } n$ for a length metric and decreases to D (near-minimal paths have compact range); transfer of `IsNormalizedWPGFF` along equality of laws; $(-2^n, 2^n)^2$ as the interior of four dyadic squares · statement form, own arguments · proposed · no.

DFB11-1..3 (P2-DFB11, 2026-10-02) · DFGPS arXiv:1905.00380, proof of Lemma 2.17 (T:1218–1282), Lemma 2.12 (T:1026–1048) · `LQGMetric/Papers/DFGPS/L2_17Core2*.lean` · (1) " $D(\cdot, \cdot; W)$ is determined by $D_{\{h,W\}}$ " (T:1262) by Lusin separation on the coupling space (`comap_intFn_le_aeClosure`) — own argument (the paper does not discuss measurability); (2) Step 3's "analog of Lemma 2.12" (T:1248) with fields varying in n (`weyl_lfpp_varying`): convergence in law of $(h, D^{\wedge \varepsilon}_h)$ does not give a fixed-field a.s. coupling in general (X uniform, Y_n the n -th Rademacher function of X), so the paper's "coupled so this occurs a.s." (T:1028) is realized by Skorokhod with varying fields — published proof not followed literally; (3) the law of (h, D_1) identified by stable convergence, since $h_r(z)$ is only a measurable function of the coordinates (`map_coupling_fst_eq`) — convention · own arguments · proposed · no.

DZZ313-a..d (P2-DZZ313, 2026-10-02) · DZZ arXiv:1807.00422, Lemma 3.13 and proof (l. 1314–1340), $r(t)$ (l. 440–446) · `LQGMetric/Papers/DZZ/S3L13{Sel,CE,Reg,Main,Cov,Asym}.lean` · (a) DZZ's one-sentence (Eq.fine-field-independent) formalized as a stopping-set argument: the box sequence is the shortest list satisfying L313Q (ties by a fixed encoding), so $\{\text{seq} = l_0\}$ is $\sigma(\mathcal{V}_\delta)$ -measurable; M' agrees with M on boxes whose region misses the fine-field region of l_0 and is 0 elsewhere; on $\{\text{seq} = l_0\}$ the partitions from M and M' coincide — own elementary formalization; (b) the conditional-law identity proved on all of $\{\text{seq} = l_0\}$, not only on $\mathcal{E}_{\{\delta, \alpha^*, u, v\}} \cap \{\text{seq} = l_0\}$ (stronger); (c) the geometric content stated as the cover condition with $\rho(s) = s \log s^{-1} + s$ (twice the bound on $r(t)$ for $t < s^2$) — reading of l. 1334–1336; (d) own elementary asymptotics (`l313_size_asym`) · own formalization, reading · proposed · no.

DV-M2M4-a..c (P2-M2M4, 2026-10-02) · GM arXiv:1905.00383v3, l. 2801, 3086–3091, 3293–3302; D83 (c), D87 (4) · `LQGMetric/Papers/GM/S5/{Event3*,Geom58T*}.lean`, Tubes58.lean · (a) the (T5) attachment proof (`exists_line_pt_sq, t5_of_segment, RadEnd`) is an own elementary argument (GM leave it implicit); (b) Lemma 5.8 proved in the form L5_8T (with (T5)) until the L5_8 statement in `Defs.lean` gets the (T5) clause (D92 application), via proof copies with the extra clauses (`window_pathsT, l58PathsT, l58GeomT`), to be folded into the originals; (c) Lemma 5.9 split into L5_9loc (GM l. 3297) and L5_9meas (GM's "by inspection") · own argument, temporary statement form · proposed · no.

DZZ312S-a..d (P2-DZZ312S, 2026-10-02) · DZZ arXiv:1807.00422, proof of Lemma 3.12 (l. 1436–1499) · `LQGMetric/Papers/DZZ/S3L12T{1..8}.lean` · (a) $C_{\{i,1\}}$, $C_{\{i,2\}}$ (l. 1468–1470) are common cells of the chain and the enclosure found on the level- N grid (D84), chosen as the last enclosure cell before C and the first after — own argument; (b) step length bound $32(\varepsilon)^{\wedge\{-2\}}$ instead of DZZ's $4(\varepsilon)^{\wedge\{-2\}}$ (l. 1457): each replacing cell contains a level- (n_C+k) square meeting C_{large} , at most $(3 \cdot 2^{\wedge k} + 1)^2$ of them — own count; (c) loops created by the splice erased with `Walk.bypass` (ψ only shrinks); (d) `c_parents` (l. 1462–1466) stated without the lt/rt/lb/rb orientation, via the outer corner $z^*(C)$ — own argument. Open: L312Core (needs closed enclosure rings? decision D93) · own arguments, constants · proposed · no.

DV-hfin, DV-L413-exit/-strict/-shrink/-sep, DV-Gy-17 (P2-M2J2d, 2026-10-02) · GM arXiv:1905.00383v3, L4.13 (l. 2071–2083), L4.14–L4.15 Steps 1–3 (l. 2097–2170), P4.12 proof (l. 2229–2276), D86 · `LQGMetric/Papers/GM/S4/P412e*.lean` · (hfin) the finite-length competitor path built by infinite concatenation (own argument), `p412e_core` uses the $4\lambda_4\varepsilon$ region; (L413-exit) smallness $\varepsilon^\wedge\chi < a^\wedge\{\chi\}$ plus a first-exit argument at l. 2079 (GM state L4.13 for all $\varepsilon \in (0, a]$; only small ε is used downstream); (L413-strict) the hypothesis is a strict dU bound at one pair of

points, so GM's auxiliary δ is not needed; (L413-shrink) GM's "shrinking X_0 " via local connectivity and an infinite concatenation; (L413-sep) the separation via θ bounded with the two arcs of $\partial\mathcal{B}$ between $P(s)$ and v plus a collar lemma, output set $\text{cl } X \setminus \mathcal{B}$, "endpoint of I " read as a point of $\text{cl } I \cap \text{cl}(\partial\mathcal{B} \setminus I)$; (Gy-17) the no-hit radius is $17\varepsilon^{\kappa r}$ instead of GM's $16\varepsilon^{\kappa r}$ (the hit of $\text{cl } B_{\{2r\}}(z_y)$ can happen exactly at time t_k) · own arguments, readings, constant · proposed · no.

P2-M2M6-a..d (P2-M2M6, 2026-10-02) · GM arXiv:1905.00383v3, proof of Lemma 5.10 (l. 3313–3333), bump functions (l. 3189–3192) · LQGMetric/Papers/GM/S5/L510C{1..8} .lean · (a) L510Diam, L510Bdy, L510Line need $0 < \gamma < 2$ (DFGPS L3.20 and Prop 4.1 hold only there; statement patch since applied (orchestrator 10:05, TASKS), proved as L510DiamG/BdyG/LineG); (b) the chaining runs through square centres inside the open tube after subdividing the squares to side $2^{-k}\varepsilon_0 r$ (where the DFGPS 3.20 bound holds w.h.p.) instead of through GM's closed squares (l. 3313–3320, 3329); connectedness of $W_r^{\wedge x}$ own elementary proof; (c) the union bound runs over the finitely many grid lines clipped to a fixed segment instead of over the sides of squares (l. 3322); the implicit step "a path near the tube's boundary has a sub-path near a single line" own argument (L510_subpath); (d) GM's Gaussian tails replaced by the first-moment bound $E|(h, \varphi)_{\nabla}| \leq ((\varphi, \varphi)_{\nabla} + 1)/2$ with Markov (sufficient); bumps as rescaled copies of fixed templates over a finite family of unit-scale tubes (l. 3331–3333) · parameter range, own arguments · proposed · no.

GMCID6-i, -ii (P2-GMCID6, 2026-10-02) · DZZ arXiv:1807.00422, (eq-LQG-positive-moment), (eq-LQG-negative-moment) (l. 684–688) · LQGMetric/Dimension/GMCID6* .lean · (i) band covariance bound $\pi \int_{\{\varepsilon^2\}^{\delta^2}} p_{\nabla} \leq \log(L/\max(\varepsilon, |u-v|)) + 1$ by an own elementary proof using $p_{\nabla} \leq p_{\mathbb{C}}$ (used implicitly by DZZ); (ii) statement forms: positive moments for open $U \subseteq \mathbb{V}$ inside a closed ball of radius ξ , negative moments for open $U \subseteq \mathbb{V}$ containing an open ball of radius ξ (ball of diameter ξ : apply with $\xi/2$, constant $\times 4^p$), both with $\delta = 2^{\{-m\}} \leq \xi$; proof via Riemann sums and a one-sided Kahane comparison against midZ (y^q convex for $q > 1$ and $q < 0$, so only an upper covariance bound is needed), following Rhodes–Vargas arXiv:1305.6221 Thm 2.11–2.12 and Berestycki–Powell arXiv:2404.16642 (DZZ write no argument) · own argument, statement form · proposed · no.

DV-M2M7-a (P2-M2M7, 2026-10-02) · GM L5.6 pigeonhole (l. 2975–2979), law modulo constants (l. 1200–1203, 3001–3004) · LQGMetric/Papers/GM/S5/{Defs, Tubes56, Tubes58, Event4*} .lean · with D92 (V before Ω) the deterministic $\mathcal{K}_r(z)$ is chosen for one whole-plane GFF and transferred to all of them, since $P[F_r(z)]$ does not depend on the field (prob_tubeEvent_eq); if no whole-plane GFF exists (a Lean-only case) V is taken from L56GeomN along $[z+\alpha r, z+r]$; L5_8 carries the (T5) clause; the ...T proof copies replaced the originals · own argument · proposed · no.

DV-D89a/b/c, DV-M2K4-1 (P2-M2K4, 2026-10-02) · GM Thm 4.2 (l. 1554–1571), L4.7 (l. 1890–1940), (4.19), L4.20 (l. 2333–2356) · LQGMetric/Papers/GM/S4/Iterate4* .lean · (D89a) stopped path reparametrized by its own length; (D89b) the λ_3 factor removed by $\text{gm_geoIterateHyp_scale}$ ($s = \lambda_3^{-1}$), then ε replaced by $\theta\varepsilon$ with $\theta \in (1/2, 1]$ at the cost $v \mapsto 2v$ and a smaller ε_0 ($\text{gm_T4_2Pair_of_one}$) — own argument; (D89c) $\lambda_3 = \lambda_4$ by enlarging λ_4 to $\min((\lambda_4 + \lambda_5)/2, \lambda_4^2/\lambda_1)$ and using every other radius ($\text{gm_T4_2Gt1E_of_strict}$) — own argument; (M2K4-1) $\theta < \varepsilon_0$ in T4.2 (D95) · own arguments, statement fix · proposed · no.

DV-T12-TRUNC, DV-T12-CONTQ, DV-T12-BC (P2-DFT12b, 2026-10-02) · DFGPS arXiv:1905.00380, proof of Thm 1.2 (T:1342–1385), Lemma 2.5 B, Lemma 2.1, Lemma 1.3 · LQGMetric/Papers/DFGPS/T12P2{A..G}, T12P3{A, B} .lean · (TRUNC) Steps 1–2 formalized with the continuous truncation $\min(d(a,b), d(a, \partial W))$ instead of the indicator at T:1368–1372, uniformly on $\bar{W}^2$; the glued metric built from truncated local limits plus the chain formula (D96); (CONTQ) measurability of $h \mapsto D_h$ (T:1349) through a continuous extension from a countable dense set (contQ) — own elementary argument; (BC) a common a.s.-convergent subsequence for countably many sequences converging in measure (exists_subseq_ae_tendsto_all, Borel–Cantelli) — own packaging of a standard argument · route change, own arguments · proposed · no.

DZZ32-1..3 (P2-DZZ32, 2026-10-02) · DZZ arXiv:1807.00422, Prop 3.2 (proof l. 1087–1207), Cor 3.3, P3.17 (l. 1505–1517) · LQGMetric/Papers/DZZ/S3P32{Low, Up, } .lean · (1) P3.2 proved as two halves: the lower half uses L3.5 at $\delta' = \delta e^{\{(\log \delta^{-1})^{0.8}\}}$ with diameter exponent $(\xi d + C_{\text{Mc}})/2$ (D71), p32_asym own elementary bookkeeping; open inputs L32BallCover (l. 1164–1206) and L32UpperCross (l. 1104–1153); $\iota = C_{\text{Mc}}/2$ (DZZ $C_{\text{Mc}}/3$) as in L3.5; (2) (Eq.boundDprime) in the L3.5 normalization $4^{\{k+2\}}(\lambda+1)$, $2R+8$ instead of DZZ's $d\lambda/\varepsilon^2 + 2\delta^{\{-l\}}\lambda$; (3) P3.17 (DZZProp317) stated for one random measure μ with the expectation as the Bochner integral of $\log \text{MinLGD}$. Open question D97: the frozen lgdDZZ allows paths and balls outside $\mathbb{V}$ (where $\mu = 0$), DZZ implicitly use paths inside $\mathbb{V}$ · own bookkeeping, normalization · proposed · no.

D-P18-S2a..d (P2-DDDF18P, 2026-10-02) · DDDF arXiv:1904.08021, Prop 18 Step 1 (tightness.tex l. 900–910) · LQGMetric/Papers/DDDF/P18S2{Site, Indep, } .lean · (a) a site is open when each of its four 3×1 rectangles has a ψ -crossing of length $\leq \ell^{\{n\}}_{\{3,1\}}(\psi, p'')$ (not when their total is small); sites = the $(3k-2) \times (k-2)$ grid shifted by $(1,1)$; gluing gives $L_{\{3k,k\}} \leq 32k^2 \cdot \ell_{\{3,1\}}^{\{n\}}$ — reading; (b) independence used only for sites at ℓ^{∞} -distance > 4 (Peierls range $r = 4$, $\theta = 1/16$, $4p'' \leq \theta^{25}$), DDDF use distance ≥ 2 (any finite range gives the same exponential bound) — parameter choice; (c) RSW with $(a,b) = (1,1)$, $(a',b') = (3,1)$, $p'' = 3p^{\{1/C\}}$; $p_0 = \min(1/4, ((1/16)^{25/12})^C)$, $C = \max(64K_p, 24/\log 2)$, $c = \log 2/2$ — parameter choice; (d) own elementary arguments: the Peierls tail $(3k-2) \cdot 2^{\{-(k-2)\}} \leq (24/\log 2) e^{\{-(\log 2/2)k\}}$; independence of finite families by induction with a

two-set split of the white noise · readings, parameters, own arguments · proposed · no.

DZZ313G-1..4 (P2-DZZ313G, 2026-10-02) · DZZ arXiv:1807.00422, Lemma 3.13 proof (l. 1314–1340) · LQGMetric/Papers/DZZ/S3L13G{1..7}.lean · (1) instead of boxes at distance $\geq \varepsilon_{\text{C}}/3$ from $\partial C_j \setminus (\Lambda_{j-1} \cup \Lambda_j)$ (l. 1327–1329), the fine boxes meeting the two segments from the door boxes to the centre of C_j , with the consequence DZZ need proved (points within $\varepsilon_{\text{C}}/10$ of such a box lie in the open cell or in a door square) — own argument; (2) first cell: $x_0 = u$ by the goodness of u ; last cell: an arbitrary chain to the box of v controlled by the goodness of v — reading of “arbitrary sequence” (l. 1329); (3) length bound $d \leq d_0(\varepsilon)^{-4}$ with $(\varepsilon)^{-4} \leq e^{(\log \delta^{-1})^{0.6}}$ (cruder than DZZ’s implicit $O((\varepsilon)^{-2})$ per cell) — own asymptotics; (4) L3.13 stated and proved for $\alpha > 0$ (L313GeomC is false for $\alpha^* \leq 0$, where $\varepsilon^* = 1$) · own arguments, readings · proposed · no.

DFB13-1..4 (P2-DFB13, 2026-10-02) · DFGPS arXiv:1905.00380, proof of Lemma 2.17 (T:1218–1282, Steps 1–4), D80, D90 · LQGMetric/Papers/DFGPS/L2_17Core3{A..J}.lean · (1) Step 3 coupling: the paper’s “coupled so the convergence is a.s.” realized by Skorokhod (Billingsley Thm 6.7) plus Lemma 2.12 with fields varying in n (weyl_of_sample); the Weyl relation proved in the direction $D_1 = e^{\{\xi \text{Fb}\}} D_2$ (T:1271), avoiding inversion of Weyl scaling; (2) Step 4 “measurable function” (T:1272) by Lusin separation on a standard-Borel Skorokhod space (comap_le_aeClosure_of_rep, comap_intFn_le_aeClosure_coupling) — own argument; (3) Step 4 limit “ W, W' increase”: each stage by conditional independence on the law of $(Jh, e^{\{-\xi h_r(z)\}} D_h)$, then the monotone limit (condIndep_iSup_of_monotone), then pull-back (own lemmas condExp_comp_map, condIndepEv_map, condIndepEv_comap_of_map) — replaces the incorrect R3 plan of handoff/P2-DFB11.md ($\sigma(\text{Fb})$ lies in the left side); (4) stage data: enumerated exhaustions dyFin, bumps from exists_bump_of_isCompact, a shift of the scales so the $\sqrt{\varepsilon}$ margins hold (T:1226–1230) — own bookkeeping · own arguments · proposed · no.

DV-D93 (P2-DEC93, 2026-10-02) · DZZ arXiv:1807.00422, proof of Lemma 3.12 l. 1462–1499 (two segments of $\mathcal{C}_{\{i, \text{cross}\}}$, “by connectivity”), Def 3.6 l. 920–939, proof of Lemma 3.7 l. 1009–1011 · LQGMetric/Papers/DZZ/S3L12U1.lean, decisions/DEC-93.md · (1) a reading: DZZ’s $\mathcal{C}_{\{i, \text{cross}\}}$ is taken to be the closed ring that their duality argument produces (the four rectangle crossings of the annulus $C_{\text{large}} \setminus C$), not an arbitrary sequence of Def 3.6; Lemma 3.12 therefore consumes a ring form (CellRing) of (eq-percolation-good-surrounding), supplied by Lemma 3.16 in the crossing form HasCross (Def 3.6’s event $\mathcal{E}_{\{6, B\}}$ is a corollary, so the statement of Lemma 3.16 is unchanged); (2) a place where we do not follow the published argument: DZZ’s “by connectivity” (l. 1476–1477), which as written assumes that each parent is visited once by the ring, is replaced by: the ring is modified so that every parent occurs exactly once (merging its zone, a rectangle of boxes inside the parent cell; separation re-derived from the crossings), and then the two arcs between $C_{\{i, 1\}}$ and $C_{\{i, 2\}}$ have disjoint parent sets, so one of them contributes at most one bad cell (own elementary counting with the zone geometry at $z^*(C)$); (3) DZZ’s requirement “interior of a segment avoids c_{parents} ” is weakened to “the replacing sequence contains at most one parent” (sufficient for (i)/(ii)); (4) the case $C \cap \partial \mathbb{V} \neq \emptyset$ (l. 1498) is read as $C_{\text{large}} \not\subset \mathbb{V}^\circ$, where at most one parent is a box of $\mathbb{V}$ and any enclosure path is used · reading, own steps · decided · D93.

DV-M2M8-1 (P2-M2M8, 2026-10-02) · GM arXiv:1905.00383v3, Lemma 5.9 (l. 3293–3302, “by inspection”), E_r (l. 3240–3266) · LQGMetric/Papers/GM/S5/Event5{Dense, C6, Link, LinkRed}.lean · measurability of E_r by own elementary arguments: conditions (4), (7), (8) via continuity of internal metrics on a countable dense subset; (5), (6), (9) via finiteness of the tube families (sqTubes), (6) coanalytic (Lusin, D30); (1)–(3) by reducing the end points to components of $U \times y \setminus B_M(0)$ ($M = (1+4p)r + 20\varepsilon_0 r < 2r$), each meeting the dense sequence, then the tubeEventB method (D48, D69/D77); uses DFGPSLem3_8 (GM S1.1) for $D_h, \tilde{D}_h \in \text{lenSet}$ a.s. · own arguments · proposed · no.

DFT12c-1..3 (P2-DFT12c, 2026-10-02) · DFGPS arXiv:1905.00380, proof of Thm 1.2 Steps 1–3 (T:1343–1385), Lemma 2.1 (T:735–738) · LQGMetric/Papers/DFGPS/T12P{4A, 4B, 5A}.lean · (1) law transfer of a.s. statements from the Skorokhod coupling to an arbitrary normalized GFF (implicit at T:1343–1356) — own elementary proof (ae_tendsto_lfppC_transfer, measurableSet_tendsto_fun, normGFFLaw_eq); (2) Weyl scaling obtained from a.s. convergence along a further subsequence plus Lemma 2.12 instead of read off the limit; no fixed-field coupling needed; (3) unbounded f in Step 3 by clamping f to a bounded $\tilde{f}$ near sqW n and comparing internal metrics (weylScale0n_congr) instead of truncations · own arguments · proposed · no.

DV-M2K5-1..3 (P2-M2K5, 2026-10-02) · GM arXiv:1905.00383v3, l. 1557, 2361–2441; GM S1.1 l. 647 · LQGMetric/Papers/GM/S4/Iterate5*.lean · (1) gm_T42_rate_abs: GM’s “ $\kappa = \varepsilon^{2\nu+o(1)}$ ” made explicit ($M_8 > 2/\zeta$, log factor absorbed by $\varepsilon^{\{\zeta/2 - 1/M_8\}}$) — own elementary proof; (2) gm_ae_len_bdd_geod: bounded D_h -balls from bounded compactness and continuity of D — own elementary proof; (3) the uniform- ε_1 restatement of the L4.19/L4.22 chain (as D75), being added as ...U versions · own arguments, statement uniformity · proposed · no.

DV-Yk, DV-L36-dyadic, DV-sel-lex (P2-M2J2e, 2026-10-02) · GM arXiv:1905.00383v3, L4.15 Steps 2–4 (l. 2139–2220), proof of Prop 4.12 (l. 2229–2276) · LQGMetric/Papers/GM/S4/P412f*.lean · (Yk) $\#y_k \leq 2\#\text{Conf}_k$ instead of GM’s equality (a connected subset of $\partial \mathcal{B}^{\bullet}_t$ has ≤ 2 endpoints, p412f_endpts_two); the factor 2 is absorbed in the union bound; (L36-dyadic) CONF L3.6 applied at a dyadic $\delta \in [17\varepsilon^\kappa, 34\varepsilon^\kappa]$ (condition 6 of $\mathcal{E}_r$ covers dyadic scales only), hence $\beta < \kappa\chi/4$ instead of GM’s $\beta < \kappa\chi/2$ — parameter range; (sel-lex) measurable selection of a point

of a random compact set by the lexicographic minimum (p412f_meas_select) — own elementary proof in place of Kuratowski–Ryll–Nardzewski · constants, parameter range, own argument · proposed · no.

DV-M2M9-a, -b (P2-M2M9, 2026-10-02) · GM arXiv:1905.00383v3, Prop 5.2 (l. 2731–2752, 3217–3373) · LQGMetric/Papers/GM/S5/P52{Card,Bilip,Main,Final,Wire}.lean · (a) GM apply $c_D \leq \tilde{D} \leq C_D$ at $h - \varphi$ without comment; proved as ae_bilipAt_subTest (Cameron–Martin transfer; the bounds survive adding a constant and form a Borel condition via a dense sequence) — implicit fact; (b) the bump family bound uniform in r , $2^{L_1^2}(2^{L_2^2})^2 + 1$ with $L_1 = \lceil 6/\varepsilon_0 \rceil + 1$, $L_2 = \lceil 8/\theta \rceil + 1$ (D87 (3); GM l. 3217 only state finiteness) — own elementary proof · implicit fact, parameter · proposed · no.

DV-D97 (P2-DEC97, 2026-10-02) · DZZ arXiv:1807.00422, $D_{\{\gamma,\delta\}}$ (l. 121–124), P3.2 (l. 1087–1207), §5 $D^\wedge A$ (l. 2268), Prop 5.1 first-exit argument (l. 2340–2350) · LQGMetric/Papers/DZZ/S3P32X.lean · the frozen lgdDZZ (paths and balls in $\mathbb{C}$, as DZZ l. 123) gives the same exponent (upper bound: frozen $\leq$ DZZ’s interior distance after a constant rescaling of δ ; lower bound: DZZ’s own first-exit argument), so no Statement change; DZZ’s §§3–5 distances are taken at the internal measure $\mu_{\text{In}} := \text{dzzWall dzzV}(M^\wedge W)$ (walled Wick chaos = DZZ’s confined $D^\wedge K$), never at qArea (L32BallCover is false at qArea: counterexample on the bottom edge); comparison lgd_sandwich_dzzWall, lgdMinSet_dzzWall_le_of_exit · reading, route · proposed · no.

DV-D101 (P2-DEC101, 2026-10-02) · DZZ arXiv:1807.00422, P3.2 upper bound (l. 1131–1137, 1149–1151), ξ -admissible pairs (l. 792), Remark 5.2 (l. 2288–2290) · LQGMetric/Papers/DZZ/S3P32UClip.lean · the upper half of P3.2 is proved at the walled internal measure μ_{In} of D97 (needed together with the lower half for one distance by Cor 3.9, P3.17 and §5); since no ball inside the closed square contains a point of ∂V , DZZ’s cover of the ring boundaries $\partial B'$ by balls of radius t_s at grid corners is replaced, for boxes touching ∂V , by the cover of $\partial(B' \cap [r, 1-r]^2)$, $r \leq t_s$, with the corner balls on ∂V moved inward by t_s ; the ends (in $V^\wedge \xi$) are never clipped; this is the content of DZZ’s “extends automatically” (Remark 5.2) for the confined distance $D^\wedge V$ · own elementary modification of the published argument · proposed · no.

DV-D93a, -b (P2-DZZ93A, 2026-10-02) · DZZ arXiv:1807.00422, Lemma 3.16 (l. 1381–1397), Def 3.6 (l. 920–939) · LQGMetric/Perc/AnnulusClipCross.lean, LQGMetric/Papers/DZZ/S3L12V{1,2,3}.lean · (a) L3.16 stated in coarse crossing form: long-way crossings (HasCross at level k) of open boxes all of whose boundary boxes at depth $\text{epsStarN} - k$ are good (HasCrossRing); Def 3.6’s enclosure a corollary (D99); (b) hasEnclosure_of_hasCross assumes $1 \leq B.n$ (for $B = [0,1]^2$ no side is active) · statement form, parameter range · proposed · no.

DG3D-1 (P2-DG3D, 2026-10-02) · DG arXiv:1807.01072, Lemma 3.1 (DG:966–974, proof DG:2243–2244), (3.1) (DG:917), RV Lemma 5.4 (DG:2146) · LQGMetric/Papers/DG/{AppA3SFub,AppA3Ker,L3_1}.lean · L3.1 for $(h^\wedge U, \hat{h})$ proved for $U = V$ and K a box with $B(z, 1/10) \subseteq V$; $h^\wedge U$ the white-noise field, its law the zero-boundary GFF on circle families via map_circVec_eq (D85); “continuous modification” read through circle averages (D100); stochastic Fubini (ae_integral_eq_wn) and the kernel identity (dgUHatKernel_ae_eq) by own elementary proofs · reading, own arguments · proposed · no.

DZZ32U-a, -b (P2-DZZ32U, 2026-10-02) · DZZ arXiv:1807.00422, crossing argument of L3.5 (l. 1054–1079), P3.2 upper-bound reduction (l. 1088–1101), (eq-B-percolation-Psi), (eq-B-good-Psi) (l. 1092–1096) · LQGMetric/Papers/DZZ/S3P32U*.lean · (a) the ball crossing (l32BallCrossing_holds, generic in μ) on the level- N grid as in DV-D84 with own elementary glue (frontiers of boxes meeting, path-connectedness of square boundaries, shrinking to rational centres); start-condition paths inside V (D97); bound normalized as $4^{k+2}(\lambda+1) \cdot d + 2R + 8$; (b) (eq-B-good-Psi) used only at $\iota = C_{\text{Mc}}/2$ · own arguments, normalization · proposed · no.

DV-D98a, -b, -c (P2-DEC98, 2026-10-02; decision D98) · GM arXiv:1905.00383v3, L4.14 (l. 2097–2102), L4.15 Step 3 (l. 2159–2163), Step 4 (l. 2189–2191) · LQGMetric/Papers/GM/S4/P412g{Trace, Centre}.lean, decisions/DEC-98.md · (a) GM choose z_y “in a manner depending only on $(\mathcal{B}^\bullet_{\cdot} \{t_k\}, h)$ ” without saying how; we make the choices inside the proof of L4.14 canonical (grid net, least rational radius, least-index maximal bounded side, rational-angle arcs) so the guard centre $c_1 = \text{gd}(K, y)$ is a Borel function, and guard $Q_k = \text{gd}(K, y_k)$ through N surely measurable frontier selections (p412g_grid_centres, own elementary measurable-code argument) · kind: a place where the paper’s implicit measurable choice is made explicit (own descriptive-set-theory argument) · (b) the guard centre is a point of $\partial \mathcal{B}^\bullet_{\cdot} \{t_k\}$ within $8\varepsilon^{\kappa r}$ of c_1 , with $(*)$ at $r = 8\varepsilon^{\kappa r}$, not GM’s z_y itself · kind: reading of the paper (any point with $(*)$ works) · (c) GM’s σ_k is a stopping time, so the event belongs to $\sigma(\mathcal{B}^\bullet_{\cdot} \{s_{k+1}\}, h)$ ” is formalized as the trace lemma p412g_inter_localSigma plus p412g_stop_inter (a.s.-completed σ -algebra, D70), with the stopping-time inputs for confSigma as an open packet · kind: own argument for an implicit step · status: (a)(b) decided (D98), (c) partly proved.

DV-DFT12d-1, -2 (P2-DFT12d, 2026-10-02) · DFGPS arXiv:1905.00380, proof of Thm 1.2 (T:1339–1386); GM (1.6) l. 300–302 · LQGMetric/Papers/DFGPS/T12P6{A,B,C,D}.lean, T12Final.lean · (1) weylScale_weylMetric ($e^{\xi f} \cdot (e^{\xi g} \cdot D) = e^{\xi(g+f)} \cdot D$), used without proof by GM (1.6) and DFGPS, by an own elementary proof (local length comparison on a Lebesgue-number partition); (2) “a.s. locally uniform convergence implies convergence in probability” (implicit at T:1386) through measurable versions lfppC and tendstoInMeasure_of_tendsto_ae on $C(\bar{B}_R \times \bar{B}_R, \mathbb{R})$ — own argument · own arguments · proposed · no.

D-L22, D-L23a..d, D-L24, D-L25 (P2-DDDF23, 2026-10-02) · DDDF arXiv:1904.08021, Lemmas 22–25 (tightness.tex l. 1022–1067, 1221–1258), (5.59) (l. 1086–1091) · LQGMetric/Papers/DDDF/{L22,L23*,L24*,L25}.lean · (L22) own elementary proof (Chebyshev around the mean, $4/p \leq 2/p^2$ for $p \leq 1/2$) instead of DDDF’s independent-copy argument (l. 1031), statement unchanged; (L23a) $\text{Var} \leq L^2$ from Gaussian concentration by expanding the mgf to second order ($e^y + e^{-y} \geq 2 + y^2$, $t \rightarrow 0$); (L23b) variance bound passed to the limit by Fatou (bounded means, subsequence) instead of dominated convergence (l. 1047–1049), avoiding the implicit square-integrability of $\sup|\varphi|$; (L23c) convergence of the discretized length by uniform continuity of φ on $[0,1]^2$ instead of DDDF’s gradient bound (l. 1044), nearest block centre on block boundaries; (L23d) bound $\xi^2 n \log 2$, which implies the stated $\xi^2(n+1) \log 2$ ($\text{Var } \varphi_{[0,n]}(x) = n \log 2$; the source is off by one); (L24) “same bound for ψ by Prop 5” as $\text{Var } A \leq 2\text{Var } B + 2E(A-B)^2$ with $E(A-B)^2 \leq \xi^2 E X_{\{1,1\}}^2$ bounded uniformly in n (Prop 5 tail + layer cake), following the blueprint’s reading of the “ $\psi \dots \psi$ ” typo; C_p chosen after W (harmless: all white noises have the same law); (L25) induction on $|a_n - n|$ proving DDDF’s unwritten lower bound (“similar”, l. 1258) as well, $C_3 = \max(3A, D)$ · own arguments, typo/off-by-one fixes · proposed · no.

DV-T12Loc (P2-DFT12e, 2026-10-02) · DFGPS arXiv:1905.00380, proof of Thm 1.2 Steps 2–3 (T:1358–1385), eqn-localized-property (T:682) · LQGMetric/Papers/DFGPS/{T12P7A,T12Blueprint}.lean · DFGPS say only that $D_h(\cdot, \cdot; O)$ “is determined by $h|_O$ ” (T:1374); the measurable $F : \text{DistOn } U \rightarrow \dots$ is built explicitly: Lusin separation ($\text{comap_le_fieldSigma_of_local}$, DV-D80) shows locF measurable for $\sigma(\text{restrictTo } U)$, then Doob–Dynkin ($\text{Measurable.exists_eq_measurable_comp}$) per pair (z, w) · standard argument made explicit · proposed · no.

DV-M202-1.4 (P2-M202, 2026-10-02) · GM arXiv:1905.00383v3, proof of Prop 6.1 (l. 3576–3650), DFGPS Prop 3.18 (T:2247–2254) · LQGMetric/Papers/GM/S6/Prop61Rest{Det, }, .lean, Assembly/M2.lean · (1) Step 4 (l. 3644): $D_h(z, w) \leq c_{re}^{\{\xi h_r(0)\} \varepsilon^{\{\chi'\}}$ replaced by $D_h(z, w) \leq S c_{re}^{\{\xi h_r(0)\}}$ on $B_r(0)$ (GM.S2.4c with $U = B_{-2}(0)$, $K = \bar{B}_{-1}(0)$), $\delta_0 = (C_* - c'_2)(b\varepsilon^{\{1+\nu\}p^{-1}})^{\{\chi'\}}/(2S)$ for one fixed small ε instead of choosing ε with $a\varepsilon^{\{(2+\nu)\chi'\}} = \delta$ — not following the published proof; (2) Step 3 (l. 3628): the mesh pair has $|z-w| \geq \beta r/2$ (grid rounding), Step 1 applied with $\beta/2$ — reading; (3) (6.2)/(6.3) (l. 3613): shrink to t' with $|P(s)-P(t')| = b\varepsilon^{\{1+\nu\}p^{-1}r}$ by the intermediate value theorem (geodesic monotonicity) so the Hölder lower bound applies — implicit fact, own proof; (4) Step 1b (l. 3601–3610) parameters $\chi = \xi(Q-2)/2$, $\chi' = \xi(Q+2)+1$, $q = ((1+\nu)\chi' + \chi)/\chi$, $\varepsilon'' = 2p^{-1}\varepsilon^{\{1+\nu\}}$, $U = B_{\{2R_2\}}(0)$ (GM.S2.4e), $K = \bar{B}_{\{2R_2\}}(0)$ — parameter range · proposed · no.

D-DDDF-T20-1.3 (P2-DDDFT20, 2026-10-02) · DDDF arXiv:1904.08021, Condition (T) (tightness.tex l. 950–960), proof of Thm 20 (l. 1069–1213), (2.27) (l. 478–487) · LQGMetric/Papers/DDDF/{T20Ind,T20,D5T}.lean · (1) Condition (T) stated with measurably selected $(1+\eta)$ -near-geodesics, $\exists \alpha, c, K_0$ before $\forall \eta \exists \gamma$, the moment as a lower Lebesgue integral (applies D-DDDF-5 / D31; DDDF’s condition with a measurable exact geodesic implies it); (2) the choice of K in (5.73) by an own elementary argument ($2\sqrt{K} \leq \varepsilon K + 1/\varepsilon$; DDDF: “take K large enough”); (3) the $\varphi \leq \psi$ direction of (2.27), $\Lambda_n(\varphi, p) \leq C \cdot \Lambda_n(\psi, p/2)$, used by DDDF at l. 1212 but stated only in the other direction (l. 486), proved the same way from Prop 5 · statement form, own arguments · proposed · no.

D-LM31-1.3 (P2-LM31, 2026-10-02) · LM arXiv:1905.00379, Lemma 3.1 (tex l. 573–590, proof l. 723–764), Lemma 3.2 (l. 604–612), Lemma 3.3 · LQGMetric/Papers/LM/L3_1{Core, ,Filt,Main}.lean · (1) l. 736–763: no stopping times k_j and no coupling; $P[\mathcal{N}(K) < bK] \leq P[\#\text{good} < \beta K] + (e^{\{\lambda \alpha\}(1-q+qe^{\{-\lambda\}})^{\{\beta K\}}}$ directly with explicit λ (part (1): $\alpha = \beta = \sqrt{b}$, Lemma 3.4 at $(a, \sqrt{b})$; part (2): $\beta = 1/2$, $\lambda = 1$, Lemma 3.4 at $(1, 1/2)$ as at l. 742) — own elementary bookkeeping; (2) Lemma 3.2 proved only modulo null sets (circleAvg is a pathwise limit, so $(h - h_r(0))_{\{r'\}}(0) = h_{\{r'\}}(0) - h_r(0)$ only a.s.): the filtration lmFiltration is the P -augmentation of $\mathcal{F}_{\{r_k\}}$; $\sigma(\cdot|_{\{B_\varepsilon(K)\}}) \rightarrow \bigcap_\varepsilon$ via a $\liminf$ over $\varepsilon = 1/(n+1)$ — modelling; (3) Lemma 3.3 enters only through its Hölder consequences (3.8) (moment exponent 2 for H , 1 for H^{-1} ; blueprint/LocalMetrics.md LM.L3.1 note), with the good events existential (only their properties are used) — statement form · proposed · no.

DV-D101 addendum (P2-DZZ32C, 2026-10-02) · DZZ P3.2 upper bound (LBM_LGDArXiv.tex l. 1098–1101) · LQGMetric/Papers/DZZ/S3P32V{Geo,Cross}.lean · the crossing with clipped rings ($\text{l32BallCrossingC_holds}$): clipped boundaries are rectangle boundaries, and meeting points of box boundaries are carried to meeting points of the clipped boundaries by the coordinatewise clamp to $[r, 1-r]$ (no side-length condition); ends as in ball_end with paths in $\mathbb{V}_{\{-r\}}$ · own elementary argument · proposed · no.

DV-K6-1.5 (P2-M2K6, 2026-10-02) · GM uniqueness-final.tex L4.7 (l. 1890–1940), L4.11 (l. 1979–1990), L4.19 (l. 2318–2327), L4.22 (l. 2527–2565), Prop 4.17 (l. 2404–2433), proof of Thm 4.2 (l. 1611–1613, 2437–2446) · LQGMetric/Papers/GM/S4/Iterate6*.lean · (1) the ε_1 of L4.19/L4.22 depends only on the numbers $\beta, \chi, \chi', a, \lambda_4, \lambda_5, \ell, \xi$ (D75), same proofs; (2) candidate-radius slack r instead of 1 (gm_h47_kS , gm_T42_fixedS ; the old $\text{gm_h47_k/gm_T42_fixed}$ with slack 1 give a count growing like r^{-2} and are not used) — Lean modelling repair; (3) Lemma 4.11 used for every $a \in (0, a_0]$ in complement form so that $a \leq \ell/4$ can be chosen; (4) in T4.2, $R.\ell = \ell/4$, events $E' = E$ on $\mathcal{R}$ and DistC elsewhere, radii rescaled by r/R (εr for $\varepsilon > \varepsilon_0$), P completed (D70), h normalized by a far test function $h(\psi_0) = 0$ — only to meet the “for all r ” hypotheses of L4.11 and P4.12; (5) rates $\delta = \max(C_8, 1)\varepsilon^p$, $p = 4M_8 + 4\nu + 8 + 2\beta + 2M$, $M_8 = 2/\zeta + 1$, $\nu_* = \beta/8$, $\zeta = \beta/4$, $\theta' = \min \theta (\beta/2)$ (GMP4_12At gives no $\theta < \beta$) — own bookkeeping of GM’s “ $\text{o}^\infty_{-\varepsilon}(\varepsilon)$ ” and “ $\kappa = \varepsilon^{\{2\nu+\text{o}(1)\}}$ ” · proposed · no.

DV-D93 (a)–(e) (P2-DZZ93B, 2026-10-02) · DZZ LBM_LGDarXiv.tex, proof of Lemma 3.12 (l. 1461–1499: parents l. 1462–1466, two segments l. 1471–1477, Case 3 l. 1489–1497, near $\partial\mathbb{V}$ l. 1498), Remark 3.15 (l. 1360–1363) · LQGMetric/Papers/DZZ/S3L12W1–W12.lean · (a) zone geometry (quadrant boxes at the outer corner, at most three parents, at most one near $\partial\mathbb{V}$, neighbours of the diagonal parent) by own elementary arguments; (b) ”by connectivity” (l. 1476–1477) replaced by D93’s counting: one of the two arcs of the ring has at most one parent; (c) $s_{\{C_{\{i,j\}}\}} \leq s_C/\varepsilon^*$ ”by maximality” (l. 1491–1492) proved via the cell after $C_{\{i,1\}}$ (before $C_{\{i,2\}}$) having side $\leq s_C$ (exists_mem_enc_of_chain, a point of each parent outside C_{large}); (d) the closed walk of fine boxes mapped to cells by cellOf with consecutive repeats removed (ring form of Remark 3.15) — own argument; (e) CellRingOfCross carries IsCell C and $1 \leq C.n$ (supplied by the wiring, $\delta_0 < 1/2$); FineGood allows boxes inside a parent cell (for arbitrary m such boxes need not have mass $< \delta^2$) · own arguments, interface design · proposed · no.

DV-MQ1..3 (P2-MQ, 2026-10-02) · Miller–Qian arXiv:1812.03913, lqg_geodesics.tex, Lemma 4.1 (l. 549–589), proof of Thm 1.2 (l. 462–506) · LQGMetric/Papers/MQ/{RNMoment,HarmGrad,Cutoff,ZBCM,Lem41,Thm12Reduce,ShiftNull}.lean · (1) l. 576–580: the harmonic gradient estimate (4.1) proved with constant $8M/s$ via Borel–Carathéodory and the Cauchy estimate (own proof of a standard fact); (2) l. 585–589: ”Jensen” in three ranges — $p \geq 1$ conditional Jensen (mathlib data processing), $0 \leq p \leq 1$ via $a^p \leq 1 + a$, $p < 0$ via $\int (d\mu/d\nu)^p d\nu = \int (d\nu/d\mu)^{1-p} d\mu$; $c = \max(2, \exp(|p(p-1)|K_0/2))$ — not literal; (3) l. 462–468: the reduction uses constant-speed geodesics on $[0,1]$ (IsGeod01) compared at rational distances $r < D(x,y)$; existence of geodesics from GM S1.1 / DFGPS Lemma 3.8 (MQThm1_2Weak asks for existence and uniqueness) — convention · proposed · no.

DV-D102 (P2-DEC102, 2026-10-02) · DZZ arXiv:1807.00422, P3.2 upper bound (l. 1131–1137), L3.5 crossing (l. 1071–1077), Remark 5.2 (l. 2288–2290) · LQGMetric/Papers/DZZ/S3P32XW.lean · supersedes the clipped rings of DV-D101 (the clipped boundary $\partial(B' \cap [r, 1-r]^2)$ of a ring box on $\partial\mathbb{V}$ is not coverable by λ balls inside the square, handoff/P2-DZZ32C.md): at the walled measure μ in the covered ring curve is the wall-interior boundary $\partial B' \cap [r, 1-r]^2$ (the sides on $\partial\mathbb{V}$ dropped, nothing added), covered by DZZ’s own corner balls of radius ts except those centred on $\partial\mathbb{V}$, with DZZ’s count $4\varepsilon/t \leq \lambda$; the level- N crossing (D84) uses only the boundary squares of a ring box that touch a side not on $\partial\mathbb{V}$ (ringSqW), for which touching rings have boundaries meeting inside $[r, 1-r]^2$ · own elementary modification of the published argument · proposed · no.

DV-LM31b-a..c (P2-LM31b, 2026-10-02) · LM arXiv:1905.00379 l. 573–764 (Lemma 3.1 and proof), l. 604–643 (Lemmas 3.2, 3.3), l. 696–706 (Lemma 3.4); MQ l. 549–610; GM l. 984–998 · LQGMetric/Papers/LM/L3_1In{Frz,Q,Rep,Scale,Asm}.lean · (a) (3.8) is used in MQ’s zero-boundary form (LMAnnulusIterInputQ: on Good M k, $M \geq M_1(p)$, $(1 - P[A|\mathcal{F}])^4 \leq C(M)(1 - P(A))$ and $P(A) \geq p \Rightarrow (p/4)^4/C(M) \leq P[A|\mathcal{F}]$) instead of LM Lemma 3.3’s comparison with the marginal law (LM l. 640–642 cite MQ for 3.3 without the marginal step; LM’s proof of 3.1 only uses $p_M \rightarrow 1$ as $p \rightarrow 1$ and $p_M > 0$ for large M); the lower bound goes through the good-event probability as in GM l. 996–998; LMAnnulusIterInputF (marginal form) is not used on this route — statement form of a cited result; (b) the good event centres $\mathfrak{h}$ by the scaled field’s unit circle average (0 a.s.) instead of $\mathfrak{h}(0)$, and reads $\mathfrak{M}$ through countably many radial pairings (GM’s goodD); $P(\mathfrak{M} > M)$ bounded uniformly by GM’s L2.7 argument at scale 1 after scaling — reading; (c) condExp_frozen, frozen_upper, indep_augSigma own elementary proofs · proposed · no.

DV-MQ4 (P2-MQ2, 2026-10-02) · Miller–Qian arXiv:1812.03913, lqg_geodesics.tex, proof of Thm 1.2 (l. 469–497) · LQGMetric/Papers/MQ/Sphere{Det,Cover,Main}.lean · own simplification of MQ’s separation step: instead of $E(R,\varepsilon,\delta)$, an $\varepsilon/2$ -net of $\partial B_h(x,r)$, the radius u and X_i, X_j , two distinct points $w \neq w'$ of $\partial B_h(x,r) \cap \partial B_h(y,s)$ give a geodesic $x \rightarrow w \rightarrow y$ missing a rational ball $B(q,3\rho) \ni w'$; for the rational bump φ (1 on $\bar{B}(q,2\rho)$, 0 off $B(q,3\rho)$) the ”tie” event holds ($D_{\{h+\varphi\}}(x,y) \leq D_h(x,y)$ and a point of $\bar{B}(q,\rho)$ on a geodesic); along $h + a\varphi$, $a \in [0,1]$, strict monotonicity of the Weyl cost (MQ l. 503) allows at most one a ; MQ’s final step (l. 505–506, D-C4 form: countable bumps, Fubini, Cameron–Martin) gives probability 0; the additive constant is fixed by a test function away from the bump (MQ l. 486); deterministic Weyl facts proved (MQ states them without proof) · own simplification, conclusion unchanged · proposed · no.

D-T20b-1..6 (P2-DDDFT20b, 2026-10-02) · DDDF arXiv:1904.08021, tightness.tex l. 1069–1213 (proof of Thm 20) · LQGMetric/Papers/DDDF/T20B*.lean · (1) half-open dyadic blocks instead of closed ones (null-set difference); the Efron–Stein sum runs over the finite nearIdx K (other blocks do not affect $[0,1]^2$ under PsiSmall); block fields read at dyadic centres so that fLen is measurable — definition change; (2) $\text{Var } \psi_{\{0,K\}} \leq \text{Var } \varphi_{\{0,K\}}$ (truncated kernel smaller) — own elementary step; (3) (5.69) with factor $e^{\{C(\log N)^{2/3}\}} = e^{\{CK^{2/3}\}}$ instead of $e^{\{CK^{1/2+\varepsilon_0}\}}$ (both subexponential, all (5.70) needs) — own computation, not the published one; (4) with near-geodesics (D-DDDF-5) each unvisited block contributes $\leq \log(1+\eta)^2$, removed by $\eta \rightarrow 0$; (5) $Q.\varepsilon_0 < 1/2$ in Step 4 / (5.70) (DDDF l. 1163; D103) — parameter range; (6) tightness for $p \leq 1/4$ (needed for $\ell \leq \lambda \leq \ell$), shape monotonicity via rectLen_rectAB_mono (D-DDDF-21) — convention · proposed · no.

DV-LM34-a,b (P2-LM34, 2026-10-02) · MQ arXiv:1812.03913 Prop 4.3 proof (l. 664–733; (4.1) at l. 677–681; decomposition l. 693–700), LM Lemma 3.4 (l. 696–706) · LQGMetric/Papers/LM/L3_4{Iter,Leaf,Incr}.lean · (a) MQ bound the waiting time to the next good scale by a geometric sum of i.i.d. Gaussian-tailed variables via an outside-

in exploration with stopping times (l. 711–733); here the oscillation at scale k is dominated linearly by $C \sum_{j \leq k} s_1^{k-j} W_j$ (W_j MQ’s independent increment oscillations, weights from MQ’s derivative estimate l. 677–681), and Chernoff on $e^{\sum W_j}$ (MQ’s Markov step l. 728–733) gives the bound — same inputs, own bookkeeping (the stopping-time exploration needs a strong Markov property the library lacks) — not following the published proof; (b) general (r_k) (LM l. 704 “exactly the same argument”): $r_{k+j}/r_k \leq s_1^j$ (ratio_le_pow), contraction $q = s_1 \cdot \text{proposed} \cdot \text{no}$.

DV-D102 addendum (P2-DZZ32D, 2026-10-02) · DZZ L3.5 crossing (l. 1071–1079), P3.2 upper-bound balls (l. 1098–1101); DEC-102 §4 · LQGMetric/Papers/DZZ/S3P32Y{Sq,Geo,Cross}.lean · l32BallCrossingW_holds proved; the meeting of touching reduced rings (frontier_inter_W_of_bdry) is an integer grid-point argument (int_coreW: a shared grid point with both coordinates in $[1, 2^N - 1]$, inside $\mathbb{V}_{\{-r\}}$ since $r < 2^{\{-N\}}$); ringSqW_conn and isPathConnected_frontier_chainW take $1 \leq b \cdot n$ per box (every cell has level ≥ 1) · own elementary geometry (DZZ do not clip) · proposed · no.

DV-LM16-1.4 (P2-LM16, 2026-10-02) · LM arXiv:1905.00379, local-metrics-final.tex, proof of Thm 1.6 (l. 776–882) · LQGMetric/Papers/LM/T1_6{Chain,Cross,Main,Thm}.lean · (1) l. 858–880: instead of Lemma 4.3 (the union of circles contains a path) plus “the triangle inequality”, the triangle inequality is run directly along the crossing sequence (for circle j go back to the last earlier circle meeting it; circles in between lie in $B_{\{r_j\}}(w_j)$ as at l. 869–871; if none meets it, $z_1 \in B_{\{r_j\}}(w_j)$) — own elementary argument; (2) l. 844–846: t_j is some exit time (intermediate value theorem), not the smallest; only the crossing property is used; (3) l. 838–840: no measurable choice of path; the bad event is covered by countably many outer-null sets indexed by $\delta = 1/(n+1)$, $R \in \mathbb{N}$ and the continuity modulus of $\tilde{D}$ at z_1, z_2 ; (4) l. 832–834: a countable dense subset of $\mathbb{C}$ replaces $\mathbb{Q}^2$; LMLem4_2 assumes (1.3) on every compact set (grid points lie in $B_\varepsilon(K)$; harmless for $U = \mathbb{C}$) · own arguments · proposed · no.

DV-stop-ae, DV-EDet (P2-M2J2g, 2026-10-02) · GM uniqueness-final.tex l. 2155–2199 (esp. 2189–2191), CONF confluence-final.tex l. 1258–1302; D98 §3 · LQGMetric/Papers/GM/S4/P412i*.lean · (stop-ae) “ σ_k is a stopping time” is used in the a.s. piecewise form (gmPieceSig, p412i_stop_inter_ae) instead of sure measurability in $\sigma(\mathcal{B}^{\wedge}_{\{\sigma \wedge s\}}, h|_V)$, which is unprovable (Axiom II holds only a.s.; confE contains the chosen version harmPart); the conclusion lands in gmAESigma $\sigma(\mathcal{B}^{\wedge}_{\{t_{k+1}\}})$ (GM state $\sigma(\mathcal{B}^{\wedge}_{\{s_{k+1}\}})$ and use $t_{k+1} \geq s_{k+1}$) — modelling; (EDet) CONF l. 1260 “ $E_r(z)$ is determined by $h|_{\{A_{\{2r,5r\}}(z)\}}$ ” read as “a.s. an event of $\sigma(h|_V)$ for every open $V \supseteq \text{cl } B_{\{5r\}}(z)$ ” (P412iEDet); the annulus version is false since the circle average $h_r(z)$ lies on $\partial B_r(z) \subset B_{\{2r\}}$; CONF’s conclusion only uses $B_{\{5r\}}(z)$ — reading; Borel measurability of σ (P412iBorel) and attainment of ρ^{n+1} (p412i_rho_attained) are own routine arguments (CONF implicit) · proposed · no.

DV-LM34b-1.3 (P2-LM34b, 2026-10-02) · MQ arXiv:1812.03913 Lemma 4.4 (l. 633–657), Remark (zero-boundary) (l. 659–667); LM l. 586–590, LM Lemma 2.1 · LQGMetric/Papers/LM/L3_4N1–N6.lean · (1) MQ bound the oscillation by the Poisson kernel on a circle, Jensen and the Gaussian law of $h(y) - h(0)$ (Gaussian-square moment); here the mean-value disc average (GM.abs_sub_le_integral_of_harmonic), Jensen and $E e^{\{tD(\varphi)\}} \leq E e^{\{tY(\varphi)\}}$ from independence give an exponential moment (all LM L3.4 needs; avoids proving the harmonic part Gaussian) — not following the published proof; (2) $h^r(0) = h_r(0)$ (implicit in LM l. 586–590): own variance argument ($Y_\rho(0) = g(0) + \Delta_\rho$, $\text{Var } Y_\rho(0) = \log(1/\rho)$); (3) a.s. uniqueness of the harmonic part (LM L2.1; IG4 Prop 2.8): own write-up (condExp_indep_eq, countable dense radial bumps) · proposed · no.

DEV-LMLOC-1.4 (P2-LMLOC, 2026-10-02) · LM arXiv:1905.00379, Lemma 1.4 (l. 253–274), Lemma 2.3 (l. 523–549) · LQGMetric/Papers/LM/Loc{CI,Equiv,Nest,Joint}.lean · (1) l. 534 “ h is determined by $h|_V$ and $h|_{U \setminus V}$ ” used without proof in LM: stated as the open node LocGermSplit; (2) l. 545: own elementary proof that the internal metric on the smaller set is a Borel function of the one on the larger set (chain formula with a countable dense subset of $W_2 \setminus W_1$; locInternalNest); (3) l. 548 “letting W increase to V ”: exhaustion $W_n = \{d(\cdot, C \setminus V) > 1/(n+1)\}$, $V = \mathbb{C}$ separately; (4) isJointlyLocal2_of_determined: shortcut for metrics determined by h , based on LM l. 233 rather than LM’s proof of Lemma 1.4 (used to bypass LMLem1_4 in GM S2.3) · proposed · no.

DV-LM42-1.5 (P2-LM42, 2026-10-02) · LM arXiv:1905.00379 l. 776–829 ((4.1), Lemmas 4.1, 4.2); blueprint/LocalMetrics.md S4.0, S4.1a, S4.2a · LQGMetric/Papers/LM/{L3_1N2,L4_Trans,L4_Meas,L4_2,L4_1}.lean · (1) l. 815 (S4.1a): LM apply L3.1 with $r_k = 2^{\{-k\}}r$, which violates L3.1’s annulus requirement; here $R_k = 4 \cdot 8^{\{-k\}}r$, $s_1 = 1/8$, $s_2 = 1/2$ (annulus exactly $A_{\{t/2,2t\}}$, $t = 8^{\{-k\}}r$), only the scales $8^{\{-k\}}r$ (one good scale suffices; no residue classes), $a = 5 \log 8$ instead of $q \log 2$ — proof route; (2) l. 813–815 (S4.0): invariance only under translation, scaling absorbed into the radii r_k — own routine argument; (3) l. 812: E_t normalized by $e^{\{-\xi \tilde{h}_{\{4t\}}(0)\}}$ (L3.1’s factor) instead of $e^{\{-\xi h_t(z)\}}$ (event invariant under a common constant factor) — convention; (4) l. 812: $\partial B_{\{t/2\}}$ lies on the boundary of the open annulus, so $D(\partial B_{\{t/2\}}, \partial B_t) = \sup_n D(\partial B_{\{\rho_n\}}, \partial B_t)$, $\rho_n \downarrow t/2$ (setDist_sphere_eq_iSup, crossing + uniform continuity) — own elementary argument; (5) l. 827 (S4.2a): (1.3) applied on $K' = \text{cthickening } 1 \text{ } K$ — reading · proposed · no.

DV-D93 (P-6a) (P2-DZZ93C, 2026-10-02) · DZZ LBM_LGDarXiv.tex, Lemma 3.12 harder case (l. 1461–1477), Remark 3.15 (l. 1360–1363), Lemma 3.16 boundary-ring refinement (l. 1381–1397), c_{parents} (l. 1462–1466); DEC-93 §2 / D99 · LQGMetric/Papers/DZZ/S3L12X1–X15.lean · DZZ’s “by connectivity” each parent is visited once (l.

1471–1477) is formalized by an own elementary construction: the four long-way crossings of HasCross are cut at the outer-corner lines of C and run straight through every half-side lying in a parent cell (zones are whole quadrant regions; not_parent_opposite), glued at common sites in the corners and walked around from the corner of C's parent box (separation as in Lemma 3.7), then refined through the boundary rings as in l. 1381–1397 and rotated at a change of cell (CoarseRing); crossing meetings via clip_corner_short/clipRect_meet (Grimmett, Percolation, 2nd ed., §11.7) · explicit proof of a step DZZ leave implicit, no change of statements · proposed · no.

D-DDDF-22..25 (P2-DDDFT20c, 2026-10-02) · DDDF arXiv:1904.08021, tightness.tex l. 1093–1172 (Step 4 of Thm 20), Prop 3 (l. 337–352) · LQGMetric/Papers/DDDF/T20C*.lean · (22) the circuit around P^K (l. 1103–1105) can leave $[0,1]^2$ for boundary blocks while $L_{\{1,1\}}$ -paths stay in the square: only the circuit pieces inside the square are used (rectangles cut across keep a crossing of the cut piece, rectangles cut along the long side are dropped; the rest with $\partial[0,1]^2$ still separates), field comparisons inside the square, each piece bounded by the full $\varphi_{\{K,n\}}$ crossing length — not following the published proof (planned for T20Step4Num); (23) X and the oscillation on $[-2,3]^2$ instead of $[0,1]^2$ (l. 1112, (5.68)), moments moved by translation invariance — convention; (24) with $(1+\eta)$ -near-geodesics (5.66) gains an additive η and the gathered bound $C \cdot 4^K \cdot \eta^2$ (removed by $\eta \rightarrow 0$); $\psi_{\{0,K\}}$ evaluated at a block $P' \in \pi^K$ near b (oscillation factor); DDDF's 1/9 and "two crossings" become one crossing and multiplicity 16 — reading; (25) 4-factor Hölder with exponents $(\alpha, 3\beta, 3\beta, 3\beta)$ for "Hölder and Cauchy–Schwarz" (l. 1168) — convention; prop3_expMoment_lintegral (Prop 3 as a lower integral) and annulus_cross own elementary steps · proposed · no.

DV-EDet', DV-P412-params, DV-D70-condExp, DV-P412-S (P2-M2J2h, 2026-10-02) · CONF l. 1133–1145, 1260; GM l. 2189–2199, 2212–2276; DEC-98 §§2–4 · LQGMetric/Papers/GM/S4/P412j*.lean · (EDet') conditions 1–2 of $E^U_r(z)$ proved (dense-sequence method of DFGPS L3.2), condition 3 modulo P412jHarmLoc; (params) GM leave the exponents implicit: $\kappa = \min(\chi/\chi', 1)/(2\chi'/\chi)$, $\omega = \kappa\alpha/4$, $\beta = \min(\kappa\chi/8, \omega\beta_{-C/4}, \chi/(2\chi'), 1/2)$, $\theta = \min(\beta/4, \omega/2)$, $N = 2\lceil \varepsilon^{\{-\omega\}} \rceil$, $\delta = 2^{\{-m\}} \in [18\varepsilon^\kappa, 36\varepsilon^\kappa]$, $C = 2$ — own elementary numerics; (D70-condExp) conditional expectations given $\sigma(h|_K)$ unchanged by completion (own routine argument); (S) Prop 4.12 at the CONF §3 parameters only (D104), as GM l. 1096–1168 · proposed · no.

DG3P-1..3 (P2-DG3P, 2026-10-02) · DG arXiv:1807.01072, metric-comparison-final.tex, proof of Prop 3.22 (DG:1729–1772), eqn-square-dist (DG:1682) · LQGMetric/Papers/DG/S3P22{Chain,Det,Main}.lean · (1) DG:1748–1767: DG's path P with exit times t_j replaced by the polygon through a shortest chain of the covering balls (intersection graph, shortest walk); makes the double count (DG:1762–1763, not literally valid for an arbitrary cover) rigorous, constant 2 instead of 3 — own argument; (2) DG:1767: the last segment $j = \mathcal{J}$ (not followed by a crossing) bounded with the global field maximum M_x — DG include it in the sum without comment; (3) DG:1742–1743, 1682: the minimum-ball-mass condition in strict form ($> \varepsilon$), the set distance $D(S, \partial S(1))$ used pointwise · proposed · no.

DV-CONF24-1..4 (P2-CONF24, 2026-10-02) · CONF arXiv:1905.00381, confluence-final.tex, Lemma 2.2 (l. 505–510), Lemma 2.3 (l. 516–527), Lemma 2.4 (l. 533–537, proof l. 548–558) · LQGMetric/Papers/CONF/L2_4{A,B,C,D}.lean · (1) l. 505–510: Lemma 2.2 at a general centre z_0 straight from MQ Thm 1.2 for each fixed pair (no translation); ratPt $q = z_0$ via the constant geodesic; (2) l. 548–556 (existence): $y_n = \varphi(t_0 \pm 1/(n+1))$ with geodesics from GM S1.1 (CONF uses rational geodesics here); (3) l. 552–555 (approximation): q_n with $D(q_n, y_n)$ below half a Euclidean-topology modulus via $D(P_q(s), y_n) \leq 2D(q_n, y_n)$ (replaces "half the distance to the clockwise arc"); "converges from the left" read through a local inverse of φ (jordan_local_inv, own proof); the statement covers every one-sided limit, so it is proved through uniqueness (CONFSideUniq); (4) l. 556: Euclidean equicontinuity for Arzelà–Ascoli from the uniform equivalence of D and the Euclidean metric on compact D-balls (euclid_modulus, own proof) · proposed · no.

DV-LM34c-i..iii (P2-LM34c, 2026-10-02) · MQ arXiv:1812.03913 proof of Prop 4.3 (l. 693–700); LM Lemma 2.1 (l. 425–429), Lemma 3.4 (l. 696–706); GMSH arXiv:1807.07511 Lemma 2.2; Janson, Gaussian Hilbert Spaces, Thm 1.7 · LQGMetric/Papers/LM/L3_4M1–M6.lean · (i) the filtration is $\mathcal{F}_j = \sigma(D_0, \dots, D_j)$ instead of MQ's $\mathcal{F}_{\{r_j\}}$; independence from orthogonality in the Gaussian space of $h(D_{\{j+1\}} \in K_j, D_i \perp K_i, K_j \subseteq K_i)$ instead of MQ's direct Markov-property argument — same mathematics, not literal; (ii) $D_{\{j+1\}} = G_{\{j+1\}} - G_j(\rho) + \lambda_j$ with λ_j a version of $\kappa_j - P_{\{K_j\}}\kappa_j$ (κ_j the vector of $h_{\{r_{\{j+1\}}\}}(0) - h_{\{r_j\}}(0)$), differing from MQ's $h^{\{r_{\{j+1\}}\}} - h^{\{r_j\}}$ by random constants that cancel in $d_j(u) - d_j(0)$ — convention; (iii) scaling invariance of the Cameron–Martin image (range_cmIso_lmScaled) proved (the paper says "by scaling") — own elementary proof · proposed · no.

DG3E-1..3 (P2-DG3E, 2026-10-02) · DG arXiv:1807.01072, Lemma 2.2 (DG:618–640), Lemma 3.1 (DG:966–974), App. A (DG:2123–2244) · LQGMetric/Papers/DG/{L3_1B, L3_1B2, L3_1B3, L2_2, L2_2B, L2_2C}.lean · (1) the pairs $(h^U, \hat{h}^{\text{tr}})$ and $(\hat{h}, \hat{h}^{\text{tr}})$ for $U = \mathbb{V}$ and a box K under the D100 circle-average reading; $(\hat{h}, \hat{h}^{\text{tr}})$ as the difference of the other two; (2) DG Lemma 2.2 by an own argument instead of IG1 Lemma 6.4 + Borell–TIS: mean value property, $E e^{\{a|h(\varphi)\}} \leq 2e^{\{a^2 \text{Var } h(\varphi)/2\}}$ (independence, Jensen), Chernoff; stated for U bounded with $U \cap \partial \mathbb{D} = \emptyset$ and $K \subseteq U$ compact; (2.5)–(2.6) not restated; (3) dg_lemma31 (all six pairs on one space) conditional on DGLem31HCoupling (Markov property on $\mathbb{V}$, which meets $\partial \mathbb{D}$, and the white-noise coupling of h^U) · proposed · no.

DV-LM31N2-1..4 (P2-LM31N2, 2026-10-02) · LM arXiv:1905.00379 Lemma 3.1 (l. 573–590), proof (l. 723–764), proof of Lemma 3.3 (l. 631–637), Lemma 2.3 (l. 534, 545); Kallenberg Prop 6.8 · LQGMetric/Papers/LM/L3_1N2P1–

P5.lean · (1) filtration $\mathcal{F}_k = \mathcal{F}_{\{r_k\}}$ with the scaled metrics on $A_0, \dots, A_{\{k-1\}}$ only (not LM's $\mathcal{F}_r$ of (3.4) with the metrics on all of $\mathbb{C} \setminus \bar{B}_r$); enough for the counting (l. 731–757); (2) (3.8) for events with metrics (l. 725–730): instead of Lemma 3.3 (RN moment bound) + Hölder, $f = P[E \mid \text{field on } A_k]$, $a = \{f \geq 1 - \eta\}$, Markov's inequality, then the $N = 0$ field bound for a — own elementary argument; (3) $e^{-\xi h_r(0)} D(\cdot, \cdot; W)$ replaced by the Borel version $e^{-\xi h_r(0)} \cdot \text{chainInf}$ (a.s. equal for length metrics), events by a.e.-equal measurable versions; (4) LocGermSplit (LM l. 534) used in the weak-union step at l. 635–636 (LM use it through form (2) of their Lemma 2.3) · proposed · no.

DV-HARMLOC-1,2 (P2-HARMLOC, 2026-10-02) · Sheffield math/0312099 §2.6 Thm 2.17 (tex l. 690–725); Berestycki–Powell arXiv:2404.16642 Thm 1.52, Lemma 1.53 · LQGMetric/Field/HarmLoc{A,B,C}.lean · (1) P412jHarmLoc via the Hilbert-space Markov property with the zero-boundary projection on U (no localisation to a neighbourhood of ∂U , since $V \supseteq \text{cl } U$); arbitrary additive constant handled by recentring at $h_1(0)$ (mean-zero pairings unchanged) plus $H_0^1(U) \perp L^2(\sigma(h|_{\mathbb{C} \setminus U}))$ from IsHarmPart — sources state it for zero-boundary or normalized fields; (2) $\int H \Delta f = 0$ for harmonic H (integral_mul_laplacian_of_harmonic) own elementary proof · proposed · no.

DV-GERM-1 (P2-GERM, 2026-10-02) · LM arXiv:1905.00379 l. 534; Sheffield §2.6 Thm 2.17 · LQGMetric/Field/GermSplitA.lean, LQGMetric/Papers/LM/T1_6Final.lean · LM l. 534 (h determined by $h|_V$ and $h|_{U \setminus V}$) proved for bounded V only (GermSplit.locGermSplitBdd); LM Thm 1.6 uses it only at the annuli A_k , so LM. l. mThm1_6 is unconditional; T1_6Final.lean copies P2-LM31N2's $N = 2$ proofs under *_bdd names with the bounded hypothesis; the unbounded case stays open as LM.LocGermSplit (needed only by l. mThm2_3_of_germ, l. mThm1_4_of, which the assemblies bypass) — parameter range · proposed · no.

DGMU-1.3 (P2-DGMU, 2026-10-02) · DG arXiv:1807.01072 DG:980–986, 994–1007, 1112–1137, 1739–1756 · LQGMetric/Papers/DG/{S3P22Grid,S3MuHat,S3MuL8}.lean · (1) $\mu_{\hat{h}}$ and $\mu_{\{\hat{h}^{\text{tr}}\}}$ defined via the coupling with $h^{\wedge V}$ (μ_{HU} = white-noise chaos on V , density $e^{-\gamma Y}$ with Y the continuous modification of $h^{\wedge V} - \hat{h}$ resp. $h^{\wedge V} - \hat{h}^{\text{tr}}$ from DG L3.1), on boxes K well inside V only (DG use $\mathbb{S}(1)$ and $\mathcal{R}_n$); Weyl relation and DG L3.2 on K proved; (2) DG L3.8 reduced to L3.8 for $\mu_{\{h^{\wedge V}\}}|_K$ (dgLem3_8_transfer, DG:l122; exponent slack $\theta = (1+\beta/\beta^*)/2$ own elementary proof), dgLem3_8 with an arbitrary S ; (3) grid indexed by gridIdx, $a = 2^{\wedge\{-n\}}$, $n \geq 1$; cell-covering own elementary proof · proposed · no.

DV-CONF-SIDE1,2 (P2-CONFSIDE, 2026-10-02) · CONF arXiv:1905.00381, proof of Lemma 2.4 (l. 552–558), Lemma 2.3 (l. 516–527), Lemma 2.2 (l. 505–510), proof of Lemma 2.5 (l. 593–600) · LQGMetric/Papers/CONF/L2_4S{1,2}.lean, LQGMetric/Papers/GM/S4/P412kBridge.lean · (1) l. 557–558: “cannot cross $P_{\{q_n^{\wedge}\}}$ ” and “lies to the right” read in the DEC-D log-cover lift order at each level $t \in (0, s)$; “it follows” formalized as TOPO-ORD + Lemma 2.3 (rat_order), interpolation through a rational geodesic (step_order), limit along the approximating geodesics (side_order); the Blueprint one-sided-limit leftmost and the DEC-D leftmost agree (isLeftmost_of_side); the D76 bridge p412k_bridge follows; (2) angle lifts of uniformly convergent paths converge (angle_tendsto), positions on a Jordan lift depend continuously on the lifted point (lift_tendsto) — own elementary arguments · proposed · no.

DV-P412-unif (P2-M2J2i, 2026-10-02) · GM uniqueness-final.tex l. 2155–2199, 2212–2276, 2216–2222, 2572–2579, 2527–2565 (D75) · LQGMetric/Papers/GM/S4/P412U{1,2,3,4}.lean, P412S.lean · the primed $\dots U$ copies of the chain $\text{gm}_L4_22 \rightarrow \dots \rightarrow \text{p412i_good}_k$ choose ϵ_1 before E, rr , the space and r (RegNum: every field of R except E , rr fixed numerically), making explicit that GM's “ ϵ small enough depending on the parameters” thresholds depend only on the parameters; same proofs — convention · proposed · no.

DV-CONF27-1,2 (P2-CONF27, 2026-10-02) · CONF arXiv:1905.00381 Lemma 2.7 (l. 617–645) · LQGMetric/Papers/CONF/L2_7A.lean · (1) l. 628–629: instead of Lemma 2.6 (winding bound) putting the lifted endpoints in one interval of length ≤ 1 , the angle lift of each leftmost geodesic is normalized at its endpoint, so the level- s position is monotone in the endpoint position and shifts by 2π with it (pos_mono, pos_shift); Lemma 2.6 not needed; the ordering is CONF l. 640–643 via step_order — not following the published proof; (2) l. 636 “by possibly re-choosing π ... each $\hat{I}$ is an interval”: arc_lift own elementary proof; the universal cover (l. 627) read as DEC-D's log-cover lift · proposed · no.

DV-D98-grid, DV-D98b, DV-D98-Yk (P2-M2J2f, 2026-10-02) · GM uniqueness-final.tex L4.14 (l. 2097–2102, proof l. 2457–2520 as repaired in DEC-86/D88), L4.15 Step 3 (l. 2155–2183), GM.S4.1 (l. 1648–1654); D98 §2 · LQGMetric/Papers/GM/S4/P412h*.lean · (grid) the canonical choices of the output function $\text{gd}(K, y)$: grid $(\epsilon/8)\mathbb{Z}^2$, net $T = \{m \mid \bar{B}_{\{3\epsilon+\epsilon/8\}}(g_m) \cap K \neq \emptyset\}$, least radius $\rho_j = 3\epsilon/2 + \epsilon/(2(j+1))$ avoiding every $|y - g_m|$, arcs indexed by $\mathbb{Z}^2 \times \mathbb{Q}$, maximal arc of least index; (b) (*) holds at every point of $\bar{B}_{\{8\epsilon\}}(\text{gd})$ (radius $r = 8\epsilon$); (Yk) $\mathcal{Y}_k$ described through the closures of two distinct arcs on the GM.S4.1 event (= p412fEndSet there); descriptive-set-theory steps (P412hBor1–3, gmAn_iInter) own arguments; the interface P412jCent res holds as stated only for metrics with $\forall g, D \, g \in \text{lenSet}$ (sure frontier condition; decision D106) · proposed · no.

D-DDDF-26,27 (P2-DDDFT20d, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex (5.65)–(5.67) (l. 1097–1149) · LQGMetric/Papers/DDDF/T20D*.lean · (26) (5.67), l. 1131–1146: one short crossing per visited block (found for-

wards or on the reversed path) and overlap factor 16 for the closed boxes $\hat{P}$ instead of two disjoint short crossings and factor 9 — constants only; (27) (5.65)+(5.66), l. 1103–1123: split at the circuit-gluing step; the ψ -to- φ comparison, the oscillation over distance $C K^{\{\varepsilon_0\}2^{\{-K\}}$ (Obig) and $\log x \leq x - 1$ proved; the gluing is the open node T20CircuitGlue (long rectangles from the fixed grid family longFam K, boundary treatment D-DDDF-22); sigma_le_range, exists_short_piece, card_filter_hatBox_le, crossLenIn_mot_ne_top own elementary arguments · proposed · no.

CONF-T1.4-a..c (P2-CONF14, 2026-10-02) · CONF arXiv:1905.00381 Thm 1.4 (l. 380–382), proof (l. 1046–1062), l. 1503 · LQGMetric/Papers/CONF/T1_4{Det, }.lean · (a) l. 1051–1054: the Lemma 2.4 rational approximation is applied directly to the leftmost geodesic Q to $\partial \mathcal{B}^{\bullet}_s$ (approximating P_q restricted to $[0, \tau']$ are unique hence leftmost, so $P_q(\tau) \in X$ finite, then Lemma 2.3) instead of the leftmost geodesic to $\partial \mathcal{B}^{\bullet}_{\{\tau'\}}$ through x — same argument without the restriction step; (b) l. 1056–1058 “ $r \mapsto \tau_r$ is continuous and surjective”: only the used consequence (for $0 < t < s$ a rational r with $t < \tau_r < s$) proved — own elementary proof; (c) Theorem 3.1 used only in qualitative form (a.s. finiteness at $\tau = \tau_r$) from Thm 3.9 + Lemma 3.8 (l. 1503) — convention; CONF L2.7 not used · proposed · no.

DV-D97a..c (P2-DZZ97A, 2026-10-02) · DZZ arXiv:1807.00422 LBM_LGDarXiv.tex l. 680–682, 1159–1213, 2342–2346, 2563; DEC-97 · LQGMetric/Papers/DZZ/S3P32W1–W15.lean · (a) (eq-cell-LQG-compare) quantified over $S \in \mathcal{S}_B$ with $S \subseteq \mathbb{V}$ (over all S it is false at $\partial \mathbb{V}$, where squares outside $\mathbb{V}$ have mass 0; DZZ’s proof only uses $S \subseteq R \subseteq \mathbb{V}$); boxes of side $\geq \delta'^{\{C_{mc}\}}$ only (boxes of smaller side lie inside cells on the Lemma 3.1 event) instead of “not empty only for $s \geq \delta'^{\{C_{Mc}\}}$ ” (l. 1185); $M_s(B) \geq \delta'^2$ instead of $>$ (matches IsCell) — reading; (b) IsChaosLimit for M^W (portmanteau, Fatou, martingale + Doob, Borel–Cantelli no-escape of mass to $\partial \mathbb{V}$); DZZ only say the limit exists a.s. — own glue; (c) P-2 heavy-ball event via a finite rational net and positive mass of rational balls — own elementary · proposed · no.

DV-D105 (P2-DEC105, 2026-10-02) · DG arXiv:1807.01072, L3.8 (DG:1112–1137; GMS arXiv:1807.07511 Lemma 2.5 cited at DG:1123), $\mu_{\hat{h}}$ and L3.3 (DG:980–988, 1010–1036; RV Thm 5.5 at DG:988), §3 squares $\mathbb{S} \subset \mathbb{S}(1/2) \subset \mathbb{S}(1)$ (DG:1098) and rectangles/annuli $\mathcal{R}_n, \mathcal{A}_n$ (DG:1189, 1614), L2.2 (DG:618–640), DZZ citations (DG:1206, 1627) · LQGMetric/Papers/DG/S3D105A.lean, decisions/DEC-105.md · (1) L3.8 lower half proved from the project’s negative moments with the explicit exponent $A = p(p-1)\gamma^2/2 - 2p = -f(p)$ and DG’s own grid union bound (DG:1137) instead of GMS Lemma 2.5, on balls $\bar{B}(u, R)$ with $\bar{B}(u, 2R) \subseteq \mathbb{V}$ (DG: $\mathbb{S}$); the optimum $q = 2/\gamma$ reproduces DG’s threshold $2/(2+\gamma)^2$ — own elementary argument; (2) $\mu_{\hat{h}}$ is characterized as the vague limit of its own dyadic circle-average approximations on interior K_0 (own glue from the definition of qAreaMeasureOn and the Weyl relation), replacing RV Theorem 5.5, and (3.6) becomes a pathwise dyadic change of variables with $(\hat{h} - \hat{h}_{\delta})(\delta \cdot + b)$ realized as the $\hat{h}$ of the rescaled white noise $W \circ U_{\{\delta, b\}}$ (dyadic $\delta, \delta \mathbb{V} + b \subseteq \mathbb{V}$ only); (3) all §3 statements at $\mathbb{V}$ via $T(z) = (z+1)/3, \mu_{\hat{h}}$ only on $K_0 = [1/10, 9/10]^2, \mathcal{R}_n, \mathcal{A}_n$ replaced by $\delta \mathcal{R}_n + b, \delta \mathcal{A}_n + b$ inside $\mathbb{V}$ (the only scales at which DG use them, DG:1535, 1696), $h^{\{\mathbb{S}(1)\}}$ -forms by affine invariance; (4) L2.2 for U meeting $\partial \mathbb{D}$ from the translate $h(\cdot + b) - h_1(b)$; the harmonic part $\hat{h}'(\cdot - b) + h_1(b)$ is not independent of the zero-boundary part (DG: independent; unused); (5) from DZZ only P3.17 + Lemma 5.3 and P3.17 + Lemma 6.1 in the “probability $\rightarrow 1$ ” form through L3.2 at $(h^{\mathbb{V}}, \hat{h}^{\text{tr}})$ and the confined measure of D97 · proposed · no.

DEV-LMC18-1.5 (P2-LMC18, 2026-10-02) · LM arXiv:1905.00379 Cor 1.8 (l. 317–329), Lemma 1.4 (l. 253–274), l. 534, l. 545–548, Lemma 5.1 (l. 914–928), l. 331 · LQGMetric/Papers/LM/C1_8*.lean · (1) l. 324–325: the copy $\bar{D}$ “individually local” — proved by showing locality depends only on the joint law of (h, D) and transferring (own routine measure theory); (2) Lemma 1.4: the l. 534 step used only for bounded V ; arbitrary V by exhaustion $V \cap B_{\{n+1\}}(0)$ and Lévy’s upward theorem (LM’s own argument for Lemma 2.3 at l. 545–548, applied to Lemma 1.4) — lmLem1_4 unconditional; (3) Lemma 5.1: the two inequalities directly from Fubini under $\kappa(x) \otimes \kappa(x)$ instead of the essential inf/sup m_h, M_h (own argument, same statement); (4) Cor 1.8’s “random continuous length metric” read on the law (LMCor1_8L); the literal a.s. form needs CopyAeLength ($\{d \mid d.\text{IsLength}\}$ not known measurable); (5) l. 331: Thm 1.7 needs $C > 1$, so C is replaced by $\max(C, 2)$ · proposed · no.

DV-CONF38-1..3 (P2-CONF38, 2026-10-02) · CONF arXiv:1905.00381 confluence-final.tex l. 1482–1500 (event $\mathcal{E}_r(a)$, Lemma 3.8 and proof) · LQGMetric/Papers/CONF/S3L38{Conds, }.lean · (1) l. 1497 conditions 1–2 (“By Axiom V”): proved tightness GMS2_4a(i), GMS2_4b at centre z plus the deterministic ball/ τ -increment comparison $GM.gm_c2_good$ at $z = w = z$ (GM’s covering argument of L4.11 condition 2, GM l. 1984) — own argument; (2) l. 1498 condition 3 at centre z : DFGPS Prop 3.18 (centre 0, $h_1(0) = 0$) transferred via the law of the rescaled metric $c_r^{-1}e^{\{-\xi h_r(z)\}}D_h(r+z, r+z)$ (GM.Tight.map_scaledField_eq) — CONF leaves it implicit; (3) l. 1497–1500: failure probability split into thirds (conditions 1–2 together), $a = \min(\text{thresholds}, 1/2)$; condition 4 by Lemma 3.5 with a dyadic union bound $\Sigma C_0 \varepsilon^2 \leq 2C_0 a$; condition 2 read with “ $+\xi h$ ” (typo note) · proposed · no.

DV-CONF-33a..c (P2-CONF33, 2026-10-02) · CONF arXiv:1905.00381 Lemma 3.3 (C:1176–1244), C:1187, C:1231; LM Lemma 2.1 · LQGMetric/Papers/CONF/S3L33{, M, CM}.lean · (a) conditioning on $\sigma((h - h_p(w))|_{\{C \setminus U\}})$ with $\partial B_p(w) \cap U = \emptyset$ (CONF’s WLOG at C:1187), since IsWholePlaneGFF allows a random additive constant and literal $\sigma(h|_{\{C \setminus U\}})$ can fail ($h = h_0 + 1_{\{\text{Diam}\}}(h_0)$) — modelling; (b) “conditional probability $\geq p$ ” stated in integrated form (summable over realizations of U for L3.6); (c) C:1231 “RN law depends only on the energy” made quantitative

by Cauchy–Schwarz, $P[G] \geq P[\text{shifted}]^2 \cdot e^{\{-E\}} (z_{b_shift_le_sqrt})$ — own elementary step; the $V_{\{\delta r/2\}}$ gap at C:1208 (corridor one square wide) pending decision D108 (since decided: DV-D108) · proposed · no.

DV-CONF-35, DV-CONF-34, DV-CONF-EMeas, DV-CONF-32 (P2-CONF3, 2026-10-02) · CONF arXiv:1905.00381 Lemma 3.2 (C:1163–1172), Lemma 3.4 (C:1262–1271), Lemma 3.5 (C:1276–1285), Lemma 2.12 (C:785–802), C:1150–1154 · LQGMetric/Papers/CONF/S3{L35, L34, L32, EMeas, Main}.lean · (35) C:1284 “ $C = \varepsilon^{\{-3/2\}}$, $q = 6$ ” gives neither $\varepsilon^{\{1/2\}}$ nor rate ε^2 : $C = \varepsilon^{\{-1/2\}}$, $q = 8$, $\eta_{\{3.5\}} = \eta_{\{3.4\}}/2$ — correction of a computation; (34) C:1270: $b = 1/2$, $a = q \log 8$, $K = \lfloor \log_8 C \rfloor$, $\eta = 1/(4 \log 8)$, $C \geq 64$ (CONF: “ $a = \log q$, $K = \lfloor \log_6 C \rfloor$ ”) — correction; (EMeas) C:1150, 1270: σ -algebra of $\mathbb{A}_{\{3r/2, 5r\}}(z)$ instead of $\mathbb{A}_{\{2r, 5r\}}(z)$ ($\partial B_{\{2r\}}$ is on the boundary of the latter), Lemma 2.12 with $S_1 = 3/2$; CONF’s claim at C:1150–1154 ($E_r(z)$ determined modulo constants) is too strong — Lemma 3.4 needs Lemma 2.12 as printed (CONFlem2_12aP), not D47’s modulo-constants form; (32) C:1168 condition 2 by uniform continuity (S2.4b), the crossing bound (S2.4a), internal = global distance below the boundary distance — own elementary argument. FOUND: Blueprint.CONFlem3_5At (hence CONFSection3) is false as stated because IsHarmPart conditions on h itself while IsWholePlaneGFF allows a random additive constant (counterexample $h = h_0 + M \cdot h_0(\varphi)$) — decision pending (P2-DEC108; since decided: DV-D108) · proposed · no.

DV-D106 (P2-DEC106, 2026-10-02; decision D106) · GM uniqueness-final.tex L4.15 Step 3 (l. 2162: “choose for each $y \in \mathcal{Y}_k$ a point $z_y \in \partial \mathcal{B}^{\bullet}_{\{t_k\}}$, in a manner depending only on $(\mathcal{B}^{\bullet}_{\{t_k\}}, h|_I)$ ”) and CONF confluence-final.tex L3.6 (l. 1312–1313: “let $x \in \partial \mathcal{B}^{\bullet}_{\tau} \dots$ be chosen in a manner depending only on ...”) · Blueprint.CONFlem3_6AtAE (Blueprint/CONFResults.lean), GM.p412m_centres, GM.gm_P4_125 (Papers/GM/S4/P412m*.lean) · the guard centres lie on $\partial \mathcal{B}^{\bullet}_{\{t_k\}}$ almost surely, not at every sample point, and CONF L3.6 is used in that form, with its property A asserted where $x \in \partial \mathcal{B}^{\bullet}_{\tau}$ (a weak LQG metric may be changed on a GFF-null set of fields, where $\mathcal{B}^{\bullet}_{\{t_k\}}$ can be all of C and has no boundary point) · kind: reading of the papers’ a.s. language; statement of a cited result (CONF L3.6) in a.s. form · proposed · no (differs only on null sets).

DV-CONF-T39a (P2-CONF310, 2026-10-02) · CONF arXiv:1905.00381 C:1452–1471, C:1570–1744 (Lemmas 3.10–3.11, Thm 3.9) · LQGMetric/Papers/CONF/S3T39*.lean · steps grouped by the dyadic level $\lfloor \log_2 n_k \rfloor$ instead of CONF’s halving epochs $[k_j, k_{j+1})$; conditioning on $\mathcal{F}_{\{k_j\}}$ becomes a split on the first time the count enters the level; (3.26) and the geometric sums (3.29)–(3.31) unchanged; constants $b_1 = 1$, $\beta = \theta/2 = \chi/16$, b_0 depending on C_1 , a , θ , N_0 ; conditional Markov step (3.25) in integrated form (t39_condMarkov); CONFlem3_7At stated in the a.s. style of D106 (I chosen measurably: $\{(\omega, y) : y \in I \omega\} \in \text{localSigma} \otimes \text{Borel}$; closed ball in ArcDisconnect vs CONF’s open ball; own α , C_0) — bookkeeping, not word for word · proposed · no.

DV-D105-1 addendum, DV-D105-4 (P2-DG105a, 2026-10-02) · DG arXiv:1807.01072 L3.8 (DG:1112–1150), L2.2 (DG:618–640); D85 law transfer · LQGMetric/Papers/DG/S3D105B.lean, L2_2D.lean · (1-add) the lower half of L3.8 for μ_{HU} from the zero-boundary GFF tail by the law transfer of ball masses (map_qArea_ball_eq_wn); DEC-105’s N1 holds only in this law form (wnField is the white-noise circle family extended by 0 off circles, not a zero-boundary GFF as a process on all measures); target sets $\tilde{B}(u, R)$ with $\tilde{B}(u, 2R) \subseteq K$ (DG: S); (4) L2.2 for U meeting $\partial \mathbb{D}$ by the translation $h(\cdot + b) - h_1(b)$, $b = |R| + 2$, so the two parts are not independent; the tail absorbs $P[|h_1(b)| > A/2] \leq 2e^{\{-A^2/(8(\text{Var}+1))\}}$ (own elementary argument) · proposed · no.

DDF6-1..7 (P2-DDDF6, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex l. 162–166, 1004–1010, 1218, 1262–1329, 1606–1638 · LQGMetric/Papers/DDDF/S6*.lean · (1) l. 1630: in (6.99) the case n or $n' = 0$ handled with the constant $|\log \lambda_0|$ ((6.100) uses (5.76), which needs $n, k \geq 1$) — own argument; (2) l. 1613: in (6.98) the uniform bound on $\sup |\varphi_{\{2^{\{-r\}}, 1\}}|$ over $r \in [0, 1]$ by Fernique + Borell–TIS (S6.low_sup_tail), comparing with the shapes (1,3), (3,1) of Cor 17 / Prop 18 instead of (1,2), (2,1) — own argument; (3) l. 1329: “with probability $\geq 1/2$, $L \leq e^{\{-C\sqrt{k}\lambda_{nL_k}\}}$ ” read as $P(L^{\{(n+k)\}} < e^{\{-C\sqrt{k}\lambda_{nL_k}\}} < 1/2$; $L^{\{(k)\}}$ bounded below by its tail (Cor 17, Λ bounded) — reading; (4) l. 1325: (5.75) stated with a constant C_1 , the oscillation through 0big on $[-2, 3]^2$ (as T20Step4Den), $n \geq 1$ — statement form; (5) lower half of (5.76) for $1 \leq k < k_1$ by iterating (6.98) with $r = 1$ (DDDF do not treat small k) — own bookkeeping; (6) $P(2^{\{-k\}} \sup \|\nabla \varphi\| \geq M\sqrt{k})$ small for large k from (2.17) on translated boxes + Markov instead of (2.16) — own argument; (7) (5.54), (5.78) stated for every $\zeta > 0$ ($\zeta = \xi \varepsilon$) and $p \in (0, 1/2]$ — reading · proposed · no.

DV-D107 (P2-DEC107, 2026-10-02; decision D107) · LM arXiv:1905.00379 Thm 1.7, proof Steps 2–3 (local-metrics-final.tex l. 1030–1089, gap at l. 1075–1084) · LQGMetric/Papers/LM/T1_7D1.lean (LMT17VarNode, LMT17LimNode) · LM’s Step 3 is incorrect as printed: the near-geodesics P_ε of $D(\cdot, \cdot; V)$ vary with ε and (5.18) compares with D where $D(\cdot, \cdot; V)$ is needed, so no uniform modulus holds near ∂V . We take $U = \mathbb{C}$ (D21), $V = B_n$, keep Steps 1–2, and conclude by $\text{Var}[D(z, w; B_n) | h, \theta] \leq C^4 E[(D(z, w; B_{\{n-1\}}) - D(z, w; B_n)) D(z, w; B_n) | h, \theta] \rightarrow 0$ as $n \rightarrow \infty$ (second near-geodesic of $D(\cdot, \cdot; B_{\{n-1\}})$), product bound in place of Cauchy–Schwarz (5.17), slack ε^3 instead of ε^2 instead of LM’s limit $\varepsilon \rightarrow 0$ at fixed V ; no measurable selection of near-geodesics (pointwise inequalities between measurable functions, Tonelli over θ); LM (5.11) typo $E[D(z, w; V) | h]$ for $E[D(u, v; V) | h]$; l. 1088 “ U is a length metric” read as “ D ” · kind: place where we do not follow the published proof (own repair of a gap; no published source found) · proposed · no (same theorem).

DV-D105-2 (P6 part) (P2-DG105c, 2026-10-02) · DG arXiv:1807.01072 (3.1) (DG:907–917), $\hat{h}$ symmetries (DG:942–946), eqn-wn-truncate (DG:948–953), DG:1010 · LQGMetric/Papers/DG/S3D105Sc{1, 2}.lean · DG’s scale invari-

ance of $\hat{h}$, $(\hat{h}_{\delta t} - \hat{h}_{\delta})(\delta \cdot) = d \hat{h}_t$, proved pathwise: $(\hat{h} - \hat{h}_{\delta})(\delta \cdot + b)$ tested on circles is the $\hat{h}$ of the white noise $W \circ U_{\{\delta, b\}}$ with $U_{\{\delta, b\}}$ an explicit L^2 isometry ($wnScale$), for every $0 < \delta \leq 1$ and every b ; the scaling of $h^{\wedge \mathbb{V}}$ ($dgHU_wnScale$) and the translation of $\hat{h}^{\wedge tr}$ ($DG:953$, $dgTr_wnScale$) the same way from the Brownian scaling of the heat and killed heat kernels — own elementary argument (DG : "immediate from the definition") · proposed · no.

DV-D108 (P2-DEC108, 2026-10-02) · CONF arXiv:1905.00381 C:347, 1154, 1187, 1308–1320, 1431 (conditioning σ -algebras of L3.3/L3.6/L3.7); C:1183, 1205, 1208, 1398 (events (3.7), (3.9), path claim, (3.21)); C:324–338 (Remark 1.2); C:1236–1243 (Step 3) · LQGMetric/Papers/CONF/S3D108A.lean, decisions/DEC-108.md · (1) statement form: the conditioning σ -algebras are read modulo additive constants ($recSigma$, $localSigma0$, $filledBallSigmaAt0$) and L3.6 carries $IsLocalSetDet0$ on $\mathcal{B}^{\wedge \bullet} \tau$; the raw forms are false for an $IsWholePlaneGFF$ with random constant (counterexample DEC-108 §1.1); (2) correction of a false step: L3.3's conclusion weakened to the fat-point event (3.9') with $V_{\{6r/3\}}$, the path claim C:1208 replaced by the square-chain claims $CONFatChainA/B$ ($K = 12$), (3.21) with the bound $(2K+4)c/100 < c$; (3) modelling: no metric object for $D_{\{\hat{h}^{\wedge U}\}}$; a measurable F on $\mathcal{D}'(V)$, $V \in U$ ($CONFZBMetric$); (4) own routine measure theory, proved: $condFKG_freeze2$ (Step 3 conditioning with $G^{\wedge U}$ depending on the frozen outside field).

DV-D105-2 (P5 part), DV-D105-2b (P2-DG105d, 2026-10-02) · DG arXiv:1807.01072 DG:984–988 · LQGMetric/Papers/DG/S3D105Mu{,2}.lean · (2) $\mu_{\hat{h}}$ characterized a.s. as the vague limit on interior K of $(2^{-k})^{\gamma^2/2} e^{\gamma \hat{h}_{\{2^{-k}\}}}$ dz, with $\hat{h}$'s circle averages through any jointly measurable version (replaces the appeal to Rhodes–Vargas Thm 5.5, DG:988): Weyl relation with $f = -Y$, uniform convergence $Y_{\varepsilon} \rightarrow Y$ on compacts, Fubini for $\hat{h}_{\varepsilon} = h^{\wedge \mathbb{V}}_{\varepsilon} - Y_{\varepsilon}$ a.s. — own glue; (2b) DG:986 implicit: $\mu_{\{h^{\wedge \mathbb{V}}\}}$ gives zero mass to Lebesgue-null compact subsets of $\mathbb{V}$ (first-moment bound with cut-offs, transfer through the circle law) — own elementary proof; DEC-105's ϕ_{iVer} hint corrected (heat-kernel regularization $\neq$ circle average) · proposed · no.

D-DDDF-28 (P2-DDDFT20e, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex l. 1097–1123 (esp. 1103–1105, 1117–1119) · LQGMetric/Papers/DDDF/T20E*.lean · the four rectangles $Q_i(P)$ of size $2^{\wedge -K}(CK^{\wedge \{\varepsilon_0\}}, 3)$ become strips of width 3 blocks around the box of $m = \lceil 2s2^{\wedge K} \rceil$ blocks around b , each crossed through an alternating chain of horizontal/vertical 3×1 grid rectangles, the sum over the distinct rectangles of a simple path in the meet graph ($reach_d0n_le_sum$); boundary (D-DDDF-22 implemented differently from its plan): a box side closer than 3 blocks to $\partial[0,1]^2$ is pushed past the boundary and its strip dropped, the other strips extended to $\partial[0,1]^2$; when the box touches the left (right) side the glued path starts (ends) at the end of a horizontal strip on that side (covers a near-geodesic starting inside the box); no rectangle is cut; T20CircuitGlue needs $\varepsilon_0 \leq 1$ (covered by Thm 20's $\varepsilon_0 < 1/2$); no published proof of the planar gluing (DDDF rely on their figure) — own route, own elementary steps, parameter range · proposed · no.

DV-D109 (P2-DG105e, 2026-10-02; decision D109) · DG arXiv:1807.01072 Lemma 3.1 (DG:966–976), proof DG:2243–2244 ("combine Lemmas 2.2, A.1, A.2") · LQGMetric/Papers/DG/L3_1C{1,2,3}.lean · DG Lemma 3.1 only asserts that a coupling exists. We take the whole-plane GFF built from the same white noise W as $h^{\wedge \mathbb{V}}$, $\hat{h}$, $\hat{h}^{\wedge tr}$ ($Field/ExistGFF$: $h_0(\varphi) = W(\ker Fun \varphi)$), normalized as $h = h_0 - h_{0_1}(0)$. Then $h_0 - \hat{h}$ is the large-time field $\sqrt{\pi} W(1_{\{t>1\}}(p_{\{t/2\}}(z, \cdot) - p_{\{t/2\}}(0, \cdot)))$, which gets its continuous modification and Gaussian tail from $exists_box_modification_tail$ (L^2 bound $\int_{-1}^{\infty} |z-w|^2/(2\pi t^2) dt$). The circle averages $h_{0_r}(z)$ are identified with $\sqrt{\pi} W(\tilde{K}_{\sigma}) + \sqrt{\pi} W(f \text{ largeKerL2 } d\sigma)$ (L^2 -Cauchy mollified kernels plus their pointwise limit). This replaces DG's Markov decomposition $h = h^{\wedge \mathbb{V}} + \mathfrak{h}$ and DG Lemma 2.2 for the pair $(h, h^{\wedge \mathbb{V}})$: we get $(h, h^{\wedge \mathbb{V}}) = (h, \hat{h}) - (h^{\wedge \mathbb{V}}, \hat{h})$ instead. Own elementary argument · kind: place where we do not follow the published proof (equivalent coupling, since the statement is existential) · proposed · no.

DV-D105-6 (P2-DG105b, 2026-10-02) · DG arXiv:1807.01072 L3.8 (DG:1112–1150, union bound DG:1131–1137, optimum (3.13)); DZZ arXiv:1807.00422 (eq-def-tilde-M), (eq-LQG-positive-moment) (l. 676–688), Lemmas 2.5, 2.6 (l. 521) · LQGMetric/Papers/DG/S3D105U{1,2}.lean · DG's $E \mu(B_{\delta}(z))^{\wedge p} \leq \delta^{\wedge \{f(p)+o(1)\}}$ replaced by the tail bound $P[\mu(B(w,s)) > t, \mathcal{B}^c] \leq C t^{\wedge \{-p\}} s^{\wedge \{f(p)-\eta\}}$, $\mathcal{B}$ the global event that the oscillation of $\hat{h}_{\delta}$ at scale δ reaches $T = O(\sqrt{\log \delta^{-1}}) + \kappa \log \delta^{-1}$ ($P(\mathcal{B}) \leq \delta$ by DZZ Lemma 2.6 in tail form, $\delta = 2^{\wedge \{-m\}} \in [s, 2s]$); via DZZ's factorization $M = cDens.\tilde{M}$, independence of $\tilde{M}$ from $\hat{h}_{\delta}$, sharp positive moment of $\tilde{M}$; the η -loss is DG's $o(1)$; the oscillation input is DZZ L2.6 for $\hat{h}_{\delta}$, not $dg_lemma34$ — own bookkeeping · proposed · no.

DV-D107a..c (P2-LM17a, 2026-10-02) · LM arXiv:1905.00379 Thm 1.7 proof (l. 1000–1089), L5.1 (l. 904–929), L5.2 (l. 937–946), (5.2) (l. 950–953), replacement argument (l. 1079–1081); DEC-107 §2–§3; D111 · LQGMetric/Papers/LM/T1_7{L1,L2,N1,G1}.lean · (a) l. 1011–1026, 1084: N-VAR stated per value g of h under $\kappa_g = condDistrib D h P g$ (LMT17VarNodeK, D111); internal metrics through $chainInf$ — modelling; (b) l. 1079–1081: the sub-path bound by the triangle inequality for $D(\cdot, \cdot; B_n)$ instead of a replacement near-geodesic, giving $\eta_n + \varepsilon^3 + \omega_n$ ($t17_near_bound$) — own elementary argument; (c) L5.2 (l. 944–946) for a fixed measurable curve and an arbitrary s -finite time measure, θ Lebesgue on $\mathbb{R}^2$ (restricts to θ uniform on $[0,1]^2$) — statement form · proposed · no.

DV-D105-N6 (P2-DG105f, 2026-10-02) · DG arXiv:1807.01072 Lemma 3.3 (DG:1014–1036), route at DG:1033–1034, $\hat{h}_t$ (3.1) DG:907 · LQGMetric/Papers/DG/S3D105Sc{3,4,5}.lean · DG Lemma 3.3 ((3.6) $ae_muHat_scale(_set)$, (3.7) ae_dgLGD_scale) proved at dyadic $\delta = 2^{\wedge \{-m\}}$ with $\delta K + c \subseteq$ interior K , on $T(\text{interior } K)$ (not all Borel $X \subseteq \mathbb{C}$),

by DG's own alternative (the circle-average approximations, DG:1034) with the D105 N4 intrinsic characterization instead of DS Prop 2.1; $\hat{h}_\delta := \text{DDDF's } \varphi_{\{\delta,1\}}(\text{hatDelta})$, its circle averages identified by stochastic Fubini ($\text{ae_circleAvg_hatDelta}$); (3.7) constants via $e^{\{-\gamma \max \hat{h}_\delta\}}$ (DG3A-1); vague-limit glue own elementary · proposed · no.

DV-D110 (P2-DEC110, 2026-10-02; decision D110) · CONF arXiv:1905.00381 C:347, 1138, 1154, 1187–1190, 1308–1320, 1425–1447; GM arXiv:1905.00383 l. 214, 1200–1205, 2164–2191 · LQGMetric/Papers/CONF/S3D110{A,B}.lean, decisions/DEC-110.md · (1) IsHarmPart conditions on $\sigma(h|_{\mathbb{C}\setminus U})$ modulo additive constants (fieldSigmaClosed0), CONF's "normalized so that $h_{\{r_0\}}(z_0) = 0, B_{\{r_0\}}(z_0) \cap U = \emptyset$ " (C:1187); the raw form is false for IsWholePlaneGFF and for the normalized field when $\partial\mathbb{D}$ meets U (DEC-110 §1) — statement form, corrects D108's "no change"; (2) CONF L3.6 is consumed by GM §4 in the raw form for the field normalized at a bump ψ_0 inside $\mathcal{B}^{\bullet}_{-\tau}$ (CONFLem3_6AtAEN, proved from the mod-constant CONFLem3_6AtAE0; on-event form CONFLem3_6AtAENE with $E_\eta = \{B_\eta(z) \subseteq \text{int } \mathcal{B}^{\bullet}_{-u}, \eta \rightarrow 0\}$, where GM conditions on the raw σ -algebra of a normalized field without this event — own routine glue (continuity from below); (3) GM l. 214: IsWholePlaneGFF keeps the arbitrary random constant (GM's field is defined modulo constants; all conclusions are constant-invariant) — convention · proposed · no.

D-P21-1..5, D-T102-1 (P2-DDDF6c, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex Prop 21 (T:970–1010; Steps 1–3 T:978–1003), (6.102)/(6.103) (T:1639–1647) · LQGMetric/Papers/DDDF/S6P21*.lean, S6Tails*.lean · (P21-1) Step 2 (T:984–990): "concatenation of straight paths in each box" made explicit by a simple path in the graph of touching blocks of π^k with one segment per block (constant $4 \cdot 2^{\{-K\}}$) — own elementary argument; (P21-2) Step 1 (T:978–981): max over $\mathcal{P}_{K^1}$ as $\sup |\varphi_{\{0,K\}}|$ over $[0,1]^2 + 3 \cdot O$ oscillation (boundary blocks outside $[0,1]^2$), $X_1 = X_{\text{big}}$ on $[-2,3]^2$, $O = O_{\text{big}}$ — as in T20; (P21-3) Step 3 (T:996–1003): one generalized Hölder with weights $(1/\rho, u/3, u/3, u/3)$, $\alpha = r = \rho = \sqrt{((\xi+2)/(2\xi))}$, $u = 1 - 1/\rho$, oscillation moment from (2.17) with $\varepsilon = 1/8$ ($e^{\{CK^{\{5/8\}}\}}$) — not literal; (P21-4) $\xi = \xi(Q-2)/2$ in (5.54), rate $c = \xi(Q-2)\log 2/4$, ψ parameters $r_0 = 1/12$, $\varepsilon_0 = 1/4$ (psiQ_1); $\xi < 2$ from $d_\gamma \geq 1$ (DG.one_le_dGamma DG.chiLeTwo) instead of DDDF's $\xi < 2/d_2$ (T:970); measurable selection from exists_nearGeodesic (D-DDDF-5) — parameters; (P21-5) prop2_expMoment_int copies prop2_expMoment keeping integrability — no change; (T102-1) (6.102)/(6.103) for $[0,1]^2$ only, via $\varphi_{\{\delta,1\}} = \varphi_{\{2^{\{-r\}},1\}} + \varphi_{\{\delta,2^{\{-r\}}\}}$ with $u = s/(4\xi)$, (6.98), Prop 18, Cor 17; small s absorbed in C — own bookkeeping · proposed · no.

D-S6P26-a..f (P2-DDDF6b, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex T:1218, 1283–1329, 337–352 (proof of (2.17)), 984–990, 1608–1612 · LQGMetric/Papers/DDDF/S6P26{Up0,Up,Mu1,Sub,Glue,Glue2,Glue3,Wire,Final}.lean · (a) T:1307: the $e^{\{C\sqrt{k}\}}$ gradient moment from DDDF's proof of (2.17) run with $a_n = a$, not from (2.17) as stated; (b) T:1307: Cauchy–Schwarz replaced by $xy \leq \theta x^2/2 + y^2/(2\theta)$, $\theta = \lambda_k e^{\{-c\sqrt{k}\}}$; T:1309: $\lambda_{\{n+k\}} \leq 3 \cdot E L^{\{(n+k)\}}$ by Markov; (c) T:1287–1306: the geodesic π_k replaced by a measurable near-geodesic within factor 2, overlap 16 instead of 9, weights $e^{\{\xi \max \varphi_{\{0,k\}}\}}$ on the 7×7 box, constants $C_1 = 12$ (gluing), 32 (ΣV_P); (d) T:1289–1294: own gluing argument (blocks clamped to $[1, 2^k - 2]^2$, circuits of $*$ -adjacent sites meet, chained along a simple path), boundary as D-DDDF-22; (e) T:1316–1328: (5.75) via a crossing within factor 2 of $\pi_{\{n+k\}}$, overlap 16; (f) $k = 1$ in Step 1 via (6.98) with $r = 1$ · proposed · no.

DV-ETAa..f (P2-DZZETA, 2026-10-02) · DZZ arXiv:1807.00422 l. 444–449, 677–688, 1116–1125, 1183–1206 · LQGMetric/Papers/DZZ/S3Eta1–8.lean · (a) $\tilde{M}_{\{\gamma,\delta,\eta\}}(B)$ defined set by set as the liminf of DZZ's approximations, proved to be the a.s. limit (DZZ l. 677–683); (b) $E \tilde{M} \leq \text{Leb}$ instead of DZZ l. 1122's "=" ("by Fubini" needs uniform integrability), sufficient for (Eq.LQG-tildeM); (c) (eq-LQG-negative-moment) proved for $\tilde{M}_\eta$ (DZZ state it for $\tilde{M}_{\{\gamma,\delta\}}$, l. 684–688, and use it for $\tilde{M}_\eta$ at l. 1202) via a ported Kahane argument (η kernel below the $\tilde{h}$ kernel); (d) l. 1204's binomial bound is wrong as displayed ($(1+e^{\{-1\}})/2 \approx 0.68 > e^{\{-4\}}$); replaced by $2^N(\beta C)^{\lfloor N/2 \rfloor}$ from l. 1202's $P(\mathcal{A}_j^c) \leq \beta C_{\{\gamma,-1\}}$ — not following the proof; (e) the comparison at l. 1193–1195 assumed only for the $K^2/4$ even subsquares used, normalisation $(h/2)^{\{-2\}}$ instead of $(\varepsilon')^{\{-2\}}$, +1 in the range condition of l. 1197 — reading; (f) parameter asymptotics own elementary · proposed · no.

DV-ES1..3, DV-G2 (P2-LM17b, 2026-10-02) · LM arXiv:1905.00379 l. 548–560, 893–896, 904–929, 937–997, 1011–1036; DEC-107 §3, §5 · LQGMetric/Papers/LM/T1_7E{1..6}.lean, T1_7G2.lean · (ES1) l. 1030–1036: Lemma 5.1 applied to the base metric $d(u,v)$ instead of $D(u,v;V)$; the bound for internal metrics on every V then follows deterministically; (ES2) l. 548–560, 992–997: Lemma 2.4 for finitely many disjoint bounded open sets through chainInf encodings; Lemma 5.4 in kernel form at fixed θ instead of under the conditional law given (h, θ) ; (ES3) l. 1017–1026: Efron–Stein under the conditional law by disintegration into a product measure, positive-part form directly (no symmetry step (5.5)); (G2) l. 941–953: $\text{len}(P \cap A; D)$ is the Stieltjes length measure of the curve (t17LenMeas) — own constructions · proposed · no.

DV-LM-C18LEN (P2-ANALYTIC, 2026-10-02) · LM arXiv:1905.00379 local-metrics-final.tex l. 323–332 (Cor 1.8 "conditionally i.i.d. samples") · LQGMetric/Papers/LM/C1_8Len.lean · the conditionally independent copy (and the conditional law given h) of an a.s. length metric is a.s. a length metric (copyAeLength , t17KernelLength): $\{d \mid d.\text{IsLength}\}$ is the projection of a Borel set (metric, countable family of paths between dense points with almost minimal length; c18_isLength_iff), hence universally measurable (Lusin, Kechris Thm 21.10, via the existing

UMeasurableSet.setOf_exists, Meas/UM.lean); from dense points to all points by infinite concatenation (GM.p412e_concat) — own elementary proof / reuse; LM take this measurability for granted · proposed · no.

D-P29-a.c (P2-DDDFP29b, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex DD:1500–1601 (Prop 29 at 1500–1506, white-noise representation 1514–1525, coupling 1527–1536, kernel terms 1555–1596, Kolmogorov/Fernique 1598–1601); DFGPS T:877–881 · LQGMetric/Papers/DDDF/S6P29*.lean, LQGMetric/Papers/DFGPS/P29Sq{A,B,C}.lean, LQGMetric/Assembly/ExistenceReducedSq.lean · (a) DDDF Prop 29 formalized only for $D = (-1,2)^2$ (DDDFProp29Sq), the only domain DFGPS uses (applied once, L2_8GffPos zb_lim_posOffDiag; DD:1500 states it for general D); the DFGPS chain primed (dfgps_existence_reduced') — parameter range; (b) φ_t (DD:1528, defined only up to law) realized as $\varphi_{\sqrt{t}}$ of the shifted noise $W \circ T_t$, so the coupling has the form DDDFProp29Coupling asks for — convention; (c) the kernel bookkeeping of DD:1514–1525, 1555–1596 (L^2 membership, Fubini, change of variable $r = (t+s)/2$, a.e. case split) own elementary proofs; DDDF's (6.96) is not uniform near ∂D , so DD:1571–1576 is replaced (open node P29FirstIncr, own route) · proposed · no.

DV-D105h-1,2 (P2-DG105h, 2026-10-02) · DG arXiv:1807.01072 L3.12 (DG:1204–1216), L3.20 (DG:1625–1636), L3.2 (DG:994–1007), grid union bound (DG:1131–1137) · LQGMetric/Papers/DG/S3L12.lean, S3L20.lean, S3L20B.lean, S3L12W.lean · (1) L3.12 stated for a square inside DZZ's $\mathbb{V}$ (not the unit square); the restricted DZZ distance bounds DG's distance on $\mathbb{S}(1)$ by geometry (DZZ's $\tilde{D}$ uses only balls in $\mathbb{V}_{\{u,v\}} \subseteq \mathbb{S}(1)$, so DZZ's wall never matters — resolves DEC-105's doubt); L3.2's C sent to ∞ after $\varepsilon \rightarrow 0$; $u = v$ from $D(u,u) \leq D(u,v)$; (2) L3.20 at DZZ L6.1's boxes; DG's unrestricted distance bounded below by the sub-path from $\partial \mathbb{S}$ to $\partial \mathbb{S}(1/2)$, the heavy-ball event (small-mass balls meeting $\mathbb{S}(1/2)$ stay in $\tilde{B}(u,1/10)$ w.p. $\rightarrow 1$), rational-centre balls, L3.2 pathwise with fixed threshold — own glue; DZZ inputs as verbatim hypotheses DZZL53Whp, DZZL61Whp · proposed · no.

DV-D110-4,5 (P2-D110a, 2026-10-02) · DEC-110 §§2–4; CONF C:347, 1154, 1187; GM l. 214 · LQGMetric/Blueprint/{M2Defs,CONFDefs,CONFResults}.lean, Field/HarmLoc{C,D}.lean, Papers/GM/S4/{P412jComp,P412mMain,P412S}.lean, Assembly/MainOpen.lean · (4) GM Prop 4.12 enters the assembly as the open node GM.P4120fL36AE0 (P4.12 from the TRUE L3.6, CONFlem3_6AtAE0) until D110 P5/P6 — temporary statement form; theorem11_open/theorem12_open gain the hypothesis $hN : GM.P4120fL36AE0$; the GM §4 theorems still built on the raw L3.6 (p412m_P4_12At, P412jMain, P412fK, P412iCond, P412kBridge, gm_P4_12S_of) are vacuous until P6; (5) HarmLoc.p412jHarmLoc for the mod-constant IsHarmPart from the raw proof by inside normalization normIn $h \varphi_0$ (C:1187) — own routine glue; old names kept as export aliases · proposed · no.

DDDF6d-1.5, D-P29c (P2-DDDF6d, P2-DDDFP29c, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex l. 1336–1376 (Prop 27), DD:1571–1576; DF arXiv:1809.02607 l. 1013–1021 · LQGMetric/Papers/DDDF/S6T20Done.lean, S6Diam*.lean, S6P29Inc{,2,3}.lean · (6d-1) chaining with DF's halves $2^{\{-k-1\}(2,1)}$ of dyadic squares (DF l. 1013) instead of DDDF's $2^{\{-k\}(3,1)}$ (l. 1343), chaining along system centres own; (6d-2) the expectation of the max over 4^K rectangles via $\max \leq (\sum X^q)^{1/q}$, Hölder, s6_moment_L31 (DDDF: tails + union bound, l. 1357–1360); (6d-3) decoupling with $\max|\varphi_{\{0,K\}}|$ at $K = k+1$, $n = 0$ via $\varphi_{\{0,0\}} = 0$; (6d-4) Step 2 (l. 1370–1374) lower bound on λ_K directly from (5.54) via $\ell_K \leq \lambda_K$; series bounded with $C\sqrt{K} \leq \beta K/2 + C^2/(2\beta)$; (6d-5) the diameter as the ENNReal diamSq, bound as $\int^-$; $\xi < 2$ hypothesis (to be discharged by $d_\gamma \geq 1$); (P29c) DDDF Prop 29 first-term increments (DD:1571–1576): L^1 argument $((\Delta F)^2 \leq 2M|\Delta F|$, Chapman–Kolmogorov, product structure of $p^{\wedge}D$, 1D shift bounds by MVT and image folding, $\int_0^1 s^{\{-1/2\}} ds = 2$) instead of the paper's split at $\sqrt{|x-x'|}$ with (6.96), which fails near ∂D — not following the published proof · proposed · no.

DV-ETA2a.d (P2-DZZETA2, 2026-10-02) · DZZ arXiv:1807.00422 l. 1105–1107, 1116–1119, 1149–1151, 1193–1195; Lemma 2.7 (l. 548–576); $\mathcal{E}_{\{\delta,\alpha\}}$ (l. 864) · LQGMetric/Papers/DZZ/S3Eta9{A..D}.lean, S3EtaB{1,2,3}.lean, S3P32Z{1..4}.lean · (a) l. 1105–1107, 1193: only the event $\tilde{E}_{\{\delta,\alpha\}}$ (DZZ intersect $\tilde{E}_{\{\delta,\alpha\}} \cap \tilde{E}_{\{\delta',\alpha\}}$), via the finer-centre nbrFineEvent with $\alpha = 128C_{mc}$ and the (eq-tilde-h-eta-assump) threshold $\sqrt{L}$; output constant $\alpha + A$ — own constants; (b) l. 1116–1119, 1193–1195: both comparisons (DZZ: "similarly", none for (eq-M-tilde-B-bound)) proved by a density comparison + portmanteau on compact/open sets, exhaustion near $\partial \mathbb{V}$ with $E \tilde{M} \leq \text{Leb}$, continuous versions matched by Fubini — own glue; (c) L32TildeMUpper for every dyadic box within 3s of c_B (a superset of DZZ's $\mathcal{B}(B,t) \cup \mathcal{B}_{\partial}(B_{\text{large}},t)$); (d) l. 1149–1151 (proposed for L32StartPhiHPC): DZZ's horizontal line cannot be covered near $\partial \mathbb{V}$ when $r \ll ts$ — modified start path (vertical escape ball, diagonal chain at corners, threshold 1/6 instead of 1/4) · proposed · no.

DV-D110 4'–6 (P2-D110b, 2026-10-02) · GM uniqueness-final.tex l. 214, 2155–2276 (esp. 2164–2191); CONF C:476–479, 1154, 1187, 1295, 1308–1320 · LQGMetric/Papers/CONF/S3D110C.lean, LQGMetric/Papers/GM/S4/P412n*.lean, LQGMetric/Assembly/MainOpenB.lean · (4') CONF L3.6 consumed in the on-event form CONFlem3_6AtAENE (hypotheses IsLocalSetDet0 and localSigma $\leq \mathcal{F}$), derived from CONFlem3_6AtAE0 by the trace lemma confD110_localSigma_inter — statement form of a cited result; (5) GM Prop 4.12 for h uses the filtration and L3.6 events of $h - h(\psi_q)$ and the events of h , linked by Weyl scaling for constants and GM.ConfRKAddConst (open), $\eta \rightarrow 0$ along bumps $\psi_q = \text{bumpTest } q \ z$ — own routine glue, departing from DEC-110 §2.3 (not transferable: the events R.E are not constant-invariant); centres made localSigma0-measurable via the trace lemma + a ball-measurable fallback (they are not functions of the ball alone); (6) deterministic scaling lemmas (P412nScale, P412nCov) own elementary

proofs · proposed · no.

DV-D105i-1..3 (P2-DG105i, 2026-10-02) · DG arXiv:1807.01072 L3.13 (DG:1282–1312), L3.14 (DG:1328–1343), L3.21 (DG:1678–1719) · LQGMetric/Papers/DG/S3L13{A, , R}.lean, S3L14.lean, S3L21{A, , R}.lean · (1) L3.13: $n_m \asymp m$ instead of $m^{\frac{1}{3/2}}$; $\varepsilon \leq 1$ (monotonicity in ζ only then; only $\varepsilon \rightarrow 0$ is used); DGLem311Scaled folds in DG:1300–1305 (scaling (3.7) + independence from $\hat{h}_s$); (2) L3.14: "shrinking ζ " is L3.13 at $\zeta/(1+\gamma)$, the union over m a geometric series; (3) L3.21: $n_\varepsilon \asymp \log \varepsilon^{-1}$ a power of 2 ($\delta_\varepsilon/n_\varepsilon$ dyadic), L3.6 at $\delta = \varepsilon^\beta$, $A = \varepsilon^\beta n/\delta_\varepsilon$, the minimum in T_S over $S(1)$ (DG:1702 writes " U = interior of S ") · proposed · no.

DV-DDDFDG-1..4 (P2-DDDFDG, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex l. 1004–1010, 1276–1281; DG arXiv:1807.01072 Thm 1.5 (DG:343–346), Prop 3.21 (DG:1603–1610), L3.7 (DG:1096–1106), L3.1 (DG:966–976); DGo arXiv:1610.09998 Prop 3.3 (DGo:604–616); DFGPS L3.6 (T:1628–1650) · LQGMetric/Papers/DDDF/S6DG*.lean · (1) (5.54) derived from DG Thm 1.5 (1.5b) second half with $K = \{0\} \times [0,1]$, $U = (-1,1) \times (-1,2)$ instead of directly from DG P3.17 for the zero-boundary GFF ((1.5b) is DG's whole-plane consequence of P3.17, DG:1593–1595, 1782–1785), compared with the whole-plane circle average instead of DGo P3.3 for the zero-boundary GFF; (2) (5.78) ("follows the same lines", l. 1278) from DG P3.21 through DFGPS L3.6 at scale 1, graph-to-crossing (polygon + two end segments $\leq \delta$) own; (3) whole-plane DG L3.7 split into DG L3.1 for $(h, \hat{h})$, the free-kernel DGo P3.3 (FreeKerBounds, open) and the oscillation bound from DDDF (2.17); countable-dense gluing own; (4) DGo P3.3 with the free heat kernel on times $(0,1]$ instead of the domain kernel, continuity from exists_box_modification_tail · proposed · no.

DEV-P2-DDDF6e-1..4 (P2-DDDF6e, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex l. 155–166, 653–662 (Prop 7), 1340–1374 (Prop 27), 1385–1490 (Prop 28), 1497–1498, 1608–1613, 1639–1648 · LQGMetric/Papers/DDDF/S6TailsAB*.lean, S6Thm11*.lean, S6Thm12*.lean · (1) l. 1639–1643 ("same argument"): shape change via Prop 7 applied to the $3 \times 1/1 \times 3$ dyadic tails; choices of $\varepsilon (C e^{-\kappa \sigma^2 / \log \sigma})$ for the right tail, $e^{-\kappa \sigma^2} + \text{contradiction}$ for the left) and the absorption of small s own elementary; (2) the coarse-field sup for rectAB a b over $[0, \max a b]^2$ with Fernique increment constant $8m$ — convention; (3) l. 1398/1450/1648 (UpperHolder)/(LowerHolder) for $\delta \in (0,1)$ read with strict inequalities — reading; (4) l. 1497 (Thm 1(2) as a corollary): tightness transferred through the modulus criterion with laws moved across coupling spaces (own argument); D-DDDF-13 ($\delta \in [1/2,1)$) by uniform Lipschitz bounds with the sup on $B(0,6)$ · proposed · no.

DV-VAR1..5 (P2-LM17c, 2026-10-02) · LM arXiv:1905.00379 l. 931–1055 (Thm 1.7 proof, Lemmas 5.1–5.4); DEC-107 §2–§3 · LQGMetric/Papers/LM/T1_7V1–V20.lean · (1) l. 977–981, 1046–1050: $\text{len}(P \cap S; D) = \text{len}(P \cap S; D')$ via BBI Prop 2.3.12(1) on the open time set $P^{-1}S$ (length measure without atoms at endpoints) — own; (2) l. 984 "Symmetrically": $F(D') \leq F(D)$ a.s. with equal laws gives $F(D) = F(D')$ a.s. via arctan — own; (3) l. 931–946, 964–1026: θ uniform on $[0,1]^2$ carried as $t17\Lambda \otimes \kappa_g$; conditioning on θ and the square metrics with one jointly measurable kernel whose θ -slices are versions of the per- θ conditional law ($t17v_slice$); square metrics jointly Borel in (θ, D) via a countable formula for $D(x, S_{-\theta^c})$ — own measurability; (4) L5.3 (l. 956–986) restricted to $F = D(z, w; B_m)$ and the finitely many squares $t17Box \in R(D107 \S 3(i))$; D' need not be a length metric for $F(D') \leq \text{len}(P; D')$; (5) D107 §3(v): $\varepsilon \rightarrow 0$ by reverse Fatou with an ε -independent dominator (near-geodesic P_2 , $\varepsilon^3 \# S \leq \varepsilon(2R+5\varepsilon)^2$), then $\kappa \rightarrow 0$ · proposed · no.

DGo-3 (P2-DGOKER, 2026-10-02) · DGo arXiv:1610.09998 Watabiki_final.tex l. 604–704 ((3.9)–(3.10) for $\mathcal{U} = \mathbb{C}$; variance split l. 618–625; $G_{\{v;2\}}$ l. 655–662; HMP Prop 2.1 cited at l. 704) · LQGMetric/Papers/DDDF/S6DGKer.lean · the fine part $G_{\{v;2\}}$ bounded by the exact Brownian scaling of the white noise ($wnScale_hatMeasKerL2$), which makes it a constant, instead of DGo's θ -integral; the HMP Prop 2.1 bound applied through the white-noise/whole-plane-GFF coupling of DG L3.1 (D109) plus the Lipschitz large-time kernel ($lipschitzWith_largeKerL2$); constants $A = 12$, $\sigma^2 = (\sqrt{\pi} \|\hat{K}\sigma_{\{0,1\}}\| + 1)^2$ — equivalent sourced route, not the published computation · proposed · no.

DV-CONF21a..d (P2-CONF21, 2026-10-02) · CONF arXiv:1905.00381 Lemma 2.1 (C:476–479), proof (C:481–490) · LQGMetric/Papers/CONF/L2_1{A, B, C}.lean · (a) [QLE, Lemma 6.8] (limits of local sets, C:483, 490) replaced by a direct argument: right-continuity of the balls at radius $t > 0$ (geodesics, $gmE_unbounded_iff$) and compacts inside an open U stay inside for slightly larger radii — own elementary; (b) optional-time form $\tau_n = 2^{\wedge\{-n\}}(|2^{\wedge n}\tau| + 1)$ instead of $2^{\wedge\{-n\}}|2^{\wedge n}\tau|$ (C:490), since $IsFilledBallStoppingTime$ is defined through $\{\tau < t\}$ — convention; (c) proved for $\tau > 0$ a.s. only (CONFLem2_1Pos): for $\tau \leq 0$ with $z_0 \notin U$ the event $\{\tau \leq 0\}$ lies only in the germ $\bigcap_{t>0} \mathcal{F}_t$ (would need a Blumenthal 0-1 law CONF neither states nor uses); GM's s4T is surely positive, so GM consumes CONFLem2_1Pos — statement issue, the Blueprint item stays as stated; (d) "immediate from the case of deterministic times" (C:483, 490) formalised as $conf21_trace_local$ (D32's dyadic hulls) — own routine measure theory · proposed · no.

DV-D110c-a..c (P2-D110c, 2026-10-02) · CONF arXiv:1905.00381 C:1154, 1187–1190, (3.16) at C:1289; LM Lemma 2.1 · LQGMetric/Field/HarmExist{A, B}.lean, LQGMetric/Papers/GM/S4/P412oRK.lean, LQGMetric/Assembly/MainOpenC.lean · (a) $fieldSigmaClosed_addConst_le$ and $\sigma(g|_{\{C \setminus U\}}) = \sigma(h|_{\{C \setminus U\}} \bmod \text{const})$ for $g = (h - h(\psi_0)) - (h - h(\psi_0))_{\rho(0)}$ (CONF uses it implicitly at C:1187) — own elementary; existence of $IsHarmPart$ for every bounded open U by two normalizations (unit bump ψ_0 outside U , then a circle average outside $cl\ U$, since

markov_normAt accepts only a circle normalization); (b) a.s. uniqueness of harmonic-part versions via the mean value property on a countable dense set (as `lmRep_harm_eq`, with `HarmLoc.unitBump`) — own routine, needed because `harmPart` is a chosen version; (c) `GM.ConfRKAddConst` (`CONF C:1154` states the invariance without proof) formalized over the countably many (k, z, T) of (3.16) · proposed · no.

DV-DZZ56-1..3 (P2-DZZ56, 2026-10-02) · DZZ arXiv:1807.00422 LBM_LGDarXiv.tex l. 2250–2427 (§5, proof of L5.3), l. 2584–2606 (L6.1); DG arXiv:1807.01072 l. 1206, 1627; DEC-105 §1 item 7, §3 N9 · LQGMetric/Papers/DZZ/S5{Defs, Sub, Glue, Exp, Geom, Adapt}.lean · (1) Hammersley’s lemma (l. 2421) own proof of the standard argument (Hammersley 1962 not in literature/); (2) DG consume only the halves `DZZLem53Upper`, `DZZLem61Lower` (same χ), and DZZ prove only the lower half of L6.1 directly (l. 2597) — statement form; (3) for $\delta^\xi > \alpha/20$ the ξ -admissible pair for L6.1 uses two singleton points instead of the box frontiers (DZZ’s admissibility of fixed sets is a small- δ condition) — reading; temporary regularity hypothesis `hreg` (`wickQArea` finite, atomless on compacts of `openSquare`) — to be proved (P2-DZZ53) · proposed · no.

DV-CONF33W-i.iv (P2-CONF33W, 2026-10-02) · CONF arXiv:1905.00381 Remark 1.2 (C:324–338), proof of Lemma 3.3 Steps 1–2 (C:1187–1234), (3.9)/(3.12) (C:1205, 1219), C:347, 1187 · LQGMetric/Papers/CONF/S3D108P1.lean, S3D108P2{B, C, D, E}.lean · (i) C:324–338: `CONFZBMetric` (D108) is false ($h = h_0 + Z$ with independent $Z \sim N(0,1)$); the normalized `CONFZBMetricN` is used and proved — statement correction; (ii) C:1229: one Axiom II functional for Y and $Y+b$ via $\Omega \times \text{Bool}$ with a fair coin (`locality_pair`) — own routine device; (iii) C:1219–1223 “By Axioms IV’ and V” made explicit as `D_h` tightness + a harmonic lower bound + Weyl comparison; Step 1 core C:1200–1205 the mirror argument — own routine; (iv) C:1205/1219, (3.9)/(3.12): `CONFlem3_3w/confFatEv` (D108) and CONF’s own (3.12) for G^U as in (3.9) are false for “pocket” configurations; G^U taken per connected component of $U_{\{\delta r/4\}}$ (`confFatEvC`) — correction of a false step, decision D112 pending (since decided: DV-D112-1..4) · proposed · no.

DV-P32F1, DV-ETA2d’ (P2-DZZ32F, 2026-10-02) · DZZ arXiv:1807.00422 LBM_LGDarXiv.tex l. 1104–1147 (eq-B-percolation-Psi), l. 1149–1153 (eq-B-good-Psi), L3.7 template l. 990–1015, 1047–1049 · LQGMetric/Papers/DZZ/S3P32F{1..8}.lean · (P32F1) parameter choice: in (eq-B-percolation-Psi) $t/\varepsilon = 2^{\ell+3}$ with $2^{\ell+3} \leq \lambda < 2^{\ell+4}$ (DZZ: $2^{\ell+3} - \lfloor L^{\ell+0.6}/\log 2 \rfloor$); Peierls with $p = e^{-u^7/100}$, $r = 3$ as in L3.7; in (eq-B-good-Psi) $t = 2^{\ell+q}$ with $2^q \leq \delta^{\ell+1/2} < 2^{\ell+q+1}$ (DZZ: $\lfloor \log_2(\delta^{\ell+1/2}/\lambda) \rfloor$); tail $\delta^{\ell+1/4}$ instead of $\delta^{\ell+1/10}$; only the inequalities DZZ use are needed ($4\varepsilon/t \leq \lambda$, the Markov bounds, $M \leq \delta^{\ell+1/4}\lambda$) — parameters; (ETA2d’) l. 1149–1151 at μ In: the start path pushes x into $[h, 1-h]^2$ by a geometric chain of balls inside $\mathbb{V}$ ($O(\log \delta^{-1})$ balls), then follows DZZ’s row; all balls of radius $\leq h/2$ (each in ≤ 4 boxes), DZZ’s threshold $s^2 e^{\ell-E}/4$ kept (`open deterministic node P32StartGeom`) — own modification · proposed · no.

DEV-P2-DDDF6f-1..4 (P2-DDDF6f, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex T:1414–1490 (Prop 28), 1497–1506, 1648; DF l. 1828–1840; DFGPS T:877–881 · LQGMetric/Papers/DDDF/S6Sup1.lean, S6P28{Fine, Sup, Up2, Low2, Wire}.lean, S6Thm12Sq2{A, B, C, Fin}.lean · (1) D-DDDF-13 ($\delta \in [1/2, 1)$): DDDF derive Thm 1(2) from Prop 29 ($t \in (0, 1/2)$ only); for $\delta \in [1/2, 1)$ the uniform sup bound on $B(0, 6)$ by variance/increment bounds + Fernique (`s6_hsup`) — own standard argument; (2) T:1648 (“the same argument”): for the δ -family the small pairs are $|x-x'| \leq \delta$ (DDDF: $< 2^{\ell-n}$), (2.11) replaced by the uniform Fernique sup tail of φ_δ on 4^n dyadic squares (DF route) — reading/own route; (3) T:1442, 1481: $\beta \leq 1$ and $\alpha \geq 1$ as hypotheses of the Step-2 lemmas, wiring with $\beta = \min(1/2, \xi(Q-2)/2)$, $\alpha = \max(1, \xi(Q+2)+1)$ — parameter range; (4) DFGPS apply DDDF Thm 1(2) only at $D = (-1, 2)^2$: `DDDFThm1_2Sq` and the Q12 chain (verbatim copy of the P29Sq chain) — specialisation of a cited result · proposed · no.

DV-DZZ53-a..c (P2-DZZ53, 2026-10-02) · DZZ arXiv:1807.00422 l. 121, 744–745, 854–857, 2411; DG arXiv:1807.01072 L3.8 (DG:1112–1150) · LQGMetric/Papers/DZZ/S5L53{Reg, Crude, CrudeP, Side}.lean · (a) no atoms of M^W on $\mathbb{V}^\circ$ (DZZ use atom-freeness and local finiteness implicitly, l. 121) derived from DG L3.8’s upper half ($P(\text{some } z \in \bar{B}(u, R) \text{ has } M\{z\} > 1/n) \leq C\varepsilon^p$ for all small ε , hence 0; rational balls cover $\mathbb{V}^\circ$) — own elementary deduction (`ae_wickQArea_reg`); (b) (eq-very-crude) for $\tilde{D}$ (l. 856, used at l. 2411) by DZZ’s covering of the segment by balls of mass $\leq \delta^2$ (l. 744–745) with DG L3.8’s upper half as ball-mass input instead of DS11 Prop 1.6’s KPZ bound, and a geometric tail sum over dyadic ε_j — own tail sum; (c) a.e.-measurability of $\omega \mapsto M^W(B(c, r))$ (`hmeas, open`) needed only for integrability of $\log \tilde{D}$ (DZZ do not address it) — modelling input · proposed · no.

DV-D105j-1..3 (P2-DG105j, 2026-10-02) · DG arXiv:1807.01072 L3.14 (DG:1328–1343), P3.9 (DG:1159–1167, proof 1346–1391), P3.22 (DG:1722–1771) · LQGMetric/Papers/DG/S3L13V.lean, S3P9{Top, Det, Det2, Ev, An, Good, }.lean, S3P22A{1..4}.lean · (1) DG:1335: the vertical rectangles take the separate hypothesis `DGLem311ScaledV` instead of DG’s rotation invariance; (2) DG:1346–1389 (P3.9): one hub point per dyadic square for DG’s #-sets; a sub-arc argument (intermediate value theorem) + `lr_tb_meet` for “ X_S connected, $X_{\{\tilde{S}\}} \cap X_S \neq \emptyset$ ”; “sum over $O(\varepsilon^{\ell-\beta})$ squares” (DG:1384) as rows and columns at the top level; DG’s gain factor $\varepsilon^{\ell-\beta(2+\gamma^2/2-2\gamma-\tilde{\zeta})/d}$ dropped with $\beta = (1/(d-\tilde{\zeta}) - 1/(d-\tilde{\zeta}))/2$ (“choosing β small”); bottom level from L3.8 upper with radius $\varepsilon^{\ell-\beta}$; paths need not be simple; $\mathbb{V}$ -scale; (3) DG:1722–1771 (P3.22): the $\hat{h}$ form at $\mathbb{V}$ -scale (`dg_prop322_hat`), L3.21 cells anchored at 0, strict mass condition via L3.8 lower at 2ε with exponent β' , max of $\hat{h}$ by L3.5 with $\zeta_3 = \tilde{\zeta}d/(2\gamma\beta)$, constant 2 instead

of 3; L3.7 and the change of scale deferred (P13 adapter) · proposed · no.

DV-CONF39-a.f (P2-CONF39, 2026-10-02) · CONF arXiv:1905.00381 C:1452–1471, 1506–1744; GM l. 2142–2144 · LQGMetric/Papers/CONF/S3T39G{1..7}.lean · (a) C:1680, 1689: radius $7\epsilon^{\{1/2\}r}$ instead of $(\epsilon + 6\epsilon^{\{1/2\}r})r$ (as GM), constant $14^\chi n^{\{-\chi/8\}}$; (b) C:1543 “the leftmost geodesic” read as “a leftmost geodesic” (D44 gives no uniqueness); (c) C:1563 $\epsilon_k \leq 2n_k^{\{-1/4\}}$ instead of the strict $<$; (d) C:1572 “alive” means $s_k < \tau_{\{3r\}}$ (strict) so $\mathcal{B}^{\bullet}_{\{s_k\}} \subseteq B_{\{3r\}}$; (e) C:1468 continuity from above of filled balls ($t39g_open_subset_filledBall$) own elementary; (f) C:1716 b_0 made explicitly uniform by swapping quantifiers ($t39g_tail_exp_unif$) · proposed · no.

DV-D112-1.4 (P2-DEC112, 2026-10-02) · CONF arXiv:1905.00381 C:1183 (3.7), C:1205 (3.9), C:1208, C:1219 (3.12), C:1396 (3.21), C:324–338, C:1133 · LQGMetric/Papers/CONF/S3D112A.lean, decisions/DEC-112.md · (1) C:1205/1208/1219: the small-diameter event $G^{\cup}$ of L3.3 is the grid-component event fatG (centres of the edge-adjacency components of the full non-removed squares, internal metric of the $\delta r/4$ -neighbourhood of the spine) — CONF’s $G^{\cup}$, (3.12) and the path claim C:1208 are false for pocket configurations (DEC-112 §1(b)) — correction of a false step; (2) C:1183/1396: L3.3’s conclusion is fatG instead of the diameter bound (3.7), and (3.21) follows from fatG , condition 2 and the grid claims CONFChainFull_4 , CONFFullConn with $(2 \cdot 4 + 3)(c/100) < c$ in place of $(c/2) + (c/100) < c$ — replacement of a false step; (3) C:324–338: CONFZBMetric is false for an IsWholePlaneGFF with an independent random constant (Weyl + Fubini); CONFZBMetricN used — statement correction (confirms DV-CONF33W-i); (4) C:1133: $\delta \leq 1/8$ (stated $\delta < 1/8$) instead of $\delta \in (0, 1)$ for the grid claims; CONF L3.2 allows any small δ — parameter range · proposed · no.

DV-CONF-212P (P2-CONF35b, 2026-10-02) · CONF C:798–800 (L2.12(1)) · LQGMetric/Papers/CONF/S3L35B*.lean (confLem2_12aP) · CONF proves L2.12 by the outward LM L3.1 argument or by inversion; we reduce to D47’s mod-constant form (own elementary argument): $h_{\{r_k\}}(\theta)$ depends only on h near $\partial B_{\{r_k\}}$, so an event of $\sigma((h - h_{\{r_k\}}(\theta))|_{\mathbb{A}_{\{S_1 r_k, S_2 r_k\}}})$ is a.s. a mod-constant event of $\mathbb{A}_{\{r_k/2, S_2 r_k\}}$; the overlapping annuli are split into residue classes $k = m + j$ ($S_2^m \geq 4S_2$), D47 is applied with $S_1' = 2$, $S_2' = 4S_2$, $a' = am$, $b' = (1+b)/2$, and a union bound over the m classes gives the rate · proposed · no.

DV-CONF-32H (P2-CONF35b, 2026-10-02) · CONF C:1170–1172 · LQGMetric/Papers/CONF/S3L35B5.lean–S3L35B8.lean · CONF: $\sup_{\{U_{\{\delta/4\}}\}} |\hat{h}^{\cup}|$ is a.s. finite plus scaling invariance of the law; we rescale to $\text{confU}_1 \delta \theta T$ ($\text{isHarmPart_affineComp} + \text{a.s. uniqueness}$) and use DG L2.2’s Gaussian tail with constants depending only on K , U and $\text{Var}[g(\psi_c)]$, which is the same for every normalized whole-plane GFF ($\text{GFFLaw.map_eq_of_normalized_ae}$); the union over T uses the fixed box $\text{sqBox } \delta$ of square indices · proposed · no.

DV-CONF36-1..6 (P2-CONF36, 2026-10-02) · CONF C:1339–1447 · LQGMetric/Papers/CONF/S3L36*.lean · (1) C:1341: grid point $z = \lfloor x/m \rfloor m$ with $m = \epsilon r/4$, so $|x - z| < \epsilon r/2$; (2) C:1393: $\tilde{G}^n$ includes the a.s. connectivity event $\text{conf36Conn}(\mathcal{B}^{\bullet}_{\tau} \text{ connected})$, property A is a sure statement; (3) C:1414: $\tilde{G}^n$ includes $\{\tilde{\rho}^n = \infty\}$ (CONF tacitly assumes $\tilde{\rho}^n < \infty$; property A vacuous there); (4) C:1396 (D108 (b)): (3.21) by a first hit of $\partial \mathcal{B}^{\bullet}_{\tau}$ along an internal-metric path, no length-space assumption; (5) C:1432: exponent $\alpha = \eta \cdot \log(1/(1-p'))$ with $p' = \min(p, 1/2)$, $C_0 = 2/(1-p')$; (6) C:1438: conditioning on $\{\epsilon = e, \dots\}$ needs countably many values of ϵ — open, decision D114 · proposed · (6) pending D114 (since decided and applied: DEC-114, P2-D114a).

DV-D105g-1..3 (P2-DG105g, 2026-10-02) · DG:1189–1278 (L3.11), 1614–1660 (L3.19), 1293–1310, 1699–1706, 1032–1036 · LQGMetric/Papers/DG/S3L11*.lean, S3L19*.lean · (1) Peierls with $p = 32^{\{-25\}}$ ($\theta = 1/32$, independence at ℓ^{∞} -distance > 4 , i.e. disjoint $S(2)$) instead of $p < 8^{\{-100\}}$ with $|P|/100$ disjoint $S(2)$ ’s; for L3.19 the enclosure lives in the site annulus $c_0 + 2 \leq \|z\|_{\infty} \leq n - 2 + c_0$ ($c_0 = \lfloor n/2 \rfloor$); the deterministic step uses only condition 1 of E_S , condition 2 is left to the caller’s events; own glue: a path visits a *-chain of sites*; (2) DG:1300–1305, 1699–1706: *the conditional stochastic domination given $\hat{h}_{\{2^{\{-m-n_m\}}\}}$ is done without regular conditional laws — the coarse factor is discretized into integer levels of $\log T$ capped at $\lfloor \log \epsilon_- \rfloor$* (DG’s monotonicity in ϵ , DG:1260), countable subadditivity plus independence of the level from the fine σ -algebra (prob_level_le); for L3.19 the top level uses the trivial bound $D \geq 1$ (DG do not treat $\epsilon \wedge \{c\sqrt{n}\} > \epsilon_-^*$); (3) $s\mathcal{R}_n + b$ (resp. $s\mathcal{A}_n + b$) is tiled by squares of side $s/16$, not s (so $S(1)$ scales onto $[13/32, 19/32]^2$ inside the box of $\mu_{\hat{h}}$, midpoints in DZZ’s $\mathbb{V}$); DG’s $e^{\{\sqrt{n}\}}$ (L3.2 with $A = c\sqrt{n}$, DG:1226) enters as the factor $e^{\{\mp(d\mp\zeta)\sqrt{n}/2\}}$ of the level plus an exceptional event Z ; vertical rectangles via the reflection swapC (DG:1335 “similarly”) · proposed · no.

DV-D114-1.5 (P2-DEC114, 2026-10-02) · CONF arXiv:1905.00381 C:1312–1316, 1438, 1313, 1167, 1393, 1414; GM l. 854 · decisions/DEC-114.md, Blueprint/CONFResults.lean (CONFLem3_6AtAE0, to be applied by the orchestrator) · (1) C:1312/1438: ϵ countably valued ($(\text{Set.range } \epsilon). \text{Countable}$): CONF’s conditioning on $\{\epsilon = e, \dots\}$ is a countable partition only then — statement form (hypothesis); (2) C:1316, GM l. 854: $\sigma(\mathcal{B}^{\bullet}_{\tau}, h|_{\{\mathcal{B}^{\bullet}_{\tau}\}})$ mod const $\leq m\Omega$ made an explicit hypothesis (tacit in CONF/GM, true for complete P) — statement form; (3) C:1313: $G^{\epsilon}_x \in \sigma(\mathcal{B}^{\bullet}_{\{\sigma^{\epsilon}\}}, h|_{\dots})$ read as a.s. equality to an event of the σ -algebra (AEEEventIn , D30), forced by the a.s. locality axiom — convention; (4) C:1167/1133: $\delta < 1/8$ built into the parameter choice of L3.2 (confCond2_prob returns $\min(b/2)(1/16)$) — parameter range; (5) C:1393/1414: DV-CONF36-2/3 (connectivity event in $\tilde{G}^n$, $\{\rho^n = \infty\} \subseteq \tilde{G}^n$) confirmed faithful · proposed · (6) of DV-CONF36 resolved by (1).

DV-D105k-1..7 (P2-DG105k, 2026-10-02) · DG:1782, 1784–1785, 1777–1779, 1515–1521, 1527–1532, 1567–1589,

1729–1731 · LQGMetric/Papers/DG/S3T15.lean, S3P18*.lean, S3P17*.lean, S3P15.lean · (1) DG:1782 ((1.5a) lower): $U = B(z, |w-z|)$, so $w \in \partial U$ and $D(z,w) \geq D(\{z\}, \partial U)$ without cutting the path (DG: any open $U \ni z, w \notin U$); (2) DG:1784–1785 ((1.5b) lower half “immediate”): via (1.5a) at two distinct points of K and a polygonal path in the connected U (own elementary, t15_exists_isDGPath); (3) DG:1777–1779 (covering K by squares with $S(1/2) \subseteq U$): own covering/chaining argument (squares $t18Sq \times s$ with $B(x, 4s) \subseteq U$ around a finite subcover and along hop chains; clopen argument in the connected U ; triangle inequality by concatenation); (4) DG:1515–1521: $D^\delta(K, \partial U)$ for $h^{\{S(1)\}}$ read as the infimum over paths in $\bar{U}$ from K to ∂U (p17SetDist), $\leq$ DG’s $D(K, \partial U)$ for a continuous whole-plane field (first exit, p17_exit); P3.17 stated for the coupling of Blueprint.DGProp3_16 (the one DG’s proof uses, DG:1527); (5) DG:1527–1532: (eqn-lfpp-lower-show) with $\delta = \varepsilon^\beta$ directly and pointwise; error term via L3.5 at $\zeta_3 = (Q-2)/4$ and $\zeta_2 = \min(\zeta/3, \xi(Q-2)/4)$; (6) DG:1567–1589: $D^\varepsilon(S_0, S_k; S(1/2)) \geq D^\varepsilon(K', \partial U')$ used pointwise (hypothesis hLb); the chain of squares (DG (3.32), implicit) via the grid graph (p17_exists_chain); (7) DG:1729–1731: L3.7 at $\mathbb{V}$ -scale with $z = w$ and polynomial rate (DGLem37V, hypothesis); L3.8 for $\mu_{\hat{h}}$ on $[1/6, 5/6]^2$ and $[1/3, 2/3]^2$ from the ball versions by a fixed cover of 441 balls of radius $1/30$ · proposed · no.

DV-DDDF28L-1.4 (P2-DDDF28L, 2026-10-02) · DDDF arXiv:1904.08021 tightness.tex l. 1455–1472, 1481, 1648 · LQGMetric/Papers/DDDF/S6P28L1*.lean · (1) DDDF’s annuli $2^{\{k\}} \leq |x-x'| \leq 2^{\{k+1\}}$ with crossings of $R_i^{\wedge} S(P)$, $P \in \mathcal{P}_{\{k+2\}}$, become crossing scale $h = 2^{\{K\}}$ with $4h \leq |x-x'| \leq 8h$, $1 \leq K < n$; the left tail needs $h \geq 2\delta$, so pairs with $\delta < |x-x'| \leq 16\delta$ are handled by the Step 2 argument (lower_mid), a range DDDF’s dyadic version does not have; (2) l. 1459: $\varphi_{\{0,n\}} = \varphi_{\{0,k\}} + \varphi_{\{k,n\}}$ replaced by $\varphi_\delta = \varphi_{\{\delta,h\}} + \varphi_{\{h,1\}}$; the coarse field is bounded on $[0,1]^2$ only (each crossing piece lies in $[0,1]^2$); “and scaling” (l. 1466) is law_mrect_phiVer; (3) l. 1462 (BoundHolderLow) becomes $\lambda_\delta \leq e^{\{2C_0\}} e^{\{C\sqrt{K}\}} \lambda_{\{\delta/h\}} \lambda_K$ via (6.98) twice and (5.76), $\lambda_K \leq C' 2^{\{K(1-\xi Q-\zeta)\}}$ from (5.78); the per-scale bound $C e^{\{ck\}} e^{\{cs\}}$ (l. 1470) realized as $2\eta r^K$ with explicit summable constants; (4) $1 \leq \alpha$ is a hypothesis (DDDF derive it at l. 1481) · proposed · no.

DV-P32G1.3, DV-P317a (P2-DZZ32G, 2026-10-02) · DZZ arXiv:1807.00422 l. 1149–1151, 1150, 1521, 1526–1530, 1627–1630, 1647 · LQGMetric/Papers/DZZ/S3P32G*.lean, S3P317*.lean · (G1) l. 1149–1151: the start path at μ_{In} is not DZZ’s horizontal segment through x but an escape chain from x to its push-in into $[h, 1-h]^2$ followed by DZZ’s row to $\partial B_{\text{large}}$ (own elementary, as DV-ETA2d); (G2) l. 1150 “each ball is covered by at most 4 boxes” proved directly (four_box, own elementary); (G3) statement correction of D102’s P-4bW cover: it needs $r \leq s_B/2$ and is false without it (not_p32StartGeom), see D115; (317a) Prop 3.17 reduced to the concentration of D' plus the crude moments (l. 1526–1530, 1627–1630); the (eq-concentration-approximate-2) rate taken as $e^{\{-c_2 L^{\{0.8\}}\}}$, reading l. 1627’s $e^{\{(\log \delta^{-1})^{\{0.8\}}\}}$ as a typo per l. 1647; DZZCrudeMoments asks the moments for admissible sets where DZZ state them for points (“the general case follows by the same proof”, l. 1521) · proposed · no.

DV-DGTRINV-1 (P2-DGTRINV, 2026-10-02) · DG:1243, 944, 953 · LQGMetric/Papers/DG/S3TrInv1–5.lean · “agrees in law” for $\mu_{\{\hat{h}^{\text{tr}}\}}/\mu_{\hat{h}}$ and their LGD events is proved as: every event of the rational ball masses has the same probability for every white noise (own elementary glue: rational-ball reduction of D^ε by compactness of the path; pathwise identification of μ_0^{fMod} ball masses with a measurable functional of (noise, Y on a dense sequence)); geometric translation enters through the noise $W \circ U_{\{\delta, c\}}$ (dgTr_translate, dgTr_wnScale), not through a vague-limit (N4) characterization of $\mu_{\{\hat{h}^{\text{tr}}\}}$ · proposed · no.

DV-D105l-1.2 (P2-DG105l, 2026-10-02) · DG:1243, 1267, 1302–1305 · LQGMetric/Papers/DG/S3L11In1–6.lean · (1) DG:1243 “the re-centered field agrees in law with $\hat{h}^{\text{tr}}$ ” formalized as the equality in law of all white noises (map_wn_eq); the tails of the chosen moduli $\text{hatMod}/\text{trMod}$ of every rescaled white noise are bounded with the constants for W (prob_sup_eq_of_circ: sup over a countable dense set, limits of circle averages), own elementary glue; (2) DG:1267, 1302–1305 “a.s. determined by W on ...” used through a.s.-equal versions (the sets a.s. equal to local events form a σ -algebra); T is a.s. a function of W on $[s^2, \infty) \times \mathbb{C}$ via the sup over a dense countable set of the continuous version phiVer, own elementary glue; the $s/16$ tiling of DV-D105g-3 is corrected by D116 · proposed · no.

DV-DZZ54-1 (P2-DZZ54, 2026-10-02) · DZZ l. 2299–2304, 2570–2577 · LQGMetric/Papers/DZZ/S5L54A–E.lean · the upper half of L5.4 (not written out by DZZ) by comparison with $\tilde{D}_\delta(u, u \pm 1/40)$ (the point lies in $\mathbb{V}$ and on $\partial \mathbb{V}_{\{u, 1/20\}}$; its tilde box sits inside $\mathbb{V}_{\{u, 1/10\}}$); the lower half follows DZZ l. 2570–2577 (Prop 3.17 at the walled measure of $\mathbb{V}_{\{u, 1/10\}}$, DZZ Remark 5.2, plus a union bound over the $4n(\delta)$ boundary segments) · proposed · no.

DV-DZZ61L-a..d (P2-DZZ61L, 2026-10-02) · DZZ l. 2553–2578, 2593–2608 · LQGMetric/Papers/DZZ/S6L61A–D.lean · (a) L6.1 “any segment L_δ ” read as: for every family $\delta \mapsto L_\delta$, the bound holds for all small δ ; (b) formal side hypotheses DZZ use without comment: probability measure, a.e. measurability of ball masses ($h\mu$), a.s. finiteness $D_\delta < \infty$ (hfin); (c) (eq-point-to-boundary-kappa), stated by DZZ for a fixed v , is used at $v_{\{L_\delta\}}$ moving with δ , so DZZL61PtBdry asks uniformity in the centre (implicit in DZZ; checked by D117); (d) own elementary glue for steps implicit in DZZ: worst/best-segment selection at each δ , completing families with two fixed points, the junk value of toNat · proposed · no.

DV-DZZ53b-a..e (P2-DZZ53b, 2026-10-02) · DZZ l. 2255–2430, 611–645, 1218–1270, 852–857 · LQGMetric/Papers/DZZ/S5L53B1–B9.lean · (a) l. 2397–2404: the bad-event bound $e^{\{-L^{\{0.22\}}\}} + e^{\{-cL^{\{0.51\}}\}}$ stated as $2e^{\{-L^{\{0.22\}}\}}$ for large k (implied by DZZ’s bound); (b) l. 2411: instead of DZZ’s Jensen bound on $E(1_{\{\mathcal{D}^c\}} \log \tilde{D})$

we use $x \leq x^2/M + M \cdot 1_{\mathcal{D}^c}$ with the second moment (eq-very-crude) for $\tilde{D}$ (lintegral_sq_log_tilde_le); (c) l. 2318–2322, 2266–2270: DZZ L2.9 applied to rotated boxes via a coupling for complex scalings a (DZZ state 2.9 for real scalings plus translations); (d) l. 1218–1233 with Remark 5.2: L3.8 used in walled form, comparison assumed only on $\mathbb{V} \cap K$; (e) l. 1266–1269, 2423: passing from high-probability comparison to exponents (tendsto_sub_div_log_of_close) and law invariance (integral_logTilde_eq) are own arguments (DZZ: Cauchy–Schwarz, implicit) · proposed · no.

DV-DG37V-1.2 (P2-DG37V, 2026-10-02) · DG:1010, 1096–1102, 1722–1731 · LQGMetric/Papers/DG/S3L37V1–2.lean · (1) the white-noise scaling by $1/3$ (wnScale with $\delta = 3$, $b = -(1+i)$, l37W) and the identity $\hat{h}_\delta[W] = \hat{h}_{\{3\delta\}}[W] \circ T^{-1} + \varphi_{\{1,3\}}[W] \circ T^{-1}$, own elementary glue as DV-D105-2; (2) DG use L3.7 on $\mathbb{S}(1/2)$ (DG:1729) although they state it on $\mathbb{S}$ (DG:1100): pending decision D118 · proposed · (2) pending D118 (since decided: DV-D118-1.3).

DV-D105m-1.2 (P2-DG105m, 2026-10-02) · DG:1593–1595, 1774–1777 · LQGMetric/Papers/DG/S3P18T1–5.lean · (1) the LFPP events are measurable in countably many rational circle averages (by chains for squares, t18_event_iff; by grid step functions plus the multiplicative comparison T18Cmp in general, t18_repr), a step DG use without comment; (2) the law of the rational coordinates is determined by IsGFFCircleAverage as a Gaussian process (t18T_measure_eq, via KilledHeat.map_eq_of_gaussian) · proposed · no.

DV-CONFCKG-1.3 (P2-CONFCKG, 2026-10-02) · CONF C:656–669, 712–752, 1236–1243 · LQGMetric/Papers/CONF/S3D108K1–K5.lean · (1) the zero-boundary field is $\mathcal{D}'(\mathbb{C})$ -valued with its extension-by-0 law fixed (IsZBExtField); perturbations are in $C(\mathbb{C}, \mathbb{R})$ (CONF: continuous on U) and the monotonicity/continuity hypotheses are required a.s. at the sample (CONF: for every distribution), because the Weyl axiom is stated for $C(\mathbb{C}, \mathbb{R})$ and holds a.s.; CONFlem2_10 is therefore slightly stronger than CONF’s L2.10 (fine for the regular domains used); (2) the S-cont-law, asserted by CONF in one sentence (C:667–669), written out (measure_level_eq_zero: scaling along a bump, Fubini in the shift parameter, absolute continuity; own elementary measure theory); (3) the conditional law given the outside field modelled as the product law $P_W \otimes P_X$, with the a.s. axioms moved onto it through the product-space field (ae_ae_weylAt, ae_ae_isLength) · proposed · no.

DV-D118-1.3 (P2-DEC118, 2026-10-02) · DG arXiv:1807.01072 L3.7 DG:1096–1105, L3.4 DG:1044–1062, P3.22 DG:1729–1731, DG:1165; DGW arXiv:1610.09998 Prop 3.3 (DG’s ”Prop 3.2”) DGW:549–554, $\mathcal{U}_\varepsilon$ DGW:163, DS reading DGW:190–192, smooth boundary DGW:161 · decisions/DEC-118.md, Blueprint/DFGPSInputsDG.lean (sqHalf, IsCircleAvgVersionSq, DGLem3_7; applied by the orchestrator) · (1) DG:1100 vs DG:1729: Blueprint. DGLem3_7 is stated on $\mathbb{S}(1/2) = [-1/2, 3/2]^2$ instead of DG’s $\mathbb{S} = [0, 1]^2$ — the statement of a cited result in the form DG use, justified by DG’s own proof (DGW Prop 3.3 holds for every $V \subseteq \mathcal{U}_\varepsilon$, here $\mathcal{U} = (-1, 2)^2$, $\varepsilon = 1/4$; DG L3.4 for every bounded domain); (2) the version/continuity clause of IsDGCoupling is on $\mathbb{S}(1/2)$ for $\delta \in (0, 1/2)$ (DS Prop 3.1 on $\{(v, \delta) : \delta < d(v, \partial \mathcal{U})\}$) in place of $\mathbb{S}$, $\delta \in (0, 1)$ — statement of a cited result (the range $\delta \in [1/2, 1)$ is dropped, unused); (3) DGW assume a smooth boundary (DGW:161), DG apply Prop 3.2 to the square $\mathbb{S}(1)$ (DG:1103–1104); kept (the proof uses only the Brownian-bridge exit bound) — reading of the paper’s text; DGProp3_16 unchanged (on $\mathbb{S}$, DG:1432–1437, P3.17 at DG:1524 uses it there only) · proposed · resolves DV-DG37V-2.

DV-D105g-3 (corrected, D116) (P2-DG105p, 2026-10-02) · DG:1269–1276 · LQGMetric/Papers/DG/S3L11*.lean, S3L19*.lean · squares of side $s/32$ (grid side $2^{\wedge}\{j\}/32$ on $32n$ cells, constant $2048 n^2$), unit-frame square $[29/64, 35/64]^2$ (side midpoints at distance $1/64 \leq 1/40$ from the centre, inside l312Vbar), PercFar 9, Peierls $p = 32^{\wedge}\{-100\}$ ($r = 9$, $\theta = 1/32$), matching DG’s $|P|/100$ and $p < 8^{\wedge}\{-100\}$; supersedes the $s/16$ tiling of DV-D105g-3 · proposed · no.

DV-D105p-1 (P2-DG105p, 2026-10-02) · DG:1226–1248 · LQGMetric/Papers/DG/S3L11R2–R3.lean · the exceptional event Z is the union over the $64n \times 32n$ index box of the comparison events of the rescaled noises at $\gamma A_n = (d - \zeta)\sqrt{n/2}$; the fine events are empty off the grid (own elementary glue) · proposed · no.

DV-DGLB-1 (P2-DGLB, 2026-10-02) · DG:1588–1589, 1393–1397 · LQGMetric/Papers/DG/S3P17L1.lean · the Step 3 lower bound $D^{\wedge}\varepsilon_{\hat{h}}(K', \partial U') \geq \varepsilon^{\wedge}\{-1/(d + \zeta)\}$ with probability $1 - O(\varepsilon^{\wedge}p)$ is obtained from DG L3.21 (annuli of all grid squares of side $\approx \varepsilon^{\wedge}\beta$, polynomial rate) plus L3.5 with small β , following the structure of DG’s lower bound of Thm 1.3 (DG:1393–1397); DG instead use DZZ L6.1 at one square transported by scaling and L3.2 — the library’s fixed-square L3.20 has no rate; stated pointwise for $y \in A$, $|y' - y| \geq a$ · proposed · no.

DV-DDDF28U-1.4 (P2-DDDF28U, 2026-10-02) · DDDF tightness.tex l. 1340–1374, 1414–1446, 1606, 1648 · LQGMetric/Papers/DDDF/S6P28U1*.lean · (1) l. 1356–1362: the maximum over the crossings at each scale is bounded by their ℓ^q norm with q -th moments from the right tail (6.102), instead of DDDF’s union bound over the tails (same deviation as S6DiamLvl); (2) l. 1414–1418: the chaining at scale K passes through a non-dyadic square of side $2^{\wedge}\{1-K\}$ with corner on the $2^{\wedge}\{-K\}$ grid (DDDF/DF: long rectangles $2^{\wedge}\{-k\}(3,1)$, constants untracked); (3) l. 1416: the last link and all pairs with $|x - x'| \leq 64\delta$ bounded w.h.p. via the field maximum ($\phi_{\text{Ver_sup_tail_unif}}$, (5.54), (6.98)), as in Step 2, instead of DDDF’s $E[2^{\wedge}\{-n\} e^{\wedge}\{\xi \sup \varphi_{[0,n]}\}]$; pairs with $|x - x'| > 1/4$ cut into 4 pieces; (4) l. 1606, 1648: the δ -family handled directly ($\varphi_\delta = \varphi_{\{0,K\}} + \varphi_{\{\delta, 2^{\wedge}\{-K\}\}}$, scaling to $\varphi_{\{2^{\wedge}K \delta\}}$, $\delta = 2^{\wedge}\{-(N+r)\}$, $r \in (0,1]$) instead of rescaling the dyadic results to $[0, 2^{\wedge}r]^2$; own elementary helpers moment_of_tail, tail_sum, rpow_ratio, four_pieces, exists_split' · proposed · no.

DV-D117-1.5 (P2-DEC117, 2026-10-02; decision D117) · DZZ l. 2271, 2318–2336, 2559–2567, 2605, 619 · LQGMetric/Papers/DZZ/S5D117.lean and the D117 packets · (1) l. 2559/2567, 2605: segments of length $\delta^{\wedge}\{2\iota\}$ with crossing cost $\delta^{\wedge}\{-\chi+\iota\}$ are valid only for $\chi \geq 1/2$; we use $\delta^{\wedge}\{K\iota\}$, $K\chi \geq 2$ (L5.4) and $\delta^{\wedge}\kappa$, $\kappa\chi \geq 2\iota$ (L6.1) — reading of the text, lemmas unchanged; (2) l. 2563: the leg from u to L_{δ} is taken inside $\text{legWall } u \wedge L \subseteq \mathbb{V}_{\sim}\{u, v\}$, not inside $\mathbb{V}_{\sim}\{u, 2\lambda\}$, the union bound repaired on DZZBigBalls (the published inequality is not justified as written); (3) l. 2271 (eq-translation-invariant) exact in our model only for the symmetries of $\mathbb{V}$ (DZZDihedralLaw); other isometries and similarities via lem-scaling-coupling (DZZSimCoupleU), as DZZ do at l. 2318–2336, 2566; (4) (eq-point-to-boundary-kappa) used uniformly in the centre v (DZZL61PtBdry), which follows from l. 619 ($C = C(\xi, \kappa_1, \kappa_2)$); (5) B8’s Cor 3.9 step for tilde distances replaced by antitonicity of lgdDZZ in δ and the existence of the per-pair limits (own elementary step) · proposed · no.

DV-CONF33G-1.3 (P2-CONF33G, 2026-10-02) · CONF C:1187–1190, 1200–1205, 1213, 1236–1243 · LQGMetric/Papers/CONF/S3D112L1–L4.lean · (1) $W_C \subseteq U_{\{\delta r/4\}}$ (confFatW_subset_innerPart) is own elementary geometry (DEC-112 §2): the open $\delta r/2$ -ball around a spine point avoids every removed square and lies in $\mathbb{A}_{\{3r, 4r\}}(z)$; (2) $G^{\wedge}U$ is the event $\{X \in \text{zbDiamSet}\}$ built from the zero-boundary metrics $F_C(\hat{h}^{\wedge}U|_{\sim}\{W_C\})$ of CONF Remark 1.2 (confZBMetricN), a function of $\hat{h}^{\wedge}U$ alone, so condFKG_freeze applies; CONF Prop 2.8 enters as the frozen hypothesis CONFFKGFrozenEU; (3) the zero-boundary part is the Markov version $X = (h - h_{\rho}(w)) - G$ (exists_isL33ZBPart_G), its law near ∂U pinned by the open node CONFZBExt0fMarkov — statement correction: IsL33ZBPart alone is too weak for FKG · proposed · no.

DV-CONF210-1.3 (P2-CONF210, 2026-10-02) · CONF C:667–669, 685–708, 717, 740–741, 1229 · LQGMetric/Papers/CONF/S3D108L1–L3.lean · (1) CONFZBShiftAC proved only for $\varphi \in \mathcal{D}(C)$ with $\text{supp } \varphi \subseteq U$, U bounded — the Cameron–Martin space, where the statement is true (it is false for general continuous φ); consumer confProp2_8_diam_test; (2) the limit in L2.10 through an increasing coarse filtration and Lévy’s upward theorem instead of the Kolmogorov 0-1 law and backward martingales; the freezing formula condExp_comp_indep is own elementary measure theory; (3) L2.9 used with a.s. continuity along measurable random sequences (AeSeqContAt), CONF’s own hypothesis form (C:717), proof as CONF C:685–708 · proposed · no.

DV-DZZCONC-a.e (P2-DZZCONC, 2026-10-02) · DZZ l. 326–348, 1538–1651 · LQGMetric/Papers/DZZ/S3ConcA–H.lean · (a) finite-net reduction in gauss_far_le (DZZ apply Lemma 2.1 to the continuum field directly), own elementary; (b) mean vs median (abs_integral_sub_median_le), own proof of l. 1607–1611; (c) a median replaces DZZ’s quantile d' ($P(\mathcal{A}') \geq 1/4$ instead of $1/2$); (d) the coarse field restricted to the η component (DZZ also include $\tilde{h}$; D' and L3.8 use only η) and to rational index points (white-noise values are only a.s. defined); (e) Part 2 constants $\ell = L^{0.9}$, $\tau = L^{0.93}$, $M = L^{1.1}$, following l. 1626 · proposed · no.

DV-D105q-1.2 (P2-DG105q, 2026-10-02) · DG:1774–1777, L2.2 · LQGMetric/Papers/DG/S3P18R1–R4.lean · (1) the law transfer between $\hat{h}^{\wedge}\{S(1)\}$ and the zero-boundary part of L2.2 through countably many circle averages (factoring through $T|_U$, uniqueness of the zero-boundary law, clamping onto the square plus $t18_repr$), own routine glue; (2) for $\delta < 1/9$ the circle averages of the harmonic part equal its values (mean value property); harmonic part bounded by $A = (\zeta/2\xi) \log \delta^{-1}$ with tail $a_0 e^{\{-a_1 A^2\}} \leq a_0 \delta$ for $\delta < \exp(-1/(a_1 \kappa^2))$, $\kappa = \zeta/(2\xi)$ · proposed · no.

DV-DGLOC-1 (P2-DGLOC, 2026-10-02) · DG:984, 1268, 1302–1305 · LQGMetric/Papers/DG/S3L11Loc1–5.lean · ”E_S is a.s. determined by the white noise on $(s\text{-neighbourhood of } S) \times (0, s^2]$ ” proved as: $\{\text{goodSq}(\mu \text{Tr } W')\}$ is a.s. equal to a $\sigma(W|_{\sim}\{(0, s^2) \times (S \cdot B_{\{1/10\}}(S(1)) + c)\})$ -measurable event — the rational-ball masses in $S(1)$ are a.s. limits of local approximations of $\mu_{\{\hat{h}^{\wedge} \text{tr}\}}$ (DG:984 restated for $\hat{h}^{\wedge} \text{tr}$ on $\text{int } S(1)$) built from local jointly measurable versions of the circle averages of $\hat{h}^{\wedge} \text{tr}$; the slice $\{s^2\} \times C$ is null · proposed · no.

DV-D114-C4 (P2-D114a, 2026-10-02; decision D114) · CONF C:1308–1320, 1425–1432; GM l. 854 · LQGMetric/Blueprint/CONFResults.lean, Papers/CONF/S3D110B–C.lean, Papers/GM/S4/P412iConf.lean, P412nStep.lean, P412nUnion.lean, P412pMain.lean, Papers/CONF/S3D114W.lean · CONFLem3_6AtAE0 per D114 (hypothesises localSigma0 $h \mathcal{B}^{\wedge}_{\sim} \tau \leq m\Omega$, $(\text{Set.range } \varepsilon).Countable$; conclusion AEEEventIn P (filledBallSigmaAt0 ...) G); in GM §4 the CONF L3.6 events G_j are a.s. events and $A_k \in \mathcal{F}(k+1)$ uses that the GM filtration is built from completed σ -algebras (p412i_filt_congr), so the GM statements are unchanged — a convention, no change of statement; parameter chain $\delta = \min(b/2)(1/16) < 1/8$ (confCond2_prob, confLem3_5At_delta) · proposed · no.

DV-T39b-1.4 (P2-CONF39b, 2026-10-02) · CONF C:1463–1471, 1556, 1570–1617 · LQGMetric/Papers/CONF/S3T39H1–H5.lean · (1) L3.7 stated with CONF’s point x (C:1467) as an explicit measurable input, one event G serving every arc disconnected by $B_{\rho}(x)$; (2) the disconnecting ball has radius $\rho < \varepsilon r$ (from a set of diameter $< \varepsilon r$, exists_frontier_ball_disconnects; CONF: $\text{diam} \leq \varepsilon r$, open ball $B_{\{\varepsilon r\}}(x)$); (3) in L3.7 A the a.s. facts CONF uses tacitly at C:1469–1470 ($\tau > 0$, geodesics from z to every point, bounded balls) are premises at ω ; (4) in T3.9, L3.7 applied with $\varepsilon'_k = \varepsilon_k$ on $\{t39gExp \ n_k \geq 1\}$ and $1/2$ elsewhere (CONF tacitly has $n_k \geq 16$, C:1556); the filtration is $\mathcal{F}_k$ plus the P-null events (conditional expectations unchanged a.s.) · proposed · no.

DV-D119-S5, DV-D119-S6 (P2-CONF39c, 2026-10-02; decision D119) · CONF C:832–894, 912–924, 1514–1744 · LQGMetric/Papers/CONF/S3T39I1–I5.lean · (S5) in the iteration of CONF Thm 3.9 the recursion $s_{\{k+1\}} = \sigma^{\wedge}\{\varepsilon_k\}_{\{s_k\}}$, (3.21') and the kill step are required almost surely, since $\sigma = \infty$ on a null set (definition change,

D119); (S6) CONF L2.14 stated with the location of X (within $C^{n^{1/2}}$ of ∂U), as CONF's proof C:862–870 provides (confL214Loc; reading of the proof, strengthens the printed statement) · proposed · no.

DV-DG317S-1..5 (P2-DG317S, 2026-10-02) · DG:1524, 1547, 1550–1565, 1586–1591 · LQGMetric/Papers/DG/S3P17S1–S5.lean · (1) DG:1547 level of L3.13 read as $m = m_\delta + 1$ (DG's " $m = \lfloor \log_2 \varepsilon^{-\beta} \rfloor - 1$ " looks like a typo), so the rectangles are $\delta_\varepsilon \times \delta_\varepsilon/2$ with corners in $(\delta_\varepsilon/2)\mathbb{Z}^2$; (2) DG:1561–1565: Y_S uses six of DG's ≤ 12 rectangles (three horizontal, three vertical), connectivity via a hub point, constant 6 instead of 12, adjacency via P2-DG105j's P39H.meet (own elementary); (3) DG:1550–1555: the m^3 term of (eqn-use-rectangle-dist) kept as a summand $m^3 \delta^{\xi(2+\eta')}$ and beaten by $\delta^{\xi/2}$ in the final algebra (DG: "the second term dominates"); (4) DG:1524, 1586–1591: $\beta = 1/(2+\gamma)^2$ (DG's example), $\varepsilon = \delta^{\xi(2+\gamma)^2}$, the small parameters explicit ($\eta = \xi/(16(2+\gamma)^2)$, $\eta' = \eta/(\xi+1)$); (5) DG:1587–1589: $K' = \text{cthickening}(r_0/3) K$, $U' = \text{thickening}(2r_0/3) K$, lower bound used pointwise (DGP317Lb, as DV-D105k-6) · proposed · no.

DV-DZZGLUE-1 (P2-DZZGLUE, 2026-10-02) · DZZ l. 2566, 2605; Maehara, Amer. Math. Monthly 91 (1984), Lemma 1 · LQGMetric/Papers/DZZ/S5Top1–3.lean · the planar gluing facts DZZ use without proof (perpendicular crossings of a rectangle intersect; four crossings of the strips of a rectangular annulus separate inside from outside and glue two crossing sets) proved from the project's loop-degree rectangle-crossing lemma (RectMeet.rect_crossings_meet, QuantumZipper), own elementary; the ring lemma statement is ours (S5D117 had none) · proposed · no.

DV-DZZCM-a..b (P2-DZZCM, 2026-10-02) · DZZ l. 744–745, 836–841, 849–857, 1521 · LQGMetric/Papers/DZZ/S3CM1–8.lean · (a) definition change (D122): the crude moments in Prop 3.17 are required only for $\delta < \delta_1$ (DZZCrudeMomentsEv replaces DZZCrudeMoments, false as $\delta \rightarrow 1$; DZZ's $O(1)$ is a small- δ statement); (b) own elementary proofs: $D' \leq (k+1)4^k$ when every level- k box has mass $< \delta^2$ (a geodesic visits distinct cells, omitted by DZZ); measurability of $D'(A,B)$ for arbitrary sets via chains of cells; DG L3.8 on $[\xi, 1-\xi]^2$ from finitely many balls by compactness · proposed · no.

DV-D105s-1..2 (P2-DG105s, 2026-10-02) · DG:1593–1595, 1774–1777 · LQGMetric/Papers/DG/S3P18S1–S6.lean · (1) DG's "by scaling" (DG:1774–1775, 1593–1595): the affine map $y \mapsto 2r \cdot y + c_0$ and the rescaled field $h(A) - h_{\{2r\}}(c_0)$; the single Gaussian $h_{\{2r\}}(c_0)$ is bounded by $(\zeta/4\xi) \log \delta^{-1}$ outside an event of probability $\leq 2\delta$; the constants $(2r)^{\{1-e\}}$ absorbed for small δ (own routine glue); (2) the lower event moved to the zero-boundary part of L2.2 as in DV-D105q-1/2, using the reversed comparison $D_{\{H-\mathfrak{h}\}} \leq e^{\xi A} D_H$, so the measurability of the harmonic part $\mathfrak{h}$ is never needed · proposed · no.

DV-D105r-1..3 (P2-DG105r, 2026-10-02) · DG:1335, 1638–1660, 1699–1706, 1032–1036 · LQGMetric/Papers/DG/S3L11R4–R6.lean, S3L19R1–R6.lean · (1) DG:1335 "similarly": the vertical inputs use the horizontal events at the reflected sites $x.\text{swap}$ with offset $b + (s/32)(1-i)$, the reflection acting only on the LGD (goodSq_map_swapC), own elementary; (2) DG:1642–1643 condition 2 of E_S in grid form with own constants (grid balls $B(w, 1/400)$ with $|w-u| \leq 1/28+3/560+1/400$ have mass $> \varepsilon$; the unit-frame " $S(3/4)$ " is $\bar{B}(u, 13/280)$), implying DG's condition 2 for balls meeting ∇_u ; condition 1 with balls restricted to $\bar{B}(u, 13/280)$ ($\geq$ DG's unrestricted distance, so DG L3.20 applies, and the event is local); DG's crossing of $\partial S(1/2)$ replaced by the smaller box ∇_u of dg_lemma320 by crossing monotonicity (l319_cross_mono); (3) the transfer from the unit frame (scaling (3.7) on $T(\bar{B}(u, 13/280))$, L3.2 lower comparison with $\gamma A = (d+\zeta)\sqrt{n}/2$, cutting the path at its first crossing) is own elementary glue · proposed · no.

DV-D120-1..5 (P2-DEC120, 2026-10-02; decision D120) · CONF C:835, 912–924, 1302, 1740–1744; GM l. 2189 · CONF Thm 3.9 files (S3L214P, S3T39J), LQGMetric/Blueprint/StoppingAE.lean · (1) C:1740–1744: the arc families $\mathcal{J}_0$ are the canonical measurable subdivision (equal-diameter refinement by measurable point selection), not arcs of harmonic measure $1/n_0$ from ∞ ; the proof uses only covering and eventual separation (definition change, harmless); (2) C:835 "prime ends" read as boundary points of the Jordan curve $\partial \mathcal{B}^\bullet_s$ (proved, D17/J1); L2.14 applied via its located form to $\Omega(K)$ (convention, exact for Jordan domains); (3) C:1302, GM l. 2189: $\sigma^\varepsilon_{\{\tau, r\}}$ and the s_k are stopping times up to null sets (IsFilledBallStoppingTimeAE); L3.6/L3.7 take this hypothesis (statement form forced by the a.s. Axiom II; D98 (b), D114 (g)); the definition lives in the new Blueprint file StoppingAE.lean rather than in M2Defs (content verbatim, avoids rebuilding M2Defs' dependents); (4) CONF's probability space is tacitly complete: CONFThm3_9RestC carries $[P.\text{IsComplete}]$ and CONFThm3_9At for general P is obtained by completion (D70) (convention); (5) C:912–924: L2.15 in the Euclidean DisconnectsFromInfy form with the side condition $C^{n^{1/2}} \leq \text{outrad}$ and the universal N_0 of the located L2.14 (D36/D119) (statement form) · proposed · no.

DV-DZZDB-1..2 (P2-DZZDB, 2026-10-02; D117 P-DIH, P-BIG) · DZZ l. 2271, 2340–2345, 2555, 2562–2563 · LQGMetric/Papers/DZZ/S5D117D1–F2.lean · (1) DZZ's "by symmetry" (exact invariance in law under the symmetries of $\mathbb{V}$, dzzDihedralLaw_dzzMuIn) proved through the white-noise isometry $f \mapsto f(\cdot, \sigma \cdot)$, invariance of the Brownian-bridge law under signed coordinate permutations, the continuous versions of $\bar{h}$, and the chaos-limit characterisation ae_isChaosLimit_wickQArea (own elementary; DV-D117-3); (2) big balls (dzzBigBalls_dzzMuIn) via a finite rational net, a.s. positivity of rational ball masses (DZZ L2.10) and continuity from above; the truncation is stated for walls containing sqBox $u(\lambda+4r)$ (lqdMinSet_cover_le_of_big) because DEC-117's legWall truncation does not follow from the first-exit argument (decision D123 pending; since decided: DV-D123-1..2) · proposed · no.

DV-DZZSIM-1 (P2-DZZSIM, 2026-10-02; D117 P-SIM) · DZZ l. 611–624, 1218–1233, 2538–2548 · LQGMetric/Papers/DZZ/S5D117A–C2.lean · DZZ lem-scaling-coupling composed with L3.8 (DZZSimCoupleU) proved for $\|a\| = 2^{-m}$ (all translations and rotations included) with the constant of L2.9 for similarities uniform in the translation (dzz_lemma29_simU, $\exists C$ before b); general $\|a\| \leq 1$ reduced to the chaos limit along $c \cdot 2^{-n}$ (DZZWickChaosAlong, DZZ l. 1209–1213) and DZZ L2.7 along $c \cdot 2^{-j}$ (DZZLemma27Along), which DZZ use without comment (Enterprise P2-DZZSIM2) · proposed · no.

DV-DZZ61K-a.b (P2-DZZ61K, 2026-10-02; D117 P-61K, P-54U) · DZZ l. 2340–2345, 2562, 2571–2578, 2593–2608 · LQGMetric/Papers/DZZ/S6L61B/D/E.lean, S5L54B.lean, S5L54F1–F3.lean · (a) L5.4 union bound with deviation $\iota' = (2K/c)^{1/2} \iota^{1/2}$ (DZZ's $C\iota^{1/2}$) and Prop 3.17 at the leg walls $\text{legWall } u \lambda L$ with a constant uniform in L (DZZProp317Leg), where DZZ use the single wall $\mathbb{V}_{\{u,2\lambda\}}$ (consequence of DV-D117-2; D123 pending; since decided: DV-D123-1..2); L6.1 segments of length δ^κ , $\kappa = 2\iota/\chi$ (DV-D117-1); (b) the truncation lemma dzzLegTrunc_of_mem is own elementary (to be re-checked under D123, cf. DV-DZZDB-1 (2)); exists_Icc_piece in S5L54B renamed l54_exists_Icc_piece (name clash with S6L61A) · proposed · no.

DV-DG317V-1 (P2-DG317V, 2026-10-02; D121) · DG:1010, 1523–1591 · LQGMetric/Papers/DG/S3P17V1–V5.lean, S3P17L1.lean (rename DGP317Lb $\rightarrow$ DGP317LbReg, name clash with S3P17S4) · P3.17's Steps run at $\mathbb{V}$ -scale via the dyadic map $T(z) = z/2 + (1+i)/4$ and the white noise $W \circ U_{\{2, -(1+i)/2\}}$ instead of D121's $(z+1+i)/3$ (grid compatibility of L3.13); transport at the level of the inputs of Steps 1/3; the m^3 term of L3.13 at level $m+1$ dominated for small δ (p17v_cub); the smooth window $\varphi_{\{1,2\}}$ absorbed in the field control (own elementary glue; scale covariance of white noise, DG:1010) · proposed · no.

DV-317K-1..2 (P2-DZZ317K, 2026-10-02; D117 P-317K) · DZZ Remark 5.2 (l. 2281–2284), l. 1088–1169, 1526–1530, 1627–1630 · LQGMetric/Papers/DZZ/S3P32K1–K5.lean, S3P317E.lean · (1) the walled P3.2/P3.17 stated only for pairs at distance $\geq \xi$ from K^c ($k\chi \leq K \xi$, DZZProp317In) — DZZ Remark 5.2 gives no range and outside it the walled P3.2 is false (parameter range); (2) the walled D' is generic in the cell family S (approxLGDSetOn S ; cellsMeeting K as in D117, cellsInside B offered for dyadic walls) (definition choice; D123 pending; since decided: DV-D123-1..2) · proposed · no.

DV-D123-1..2 (P2-DEC123, 2026-10-02; decision D123) · DZZ l. 2559–2563, Remark 5.2 (l. 2281–2284) · LQGMetric/Papers/DZZ/S5D117.lean (legWall redefined, stadium kept as legWallC), S5D123.lean, S5L54F2–F3.lean · (1) the leg from u to $L_\delta \subseteq \partial \mathbb{V}_{\{u,\lambda\}}$ on side j is measured with the wall $\mathbb{V}_{\{u,2\lambda\}} \cap \mathbb{V}_{\{u,u+\lambda e_j\}}$ (an axis-parallel square, legSq), i.e. DZZ's $\bar{D}^{\{u,2\lambda\}}$ restricted to $\bar{D}(u,v)$ -chains; replaces DV-D117-2's stadium (DZZ's inequality l. 2563 is not justified as written); (2) P3.17 for walled distances is used only for pairs at distance $\geq \xi$ from the complement of the wall (DZZProp317In), at finitely many fixed walls per consumer (DZZ do not state the restriction; the statement over all of $\mathbb{V}^\xi$ is false) · proposed · no.

DV-CONF33H-1..2 (P2-CONF33H, 2026-10-02) · CONF C:1187–1190 · LQGMetric/Papers/CONF/S3D112M1–M6.lean · (1) statement correction: CONFZBExtOfMarkov is false as stated (not_confZBExtOfMarkov: the law of the zero-boundary part is pinned only up to a deterministic mean supported on ∂U , e.g. a Dirac mass on ∂U); replaced by the existence form of the canonical Markov decomposition (exists_isL33ZBPart_G_ext), consumed by l33Gen_fatG_ext; (2) the comparison $\text{recSigma} \leftrightarrow \text{fieldSigmaClosed0}$ holds only off the null set where the mollified circle averages diverge (astronglyMeasurable_fieldSigmaClosed0_of_recSigma, own elementary; used implicitly by CONF) · proposed · no.

DV-D124-1..4 (P2-DEC124, 2026-10-02; D124) · DZZ l. 1555–1594, 1631–1645 · LQGMetric/Papers/DZZ/S3D124.lean, S3ConcF–H (after packet I0) · (1) $\mathcal{X}_\delta$ indexed by the dyadic boxes of side $\geq \delta^\kappa$ (η) and the rational points of $\mathbb{V}$ at the scales $2^{-n} \geq \delta^\kappa$ and δ^κ ($\bar{h}$) — DZZ: all $v \in \mathbb{V}$, all $\varepsilon \geq \delta^\kappa C_{mc}$; this countable set is what Lemma 2.1 and (eq-distance-Lip) read (l. 1557, 1584); (2) DZZ's conditioning on $\mathbf{X}_\delta$ (l. 1579–1582) realised on $\Omega \times \Omega$ by the mixing white noise wnMix , " $P(\mathcal{E}^* \mid \mathbf{X}_\delta) \geq 0.9$ on $\mathcal{A}$ " = Fubini slice $\leq 1/10$ after Markov; event probabilities of the mixing noises read off the canonical white noise by law uniqueness (wnLaw_eq); (3) $\mathcal{E}^*$ taken with $\delta'' = \delta$ only, with explicit finiteness / $D' \geq 1$ clauses; probability polynomial (δ^c , c independent of ι) from Prop 3.2 at $\delta \varepsilon^{\{\pm r\}}$ (admissibility at the larger scale with $\xi d = \xi/(1 - \iota/\alpha)$) and L3.5 twice — DZZ cite "Prop 3.2, Lemmas 3.1 and 3.5" without the scale bookkeeping; (4) (eq-distance-Lip) obtained with $(\iota/2) \log \delta^{-1}$ (DZZ: $\iota \log \delta^{-1}$); part 2 with $\alpha = 2/\gamma$, $\tau = (\log \delta^{-1})^{\{0.91\}}$, $\ell = (\log \delta^{-1})^{\{0.9\}}$, Lipschitz bound $(\log \delta^{-1})^{\{0.93\}}$ (DZZ absorb lower powers, l. 1626). Status: readings of the paper's proof (1–3) and a convention (4).

DV-DZZSIM-1 (update) (P2-DZZSIM2, 2026-10-02) · DZZ l. 548–576, 611–624, 638–641, 680–682, 1209–1213 · LQGMetric/Papers/DZZ/S5D117G1–G3.lean · DZZSimCoupleU proved for all $0 < \|a\| \leq 1$ (dzzSimCoupleU_of_norm_le); DZZ use L2.7 and the chaos limit along non-dyadic scales without comment — we prove both following DZZ's L2.8 scaling argument (dzzLemma27Along_of) and the martingale limit of l. 680–682 along the merged scale sequence (dzzWickChaosAlong_of, Doob) · proposed · no.

DV-HEAT1-1..2 (P2-HEAT1, 2026-10-02) · DGo:479–515; DDDF DD:1514–1525 · LQGMetric/Papers/DGo/HeatDir*.lean · (1) DGo use the square integrability of the circle-averaged Dirichlet heat kernel without proof; proved with $\int_0^\wedge T p_s(x,y) ds \leq 1/\pi + (2\pi)^{-1}(|\log T| + 2|\log|x-y||)$, the circle integrability of $\log| \cdot - a|$ (mathlib's

Jensen-type identity) and domination of p^D by p plus a constant ($s \leq 2$) and by $K e^{-cs}$ ($s > 2$) (own elementary); (2) DGo L3.1's circle average $h^D_\delta(v)$ is read as the $L^2(P)$ limit of $h^D(\text{heatCirc}_{\{v,\varepsilon\}})$ as $\varepsilon \rightarrow 0^+$ (the standard extension of the GFF to finite-energy measures, Berestycki–Powell §1.2), the zero-boundary GFF being a process on bounded densities (D39); both the law form (any zero-boundary GFF, `dgo_lemma31_sq`) and the coupling form (`dgo_lemma31_coupling`) proved; the pointwise $\pi f^p D = G_D$ was not needed (covariances via `zeroGFFTestCov_sqOpen_eq_heat_gen` and Chapman–Kolmogorov) · proposed · no.

DV-CONF36N-1..3 (P2-CONF36N, 2026-10-02) · CONF C:1350, 1393, 1425–1447 · LQGMetric/Papers/CONF/S3D114NA.lean, S3D114S1–S7.lean · (1) the filled ball is closed for every ω (own elementary: the unbounded components of the open complement are open); (2) the trace lemma in a variant for U with compact complement, removing an a.s. boundedness hypothesis (own adaptation of `conf21_trace_local`); (3) connectivity of the filled ball gives `conf36Conn` a.s. (connected image under $|\cdot - z|$, DEC-114 §1(d)) · proposed · no.

DV-DGWIRE-1 (P2-DGWIRE, 2026-10-02) · DG:1587–1589, 1722–1772 · LQGMetric/Papers/DG/S3Wire1–3.lean, Assembly/MainOpenI.lean · the box margins in the proofs of P3.22 and P3.17 Step 3 are halved to $1/12$, so DG L3.19 is used inside $[1/6, 5/6]^2$ (the old h319 on $[1/12, 11/12]^2$ of S3P18C/D/E/R4/S6 and S3P17V3/V5 is unsatisfiable for μ_{Hat} , which lives on $K_0 = [1/10, 9/10]^2$: an annulus in the strip $[1/12, 1/10)$ has zero mass); the mathematics is unchanged since LFPP distance is monotone in the domain (`wire_dg*_mono`) · proposed · no.

DV-DZZ53T-1 (P2-DZZ53T, 2026-10-02) · DZZ l. 2421–2423, 611–624 · LQGMetric/Papers/DZZ/S5L53B11.lean, S5L53B11A.lean, S5L54G0.lean · DZZ l. 2423 (χ independent of u, v): the δ -shift $e^{\pm\lambda} \cdot |a|$ of the scaling coupling is absorbed by our own elementary argument (a.s. monotonicity of $\log \tilde{D}_\delta$ in δ , existence of the per-pair limit from L5.3 part 2, Markov), not by DZZ's Cor 3.9 for tilde distances (DEC-117 §2(d)); the longer pair is mapped onto the shorter one ($\|a\| \leq 1$), the other order by symmetry · proposed · no.

DV-DZZ61G-1..2 (P2-DZZ61G, 2026-10-02) · DZZ l. 2455, 2474, 2553–2570, 2605 · LQGMetric/Papers/DZZ/S6L61F1–F2.lean, S6L61G1.lean · (1) the gluing of Fig. glue uses 40 tilde pairs at a dyadic scale $d \in [\delta^\kappa/2, \delta^\kappa]$ plus a connector chain from v to the contour (`ring_glue_lgd`, own elementary); (2) the short crossings need a scale-uniform coupling (`DZZSimCoupleScale`, factor $\|a\|^{1+\gamma^2/4}$, tail $C e^{-\lambda^2/(C(m+1))}$) instead of `DZZSimCoupleU`, whose constant depends on a (statement of a cited result, DZZ l. 2474; decision D125 pending) · proposed · (2) pending D125 (since decided: DV-D125-1..2).

DV-CONF-HLZ (P2-CONFHLZ, 2026-10-02) · CONF C:1169–1172, 1217–1234; DG:618–640 · LQGMetric/Papers/CONF/S3D112N1–N4.lean · the harmonic lower bound of CONF L3.3 Step 2 (`confHarmLowZB`) by our adaptation of DG L2.2's argument (`dg_tail_unif`): $|\mathfrak{h} - h_r(z)|$ on $rV_0 + z$ is bounded by bounding the whole-plane piece and the zero-boundary piece separately, the latter through $\text{Var} = \|\psi\|_{H^{-1}(U)}^2$, monotonicity in the domain and the square bound; $\mathfrak{h}$ is not identified with the harmonic part of h , so the bound holds for every version of $\mathfrak{h}^U$ without uniqueness of the Markov decomposition (CONF: $\mathfrak{h}^U$ continuous plus scale invariance) · proposed · no.

DV-D124-add (P2-DZZI0, 2026-10-02) · DZZ l. 1555–1569, 1600 · LQGMetric/Papers/DZZ/S3ConcF–G.lean (re-based on S3D124), S3ConcI0–I2L.lean · the I1 kernel identities assume $\delta > 0$ (needed for `rpow`); $\mathcal{A}_\delta \supseteq \text{cellSizeEvent}$ (`coarseGood_of_cellSizeEvent`) is own elementary · proposed · no.

DV-CONF-FKGB-1..5 (P2-CONF-FKGB, 2026-10-02) · CONF C:667–669, 717–738, 1236–1243 · LQGMetric/Papers/CONF/S3D108M1–M11.lean · (1) C:736–737: the frozen continuity proved under the law of S (own Fubini argument); (2) `chainInf`-based events replace `internal`-based ones (Lean measurability device; the events agree a.e. since every perturbed field is a length metric under Axiom III); (3) C:1240–1241 (“a similar justification”): the S -cont-law for E^U condition 2 proved without conditioning after normalizing the field on $\partial B_r(z)$, then made conditional by Fubini; (4) C:1238: condition 1 outside-measurable by a $\liminf$ over neighbourhoods $B_{\{1/(n+1)\}}(\text{cl } \mathbb{A}_{\{2r, 3r\}})$ (own); (5) statement correction: `CONF-FKGFrozenEUC` adds the countable- A hypothesis (the consumer uses only the finite sets K_C); note `l33Gen_fatG_of_lem2_10` takes the false `CONF-ZBExtOfMarkov` and must not be used (P2-CONF210B rewires) · proposed · no.

DV-CONF-J1-1..2 (P2-CONF-J1, 2026-10-02; D120 J1/J2) · CONF C:832–924, 1586 · LQGMetric/Papers/CONF/S3L214P1–P2.lean, S3T39J1.lean, S3T39J1b.lean · (1) connected subsets $J_i = \varphi^{-1}(u(I_i))$ of the circle placed between an open arc and its closure (`t39p_circle_preconnected`, own via `arg`); one-point arcs always good (`t39i_arc_loc`, $\ell \rightarrow 0$); L2.14 applied to the positive-length subfamily with rescaled $C' = C(n/n')^{-1/2}$ (reading of “arcs”, harmless); (2) C:1586: the centre $x_{\{k,i\}}$ selected deterministically (lexicographic minimum of the closure of the centres $c \in \partial \mathcal{B}^\bullet$ with $\bar{B}_{\{\varepsilon r/2\}}(c)$ disconnecting; open-ball radius $\rho = 3\varepsilon r/4 < \varepsilon r$) — convention (CONF only asserts existence) · proposed · no.

DV-HEAT2-1..2 (P2-HEAT2, 2026-10-02) · DGo:618–700 · LQGMetric/Papers/DGo/HeatDirR2a–c.lean · (1) DGo:661–668 (G2): instead of DGo's θ -integral bound $O(\sqrt{s/\delta})$ we integrate in s first with the sharp $\int_0^{\delta^2} p_s \leq 1/\pi + (2\pi)^{-1} 2 \log(2\delta/r)$, then Jensen's formula on the circle (`mathlib_circleAverage_log_norm_sub_const_eq_log_radius_add_posLog`); (2) DGo:630–682 (G1, G3) without Brownian bridges: the image-series bounds `abs_sqDirKernel_le_exp` ($s \geq 2$) and `abs_sqDirKernel_sub_le` ($s \leq 2$, at margin $\varepsilon/2$), the part of G3 outside D bounded

directly from the Gaussian kernel; G4 in function form by Cauchy–Schwarz in θ plus DGo L3.2 (as dgo_varG4_le) · proposed · no.

DV-D125-1.2 (P2-DEC125, 2026-10-02; D125) · DZZ l. 2455–2503, 2537–2545, 2566–2568 · LQGMetric/Papers/DZZ/S6L61G1.lean (DZZSimCoupleScale, after the step-0 edit), S5D125A–C (packet P-125) · (1) the scale-uniform coupling is stated with the factor $\|a\|$ on δ (DZZ’s a^{-2} on the masses, l. 2502), not the sharp Wick factor $\|a\|^{1+\gamma^2/4}$ of P2-DZZ61G’s draft, which is true but needs a lower bound on the killed heat kernel diagonal (the band variance $\geq \log \|a\|^{-1} - B$) that neither the library nor DZZ provide; consumers’ exponent conditions become $\kappa < 1$ (P-61G) and DEC-117’s $K_X \geq 2$ (P-54C); (2) DZZ’s “whp with respect to $P(\cdot \mid \mathcal{F}^*)$ ” with errors $(\log \delta^{-1})^{\{0.9x\}}$ is replaced by the unconditional tail $C e^{-\lambda^2/(C(m+1))}$ for dyadic $\|a\| = 2^{-m}$ (the $(m+1)$ is DZZ’s $\text{Var } \hat{h}_{s_i}^{-1} \leq 0(1) \log \delta^{-1}$, l. 2541), used at $\lambda = (m+1)^{3/4}$. Status: (1) a weakening of an input to the paper’s own form; (2) a convention.

DV-DG316-1 (P2-DG316, 2026-10-02) · DG:1432–1441 (Prop 3.16), 1452–1456, 1500–1512 · LQGMetric/Papers/DG/S3P16A–C.lean · **DG’s printed Prop 3.16 is false:** `not_dgProp3_16 : ¬ Blueprint.DGProp3_16` (proved in Lean; z, w at distance 2ε on either side of a shared side of two grid squares: $D^\delta(z, w) \leq 2\varepsilon e^{\xi \max h_\delta} \rightarrow 0$ while $\hat{D}^\delta - \delta e^{\xi \hat{h}(v_{S_z})} \geq \delta e^{\xi \hat{h}(v_{\{1,0\}})} > 0$, so the lower inequality fails with probability 1; DG’s proof breaks at DG:1500–1512, where the “largest $j' \leq j$ with a long piece” need not exist for the first squares); the upper half is true and proved (`p16_upper_det`, `p16_upper_of_lem37`, via the polygon through the midpoints of shared sides, DG:1452–1456); the corrected lower half is decision D126 · proposed · pending D126 (since decided: DV-D126-1.2).

DV-CONF210B-1.2 (P2-CONF210B, 2026-10-02) · CONF C:722–738, 1238–1241 · LQGMetric/Papers/CONF/S3D108Q1–Q3.lean · (1) CONF builds the splitting on the white-noise space; `CONFlem2_10` holds for any realisation via an own coupling argument (conditional kernel of a standard-Borel model sample given X_0 on $\Omega \times \Omega_0$, `cov_nonneg_of_model`), needed because `CONFlem2_10` is stated for every realisation; (2) Step 3 of L3.3 uses the countable-set variant `CONFCKGFrozenEuC` (the sets K_C are finite); `l33Gen_fatG_lem2_10` supersedes the vacuous `l33Gen_fatG_of_lem2_10` · proposed · no.

DV-D126-1.2 (P2-DEC126, 2026-10-02; D126; supersedes the “pending D126” status of DV-DG316-1) · DG:1432–1441 (Prop 3.16), 1445–1449 ((3.34)), 1458–1512 (proof), 1523–1532 (use in Prop 3.17) · LQGMetric/Blueprint/DFGPSInputsDG.lean (DGProp3_16), Papers/DG/S3P16A.lean (DGProp3_16Printed, `not_dgProp3_16Printed`), Papers/DG/S3P17.lean (DGProp316For, `dg_prop317_of_coupling`), packet P-126 Papers/DG/S3P16D1–D3.lean · (1) **statement of a cited result — DG’s printed Prop 3.16 is false:** the lower bound $\delta^\zeta (\hat{D}^\delta(z, w; S) - \delta e^{\xi \hat{h}_\delta(v_{S_z})}) \leq D^\delta(z, w; S)$ of DG:1436 fails with probability 1 for z, w at distance $2\varepsilon \rightarrow 0$ on either side of a side shared by two grid squares ($\hat{D}^\delta \rightarrow 0$, every chain contains both squares; Lean: `not_dgProp3_16Printed`); the gap in DG’s proof is at DG:1500–1512 (the short-crossing squares before the first long crossing have no J_j); DG’s own counting argument (DG:1505–1508) bounds these by at most 4 squares with centres within 3δ of v_{S_z} , whose weight on (3.34) is $\leq 4 \delta^{1-\zeta/2} e^{\xi \hat{h}_\delta(v_{S_z})}$, so the repaired proof gives $\delta^\zeta (\hat{D} - \delta^{1-\zeta} e^{\xi \hat{h}_\delta(v_{S_z})}) \leq D$ (i.e. $\delta^\zeta \hat{D} \leq D + \delta e^{\xi \hat{h}_\delta(v_{S_z})}$), which is now `Blueprint.DGProp3_16`; DG’s only use, Prop 3.17 (DG:1528–1532, “the error term is at most $\varepsilon^{\{\beta(1-2\gamma/d)\}}$ and $1 - 2\gamma/d > 1 - 2/d - \gamma^2/(2d)$ ”), absorbs the extra $\delta^{1-\zeta}$ by taking $\zeta_1 \leq \xi(Q-2)/8$ in `dg_prop317_of_coupling` · kind: statement of a cited result, corrected · status: decided (D126) · could this change a main theorem: no (the hypothesis of `theorem11_openI` becomes a true statement of the same strength for its consumer). (2) **place where we do not follow the published proof** (packet P-126): the lower half is proved by a covering argument (sampled squares along the path, `Walk`.bypass chain, exit times from the $2a$ -neighbourhoods of the squares, multiplicity ≤ 36 , regularity (3.34) at distance 6δ instead of 4δ) rather than DG’s loop erasure (DG:1458–1476) with the short-piece bookkeeping (DG:1493–1512); same statement and same inputs (DG L3.7, L3.4) · status: proposed with the packet · no.

DV-ADAPT-1 (P2-DZZADAPT, 2026-10-02; D123 P-317K-ADAPT) · DZZ Remark 5.2 (l. 2281–2284), Prop 3.17 (l. 1505–1517) · LQGMetric/Papers/DZZ/S5Walls0–3.lean; in-place `S5Defs`, `S5Adapt`, `S5L53Reg`, `S5L54D/E/F1/F2`, Papers/DG/S3Wire3.lean · every consumer of the walled P3.17 now takes `DZZProp317In ... K ξ` (pairs in $kX_i K \xi$) and DG.DZZInputsDG takes `DZZProp317Walls ... dgWalls` (the false all-pairs form is gone from the repository); u and v lie in $\mathbb{V}_{\{u,v\}}^\xi$ when $\xi \leq |u-v|/2$ ($B(u, |u-v|/2) \subseteq B((u+v)/2, |u-v|) \subseteq \mathbb{V}_{\{u,v\}}$, own elementary), so the DG L3.12 adapter takes $\xi \leq |u-v|/2$ (as DV-D123-2) · proposed · no.

DV-D124-I4 (P2-DZZI4, 2026-10-02) · DZZ l. 1557, 1584–1594 · LQGMetric/Papers/DZZ/S3ConcK.lean · the regularity of the coarse field $\mathcal{X}_\delta$ (DZZCoarseReg), implicit in DZZ, by an own elementary argument: uniform continuity of a continuous version of $\hat{h}_\varepsilon$ on $\mathbb{V}$ for each of the finitely many coarse scales, a floor-grid projection, and the finiteness of the boxes of side $\geq \delta^\kappa$ · proposed · no.

DV-HEAT3-1.3 (P2-HEAT3, 2026-10-02) · DGo:549–556, 702–704; DG:1103–1105 · LQGMetric/Papers/DGo/HeatDirR3a–R4c.lean, Papers/DG/S3L37P1.lean · (1) DGo use (3.9) on a general domain citing Hu–Miller–Peres Prop 2.1; here from the unit-square Green-function bound plus an own Brownian-scaling computation (`sqDirKernel_affT`); (2) R4 and Prop 3.3 on boxes with the field taken through the clamp map onto the box (DGo: general V); the compact- V form of R4 is a corollary; (3) DG L3.7 reduced to `DGZBRealization` (DGo (3.1)/L3.1 real-

ized as a DistC field) · proposed · no.

DV-D124-I2, DV-D124-I3a..b (P2-DZZI2, P2-DZZI3, 2026-10-02) · DZZ l. 1218–1233, 1572–1594, 1631–1645 · LQGMetric/Papers/DZZ/S3ConcJ1–J3.lean, S3ConcL1–L5.lean · (I2) Prop 3.2 and L3.5 used at $\delta\epsilon^{\pm r}$ with the fixed diameter exponent $\xi d = (\xi + C_Mc)/2$ (not $\xi/(1 - \iota/\alpha)$, which would make the constants depend on ι), the window restricted to $r \leq \min(1/2, (C_Mc - \xi)/(C_Mc + \xi)) \log \delta^{-1}$; c independent of ι, α and the white noise (law transfer through $wnCanon$); (I3a) part 2 uses DZZEStar2Le ($\alpha \leq 2/\gamma$; the plan's DZZEStar2 with "some α " is too weak, internal node); (I3b) (eq-concentration-approximate) through DZZCrudeMomentsEv (D122) via copied proofs · proposed · no.

DV-CONF36-325 (P2-CONF36P, 2026-10-02; D114/D120) · CONF C:1150–1154, 1167, 1362–1368, 1425–1447 · LQGMetric/Papers/CONF/S3D114T1–T3.lean; the D120 C-edit (Blueprint CONFLem3_6AtAE0 with IsFilledBallStoppingTimeAE, S3D110B/C, S3L36H, S3T39H2/H4, S3T39I4/I5, S3D114W) · (3.25) proved for $0 < \delta < 1/8$ (CONF's δ small, C:1167; D114 §2 (a)); the locality facts DL1/DL2 are read with h normalized on $B_{\{5r\}}(\zeta)$ and renormalized at $h_r(\zeta)$, using the a.s. identity $(h+a)_r = h_r + a$ · proposed · no.

DV-D125-3.4 (P2-DZZ125, 2026-10-02; D125) · DZZ l. 611–624, 2474, 2501–2503, 2537–2545, 2562–2568, 2605 · LQGMetric/Papers/DZZ/S5D125A–C.lean, S6L61H1–H3.lean · (3) own elementary asymptotics $tendsto_exp_rpow34_sub$, $tendsto_scaleTail$ and the exponent arithmetic $glue_arith$ for $\lambda = (m+1)^{3/4}$ and $2^{-m} \in [40\delta^\kappa, 80\delta^\kappa]$; (4) $dzzL61GlueK_of_scale$ takes DZZProp317In at the fixed pair (u_0, v_0) and $\xi \leq 1/80$, as $dzz_lem53_upper_whp$ requires in its D123 form · proposed · no.

DV-D126-2 (implemented) (P2-DG126, 2026-10-02; D126) · DG:1443–1512 · LQGMetric/Papers/DG/S3P16D1–D3.lean · the lower half of the corrected P3.16 is proved by covering (sampled squares, Walk.bypass, exits from 2a-neighbourhoods, multiplicity 36, constant 72) instead of DG's loop erasure (DG:1458–1476); L3.7 and L3.4 used at $C = 6$ (DG uses 4 δ) · implemented · no.

DV-CONF36Q-1.3 (P2-CONF36Q, 2026-10-02) · CONF C:1295–1302, 1312, 1341–1368 · LQGMetric/Papers/CONF/S3D114U1–U3.lean · (1) statement form: the Step 1 (i) node Conf36MeasNodeAEiX takes $x \in \partial \mathcal{B}^{\bullet}_{\sigma^{\epsilon}} \tau$ a.s. (C:1312), which the C:1341 choice of z needs (the node without it is false: for x far away G^{ϵ}_x depends on the field outside $\mathcal{B}^{\bullet}_{\sigma^{\epsilon}}$; project slip in DEC-114 §4 C3, not a paper error); (2) C:1366 $B_{\{5\tilde{p}^n\}}(z) \subseteq \mathcal{B}^{\bullet}_{\sigma^{\epsilon}}$ and the a.s. boundedness of $\mathcal{B}^{\bullet}_{\sigma}$ proved from DFGPS L3.8 (a.s. geodesics, bounded balls) for $\gamma \in (0, 2)$; (3) $E^{\wedge}U$ and $fatG$ events (determined by $h - h_r(z)$ on $B_{\{5r\}}(z)$) moved to $\sigma(h|_{\{B_{\{5r\}}(z)} \bmod \text{constants by normalising at a bump inside the ball (own, a.s. via } ae_circleAvg_addConst) \cdot \text{proposed} \cdot \text{no.}$

DV-DGZB-1 (P2-DGZB, 2026-10-02) · DGo:491–515, (3.9); DDDF DD:1514 · LQGMetric/Papers/DGo/ZBField.lean, ZBSFubG.lean, ZBSFub.lean, ZBDist.lean, ZBCirc.lean, ZBCirc2.lean, Papers/DG/S3L37Q1.lean · DG L3.7 coupling: DGo's $h^{\wedge}U = \sqrt{\pi} W(K)$ made a random distribution as in D109 (continuous antiderivative field, antiDist, stochastic Fubini, null-set modification for vanishing off cl D for every ω); circle averages identified through circleAverage_pairing, the mixture identity $zbKer(\sigma_{\{x, \delta\}} * \psi_n) = \int \psi_n(z) K_{\{\delta, x+z\}} dz$ and DGo (3.9), with an $O(2^{\wedge\{-n\}})$ rate and Borel–Cantelli (own elementary; DGo take the field for granted); the statement is unchanged · proposed · no.

DV-DZZ61P-1.2 (P2-DZZ61P, 2026-10-02) · DZZ l. 611–624, 849–857, 2340–2345, 2593–2596 · LQGMetric/Papers/DZZ/S6L61P1–P4.lean · (1) DZZ's tacit step from "with high probability" to "in expectation" in (eq-point-to-boundary-kappa) made rigorous with $Y \leq t + Y^2/t$, $t = M \log \delta^{-1}$, and the walled second moments (own elementary); the uniformity in the moving centre (DV-D117-4) is now proved; (2) λ in the scaling coupling fixed from ϵ (shifts $\log \delta^{-1}$ by a constant) instead of $\lambda = (\log \delta^{-1})^{\wedge\{0.6\}}$ (convention) · proposed · no.

DV-CONFZBM-1 (P2-CONFZBM, 2026-10-02) · CONF C:722–727 · LQGMetric/Papers/CONF/S3D108R1–R2.lean · the killed-heat white-noise field of CONF L2.10 on any bounded open U ($uKerL2$, $zbProcU$, the coarse/fine split, indep_fine_coarse); the L^2 bound and the Chapman–Kolmogorov inner-product bookkeeping are own elementary Fubini–Tonelli arguments; the identity $G_U = \pi \int p_U$ (ZBHeatRepr, D39 item 1) is open outside squares (decision D127 pending; since decided: DV-D127-1.4) · proposed · no.

DV-DZZ54C-a..d (P2-DZZ54C, 2026-10-02) · DZZ l. 2271, 2553–2578, 611–624 · LQGMetric/Papers/DZZ/S5L54G1–G2.lean, S5L54H1–H3.lean, S5L54I1–I4.lean, S5L54J1.lean · (a) "by symmetry" (l. 2555) realised by the reflection in $x = 1/2$ at the single configuration $u_{54} = 19/40 + i/2$ (DZZ use the corners of $\mathbb{V}$); the other sides at a general u through the isometric coupling with $a = \text{conj } e_j \in \{\pm 1, \pm i\}$ (own elementary); (b) uniformity over the segment family by contradiction along a chosen sequence of segments (reading of the text); (c) the coupling's δ -shift $\delta' = \delta\epsilon^{\wedge\lambda} = \delta^{\wedge\{1-\epsilon\}}$, $\lambda = \epsilon \log \delta^{-1}$, $\epsilon = \min(1/4, \iota/\chi)$; from w.h.p. to the mean via P3.17's concentration for the leg (own elementary); (d) symmetry and the triangle inequality of $\lgdDZZ$ (own elementary); $\chi \leq 0$ handled separately (DZZLem54SegQ needs $K\chi \geq 2$) · proposed · no.

DV-DZZCONCW-1 (P2-DZZCONCW, 2026-10-02) · DZZ l. 1519–1652, 2281–2284 · LQGMetric/Papers/DZZ/S3ConcW1–W6.lean · DZZ Prop 3.17 walled at a dyadic wall ($dzzProp317In_dzzMuIn_inside$): the walled (Eq.boundDprime) is taken for every white noise and every $\xi d < C_Mc$ (L32UpperCrossInside), because of the law-transfer form of DZZ's conditioning (D124-4) and the fixed $\xi d = (\xi + C_Mc)/2$ (DV-D124-I2) — statement of a

cited result · proposed · no.

DV-DZZSIMW-a..b (P2-DZZSIMW, 2026-10-02; D123 P-317K-SIM) · DZZ l. 611–624, 807–812, 1235–1244, 1505–1530, Remark 5.2 · LQGMetric/Papers/DZZ/S5WallSim1–3.lean · (a) $\text{wsim_abs_sub_integral_le}$ own elementary (Chebyshev plus AM–GM); (b) the walled P3.17 at non-dyadic walls is obtained by running DZZ’s P3.17 proof (walled P3.2, concentration, walled Cor 3.9 cor39_bound0n) through the scaling coupling, not by “the same proof” in a scaled box (Remark 5.2): the P3.17 window $(\log \delta^{-1})^{\{0.95\}}$ has no slack for the scale shift $\delta \mapsto \delta e^{\{\pm \lambda\}/\|a\|}$, so the P3.17 conclusion at a dyadic box cannot be transported · proposed · no.

DV-D127-4 (P2-HEATB, 2026-10-02; D127 P-127B N2) · BP arXiv:2404.16642 Lemma 1.38 (txt l. 1659–1683) · LQGMetric/Papers/CONF/S3D127B1.lean · BP apply Cauchy–Schwarz to $\int \nabla g \cdot \nabla (G_D \rho)$, which needs $G_D \rho$ to be regular (their Lemma 1.37). Following DEC-127 §3 N2, Cauchy–Schwarz is applied instead to the form $B(a, b) = \pi \langle K a, K b \rangle$ (killedGreenForm_eq_inner), positive semidefinite by Chapman–Kolmogorov, and $B(-\Delta g, -\Delta g) = (2\pi)^2 E(g)$ is computed from N1 and Green’s first identity, so no regularity of $G_U \rho$ is needed; own reordering of BP’s argument · proposed · no.

DV-J3-1 (P2-CONFJ3, 2026-10-02; D120 packet J3) · CONF C:1740–1744 · LQGMetric/Papers/CONF/S3T39J2b.lean · T39JArcChoice.subset reads $\text{arcs } m \ i \ \Gamma \subseteq \text{closure } \Gamma$ instead of DEC-120 §4’s $\Gamma \subseteq \text{closure } \Gamma$: every effrosSigma-measurable set is closure-invariant, so $\text{meas forces closure } (\text{arcs } m \ i \ \Gamma) = \text{closure } (\text{arcs } m \ i \ (\text{closure } \Gamma))$, and with $\Gamma \subseteq \Gamma$ for all Γ a Jordan curve could have no one-point arc (take $\Gamma = K \setminus \{p\}$); the consumer applies it only to closed $\Gamma = \partial \mathcal{B}^{\bullet}_{\tau}$, where both forms agree · accepted by the orchestrator · no (project-internal interface).

DV-D120-1 addendum (P2-CONFJ3) · C:1740–1744 · S3T39J2.lean, S3T39J2b.lean, S3T39J3.lean · the canonical subdivision cuts Γ at an Effros-measurable dense sequence of points (t39jPt , via $\text{GM.p412f_meas_select}$ on $\Omega := \text{Set } \mathbb{C}$); the arcs are the cut points and the components of $\Gamma \setminus \text{cut}$, described by rational chains (t39jClass), in place of DEC-120’s recursive diameter subdivision; eventual separation from density and t39j_jordan_split . Own elementary construction (no published proof covers the measurable selection) · proposed · no.

DV-DZZ53E-a..c (P2-DZZ53E, 2026-10-02; P-53E) · DZZ l. 2389–2410, 2429, 2495–2502, 2504–2530, 1946–2000 · LQGMetric/Papers/DZZ/S5L53E1–E7.lean · (a) the argmin choice of x_i (l. 2527) is replaced by any point of Λ^*_i at good distance (l53_chain_select); only Λ_u is required measurable; reading. (b) The (eq-z-open) counting (l. 2495–2502, l53_far_count , Tonelli + Markov), the open-box chaining (LGD analogue of l. 1946–2000, l53_open_chain) and the joint measurability of $\tilde{D}$ ($\text{measurable_lgdTilde}$): own elementary proofs. (c) Open boxes use $\mathcal{L}_1(\Lambda^*) \geq (1-10^{-5}) \mathcal{L}_1(\partial B)$ (erratum to l. 2429, ERRATA DZZ) · proposed · no.

DV-DZZUPW-1 (P2-DZZUPW, 2026-10-02; P-317K-UP) · DZZ l. 1088–1153, Remark 5.2 (l. 2281–2284) · LQGMetric/Papers/DZZ/S3P32UW1–UW10.lean · P3.2 (Eq.boundDprime) for walled distances at dyadic walls is proved by mapping $\mathbb{V}$ onto the wall box with a similarity ($\text{wHom/wEmb, lgdDZZ_map_similarity}$): the $\mu \ln$ proof (D102 chain) is run on the dyadic grid of the wall box with the masses and site events of the real sub-boxes. Same published proof; only a change of coordinates · proposed · no.

DV-CONFJ6-1..2 (P2-CONFJ6, 2026-10-02; D120 packet J6) · CONF C:1258–1302, 1295, 1506 · LQGMetric/Papers/CONF/S3T39J6–J10.lean · (1) C:1295: σ^ϵ is a.s. finite under L3.5 ($\text{t39j6_confSigma_ae_ne_top}$, own short argument from L3.5 with $K = \{0\}$), and $s_{\{k+1\}} := \text{toReal } \sigma$. (2) C:1506: the iteration assumes that τ gives a local set modulo constants ($\text{IsLocalSetDet0, CONFThm3_9RestCL}$), which does not follow from $\text{IsFilledBallStoppingTime}$ under our IsWholePlaneGFF (random additive constant); decided by D130 (IsLocalSetDet0 hypothesis on τ , DV-D130-1) · proposed · (2) changes CONFThm3_9At (D130).

DV-D120-1 extension (P2-CONFJ3b, 2026-10-02; D120 packet J3) · CONF C:1514, 1740–1744 · LQGMetric/Papers/CONF/S3T39J3b–J3e.lean · the arcs of equal harmonic measure (C:1740) are replaced by a canonical subdivision: cut points are the dense measurable points $\text{t39jPt } j$ with index below a Γ -dependent level $\text{t39jLev } m \ \Gamma = \text{Nat.findGreatest } (\text{t39jQ } \cdot m \ \Gamma) \ m$; the arcs are those cut points (one-point arcs) and the components of the rest (t39j_class_eq : chain classes = $\text{connectedComponentIn}$). DEC-120’s recursive diameter subdivision and its posLift argument are not used; CONF’s proof uses only covering and eventual separation. Own elementary construction · proposed · no.

DV-D129-1..5 (P2-DEC129, 2026-10-02; D129) · DZZ l. 126–139, 748–759, 1505–1517, 2306–2350 · LQGMetric/Papers/DZZ/S5Chi*.lean (packets P-129A–D) · (1) DZZ’s “with high probability” steps carry explicit rates ($\text{HasRate: } C(\delta^c + e^{\{-(\log \delta^{-1})^{\{0.7\}}\}})$), since Thm 1.1 is an a.s. statement and a.s. convergence needs summable tails along dyadics (convergence in probability plus monotonicity in δ is not enough); convention/gap. (2) l. 2311: the chain step is chosen so that every tilde box lies in $\mathbb{V}^\xi$ (DZZ’s $2\xi/\sqrt{5}$ gives only $\subseteq \mathbb{V}$); reading. (3) l. 2333–2350: the first-exit identity (eq-geodesic-range) is used one-sidedly ($\leq$), with the wall $\mathbb{V}_{\{u, 2\lambda\}}$ kept ($\text{lgd_sandwich_dzzWall_lower_K}$), for any $\lambda \leq 1/20$; reading. (4) l. 748–759: for $\chi > 0$, DZZ’s ball-size input ((eq-max-white-noise-process), (eq-LQG-negative-moment), union bound) is replaced by L3.1 (dzz_lemma31) and the cell cover of the P3.2 proof ($\text{l32BallCover_of_cellCompare}$); route. (5) The leaf DZZChiIdentW takes the walled P3.17 at dgWalls as a hypothesis; DZZChiIdent follows (dzzChiIdent_of); statement form, no weakening · pro-

posed (decision D129) · no.

DV-D127-A1 (P2-HEATA, 2026-10-02; D127 P-127A N1) · BP arXiv:2404.16642 Def. 1.11, Prop. 1.18(3) eq. (1.21), sketch txt l. 1175–1186 · LQGMetric/Papers/CONF/S3D127A1–A6.lean · N1 (killedGreen_lap, every bounded open U , every y) is proved by Dynkin’s formula on dyadic grids for planar BM (own elementary proof, copying QZ Dynkin.step_expect/dynkin_additive); the grid-staying events are taken on the closed inner sets innerSet U_k , then $k \rightarrow \infty$ by compactness of the path image (the square-specific stayGrid_ae_eq is false for general U , e.g. a disc minus a segment). No strong Markov property, no Itô formula; BP’s sketch is informal and assumes D regular and simply connected · proposed · no.

DV-DZZ53F-a.c (P2-DZZ53F, 2026-10-02; L5.3 node 1) · DZZ l. 2361–2393, 2504–2530, L3.12, Def. 3.11 · LQGMetric/Papers/DZZ/S5L53F1–F4.lean · (a) $\mathcal{D}_1$ uses the threshold $E X_k + L^{\{0.97\}}$ instead of $L^{\{0.95\}}$ (dzzLem53Exp_dzzMuIn_of_desirable97): the errors of (Eq.calD1) exceed $L^{\{0.95\}}$ (erratum); (eq-040418) still holds since $L^{\{0.97\}} + 4L^{\{0.98\}} \leq 5L^{\{0.98\}}$; parameter range. (b) The tilde partition is the global $\mathcal{V}_\delta$ restricted to the nine walls $\mathbb{V}_{\{w_{i-1}, w_i\}}$ (D117 §3), so the nine chains concatenate without surgery (l53_concat_chain); definition change. (c) eventRegularIn leaves out the good-point clauses of eventRegular (L3.12 as used here); reading · proposed · no.

DV-D130-1 (P2-DEC130, 2026-10-02; D130) · CONF C:1506–1509, 1431 · Blueprint/CONFResults.lean (CONFThm3_9At) · statement form: the stopping time τ of Theorem 3.9 is also assumed to have a filled ball that is a local set of h modulo additive constants (IsLocalSetDet0), as in CONFLem3_6AtAE0 (D108); this is what CONF’s proof uses at C:1431 (L2.1 for the field viewed modulo constants, C:1154), and it holds for every stopping time CONF and GM use (p412n_isLocalSetDet0_s4T'). Under our IsWholePlaneGFF (random constant) the raw hypothesis alone makes the statement equivalent to T3.9 for arbitrary random times (DEC-130 §1) · decided · yes (Blueprint hypothesis added; Statement layer unchanged) Applied by P2-D130W (2026-10-03): GM supplies the hypothesis at $\min(s4S \dots k, \tau_{\{2r\}})$ via the new GM.p412b_isLocalSetDet0_tauMin (copy-and-adapt of p412n_isLocalSetDet0_s4T'; DEC-130 §5 G-1 named the wrong time), and CONF T1.4 (t14_rat_finite, which DEC-130 missed) at τ_R ; CONFThm3_9RestC (false for invalid p) deleted, replaced by CONFThm3_9RestCL (S3T39J11).

DV-D130-2 (P2-DEC130) · CONF C:1506 · not formalized · the printed statement (raw stopping times, and even arbitrary random times in $[\tau_r, \tau_{\{2r\}}]$) remains true by a grid-of-hitting-times reduction (DEC-130 §2: $\tau_{\{r(1+j\theta)\}}$, Hölder spacing on $\mathcal{E}_r(a)$, isLeftmost_restr, L2.4 uniqueness, union bound); recorded as the reason the Blueprint change restricts hypotheses rather than repairing a false statement · own argument, documented only · no.

DV-DZZSIMW2-1 (P2-DZZSIMW2, 2026-10-02; D123 P-317K-SIM) · DZZ l. 1505–1530, 1627–1630, 611–624, 1235–1244, Remark 5.2 (l. 2281–2284) · LQGMetric/Papers/DZZ/S5WallSim4–S5WallSim6E.lean · DZZ use P3.17 at non-dyadic walls (Remark 5.2) without proof; it is proved by running DZZ’s P3.17 proof (l. 1526–1530) at θB_0 through the similarity coupling, with the dyadic inputs at the fixed box $wsimB_0 = [1/4, 1/2]^2$ (every wall of dgWalls is $\text{simMap } a \ b \ '' \ wsimB_0.\text{closedBox}, \|a\| \leq 1$); concentration admissibility uses $\xi_d = 2\xi$ at the scale $\delta e^{\{\lambda\}}/\|a\|$, $\lambda = (\log \delta^{-1})^{\{0.6\}}$ (hence the hypothesis $2\xi < C_{Mc}$). Geometry and asymptotics are own elementary proofs · proposed · no.

DV-D129-1/2 implementation (P2-CHIA, 2026-10-02; P-129A) · DZZ l. 2308–2331 (Eq.boundfortildeD l. 2318–2325), 611–624 · LQGMetric/Papers/DZZ/S5ChiA1–A3.lean · (1) rates are HasRate; the coupling uses $\lambda = \iota' \cdot \log \delta^{-1}$ (DZZ: a fixed large λ ; DEC-129 suggested $\sqrt{(\log \delta^{-1})}$), giving a polynomial tail $C \delta^{-\{\iota'^2/C\}}$, and the coupling scale $\delta/(\|a\|e^\lambda) \geq \delta^{-\{1+\iota'\}}$ is absorbed into ι by monotonicity in δ (lgdDZZ_antitone). (2) The chain length is $\iota = \lceil (2\theta + 4/m) |v-u| \rceil + 1$, m the margin of u, v inside the open square, so every tilde box lies in $\mathbb{V}_{\{\min(m/2, 1/4)\}}$ and each step is $\leq 1/2\theta = |\tilde{v} - \tilde{u}|$ (reference pair $1/2 \mp 1/4\theta + i/2$) · proposed · no.

DV-D127-1 extension (P2-HEATE, 2026-10-02; D127 P-127E N3) · BP arXiv:2404.16642 Lemma 1.37 (txt l. 1640–1658), Lemma 1.38 · LQGMetric/Papers/CONF/S3D127E1–E4.lean · the lower bound uses the truncations $\psi_\varepsilon \circ G_U \rho$ with $\psi_\varepsilon(s) = \int_0^s \text{smoothTransition}(t/\varepsilon - 1) dt$ in place of BP’s $f \in C_c^\infty(D)$; bounded ρ vanishing off U are reached by cutting off to compacts and mollifying, with $B(\rho - \rho_n) \rightarrow 0$ by dominated convergence on $\mathbb{C} \times \mathbb{C}$ (dominated by $C^2 \cdot 1_U \cdot 1_{U \cdot G_U}$) and the seminorm $\sqrt{B}$ (D127 addendum). Own elementary steps · proposed · no.

DV-P127C-1.2 (P2-HEATC, 2026-10-02; D127 P-127C N4a/N4c) · DEC-127 §3 N4(a), N4(c) · LQGMetric/Papers/CONF/S3D127C1–C7.lean · (1) N4(a), ”exterior cone condition $\Rightarrow E^x \tau_U \rightarrow 0$ ”, by an own iteration over scales instead of the standard Blumenthal 0-1 law argument (not available): survive scale R_j , then from within R_j of p hit an exterior ball $B(c, R_j/4)$ by time R_j^2 with probability $\geq e^{\{-81/32\}}/32$, plus a Gaussian tail; $P^x(\tau > T_n) \leq (1-\gamma_0)^{n+1} + 2n e^{\{-\Lambda^2/(16(n+1))\}}$, combined with $E^x \tau \leq 2h + P^x(\tau > h) \cdot 2R^4/h$; the cone condition is stated as an exterior corkscrew condition (ExtCorkscrew, balls instead of cones/quadrants; extCorkscrew_confU). (2) N4(c): the bridge-touching estimate is not proved (no consumer uses it); the continuity it was meant to give is proved directly for every bounded open U (continuous0n_killedConv, continuous0n_greenPot') by the strong-Feller approximation $|P^U \varepsilon g - P_\varepsilon g| \leq \|g\| \cdot 104\varepsilon/d^2$ and dominated convergence in s , with no boundary regularity · proposed · no.

DV-DZZ53G-1..5 (P2-DZZ53G, 2026-10-02; L5.3 node 2) · DZZ l. 2452–2502, 2474, 2490–2492, 2531–2547 · LQGMetric/Papers/DZZ/S5L53G1–G6.lean · (1) $\mathcal{E}_{\{B, \text{open}\}}$ is the complement of the explicit counting event $\mathcal{L}53Z\text{Bad}$ (DZZ only say “there is an event”); definition. (2) Conditioning on $\mathcal{F}^*$ is done through an event A_0 of the white noise in a region R_0 and $P[\cdot | A_0]$ ($\mathcal{L}53_cond_biInter$), not through conditional expectation; reading. (3) $B^* = \text{sqBox } c(5t)$, which equals DZZ’s $\{\|x - \partial B\|_\infty \leq 2t\}$; notation. (4) The (u,v) step of the far bound applies concentration at $\delta' = \delta e^{-L^{0.95}}$ and compares the means in two regimes (crude bound when $\log 2 \leq L^{0.96}$), a positive-probability intersection with the walled Cor 3.9 otherwise) ($\mathcal{L}53_uv_far$), filling the gap at l. 2490–2492 (ERRATA); own argument. (5) Rational-ball measurability variants of E6 and a null-measurable variant for μ_{In} ’s ball masses; technical · proposed · no.

DV-DZZ53I-1 (P2-DZZ53I, 2026-10-02; L5.3 hreg) · DZZ L3.12 (l. 1300–1302), proof l. 1428–1499, Remark 5.2 (l. 2281–2284), tilde partition l. 2275 · LQGMetric/Papers/DZZ/S5L53I1–I5.lean · walled DZZ L3.12 for D117’s restricted partition: the good sequence’s cells lie within $8\delta^{C_Mc}$ of K (eventRegularNear , dzz_lemma312Near , $\mathcal{L}53_hregN$), not in $\text{cellsMeeting } K$: the surgery reroutes around a bad cell through its enclosure, whose cells near ∂K can miss K (DZZ partition $\mathbb{V}_{\{u,v\}}$ itself, D117 §3 uses the global $\mathbb{V}_\delta$ restricted to the wall — modelling gap, under review in D131). The surgery starts from the $D' \wedge K$ geodesic; keeping the cells near K ($\mathcal{L}312\text{StepLoc}$ invariant: each bad cell b within $4s_b$ of K) is own elementary argument. Consumers $\mathcal{D}_1$ and hbad use the thickened region ($\mathcal{L}53D1\text{EventR}$, $\mathcal{L}53\text{BadR}$; new end-to-end $\text{dzzLem53Exp_dzzMuIn_of_hd_hbadN}$) · proposed · no.

DV-D130-J6a (P2-T39K1, 2026-10-02; D130 J6a) · CONF C:1506–1566, C:1154, C:1559–1561; GM (4.9) l. 1650–1654 · LQGMetric/Papers/CONF/S3T39K1.lean · own elementary proof of “ $\mathcal{B}^\bullet_\tau$ local modulo constants $\Rightarrow \tau$ is an a.s. stopping time of the mod-constant filtration” ($\mathcal{L}0_t39k_ae_of_isLocalSetDet0$, piece form $\text{t39k_ae_le_localSigma0}$), and of hmono (t39k_hmono_seq) as the mod-constant copy of $\text{GM.gm_localSigma_le_aeSigma}$ (hit events of the smaller set are piecewise local for the larger one); CONF uses both tacitly · proposed · no.

DV-D130-L2 (P2-T39K2, 2026-10-02; D130 J6b) · CONF L2.1 (C:476–479), C:1431–1432, C:1154 · LQGMetric/Papers/CONF/S3T39K2.lean · CONF L2.1 modulo constants is formalized for stopping times covariant under $h \mapsto h - h(\psi)$ ($\text{t39k2_isLocalSetDet0_of_cov}$). The D130 §4 L2 form (every a.s. mod-constant filled-ball stopping time gives a mod-constant local set) is false — every deterministic time satisfies the hypothesis (t39k2_ae0_const), while $\mathcal{B}^\bullet_\tau(1(D_{\{h_0+Z\}})) = \mathcal{B}^\bullet_{\{e^{-\xi Z}\}(D_{\{h_0\}})}$ is not local modulo constants for an independent random constant Z ; the route of the iteration is re-decided in D130B · statement form · no.

DV-DZZ53H-1 (P2-DZZ53H, 2026-10-02; L5.3 hd) · DZZ l. 2380–2391, 1159–1169, 919–923, 1235–1244, 611–624 · LQGMetric/Papers/DZZ/S5L53H1–H5.lean · d_i comparison: instead of the walled P3.2 at the non-dyadic box K_i (which would need the walled L3.5 at K_i , proved only at dyadic walls), we use (eq-280318b) for the walled $D' \wedge K$ plus the ball cover $\mathcal{L}32\text{BallCoverIn_dzzMuIn}$ at $\delta_1 = \delta e^{-2L^{0.8}}$; Cor 3.9 is applied at the dyadic box $\tilde{B}_0 = [1/4, 1/2]^2$ and connected to K_i and $\mathbb{V}_{\{u,v\}}$ by two similarity couplings ($\lambda = L^{0.6}$). Scale bookkeeping own elementary proof; same conclusion and threshold $L^{0.96}$ · proposed · no.

DV-D129-4 implementation (P2-CHIC, 2026-10-02; P-129C) · DZZ L2.12 lower half (l. 740–759), (eq-Euclidean-Ball-covering) l. 1164–1168, (Eq.lowerboundforDprime) l. 1080, L5.3 l. 2292–2297 · LQGMetric/Papers/DZZ/S5ChiC1–C2.lean · the ball-size input of $\text{lem-obvious-bounds}$ is L3.1 at δ' ($\text{highProb_cellSize_p32Up}$) plus the ball cover ($\mathcal{L}32\text{BallCover_dzzMuIn_eta}$), giving “w.h.p. every ball of μ_{In} -mass $\leq \delta^2$ has radius $\leq 32\delta^{C_Mc/2}$ ” ($\text{radius_le_of_ballCover}$, $\text{prob_light_ball_le}$); the step from this w.h.p. statement to the polynomial tail BallMassLowerTail ($\delta = \sqrt{t}$, three cases) is own elementary bookkeeping · proposed · no.

DV-D129-3 implementation, DV-D129-1 addendum (P2-CHIB, 2026-10-02; P-129B) · DZZ Prop 5.1 proof l. 2333–2350, L5.4, $\text{lem-scaling-coupling}$ l. 611–624 · LQGMetric/Papers/DZZ/S5ChiB1–B2.lean · (3) the first-exit identity (eq-geodesic-range) is used one-sidedly ($\leq$) with the wall $\mathbb{V}_{\{u, 2\lambda\}}$ kept ($\text{lgd_sandwich_dzzWall_lower_K}$, $\text{ae_lgd_sandwich_wick_K}$), for any $\lambda \leq 1/2\theta$ with $\mathbb{V}_{\{u, 2\lambda\}} \subseteq \mathbb{V}^\circ$. (1, addendum) the box scaling uses $\lambda' = \iota' \log \delta^{-1} + \log(2\theta\lambda)$ instead of DZZ’s $\sqrt{(\log \delta^{-1})}$, so the reference scale is exactly $\delta^{\{1-\iota'\}}$; the reference lower tail is kept in polynomial AlphaHighProb form, since HasRate is not preserved by $\delta \mapsto \delta^p$ for $p < 1$ (only P3.17’s first concentration bound $\delta^{\{c\}^2}$ is needed there) · proposed · no.

DV-D127-H1 (P2-HEATH, 2026-10-02; D127 P-127H (L3)) · CONF C:722–731 (Rhodes–Vargas Lemma 5.4 white-noise representation) · LQGMetric/Papers/CONF/S3D127H1–H4.lean · exists_zbDistU generalizes DGo.ZB . exists_zbDist from $\text{sqOpen } a$ to every bounded open U (no boundary regularity needed; copy-and-modify with credit). $\text{exists_version_of_supportedIn}$ (measurability of the field for the coarse filtration, which CONF does not address) is own elementary argument: truncate g to $g \cdot 1_{\{(t_n, \infty) \times \mathbb{C}\}}$, dominated convergence in L^2 , the $\text{It}\delta$ isometry, and closedness of the m' -a.e.-measurable classes in $L^2(P)$ · proposed · no.

DV-P127G-1..3 (P2-HEATG, 2026-10-03; D127 P-127G (L2)) · CONF C:725–727 (“easily checked using Kolmogorov”); DEC-127 §3 (L2), $\text{handoff/P2-CONFZBM}$ (L2) · LQGMetric/Papers/CONF/S3D127G1–G6.lean · (1) Hölder continuity of the coarse kernel up to ∂U by the semigroup step ($\text{G1_sq_norm_wndKernelL2_sub_le}$) plus the L^1 increment bound (G2); own elementary argument. (2) Power-law decay of the survival probability under the exterior corkscrew condition ($\text{killedSurv_le_rpow_of_step}$, survG_decay_confU) by a multi-scale induction

using only fixed-time Markov steps, replacing the textbook strong-Markov proof; own argument. (3) The planned Lipschitz form (coarseKer_lip_confU) is replaced by a Hölder form (coarseKer_holder_confU, exponent $\beta/4$) fed into the generalized Kolmogorov–Čentsov lemma (exists_continuous_modification_of_kernel_holder), giving exists_continuous_coarseField_confU: the plan’s route fails at confU (BridgeShellBound only covers balls centred at the bridge’s start; the reflection argument needs the domain on one side of each line) · proposed · no.

DV-D130B-1 (P2-DEC130B, 2026-10-03; D130B) · CONF C:1292–1302 (“if τ is a stopping time for $\{(\mathcal{B}^\bullet_t, h|_{\mathcal{B}^\bullet_t})\}$, then so is $\sigma^\varepsilon_{\tau,r}$ ”) read with h modulo additive constants (C:1154), C:1431 · Papers/CONF/S3T39K6–K7 · own argument (standard): the locality modulo constants of $\mathcal{B}^\bullet_{s_{k+1}}$ is propagated by the Weyl covariance of the iteration — for every normalization $h' = h - h(\psi)$ ($\int \psi = 1$) the radii satisfy $s'_k = e^{-\xi h(\psi)} s_k$ a.s. (arcs, R^ε_r and σ^ε covariant, Axiom III) — together with CONF L2.1 for the normalized field (t39k2_isLocalSetDet0_of_cov), exactly as L2.1 is applied at C:1431 (p412n_isLocalSetDet0_s4T'). CONF needs none of this because its field is normalized (C:347). The D130 §4 route L2 is false in our model (t39k2_ae0_const) · decided · no.

DV-D130-J6c (P2-T39K3, 2026-10-03; D130 J6c / L1) · CONF C:1292–1302, C:1150–1154, 1260 · LQGMetric/Papers/CONF/S3T39K3–K4.lean · CONF’s claim that $\sigma^\varepsilon_{\tau,r}$ is a stopping time is proved in a.s. form for the mod-constant filtration (t39k_confSigma_ae0) through the stopped- σ -algebra inclusion $\sigma(\mathcal{B}^\bullet_s, h|_{\mathcal{B}^\bullet_s} \bmod \text{const}) \subseteq \mathcal{F}^\circ_t$ on $\{s < q\}$ (t39k_localSigma0_le_trAE), using right-continuity of filled balls and dyadic-hull pieces inside $\mathcal{B}^\bullet_{\{(q+t)/2\}}$, and the mod-constant locality of $E_r(z)$ (t39k_confE_ball_ae); CONF uses the inclusion tacitly. Own routine argument · proposed · no.

DV-D131-1.4 (P2-DEC131, 2026-10-03; D131) · DZZ l. 2363–2378, 2370, 2433–2438, 2455, 2474, 2490–2502, 2533, Remark 5.2; D117 §3 · LQGMetric/Papers/DZZ/S5L53* (D131 packets, assignment in DEC-131 §7 orchestrator table) · (1) the proxy field $\eta^\wedge B$ (heat kernel killed outside B^{**}) is replaced by the time-truncated field $\eta^\wedge \{s'\}$ ($\eta^\wedge W \varepsilon s', s' = s_B / (K 2^\ell), 2^\ell \approx 4C_{Mc} L$) and its chaos fineChaos; locality (range $6s_B/K$ instead of $3s_B/K$), domination with the same factor $6^2 s_B^{-2} e^{\{L^\wedge \{0.91\}\}}$, and the coupling with $\eta(\theta \cdot)$ on $\mathbb{V}_{\{u,v\}}$ are proved for it; the band $\eta^\wedge \{s_B\}_{s'}$ is controlled by DZZ’s own band-sup argument (l. 2537–2545); same mathematics, no sub-box killed kernels. (2) The box chain of $\mathcal{E}^*$ is selected by selMin over L53Q (shortest, then least key), as l313Sel; DZZ only say “measurable with respect to $\mathcal{F}^{**}$ ”. (3) Pairs $z, z' \in \partial B$ with $|z - z'| < s_B K^{-5}$ are declared far (contribution $\leq 2s_B K^{-5} \mu^{H^1}(\partial B)$ to the Tonelli bound); DZZ tacitly assume $|z - z'| \asymp s_B/K$ (l. 2474, 2533). (4) The nine partitions are the global $\mathcal{V}_\delta$ restricted to cells near the walls (margin $86^\wedge \{C_{Mc}\}$, DV-DZZ53I-1), not DZZ’s tilde partitions; node 1 is restated on DZZ’s own $\mathcal{E}^*$ (L3.13 boxes of side $\varepsilon^{*2} s_C$) so that the conditioning event is a white-noise event of a region disjoint from the sub-box regions (R3) · proposed (decision D131) · no.

DV-D127-I1 (P2-HEATI, 2026-10-03; D127 P-127I (L4)) · CONF L2.10 C:719–742, the white-noise split C:722–731, Kolmogorov C:725–727 · LQGMetric/Papers/CONF/S3D127I1–I5.lean · the Kolmogorov step for $h_{\{t,\infty\}}$ is run on the σ -algebra of the white noise at times $> t$ (trimmed measure, exists_coarseVersion), so that the coarse field is $\mathcal{F}_n$ -measurable as IsCoarseSplit.measS / cov_nonneg_of_model need; stochastic Fubini against any test function (integral_tmul_inner_gen, copied from DGo integral_d12_mul_inner_gen); the parameters $r \leq 0$ or $\delta \leq 0$ of CONFLem2_10AtConfU, which CONF does not consider, are reduced to $U = \emptyset$ and to the annulus (confU_eq_empty, confU_eq_annulus, zbHeatRepr_empty). Own routine measure-theoretic steps · proposed · no.

DV-D127-1.3 (P2-DEC127, 2026-10-02; D127; entries added after AUDIT-2026-10-03-L B1) · CONF L2.10 C:712–742; BP arXiv:2404.16642 Lemma 1.37 · LQGMetric/Papers/CONF/S3D127*, S3Sec3W2 · (1) the lower bound of the dual norm uses the smooth truncations $\psi_\varepsilon \circ (G_U \rho)$ in place of BP’s hypothesis $f \in C_c^\infty(D)$ (implemented in S3D127E1–E4, DV-D127-1 extension). (2) Continuity of $h_{\{t,\infty\}}$ up to ∂U (C:725–727, “easily checked using Kolmogorov”) was planned via the radial bridge-shell bound; it is proved instead in Hölder form (DV-P127G-1.3). (3) CONFLem2_10 is discharged only at $U = \text{confU } r \delta \geq T$ (CONFLem2_10AtConfU, the only U CONF uses; $r \leq 0$ or $\delta \leq 0$ reduced to $U = \emptyset$ and the annulus, DV-D127-I1); the published all- U statement remains an unused open definition · decided · no.

DV-D128-1 (orchestrator, 2026-10-02; D128; refinement of DV-D123-2, AUDIT-2026-10-03-L L8) · DZZ Remark 5.2, L5.4 (l. 2281–2284, 2557–2563) · LQGMetric/Papers/DZZ/S5Walls2.lean · dgWalls contains only the side-1/10 squares centred within sup-distance 1/20 of $(1/2, 1/2)$ and the tilde boxes of pairs of $\mathbb{V}$ (the walls DZZ use), not every side-1/10 square inside $\mathbb{V}$ · decided · no.

DV-DZZ53V-1 (P2-DZZ53V, 2026-10-03; D131 P-131V) · DZZ l. 2480–2481 (“straightforward computation”), L2.7 proof l. 554–559 · LQGMetric/Papers/DZZ/S5L53V1.lean · own elementary proof of etaVar_band_ge: reduction to the integral of the killed heat kernel, bounded below by the free kernel minus the exit probability from the ball of radius $\min(\sqrt{t} \log t^{-1}/4, 1/10, \xi)$ (reflection principle, as in DZZ’s L2.7); loss $0_\xi(1)$ · proposed · no.

DV-D127-F1 (P2-HEATF, 2026-10-03; D127 P-127F N5) · DEC-127 §3 N5; QZ zeroGFFTestCov/dualCov · LQGMetric/Papers/CONF/S3D127F1.lean · ZBHeatRepr U for every bounded open U with GreenPotReg U (zbHeatRepr_of_reg) by polarization: dualCov of a $dx, b dx$ ($a, b \geq 0$ bounded measurable, vanishing off U) equals killedGreenForm U a b (three applications of dualNormSq_eq_killedGreen), then bilinear expansion over

(U.indicator ϕ)[±](zeroGFFTestCov_eq_killedGreenForm); GreenPotReg (confU ...) from greenPot_confU_props (C7). Routine · proposed · no.

DV-D131-1 amendment (P2-DZZ53K, 2026-10-03; D131 P-131C, DEC-131 §9) · DZZ l. 2433–2459, 2474–2503, (eq-def-tilde-M) l. 677–683 · LQGMetric/Papers/DZZ/S5L53K1–K2.lean · the proxy for η^B is $c_B \cdot \tilde{M}_{\{Y, s', \eta\}}$ (DZZ's η -chaos etaChaos, set by set) as a rational-ball mass map (fineMass, proxyMass), not dzzMuIn · fineDens⁻¹ (which is not local: it is the chaos of $(\tilde{h}_\varepsilon - \eta_\varepsilon) + \eta^{\{s'\}}_\varepsilon$); locality region $(0, s^2) \times \text{sqBox}(c_B, 7t)$ (l53_proxyMass_local); domination is an inequality through the density comparison (ae_dzzMuIn_ball_le_proxyMass), not an identity; the coupling uses the mass-map core lgdRat_sim_upper · proposed · no.

DV-DZZ53L-1 (P2-DZZ53L, 2026-10-03; D131 P-131A) · DZZ l. 2364–2379, 2427–2453, 1314–1340, 1700–1740 · LQGMetric/Papers/DZZ/S5L53L1–L5.lean · L5.3's $\mathcal{E}^*$ is built on the global $\mathcal{V}_\delta$ refined to L3.13 boxes (DZZ's own chain, l53Sel/l53Chain by selMin over L53Q); R3 (measurableSet_l53Chain_eq_wn: the chain event is a white-noise event of $(\text{fineReg } c_\theta)^c$) via a stopping-set principle generalizing ae_condExp_fineField_l313Sel (measurableSet_wnSigma_compl_of_cellSigma). The cell-chain hbadN of S5L53I5 is superseded (node l's global cells cannot give R3: neighbouring smaller cells read the white noise at small times). Also an alternative box-chain adapter dzzLem53Exp_dzzMuIn_of_hd_hbadBox (S5L53L2–L4) · proposed (D131) · no.

DV-J6d-1 (P2-T39K5, 2026-10-03; D130 J6d) · CONF C:1559–1561 (" $\mathcal{J}_k$ is determined by $(\mathcal{B}^\bullet_{\{s_k\}}, h|_{\mathcal{B}^\bullet_{\{s_k\}}})$ ", stated without proof), C:1154 · LQGMetric/Papers/CONF/S3T39K5–K5i.lean · own elementary arguments: scale invariance of the D-D1 leftmost geodesics and arcs (t39k5_gArc_smul), the hit-pattern factorisation of the Effros σ -algebra (t39k5_effros_eq_comap), joint membership measurability of the canonical arcs (t39k5_arcs_mem_meas), saturated events of $(h - h_r(z), Z)$ as a.s. fieldSigma00n h A $\sqcup$ G-events (t39k5_piece_sat, from GM exists_fieldSigma_piece/aeEventIn_of_saturated); fields 4–6 at step k (t39k5_fields_at modulo hB (universal measurability of the saturated sets, D132) and hCtr/hGood (deterministic Effros-measurability on the Jordan class) · proposed · no.

DV-J6e-1 (P2-T39K6/P2-T39K7, 2026-10-03; D130B J6e) · CONF C:1292–1302, C:1431–1432, C:1543–1566 · LQGMetric/Papers/CONF/S3T39K6–K7b.lean · implementation of DV-D130B-1: scaling of the iteration under normIn (t39k6_iter_normIn), the confRK adapter for $\varepsilon = 1$ (copied from p412o_confRKAddConst, which required $\varepsilon < 1$), L2' (t39k7_isLocalSetDet0_succ), the induction t39k7_rest, and the reduction confThm3_9RestAll_of_K7 to the J6d fields (T39K5FieldsJ3); S3T39KC1 (P2-T39KC) discharges the context inputs hs, hgood, hthit, hle of t39k5_fields_at and proves Measurable s_k by induction (copied from t39j9_restData_of), giving confThm3_9RestAll_of_meas (AUDIT-O O10) · proposed · no.

DV-D131-5 (P2-DZZ53P, 2026-10-03; D131 P-131B) · DZZ l. 2363–2391, 1314–1316 · LQGMetric/Papers/DZZ/S5L53P1–P2.lean · the box chain of $\mathcal{E}^*$ lies in the $2\delta^{\{C_{Mc}\}}$ -thickening of the cell-chain region (L53RefineB), because boxes in cells meeting R need not meet R (DEC-131 §2's refinement with the same R is false as written); $\mathcal{D}_1$ on cells is used at EX_k + L^{0.965} and on boxes at EX_k + L^{0.97}, leaving room for the $(\varepsilon^*)^{\{-4\}} \leq e^{\{L^{\{0.6\}}\}}$ length factor; one α_s serves both regularity and good points ($\alpha_s = \max \alpha_r \alpha_g$, monotonicity in α^*) · proposed · no.

DV-D131-A (P2-DZZ53LR, 2026-10-03; D131 P-131A remainder) · DZZ l. 2366–2379, 1327–1336 · LQGMetric/Papers/DZZ/S5L53L6–L7.lean · in the refinement l53D1EventR $\subseteq$ l53D1EventB (l53D1EventR_subset_l53D1EventB_gen, l53RefineB_of_pos) the box region is cthickening ($2\delta^{\{C_{Mc}\}}$) R rather than R: the L3.13 boxes lie in cells meeting R, not in R (counterexample R = {u, w, v}); same correction as DV-D131-5; definition change forced by the statement · proposed · no.

DV-DZZ53M-1.2 (P2-DZZ53M, 2026-10-03; D131 P-131U) · DZZ l. 2393, 2495–2502, 2516–2522 · LQGMetric/Papers/DZZ/S5L53M1–M5.lean · (1) (eq-Lambda-i-not-small) for the end boxes is an own elementary proof (l53_box_side_good, l53_iface_ge_min_of_nest; DZZ state it without proof, and for the L3.13 boxes it needs u to be a good point). (2) Node 4 uses a union bound over the fixed candidate boxes boxAt n w, n ≤ N, instead of conditioning on $\mathcal{F}^*$ (l53UVBadBox_le); the slot l53BadBox_subset : l53BadBox $\subseteq$ l53UVBadBox $\cup$ l53CellsBadBox · proposed · no.

DV-D131-6 (P2-DZZ53Z, 2026-10-03; D131 P-131R) · DZZ l. 2440–2455, (eq-M-A-upper-bound-bis) l. 2455 (no proof in DZZ) · LQGMetric/Papers/DZZ/S5L53Z1–Z4.lean · DZZ's $\mathcal{E}_4$ is realised by its field part only (l53E4: (eq-tilde-h-eta-assump), the a.s. limit steps l53ProxyGood, and the η -comparison between the cell scale s_C and the proxy scale, nbrFineGen with Y = e^{{L^{0.55}}}, T = L^{0.8}); its probability bound is for $\mathcal{E}_4^c$ itself (l53_E4_prob, e^{{-L^{0.23}}}). The mass clause of $\mathcal{E}^*$ is replaced by the cell mass $M_{\{Y, s_C\}}(C) \leq \delta^2$ (approxLQG C $\leq \delta^2$) of the cell containing the chain box; the factor $\delta^2 s_C^{\{-2\}}$ is $\leq$ DZZ's $\delta^2 s_b^{\{-2\}}$. The domination (l53_domination, l53_domination_chainBox) is the density comparison (l53_dens_le, generalizing l32TildeMUpper_wickQArea); the scale estimate l53_two_pow_j_le is own elementary proof · proposed · no.

DV-DZZ53W-1 (P2-DZZ53W, 2026-10-03; D131 P-131W) · DZZ L5.3 l. 2361–2548 · LQGMetric/Papers/DZZ/S5L53W1–W2.lean, Assembly/MainOpen0.lean · the DZZ L5.3 leaf is restated as DZZLem53ExpAll ($\exists \chi$, DZ-

ZLem53Exp, for every white noise and $\gamma \in (0,2)$), which is the only thing the proved route uses (AUDIT-L L6); it is implied by DZZInW.DZZLem53EventAll (dzzLem53ExpAll_of_eventAll) and reduced to L53HbadBAll (dzzLem53ExpAll_of_hbadB, with l53RefineB_of_pos); the fibre bookkeeping (l53BadB_le_split) is own elementary argument · proposed · no.

DV-D132-1 (P2-DEC132, 2026-10-03; D132) · CONF C:1559–1561, C:1543; GM l. 1705–1708 · Papers/CONF/S3T39K9a, S3T39KBb, amended S3T39K5h–K5i · the locality of the arc hit events is proved for the P-a.s.-equal events built from GM’s analytic arc relation (confPts/arcOf, gm_arcRelAn); t39gArc is kept, and the two leftmost-geodesic notions agree off a P-null set (t39k5_hit_iff_arcOf, CONF L2.4); hB is replaced by hU, harcs, harcsM (proved for t39jArcs) · decided · no — APPLIED: P-132A (S3T39K9a), P-132B (S3T39KBb, K5h/K5i amended), P-132C (S3T39KC2) are committed; KC1 passes harcs/harcsM (unused hB binder kept).

DV-131S-1..2 (P2-DZZ53X, 2026-10-03; D131 P-131S) · DZZ (eq-scaling-invariance-approximate) l. 2474, proof l. 2531–2548; L2.7 l. 548–576; L2.8 l. 593–610 · LQGMetric/Papers/DZZ/S5L53X2–X4.lean · (1) fineChaos_sim_couple holds for $2^{-m} \leq \|a\|$ with tail $C(\|a\|^{2m})^2 e^{-\lambda^2/(C(\log(\|a\|^{2m})+1))}$ (the coarse band $\hat{h}^1_c$, $c = 2^{-m}/\|a\|$, has variance $\log(\|a\|^{2m})$ and must be controlled on all of K; DEC-131 §3’s tail $\log_2 \|a\|^{-1} + 2$ is false for large m); the small side is wallMass (0K) (fineMass $W_2 \gamma m$) (DEC-131 §9); L2.7/L2.8 are made uniform in the starting scale (dzzLemma27Along_unif, dzz_lemma28_unif). (2) The grid union bound for $\hat{h}^1_c$ on $\mathbb{V}^\xi$ (hat_sup_tail_grid) is own elementary bookkeeping · proposed · no.

DV-D131-3 Lean form / DV-DZZ53Y-1 (P2-DZZ53Y, 2026-10-03; D131 P-131F) · DZZ l. 2452–2502 (Tonelli l. 2495–2500), l. 2474, 2490–2493 · LQGMetric/Papers/DZZ/S5L53Y1–Y4.lean · pairs with dist $z \cdot z' < r$ are declared far (l53FarQ', l53ZBadQ'), at the cost $a \mapsto a - 8r$ in the Tonelli bound (l53ZBadQ'_le), which needs $16r \leq K^{-1}t$; the result is $P[\text{bad} \mid A_0] \leq 2pK^2$ (l53_bad_cond_proxy); the per-pair bound (l53_far_bound_dy, $2K^{-4}$) uses the coupling in X4’s form for $\rho = \|a\|^{2m} \in [1, e^{\{L^{\{0.52\}}\}}]$ (scale choice $\ell = 4\kappa + \ell_0$, DEC-131-IF) with the asymptotic lemma l53_far_eventually (own elementary) · proposed · no.

DV-DZZ53HB-1 (P2-DZZ53HB, 2026-10-03; D131 P-131W first half) · DZZ l. 2440–2443, 2504–2522 · LQGMetric/Papers/DZZ/S5L53HB2–HB3.lean · DZZ’s “conditionally on $\mathcal{F}^*$ ” is formalized as conditioning on the countable partition $\{l53Chain = c_0\}$ (l53BadB_le_split'); the removed event is $E = \mathcal{E}_4 \cap$ (cell sizes of L3.1) (l53_E_prob); node 4 enters unconditionally (l53_uv_selected, U’s union bound over candidate boxes). Own bookkeeping · proposed · no.

DV-T39K8-1 (P2-T39K8, 2026-10-03; D130 J6d O2) · CONF C:1586 (centres and the disconnection event, no measurability argument in CONF); Pommerenke, *Boundary Behaviour of Conformal Maps*, Thm 2.6 with Prop. 2.3 · LQGMetric/Papers/CONF/S3T39K8–K8g.lean · Effros-measurability of t39jCtr and t39jGoodAct (k8_hCtr, k8_hGood) on closed bounded K with Jordan frontier and $J \subseteq \partial K$, by own elementary arguments (chains of rational balls, a countable form of DisconnectsFromInfy k8_disc_iff, measurability of the lexicographic minimum); the only topological input is uniform local accessibility of Jordan curves from the unbounded side (T39K8LocAccess), restricted to K with nonempty interior (the empty-interior case would need the Jordan curve theorem’s separation half, which the library lacks; every K used is a filled ball containing z_0 in its interior) · proposed · no.

DV-T39INT-1 (P2-T39INT, 2026-10-03; D130 J6d O2) · CONF C:1559–1561, C:1586 · LQGMetric/Papers/CONF/S3T39K5j.lean · the deterministic Effros-measurability of the centre and the good event (hCtr/hGood) is required only for closed bounded K with NONEMPTY INTERIOR and Jordan frontier (t39k5_fields_of_hitI, t39k5_fields_atI); this suffices because every use is a filled ball $\mathcal{B}^\wedge \cdot s(z_0)$, $s > 0$, with z_0 in its interior (t39k5j_mem_interior_filledBall), and avoids the separation half of the Jordan curve theorem; Lean modelling restriction, the paper is unaffected · proposed · no.

DV-DZZ53GA-1 (P2-DZZ53GA, 2026-10-03; D131 glue) · DZZ L3.13 l. 1314–1340 (tacit), DV-D131-4 · LQGMetric/Papers/DZZ/S5L53GA1–GA2.lean · (1) consecutive L3.13 chain boxes have side ratio $\geq \varepsilon^2$ (l53_side_ratio, from L313Q’s explored-box clause); (2) the $|u-v|/3$ -neighbourhood of $\bigcup \mathbb{V}_{\{i-1, w_i\}}$ lies in $\mathbb{V}_{\{u,v\}}$ (l53_thickening_region_sub_tildeBox, l53_sqBox_five_subset_tildeBox); (3) $\mu^{H^1}(\partial B) = 4 \cdot \text{side}$; (4) $EX_k \leq C \cdot L$ from the crude moments; (5) the common proxy mass map l53MB with the domination l53MB_dom at $\ell \leq \alpha\kappa + \log(C'L)$. Own elementary proofs · proposed · no.

DV-DZZ53GB-1 (P2-DZZ53GB, 2026-10-03; D131 glue) · DZZ l. 2425–2514, 2516–2522 · LQGMetric/Papers/DZZ/S5L53GB1–GB5.lean · J6’s Peierls bound in_on E form (l53_cell_desirable_prob_on), the node-4 cut-off variant (l53WBadQ_le_cut, DV-D131-3 applied to node 4), and the per-chain-box desirability on the l53Sub grid with range $r = 7$ (l53_box_desirable_prob(_proxy)); the sub-box grid count Kt must satisfy $40/\varepsilon^2 \leq Kt$; the earlier claim “ $Kt \ll K_L$ (with $Kt = K_L$ the binomial estimate fails)” came from l53_box_rhs_le using $N+1$ where the Peierls bound has $N-n+2$ — with the sharp count $Kt = K_L$ works (AUDIT-P P6, l53_box_rhs_le') · proposed · no.

DV-DZZ53J-1..3 (P2-DZZ53J, 2026-10-03; L5.3 node 3) · DZZ (eq-par) l. 1900–2000, l. 2425–2514 · LQGMetric/Papers/DZZ/S5L53J1–J13.lean · (1) DZZ’s planar duality and enumeration (eq-enumeration) is replaced by the existing annulus-enclosure Peierls bound (perc_annulus_peierls) plus a bad-column count (l53_count_bad), giv-

ing one cluster for all interfaces at once (l53_perc_cluster, l53_cluster_chain); own combinatorial argument. (2) The even $K \times K$ grid is handled by $\text{Box} = [-N, N+1]^2$ with the annulus on $[-N, N]^2$ (l53EvenBox). (3) The independence range is a general r , used with $r = 7$ for the 7x-boxes (the “ ≥ 2 sub-box sides” of DZZ l. 1993/2512 is too small: erratum) · proposed · no.

DV-UF-1..2 (P2-DZZ53UF, 2026-10-03; L5.3 node 4) · DZZ l. 2516–2522 (with l. 2474, 2490–2502, 2433–2455) · LQGMetric/Papers/DZZ/S5L53UF1–UF5.lean · (1) DV-D131-3 at the point level: points $x \in \partial B$ with $|x-w| < \varepsilon_{\text{B}}/1600$ are declared far ($\mu H^1 \leq 0.005\varepsilon_{\text{B}}$), so the Markov constant is $800/\varepsilon^*$ instead of $400/\varepsilon^*$ (l53WBadQ_le_cut). (2) The node-4 proxy of the box $B \ni w$ uses the η -chaos at scale $\varepsilon^*s_{\text{B}}/4096$ with no sub-boxes ($K = 1$), factor $\delta^2s_{\text{B}} \cdot 2e^{\{L^{\{0.91\}}\}}$, region $\mathbb{V}_{\{c_{\text{B}}, 5s_{\text{B}}\}}$ (l53ufM); no locality is needed because node 4 is unconditional. UF’s headline l53UVBadBox_node4 is off the route (audit N6); its components feed l53fn_node4 (S5L53FN1) · proposed · no.

DV-DZZ53NUM-1 (P2-DZZ53NUM, 2026-10-03; D131 scales) · DZZ l. 2427–2502 with the DEC-131-IF parameters · LQGMetric/Papers/DZZ/S5L53NU1–NU3.lean · the real-number side conditions at DZZ’s scales (l53_scales_ok: $K = 2^{\{L^{\{0.51\}}\}}$), proxy level $m = b.n + \kappa + (4\kappa + \ell_0)$, cut-off $r \in [B.\text{side} \cdot K^{-4}, B.\text{side}/(16K)]$, $\rho \geq 1$ for non-near pairs) and the conditional sub-box bound with them discharged (l53_bad_cond_proxy_scales); own elementary asymptotics · proposed · no.

DV-DZZ53FIN-1 (P2-DZZ53FIN, 2026-10-03; D131 closing) · DZZ l. 2516–2522 (node 4), 2523–2548 · LQGMetric/Papers/DZZ/S5L53FN1–FN3.lean · node 4 on the selected chain uses $\varepsilon = \varepsilon^2$ (side ratio, $G-A1$) with the proxy level $n_{\text{B}} + 2n_{\{\varepsilon\}} + 12$ and Markov factor $800/\varepsilon^{*2}$ (l53fn_node4, l53fnM, l53fn_hbad), where DZZ use ε^* at good points; the combination l53HbadBall_holds (h4 : l53Node4PartAll) (h3 : l53Node3PartAll) with the numerics conjunct l53fn_numerics ($\beta \leq e^{\{-L^q\}}$, $q > 1$) · proposed · no.

DV-T39K8L-1..2 (P2-T39K8L, 2026-10-03; J6d O2 topological input) · Pommerenke Thm 2.6 (Carathéodory); Burckel Ex. 4.37(i) (Eilenberg); QZ T4(b) · LQGMetric/Papers/CONF/S3T39K8L1–L3.lean (an independent insurance proof of the same restricted form, off the main route, is S3T39LA1–LA2.lean: la2_locAccessI, P2-T39LA2) · (1) T39K8LocAccess (hence hCtr/hGood) is proved only for K with nonempty interior (t39k8_locAccessI, t39k8_hCtrI, t39k8_hGoodI); every K used is a filled ball containing z_0 in its interior; the empty-interior case would need the separation half of the Jordan curve theorem (see DV-T39INT-1). (2) $\partial W = \partial K$ for the unbounded component W (k8l_frontier_cc) is own short argument from the non-separation of arcs (Newman Ch. V specialized), and the path construction from Carathéodory’s theorem (jm_jordan_closedDisc_extension' on the inverted $\hat{K}$) is own argument · proposed · no.

DV-DZZ53O1-1 (P2-DZZ53O1, 2026-10-03; L5.3 node 3 (O1)) · DZZ l. 2393 (Λ_i), l. 2425–2430 (the K^2 dyadic sub-squares), l. 1946–1952 (coverage) · LQGMetric/Papers/DZZ/S5L53O1A.lean · the fact that the interface of consecutive chain cells is a union of whole sub-box sides of the K -grid (l53_l313_iface_sidePiece, l53_iface_eq_sidePiece) is implicit in DZZ (“clearly”); own elementary proof by dyadic arithmetic (l53O1_grid), using $K \geq \varepsilon^{*-2}$ · proposed · no.

DV-DZZ53YC-1 (P2-DZZ53YC, 2026-10-03; D131 F closure) · DZZ l. 2490–2493 (Cor 3.9 + P3.17), l. 1235–1244, l. 611–624 · LQGMetric/Papers/DZZ/S5L53YC1–YC2.lean · Cor 3.9 is not applied at the wall $\mathbb{V}_{\{u,v\}}$ (the walled L3.5 is only proved at dyadic walls, so DEC-131 l. 270’s “hcor exactly as H4/H5” is not available); the mean comparison it gives (l53yc_mean) is transported from the dyadic wall $\bar{B}_0$ through two scaling couplings ($\lambda = L^{\{0.6\}}$), giving l53_uv_far' without hcor and the closed far bounds l53_far_bound_dy', l53_bad_cond_proxy'. Proof route differs from the source; statement unchanged · proposed · no.

DV-AUDITP-gaps (orchestrator, 2026-10-03; AUDIT-2026-10-03-P P12) · one line each for files without an entry: (KC2/KC3) the T3.9 closing wiring (confThm3_9RestAll_holds, dfgpsLem3_8_of_exp, theorem1x_openR) is routine composition (copies of KC1 with D132’s harscs/harcsM and the interior class); (ZR1) l53_domination_chainBox'/_gen copy Z4 at the larger proxy scale $\ell = 4\kappa + \ell_0$ (own elementary scale estimate); (NN1) the non-nesting of consecutive L3.13 boxes (l53_l313_nonnest) copies eq_of_safe_subset; (N4F/UFC1) the node-4 domination l53n4_dom/l53ufc_hdom apply Z3 at $j = 4n_{\{\varepsilon\}} + 12$ (resp. $3n_{\{\varepsilon\}} + 12$) with $n_{\{\varepsilon\}} \geq 2$ eventually for $\alpha > 0$ (DZZ’s α^* is positive); (KB1–KB4) route-(b) insurance for hB, unused after D132. DV-D132-1’s status: applied (S3T39K9a, S3T39KBb, K5h/K5i, KC2 committed) · proposed · no.

DV-DZZ53O2-1 (P2-DZZ53O2, 2026-10-03; L5.3 node 3 (O2)) · DZZ l. 1946–1952, 2510–2514 · LQGMetric/Papers/DZZ/S5L53O2A.lean · the covering inequalities of the Peierls step (l53_node3_cover) via explicit constants $20(2(N-n)+3+128j) \cdot 2^{\{-\kappa\}} < \varepsilon^2$ (l53_node3_hpar, $j = \lfloor \varepsilon^2 K/5120 \rfloor$) and $Kt \geq 80$ (the paper: “interfaces $\gg L^2 t$ ”); own elementary arithmetic · proposed · no.

DV-DZZ53N4F-1 (P2-DZZ53N4F, 2026-10-03; L5.3 node 4) · DZZ l. 2453–2456 (eq-M-A-upper-bound-bis), l. 2516–2522 · LQGMetric/Papers/DZZ/S5L53N4F1–N4F3.lean (an off-route duplicate of the hcor-free far and bad-event bounds is S5L53UFH1.lean, P2-DZZ53UFH, superseded) · node 4 closed (l53Node4PartAll_holds); domination at every chain box by Z3 at $j = 4n_{\{\varepsilon^*\}} + 12$ (l53n4_dom, own scale estimates l53n4_two_pow_le, l53n4_sq_sub), the

far bound and bad-event bound without hcor (l53n4_far_bound, l53n4_hbad: UF1/FN2 copies with YC1's l53_uv_far'), numerics at ε^2 (UF4 at $\alpha' = 2\alpha+1$, $Z = \varepsilon^{*-2}$) · proposed · no.

DV-DZZ53N3P-1 (P2-DZZ53N3P, 2026-10-03; L5.3 node 3 parameters) · DZZ l. 2452–2514; AUDIT-N N11, AUDIT-P P6 · LQGMetric/Papers/DZZ/S5L53N3P1–N3P3.lean · node 3 takes $j = \lfloor L^2 \rfloor$ instead of $j \asymp \varepsilon^{*2}K/400$ and the cut-off = grid count = $K_L = 2^{\lfloor L^{0.51} \rfloor}$ (l53_node3_params); parameter-range choice with the same bound $e^{\lfloor -L^{1.5} \rfloor}$; the ℓ_0 choice l53_ell0_exists and the sub-box containment l53_sub_sqBox_five_subset_dzzVXi are own elementary bookkeeping · proposed · no.

DV-DZZ53N3-1 (P2-DZZ53N3F / P2-DZZ53N3B / P2-N3COMPOSE, 2026-10-03; L5.3 node 3) · DZZ l. 2425–2548 · LQGMetric/Papers/DZZ/S5L53N3F1–F3.lean (primary l53_node3_part), S5L53N3B1–B3.lean (independent backup l53_node3_partB), S5L53N3C1.lean (composition check l53Node3PartAll_of_core) · node 3 is assembled per fibre $\{\text{chain} = c_0\}$ (trivial when $P A_0 = 0$ or $A_0 \cap E = \emptyset$; otherwise the chain properties come from a point of $A_0 \cap E$), with the Peierls step GB5 on $E \cap A_0$, the per-site bound from YC2 + NU2 (cut-off $K_L/2$ resp. K_L), the domination GA2 at $\ell = 4\kappa$, the fibre independence from R3, and the interface covering O2A; the numerics ($\kappa \geq 128$, the threshold $L^{\{0.97\}} + \log(K_L^2+1) \leq 2L^{\{0.98\}}$) are own elementary proofs · proposed · no.

3.4 Records from the GMLEM audits (2026-10-03) Departures found by the six statement audits audits/AUDIT-2026-10-03-GMLEM-A..F.md without an earlier entry (GMLEM-REC, 2026-10-03). None changes the content of a main theorem.

DV-GMLEM-A1 (AUDIT-GMLEM-A issue 1) · GM Prop 6.1, l. 3571: “there exists $c'' > c_*$, depending only on the values of c_* and C_* ” · GM.P6_1 (LQGMetric/Papers/GM/S3/Defs.lean): $\exists c'' > c_*$ comes after PairSetting γ $D D' c$ and RatiosAre $D D' c s$, so c'' may depend on $(\gamma, D, \tilde{D}, c)$ as well · statement reading (weaker dependence clause) · this is exactly how GM use it in the proof of Thm 1.9 (l. 3670: “so that c'' depends only on the choice of metrics D and $\tilde{D}$ ”); c'' is still chosen before $\beta, \tilde{\beta}, \eta$ · proposed · no.

DV-GMLEM-B2 (AUDIT-GMLEM-B issue 2) · GM Thm 2.15 (= CONF Thm 1.4, l. 1142–1144) and GM Thm 2.16 (= CONF Thm 3.9), both “leftmost D_h -geodesics” (CONF Lemma 2.4) · two formal readings are on the route: Blueprint.CONFThm1_4 and Blueprint.hitSet (GM §4's confPts, arcOf) use Blueprint.IsLeftmostGeod (= IsSideGeod true, the one-sided limit of geodesics to boundary points approaching from the left, the construction in CONF's proof of L2.4, CONF l. 548–556; LQGMetric/Blueprint/CONFDefs.lean), while Blueprint.CONFThm3_9At counts hitSetLM, built on CONF.DD.IsLeftmostGeod (CONF L2.4's order characterization “every geodesic to the endpoint lies weakly to its right”, D44, switched for T3.9 by D76; LQGMetric/Papers/CONF/Lift.lean). The two are connected by CONF.isLeftmost_of_side (LQGMetric/Papers/CONF/L2_452.lean:212, “the D76 bridge”, CONF L2.4 proof l. 557–558: one-sided-limit leftmost $\Rightarrow$ order-leftmost, given bounded compactness, existence of geodesics and uniqueness of geodesics to rational points), used on the route by GM.p412k_bridge (P412jBridge, S4/P412kBridge.lean) for GM Prop 4.12 and by CONF.t39j_gArc_eq (S3T39J1b.lean); CONF.confThm1_4 itself derives the one-sided-limit statement from the finiteness of hitSetLM by CONF's own approximation argument (t14_core, T1_4Det.lean, CONF l. 1050–1062) · definition reading (two readings of one notion, bridged by a proved lemma; D76 lists CONFThm1_4 as kept on the old reading, this entry records the pair) · proposed · no.

DV-GMLEM-B3 (AUDIT-GMLEM-B issue 3) · GM L2.14 property A (l. 1134): “no D_h -geodesic from z to a point in $C \setminus \mathcal{B}^{\wedge}_{\sigma^{\wedge}_{\tau, r}}\}$ can enter $B_{\{\varepsilon_r\}}(x) \setminus \mathcal{B}^{\wedge}_{\tau}$ ”; CONF L3.6 (C:1316): no geodesic “to a point in $C \setminus B_{\{R^{\wedge}_{\varepsilon_r}}(\mathcal{B}^{\wedge}_{\tau})\}}(\mathcal{B}^{\wedge}_{\tau})$ ” · Blueprint.CONFLem3_6AtAE0 (LQGMetric/Blueprint/CONFResults.lean) states CONF's form (targets $y \notin \text{enbhd } R^{\wedge}_{\varepsilon_r} \mathcal{B}^{\wedge}_{\tau}$); since $B_{\{R^{\wedge}_{\varepsilon_r}}(\mathcal{B}^{\wedge}_{\tau})\}} \subseteq \mathcal{B}^{\wedge}_{\sigma^{\wedge}_{\tau, r}}\}$ by the definition of σ (GM l. 1122), CONF's property A forbids more geodesics, so GM's restatement is weaker and is implied · statement of a cited result (the source's stronger form) · proposed · no.

DV-GMLEM-C1 (AUDIT-GMLEM-C issue 1) · GM Prop 3.3, l. 1236: “ $\underline{\beta} = \underline{\beta}(\mu, \nu) \in (1/2, 1)$ ” · GM.P3_3 (LQGMetric/Papers/GM/S3/Defs.lean) has $\beta_l \in \text{Ioo } 0 \ 1$ · parameter range (corrected statement) · GM's proof (P3.5 $\Rightarrow$ P3.3 by the $D \leftrightarrow \tilde{D}$ symmetry of S3.3) gives only $\underline{\beta} = 1 - \alpha \in (0, 1/2)$, and the only use (proof of Thm 1.9, l. 3671) restates $\underline{\beta} \in (0, 1)$; ERRATA row “P3.3, l. 1236” · proposed · no.

DV-GMLEM-C5 (AUDIT-GMLEM-C issue 5) · GM (1.21), l. 660–664: $c_* = \inf_{u \neq v} \tilde{D}_h(u, v)/D_h(u, v)$, $C_* = \sup_{u \neq v} \tilde{D}_h(u, v)/D_h(u, v)$ · GM.lowerRatio, GM.upperRatio (LQGMetric/Papers/GM/S3/Defs.lean) are the real-valued $\prod/\prod$ over $\{(u, v) : u \neq v\}$; in Lean a $\prod$ over a set unbounded above is the junk value 0 (the $\prod$ of these nonnegative ratios is always the true infimum) · definition convention · harmless: before any use, GM.gm_S1_23 (from GM Prop 2.2) gives a.s. $C^{-1} \leq c_* \leq C_* \leq C$ with a deterministic $C > 0$, so a.s. both agree with (1.21) · proposed · no.

DV-GMLEM-D2 (AUDIT-GMLEM-D issue 2) · GM Prop 4.3, l. 1586–1588: “parameters $b, \rho \in (0, 1)$ depending only on μ, ν ”, “ $c'' = c''(c_1', \mu, \nu)$ ”, “ $\varepsilon_0 = \varepsilon_0(\beta, c_1', c_2', \mu, \nu)$ ”, “ $\Lambda = \Lambda(c_1', c_2', \mu, \nu)$ ” · GM.P4_3 (LQGMetric/Papers/GM/S3/Defs.lean) puts $\exists c''$ after $\forall \rho$ (and after the selector and c_1, c_2) and $\exists \varepsilon_0 \ \Lambda$ after $\forall \beta$, so c'' may depend on ρ and Λ on β · quantifier order (weaker dependence) · harmless: ρ is fixed by Thm 4.2 before Prop 4.3 is invoked, c'' is still

chosen before β (as Prop 6.1 needs), and the rate of Thm 4.2 already depends on $\varepsilon_0(\beta)$ · proposed · no.

DV-GMLEM-D3 (AUDIT-GMLEM-D issue 3) · GM Lemma 4.11, l. 1980: "a parameter $a \in (0,1)$, depending only on U, ν, ℓ, p " · GM.gm_L4_11A (LQGMetric/Papers/GM/S4/Iterate6L411.lean) chooses a_0 after ε_0 (and after γ, D, χ, χ' , the CONF parameters) and proves the bound for every $a \in (0, a_0]$ · quantifier order (weaker dependence) · GM's dependence list is incomplete: condition 5 of $\mathcal{E}_r$ needs $a \leq \varepsilon_0$ · proposed · no.

DV-GMLEM-E6 (AUDIT-GMLEM-E issue 6) · GM Lemma 4.13, l. 2071–2085 (everything lies in $B_{\{3\ell r\}}(z)$, left implicit) · GM.p412e_GML4_13 (LQGMetric/Papers/GM/S4/P412eL413c.lean) takes the explicit location hypotheses $hKreg$ (the εr -thickening of $\mathcal{B}^{\bullet}_s$ lies in $regRegion\ R\ r$) and $hreg$ (P stays in that region for D -time $\leq c_{re}^{\{\xi_{h_r}(0)\}\varepsilon^{\chi}}$ after s), because condition 3 of $\mathcal{E}_r$ is stated on a region; both are discharged where the lemma is applied (module docstring) · hypothesis made explicit · proposed · no.

Errors found in the source papers

Source: ERRATA.md.

This file records every problem the formalization found in a source paper. Each entry gives the location, the kind of problem, how it was established, and what the formalization uses instead. New findings are added here as soon as an agent reports them (user, 2026-10-02: "please make sure you record everything you are finding wrong in every source"). Full details are in the cited DEVIATIONS.md / DECISIONS.md entries.

Kinds:

- **FALSE**: the printed statement is false (counterexample).
- **GAP**: a proof step is wrong or unjustified as written.
- **WEAK**: the statement is true but weaker than what the paper later uses.
- **TACIT**: an assumption is used without being stated.
- **TYPO**: misprint or citation slip.

"Lean" means the problem was established by a kernel-checked proof in this repository.

GM — Gwynne–Miller, uniqueness of the LQG metric (arXiv:1905.00383, uniqueness-final.tex)

where	kind	problem	fix (ref)
L4.18, l. 2284–2318 (eqn-uncond-to-cond)	FALSE	off by one: $K = 1, m = 1, \alpha = 0.99, \delta = 0.98$ with independent events of probability 0.98 gives 0.9604 > $e^{-1.92}$	two corrected forms proved, <code>gm_uncond_to_cond</code> , <code>gm_uncond_to_cond_K</code> (D29, DV-BINOM-2)
L4.18, l. 2286	TYPO	" $\mathcal{F}_1 \subset \mathcal{F}_1$ "	the filtration starts at $\mathcal{F}_0$ (AUDIT-2026-10-02-K C1)
§2, S2.4	GAP	the uniform tightness facts S2.4 are used without proof; S2.4e (geodesic confinement) does not follow from Axiom V alone	proved from DFGPS T1.5 + LM L3.1 (D14)
L5.6, l. 2945 (tube $V_r(z)$)	FALSE	the tube built from squares of $\mathcal{S}(B_{\{2r\}}(z))$ need not contain $z \pm 2r$ when $z - 2r$ lies on a grid line	restated geometry (D69, P2-SEPDEC-2)
proof of L4.14, l. 2482, 2485, 2501	GAP	"every unbounded connected subset of $\mathbb{C} \setminus \mathcal{K}$ whose closure contains y meets X " fails for $y \in \text{cl } X$; "one such arc Y_B " fails for $y \in \partial B$	repaired proof (D86, DV-D86a)
S3.3 footnote, l. 1222	FALSE	"the event (A) is contained in $\bar{G}_r(C', 1-\alpha)$ " fails as written ($v \in \partial B_r(0)$ is not in the open ball)	P3.4 applied with $C_1 = (C' + C_*)/2$, v moved radially inward (GA-S3.3)
L3.9, l. 1325–1404	FALSE	fails for unbounded U and for points near $\partial(rU)$	U bounded, w within $\epsilon^{\{1+\nu\}r/100}$ of x (GA-6a-e)
(4.11), l. 1692–1697	TACIT	the avoiding geodesics are taken to have finite length	finiteness proved on $\mathcal{E}_r$ (DV-D86c)
l. 2555	TACIT	P4.12, L4.22, L4.16 need $\lambda_4 > 1$ ("since $\lambda_4 \geq 1$ ")	parameter range $\lambda_4 > 1$ (DV-M2J-1)
l. 2189–2191, CONF	TACIT	σ_k is a stopping time only almost surely (Axiom II is a.s.)	a.s. stopping times (D49, DV-stop-ae, D120)
l. 1302	TACIT	"small η " must satisfy $2c s^{-1} C_s \eta < 1$	explicit hypothesis (P2-M2M2-1, D83)
condition 5 of $\mathcal{E}_r$, l. 1948–2010	TYPO	printed " $(0, ar] \cap \{2^{\{-n\}}\}$ "	$\epsilon \in (0, a] \cap \{2^{\{-n\}}\}$ (DV-M2H)
l. 223 (definition of α_ϵ) vs l. 587–590	TACIT	α_ϵ is the median of the <i>whole-plane</i> side-to-side distance $D^\wedge \epsilon_h(\text{left}, \text{right})$, while the original sources normalize by the crossing <i>along paths which stay in the unit square</i> (DFGPS l. 285; DDDF Thm 1.1(2)); l. 587–590 then apply DFGPS Thm 1.2 with GM's α_ϵ as if the two agreed. They differ by a factor bounded above and below, so the limit of Thm 1.1 changes only by a deterministic positive constant and Thm 1.2 is unaffected	Lean follows GM's α_ϵ (statement layer, D7); the bridge to the DFGPS normalization is decision D-A2 (decisions/DEC-A.md, DA3, DA4); found by P2-DEFAUDIT-B (audits/AUDIT-2026-10-03-DEFS-B.md)
l. 588–590 (derivation of Thms 1.1/1.2)	TACIT	"the median D_h -distance between the left and right boundaries of the unit square is 1. Therefore, the multiplicative constant factor is 1": this needs the median of $D_h(L, R)$ to be unique (no gap in its law); not stated or proved (the same gap is in DFGPS L2.14)	no-gap lemma <code>gm_S1_18</code> (Cameron–Martin + Weyl + IVT, adapting Gwynne arXiv:1909.08588 L2.5) (DA4, D13; AUDIT-2026-10-03-GMLEM-A issue 3)
l. 914 (before T2.5)	TYPO	"By Axiom (Weyl scaling), it follows that our metrics D_h and $\tilde{D}_h$ are jointly local for h ": Weyl scaling gives ξ -additive (joint locality comes from Axiom II)	read as " ξ -additive" (M2A-1; AUDIT-2026-10-03-GMLEM-B issue 5a)
L2.10, l. 1047–1050, used at l. 1071	WEAK	stated for "a compact set which is either a line segment or an arc of a circle"; the proof of L2.11 applies it with $L = \partial D$, a whole circle	cite DFGPS Prop 4.1, which includes whole circles (GA-7; AUDIT-2026-10-03-GMLEM-B issue 5b)

where	kind	problem	fix (ref)
event $\mathcal{E}^{\wedge}_{\mathbf{z}_r}(\mathbf{a})$, conditions 1–2, l. 1155–1156	TYPO	the centre- $\mathbf{z}$ event prints $\mathbf{c}_{re}^{\wedge}\{\xi_{hr}(0)\}$, " $\mathbf{c}_{re}^{\wedge}\{-1\}e^{\wedge}\{-\xi_{hr}(0)\}$ " and " $\mathbf{u}, \mathbf{v} \in \mathbf{B}_{\{4r\}}(0)$ "; $\mathbf{z}$ and $\mathbf{r}$ are meant (CONF L3.8 transfers to centre $\mathbf{z}$ only so)	$\mathbf{h}_r(\mathbf{z})$ and $\mathbf{B}_{\{4r\}}(\mathbf{z})$ throughout (Blueprint.confReg at centre $\mathbf{z}_0$; AUDIT-2026-10-03-GMLEM-B issue 4)
P3.3, l. 1236	TYPO	"there exists $\beta = \beta(\mu, \nu) \in (1/2, 1)$ ": the proof (P3.5 $\Rightarrow$ P3.3 via S3.3, $\beta = 1 - \alpha$ with $\alpha \in [\alpha_*, 1) \subset (1/2, 1)$) gives only $\beta \in (0, 1/2)$; the only use (l. 3671) restates $\beta \in (0, 1)$	stated and proved with $\beta \in (0, 1)$ (GM.P3_3; DV-GMLEM-C1; AUDIT-2026-10-03-GMLEM-C issue 1)
S3.5 (P3.6 $\Rightarrow$ P3.4), l. 1303–1306	FALSE	"note that $\#\mathcal{K}^{\wedge}_{\varepsilon_r} = \lfloor \nu \log_8 \varepsilon^{-1} \rfloor$ " fails ($\varepsilon = 1/8$, $\nu = 1/2$: $\#\mathcal{K} = 1$, $\lfloor \nu \log_8 \varepsilon^{-1} \rfloor = 0$), and the next step "fewer than $\mu \log_8 \varepsilon^{-1}$ good $\Rightarrow$ at least $(\nu - \mu) \log_8 \varepsilon^{-1}$ bad" needs $\#\mathcal{K} \geq \nu \log_8 \varepsilon^{-1}$, which also fails ($\log_8 \varepsilon^{-1} = 10.1$, $\nu = 1/20$: $\mathcal{K} = \emptyset$)	$\#\mathcal{K} \geq \nu \log_8 \varepsilon^{-1} - 1$; P3.6 proved with $(\nu - \mu)/2 \cdot \log_8 \varepsilon^{-1}$ bad radii, $\varepsilon \leq 8^{\wedge}\{-2/(\nu - \mu)\}$ (GA-5; AUDIT-2026-10-03-GMLEM-C issue 2)
proof of L3.7, l. 1347	TYPO	"subtracting $\mathbf{h}_{\{4r\}}(0)$ from $\mathbf{h}$ "	$\mathbf{h}_{\{4r\}}(\mathbf{z})$ (no effect; AUDIT-2026-10-03-GMLEM-C issue 4)
proof of L3.7, l. 1351	TYPO	"the $\mathbf{D}_{\mathbf{h}}$ -ball of radius $\mathbf{D}_{\mathbf{h}}(\mathbf{u}, \partial \mathbf{A}_{\{r/2, 2r\}}(\mathbf{z}))$ centered at $\mathbf{v}$ "	centred at $\mathbf{u}$ (no effect; AUDIT-2026-10-03-GMLEM-C issue 4)
L3.8, l. 1370 and proof l. 1377	TYPO	the scales are printed as $[\varepsilon^{\wedge}\{1 + \mu\}r, \varepsilon r]$; the radii $r_1, \dots, r_K$ come from (B) and L3.9, which live in $[\varepsilon^{\wedge}\{1 + \nu\}r, \varepsilon r]$	read as $[\varepsilon^{\wedge}\{1 + \nu\}r, \varepsilon r]$ (GA-6, GA-6a–e; AUDIT-2026-10-03-GMLEM-C issue 3)
proof of P3.6, l. 1457	TYPO	" $\mathbf{P}(\mathbf{s}_j) \in \partial \mathcal{B}_{\{\alpha_r\}}(\mathbf{w}_j)$ " uses the metric-ball symbol for the Euclidean circle	$\partial \mathcal{B}_{\{\alpha_r\}}(\mathbf{w}_j)$ (no effect; AUDIT-2026-10-03-GMLEM-C issue 4)
Thm 4.2, l. 1556 (also P4.3, l. 1587)	TYPO	"a deterministic set of radii $\mathcal{R} \subset (0, \varepsilon_0)$ " contradicts condition (1), which takes radii in $[\varepsilon^{\wedge}\{1 + \nu\}r, \varepsilon r] \cap \mathcal{R}$ for $\varepsilon \in (0, \varepsilon_0]$ at base scale $r \neq 1$	$\mathcal{R} \subset (0, \varepsilon_0 r]$ (DV-D79 (2); AUDIT-2026-10-03-GMLEM-D issue 1e)
l. 1570	TYPO	"Theorem 4.1 is the special case of Theorem 4.2 where $\mathcal{R} = (0, \infty)$ " contradicts $\mathcal{R} \subset (0, \varepsilon_0]$	none needed: Thm 4.1 is not used in any proof (AUDIT-2026-10-03-GMLEM-D issue 1g)
L4.7, l. 1783 (also l. 1923, 1930)	TYPO	the event $\{\mathcal{B}^{\wedge}_{\bullet} \cdot \{t_k\} \subset \mathbf{B}_{\{\varepsilon^{\wedge}\{-M\}}(\mathbf{z})\}$ lacks the factor r	$\mathbf{B}_{\{\varepsilon^{\wedge}\{-M\}}(\mathbf{z})$, as at l. 1801–1830 (AUDIT-2026-10-03-GMLEM-D issue 1f)
condition 6 of $\mathcal{E}_r$, l. 1969	TYPO	prints the grid $(\varepsilon^{\wedge}\{1 + \nu\}r/4)\mathbb{Z}^2$, but L2.13 (CONF L3.5) controls $\rho_{\{r, \varepsilon\}}$ only on $(\varepsilon r/4)\mathbb{Z}^2$, and condition 6 is used only through Remark 4.10, which needs that grid	grid $(\varepsilon r/4)\mathbb{Z}^2$ (DV-M2H-1.4 (3); AUDIT-2026-10-03-GMLEM-D issue 1d)
Remark 4.10, l. 1975–1977	FALSE	"the event $\mathcal{E}_r \dots$ is contained in the event $\mathcal{E}^{\wedge}_{\mathbf{z}}\{\ell r\}(\mathbf{a})$ " is not literally true: $\mathcal{E}^{\wedge}_{\mathbf{z}}\{\ell r\}(\mathbf{a})$ normalizes by $\mathbf{c}_{\{\ell r\}}e^{\wedge}\{\xi_{hr}\}(\mathbf{z})$ and uses $\rho_{\{\ell r, \varepsilon\}}$, $\mathcal{E}_r$ by $\mathbf{c}_{re}^{\wedge}\{\xi_{hr}(0)\}$ and $\rho_{\{r, \varepsilon\}}$; their ratio is random (argued, no Lean counterexample)	$\bigcap_{\mathbf{z} \in \mathbf{r}\mathbf{U}} \mathcal{E}^{\wedge}_{\mathbf{z}}\{\ell r\}(\mathbf{a})$ added to $\mathcal{E}_r$ as condition 7, its probability bounded in L4.11 as GM announce at l. 1161 (DV-B4, DV-M2H-1.4 (4); AUDIT-2026-10-03-GMLEM-D issues 1a, 6)
proof of L4.11, l. 1987	GAP	applies L2.6 (= LM L3.1) to the radii of Thm 4.2 (1), but L2.6 needs annuli with $R_{\{k+1\}}/R_k \leq s_1 = \lambda_1/\lambda_5$, while condition (1) only gives $r_{\{k+1\}}/r_k \leq \lambda_1/\lambda_4$	use every m -th radius, $m = m(\lambda)$ (DV-B2; AUDIT-2026-10-03-GMLEM-D issue 1b)
proof of P4.3, l. 2694	GAP	"Condition (1) of Thm 4.2 (with $\mathcal{R} = \rho^{-1}\mathcal{R}_0$) is true by the definition of $\mathcal{R}_0$ and P3.4": consecutive radii $8^{\wedge}\{-k\}$ have ratio $8 < \lambda_4/\lambda_1 = 16$ ($\lambda_1 = 1/4$, $\lambda_4 = 4$, l. 2844)	keep every other radius and apply Thm 4.2 with $\mu/2$ (DV-B3; AUDIT-2026-10-03-GMLEM-D issue 1c)
L4.14 statement, l. 2097–2101	FALSE	"there is a compact connected set $\mathbf{Y}^{\wedge}_{\varepsilon_y} \subset \mathbb{C} \setminus \mathcal{K} \dots$ such that $\mathcal{C}^{\wedge}_{\varepsilon_y}$ is contained in the prime end closure of a single bounded connected component of $\mathbb{C} \setminus (\mathbf{Y}^{\wedge}_{\varepsilon_y} \cup \mathcal{K})$ " is impossible: $\mathcal{K}$ the closed unit disc, $y = 1$; $\mathcal{C}^{\wedge}_{\varepsilon_y}$ accumulates at y , while for compact $\mathbf{Y} \subset \mathbb{C} \setminus \mathcal{K}$ of diameter $< \text{diam } \mathcal{K}$ the closure of every bounded component of $\mathbb{C} \setminus (\mathbf{Y} \cup \mathcal{K})$ is disjoint from $\mathcal{K}$	GM's own construction (l. 2516): an open arc $\mathbf{Y}^{\wedge}_{\varepsilon_y}$ of a circle minus $\mathcal{K}$ (GM.p412h_L414B; DV-L414-compact; AUDIT-2026-10-03-GMLEM-E issue 4)
proof of P4.17 (L4.20 input), l. 2349	GAP	"Since $\mathbf{E}_r(\mathbf{z})$ is determined by $\mathbf{h}_{\{\mathbf{B}_{\{\lambda_4 r\}}(\mathbf{z})\}}$ ": by Thm 4.2 (2) (l. 1559) $\mathbf{E}_r(\mathbf{z})$ also depends on the circle average $\mathbf{h}_{\{\lambda_5 r\}}(\mathbf{z})$, $\lambda_5 > \lambda_4$ (minor: fine for GM's $\lambda_4 = 4$, $\lambda_5 = 5 < 2\lambda_4$)	the ball part of $\mathbf{F}_k$ uses radius $(2\lambda_4 + \lambda_5)\varepsilon r$ (D81 (b), DV-D81; AUDIT-2026-10-03-GMLEM-E issue 5)
proof of L5.4, l. 2826 (with $\Lambda := e^{\wedge}\{\Lambda_0\}$, l. 2770)	GAP	"Property (subtracting a bump function) says that the main condition (eqn-frkE-dist) ... is satisfied with $\mathbf{h} - \varphi$ in place of $\mathbf{h}$ " is not literal, and $\Lambda = e^{\wedge}\{\Lambda_0\}$ does not bound $\exp(-(\mathbf{h}, \varphi)_{\nabla} + 1/2(\varphi, \varphi)_{\nabla})$ as needed	$\Lambda := \exp(\Lambda_0 + 2 \max_{\{\varphi \in \mathcal{G}_r\}}(\varphi, \varphi)_{\nabla})$, and in Lean $\Lambda = (N+1)e^{\wedge}\{3\Lambda_0\}$ from the union bound over $\mathcal{G}_r$ (DV-B6, DV-M2N2-a; AUDIT-2026-10-03-GMLEM-F issue 1a)
L5.8 statement, l. 3045	FALSE	" $\mathbf{U}^{\wedge}\{\mathbf{x}, \mathbf{y}\}_r$ is the interior of a finite union of squares in $\mathcal{S}_{\{\varepsilon_0 r\}}(\mathbf{A}_{\{r/2, 2r\}}(0))$, $\mathbf{x}, \mathbf{y} \in \mathbf{U}^{\wedge}\{\mathbf{x}, \mathbf{y}\}_r$ ": with the open annulus, a point $\mathbf{x} \in \partial \mathbf{B}_{\{2r\}}(0)$ at a grid corner is not in $\mathbf{U}$ (the twin of the L5.6 row above)	squares of the closed annulus $\{r/2 \leq \mathbf{w} \leq 2r\}$ (D69 (ii), P2-SEPDEC-2; AUDIT-2026-10-03-GMLEM-F issue 1c)
§5.4, l. 3185	TYPO	"with $p = 1 - (1 - p)/2$ ": the proof of L5.10 uses $p = 1 - (1 - p)/100$ (l. 3309)	$p = 1 - (1 - p)/100$ (P2-M2M-1..5 (3); AUDIT-2026-10-03-GMLEM-F issue 1g)
condition (10) of $\mathbf{E}_r$, l. 3266	TYPO	" $(\mathbf{h}, \varphi)_{\nabla} + 1/2 (\varphi, \varphi)_{\nabla} \leq \Lambda_0$ " bounds $(\mathbf{h}, \varphi)_{\nabla}$ only from above, while (eqn-frkE-dirichlet), l. 2768–2770, needs $\exp(-(\mathbf{h}, \varphi)_{\nabla} + 1/2(\varphi, \varphi)_{\nabla})$ bounded, i.e. a bound on $-(\mathbf{h}, \varphi)_{\nabla}$	read with $ (\mathbf{h}, \varphi)_{\nabla} $ (DV-B6; AUDIT-2026-10-03-GMLEM-F issue 1b)
proof of L5.15, entry step, l. 3477–3489	GAP	" $\mathbf{P}^{\wedge}\varphi$ must have a segment ... so $\mathbf{P}^{\wedge}\varphi$ must enter $\mathbf{O}_{\mathbf{u}}$. Similarly, $\mathbf{P}^{\wedge}\varphi$ must enter $\mathbf{O}_{\mathbf{v}}$ (and must do so at some time after it enters $\mathbf{O}_{\mathbf{u}}$): the first point of $\mathbf{P}^{\wedge}\varphi$ need not be near $\mathbf{x}'$, and condition (2) is symmetric in $(\mathbf{u}, \mathbf{x}) \leftrightarrow (\mathbf{v}, \mathbf{y})$, so $\mathbf{P}^{\wedge}\varphi$ may enter $\mathbf{O}_{\mathbf{v}}$ first	first/last hitting times of $\text{cl } \mathbf{B}_{\{3r\}}(0)$ with condition (4) and L5.12–5.13; the bound at $\mathbf{v}$ added to condition (1) of L5.6/L5.8 (D83 (a)–(b), DV-D83 (1), P2-M2M3-1..3; AUDIT-2026-10-03-GMLEM-F issue 1d)

where	kind	problem	fix (ref)
proof of L5.15, l. 3480–3483	TACIT	"the connected component of $(U \cup W) \setminus O_u$ which contains x' lies at Euclidean distance at least $\epsilon_0 r$ from the union of the other connected components" uses that U is attached to x, y near those points, which L5.8 does not state (true for GM's U built along $\tilde{L}_x, \tilde{L}_y$)	clause (T5) of IsTubeFam, supplied by L5.8 with radial stubs at x, y (D83 (c), DV-D83 (2); AUDIT-2026-10-03-GMLEM-F issue 1h)
proof of L5.15, l. 3484	FALSE (harmless)	" $B_{\{2\zeta r\}}(O_u) \subset O_u \cup (B_{\{2\zeta r\}}(\partial U) \setminus W)$ " fails at the outer rim of O_u (points of U just outside $B_{\{20\epsilon_0 r\}}(u)$, far from ∂U ; argued, no Lean counterexample); the argument uses it only off the two sides, where it holds	not used in that form (D83, DV-D83 (2); AUDIT-2026-10-03-GMLEM-F issue 1e)
proof of L5.15, l. 3505	GAP	the display compares $\tilde{D}_{\{h-\varphi\}}(P^\wedge\varphi(s))$ through $\tilde{D}_{\{h-\varphi\}}(u, v)$ and then bounds by $\tilde{D}_{\{h-\varphi\}}(P^\wedge\varphi(s), \partial B_{\{3r\}}(0))$, conflating distances from u and from $P^\wedge\varphi(s)$	triangle inequality, which needs $1 + 3\eta \leq C_-/c_-$ (added to EtaChoice) (P2-M2M2-1.5 (2), DV-D83 (3); AUDIT-2026-10-03-GMLEM-F issue 1f)

CONF — Gwynne–Miller, confluence of geodesics (arXiv:1905.00381, confluence-final.tex)

where	kind	problem	fix (ref)
L2.14, C:832–894 (C:855)	FALSE	false for small n : dropping the " ≤ 8 long arcs" at C:855 costs an additive constant (a disc with a long thin fjord, $n = 1$); C:861 should read "contained in $\varphi(H_1)$ "	$\#good \geq (1 - A C^{-2} \text{area}(U)) n - N_0$ (D36, DV-CONF-214b)
L2.14 statement, C:832–838	WEAK	as printed the conclusion is trivially true (a small circle around 0 disconnects), too weak for L2.15 by inversion	located L2.14, X within $C n^{-1/2}$ of ∂U , as the proof gives (D119, $\text{t39h_disconnectsIn_trivial}$, Lean)
L3.3, (3.9)/(3.12) and $G^\wedge U$, C:1205, 1208, 1219	FALSE	"pocket" configurations (two columns of removed squares) disconnect $U_{\{8r/4\}}$ while U is connected, so the event is empty	grid-component event fat6 (D112, DV-D112-1)
L3.5, C:1284	GAP	the constants " $C = \epsilon^{-3/2}$, $q = 6$ " do not give the claimed bound	own constants (DV-CONF-35)
L3.6 Step 3, C:1425–1433 (conditioning at C:1438)	GAP	conditioning on $\{\epsilon = e, z = z, \tilde{\rho}^n = r, \tilde{U} = \mathcal{U}\}$ is a countable partition only for countably valued ϵ	hypothesis (Set.range ϵ).Countable; every use is countably valued (D114)
L3.6, C:1414	TACIT	assumes $\tilde{\rho}^n < \infty$	$\tilde{G}^n$ includes $\{\tilde{\rho}^n = \infty\}$ (DV-CONF36-3)
L3.7, C:1460 vs C:1600–1602	WEAK	property A only excludes geodesics to targets outside $\mathcal{B}^\wedge \cdot_\sigma$, but is applied to $I' \subseteq \partial \mathcal{B}^\wedge \cdot_{\{s_{k+1}\}} \subseteq \mathcal{B}^\wedge \cdot_\sigma$	targets outside $\text{int } \mathcal{B}^\wedge \cdot_\sigma$ (D113)
T3.9, C:1556	TACIT	uses $n_k \geq 16$	$\epsilon'_k = 1/2$ when $\text{t39gExp } n_k = 0$ (DV-T39b-4)
T3.9	TACIT	the probability space is complete; the iterated radii are stopping times only a.s.	$[P.\text{IsComplete}] + \text{completion transfer}$, $\text{IsFilledBallStoppingTimeAE}$ (D120)
C:835, 1550	TACIT	prime ends of $\partial \mathcal{B}^\wedge \cdot$ used without justification	$\partial \mathcal{B}^\wedge \cdot$ is a Jordan curve (proved, $\text{GM.gm_filledBall_frontier_isJordanCurve'}$), so prime ends are points (D120)
L3.3 Step 3, C:1240–1241	GAP	"a similar justification" for the S-cont-law of condition 2	proved without conditioning, then Fubini (DV-CONF36Q-3)
L2.10, C:722–731	GAP	the white-noise approximation requires $h_{\{t, \infty\}}$ extended by 0 to be continuous, which can fail for irregular domains	used only for $U = \text{confU}$ (an annulus minus finitely many squares) (P2-CONF210B; proved there, D127)
L3.6 Step 1, C:1300, 1366 (also C:1468)	TACIT	"so that $\mathcal{B}^\wedge \cdot_\sigma$ contains $B_{\{6\rho(z)\}}(z)$ " assumes an open set inside every $\mathcal{B}^\wedge \cdot_{\{s'\}}$, $s' > \sigma$, lies in $\mathcal{B}^\wedge \cdot_\sigma$; this needs a.s. geodesics and bounded balls	from DFGPS L3.8 (DV-CONF36Q-2, $\text{t39g_enbhd_subset_interior}$)
C:1187	TYPO	"LM Lemma 2.2"	LM Lemma 2.1 (DEV-LM-h)
event $\mathcal{E}_r(a)$, condition 2, C:1488	TYPO	prints $e^\wedge\{-\xi h\}$; with that sign CONF L3.8 is false	$e^\wedge\{+\xi h\}$ (AUDIT-2026-10-02-K C1)
C:1128	TYPO	$\text{dist}(z, \partial U)$	$\text{dist}(u, \partial U)$ (AUDIT-2026-10-02-K C1)
T3.9, C:1506–1509 with L3.6 Step 3, C:1431–1432	TACIT	"stopping time for $\{(\mathcal{B}^\wedge \cdot_s, h _{\{\mathcal{B}^\wedge \cdot_s\}})\}$ " is used (via L2.1 with h modulo constants, C:1154) as if $\mathcal{B}^\wedge \cdot_\tau$ were determined by h modulo additive constants; for the normalized field, $\tau := \tau_r$ on $\{B_{\{r/10\}} \subseteq \mathcal{B}^\wedge \cdot_{\{\tau_r\}}, h(\psi) > 0\}$ ($f/\psi = 1$), $\tau_{\{2r\}}$ otherwise, is a stopping time whose ball is not $(h(\psi) = \tilde{h}(\psi - \rho_1))$ is not local modulo constants). The conclusion stays true for such τ (grid reduction, DEC-130 §2); all stopping times CONF/GM use have the tacit property	hypothesis $\text{IsLocalSetDet0 } P \text{ h } \mathcal{B}^\wedge \cdot_\tau$ (D130, DV-D130-1)
T3.9 proof, C:1559–1561	TACIT	" $\mathcal{J}_k$ is determined by $(\mathcal{B}^\wedge \cdot_{\{s_k\}}, h _{\cdot})$ " is asserted and used for the filtration (C:1567–1568) with no measurability argument, while the arcs are defined via "the leftmost geodesic", unique only a.s., and the sets are a priori Σ^1_2	a.s. transfer to GM's analytic arc0f relation, whose measurability GM supply at l. 1705–1708 (D132, DV-D132-1)

DG — Ding–Gwynne, the fractal dimension of LQG (arXiv:1807.01072, metric-comparison-final.tex)

where	kind	problem	fix (ref)
Prop 3.16, DG:1432–1441; proof DG:1500–1512	FALSE	lower bound fails with probability 1: z, w at distance 2ε on either side of a shared grid side give $D^\delta(z, w) \rightarrow 0$ while $\tilde{D}^\delta - \delta e^{\xi h(v_{S_z})} \geq \delta e^{\xi h(v_{(1,0)})}$; the proof's "largest $j' \leq j$ with a long piece" need not exist	not_dgProp3_16Printed (Lean); corrected statement (error term $\delta^\alpha \{1 - \xi\} e^{\xi h}$), from DG's own count DG:1505–1508) is the cited Blueprint. DGProp3_16 (D126); both halves proved from DG L3.7 (dgProp3_16_of_lem37, DV-DG316-1, DV-D126) cited on $\mathbb{S}(1/2)$ (D118)
L3.7, DG:1096–1102 vs DG:1729–1731	WEAK	stated on $\mathbb{S} = [0, 1]^2$, used on $\mathbb{S}(1/2) = [-1/2, 3/2]^2$ (its proof via DGW Prop 3.2 gives that)	$\tilde{T} = \delta^\alpha \{-2 - \gamma^2/2\} e^{\alpha \{-\gamma \max \tilde{h}_\delta\}}$ (DG3A-1)
(3.7), DG:1032–1036 DG:1547	TYP0	$\exp(-\max \tilde{h}_\delta)$ is missing the γ produced by (3.6)	$m = m_\delta + 1$ (DV-DG317S-1)
L3.19 proof, DG:1614–1660	TYP0 GAP	" $m = \lfloor \log_2 \varepsilon^{\alpha \{-\beta\}} \rfloor - 1$ " the case $\varepsilon e^{\alpha \sqrt{c} \sqrt{n}} > \varepsilon_-^*$ is not treated	trivial bound $D \geq 1$ at the top level (DV-D105g-2)

DZZ — Ding–Zeitouni–Zhang, heat kernel / Liouville graph distance (arXiv:1807.00422, LBM_LGDaXiv.tex)

where	kind	problem	fix (ref)
L5.4, L6.1, l. 2559, 2567, 2605	GAP	segments of length $\delta^\alpha \{2t\}$ with crossing cost $\delta^\alpha \{-\chi + t\}$ work only for $\chi \geq 1/2$	$\delta^\alpha \{Kt\}$, $K\chi \geq 2$ (L5.4); $\delta^\alpha \kappa$, $\kappa\chi \geq 2t$ (L6.1) (D117, DV-D117-1)
L5.4, l. 2563	GAP	the leg inequality is not justified: the leg lives in $\mathbb{V}_{\{u, 2\lambda\}} \not\subseteq \mathbb{V}_{\{u, v\}}$	leg wall $\mathbb{V}_{\{u, 2\lambda\}} \cap \mathbb{V}_{\{u, u+\lambda e_j\}}$ (D117, D123, DV-D123-1)
L6.1, l. 2605	TYP0	"the geodesic between v_L and $\partial \mathbb{V}_u$ "	a tilde geodesic from v_L to the ring (DV-DZZ61L, D117)
L6.1, l. 2593–2596	TACIT	(eq-point-to-boundary-kappa) is applied at centres moving with δ , which needs uniformity in the centre	uniform version from l. 619 (DV-DZZ61L-c, DV-D117-4)
Remark 5.2	TACIT	the walled P3.2/P3.17 are asserted with no range of pairs; read literally they fail for pairs near or outside the wall, and DZZ only use pairs inside	pairs in $k\mathbb{X} \ K \ \xi$ (DZZProp317In) (D123, DV-317K-1)
(eq-concentration-approximate-2), l. 1627	TYP0	rate $e^{\alpha \{(\log \delta^{-1})^\alpha \{0.8\}\}}$	$e^{\alpha \{-c(\log \delta^{-1})^\alpha \{0.8\}\}}$, as l. 1647 (DV-P317a)
L2.9 used at l. 2266–2270, 2318–2322	TACIT	stated for real scalings and translations, used for rotated boxes	coupling for complex a (DV-DZZ53b-c)
L5.3 part 3, l. 2423	TACIT	Cor 3.9 for tilde distances (Remark 5.2) used without proof	own argument (DV-DZZ53T-1)
scale couplings, l. 611–624	TACIT	L2.7 and the chaos limit used along non-dyadic scales without comment	proved along $c \cdot 2^\alpha \{-n\}$ (DV-DZZSIM-1)
l. 1122	GAP	" $E \tilde{M} = \text{Leb}$ by Fubini": only $\leq$ holds as argued	$E \tilde{M} \leq \text{Leb}$ (DV-ETAa.f (b))
(eq-cell-LQG-compare)	FALSE	over all squares S it fails at $\partial \mathbb{V}$, where squares outside $\mathbb{V}$ have mass 0; the proof only uses $S \subseteq R \subseteq \mathbb{V}$	quantified over $S \subseteq \mathbb{V}$ (DV-D97a)
P3.2, covering balls l. 1131–1137	GAP	the covering balls of radius t s at grid corners stick out of $\mathbb{V}$	in-box measure and clipped covers (D101, D102)
L3.12, l. 1471–1477	GAP	splitting the enclosure $\mathcal{C}_{\{i, \text{cross}\}}$ into two segments presumes a closed ring	route of D93
L5.3 part 1, l. 2517, 2521	TYP0	$\log E \tilde{D}_{\{y, \delta; \eta\}}(u, v)$	$E \log \tilde{D}$, as in (eq-start-end) and (eq-union-bound-distance) (P2-DZZ53E)
L5.3 part 1, l. 2429 (open box)	TYP0	$\mathcal{L}_1(\Lambda') \geq (1-10^{-5}) s_{-i}/K$ is too weak ($\mathcal{L}_1(\partial B) = 4s_{-i}/K$, so Λ' need not meet every side)	$\mathcal{L}_1(\Lambda') \geq (1-10^{-5}) \mathcal{L}_1(\partial B)$, which the counting at l. 2495–2502 gives (l53_open_of_not_bad, DV-DZZ53E-c)
L5.3 part 1, l. 2490–2492	GAP	"Cor 3.9 and Prop 3.17 give $O(K^{-4})$ ": at $\tilde{\delta} = 2^\alpha \{-l\}$ with l small (subadditivity needs all $l \geq 1$), Cor 3.9 holds only with probability $1 - \delta^\alpha c$ and P3.17 with $1 - e^{\alpha \{-(\log \tilde{\delta}^{-1})^\alpha \{0.7\}\}}$, neither of which is $O(K^{-4})$	concentration at $\tilde{\delta} e^{\alpha \{-L^\alpha \{0.95\}\}}$ + two-regime mean comparison (l53_uv_far, DV-DZZ53G-1..5 (4)); confirmed in D131 (SE-1)
L5.3 part 1, l. 2437–2438	FALSE (harmless)	$\max \eta^\alpha \{B, s_{-i}\}_{\{\varepsilon'\}} - \eta^\alpha \{s_{-i}\}_{\{\varepsilon'\}} = 0(\sqrt{\log 6^{-1}})$ "similarly to (eq-assumption-in-lower-heat-kernel)": for ε' between s_{-i}/K and s_{-i} the bridge exits B^{**} with probability ≈ 1 , so the difference field has variance $\asymp \log K \asymp L^\alpha \{0.51\}$ and $\sup \asymp L^\alpha \{0.76\}$ over B^* , not $O(\sqrt{L})$; the slack $e^{\alpha \{L^\alpha \{0.91\}\}}$ absorbs it. Same mechanism makes l. 1808 only marginally $O(\sqrt{L})$	not used: fine η -chaos proxy (D131, DV-D131-1)
L5.3 part 1, l. 2474, 2533	TACIT	$a \asymp s_{-i}^{-1} K$ assumes $ z - z' \asymp s_{-i}/K$; close pairs are not covered	close pairs declared far (D131, DV-D131-3)
L5.3 part 1, l. 2361	TYP0	$w_i = u + (i/9) u - v $	$w_i = u + (i/9) (v - u)$ (P2-DZZ53F)

where	kind	problem	fix (ref)
L5.3 part 1, l. 2516–2522	GAP	u and v desirable only “with probability tending to 1”, but the completion step (l. 2414, $E(1_{\{\mathcal{D}^c\}} \log \bar{D}) \leq (\log \delta^{-1})e^{-(\log \delta^{-1})^{0.1}}$) needs a rate	the Markov bound gives $O(L\epsilon^{*-1}K^{-4})$ (P2-DZZ53M)
(eq-par) l. 1993, reused at l. 2512	TYPO (constant)	boxes at ℓ_∞ -distance $\geq 2s_i/K$ are treated as independent, but with $B^{**} =$ the $3s_i/K$ -neighbourhood of ∂B (l. 2433) their white-noise regions overlap; independence needs a gap $> 6s_i/K$ (L/25 should be about $L/64$); result unaffected	range $r = 7$ in the Peierls step (P2-DZZ53J, DV-DZZ53J-1..3)
L5.3 part 1, l. 2516–2522 (“similar” to l. 2490–2502)	TACIT	assumes $ x - w \asymp s_B$ for $x \in \partial B$, but w may lie on ∂B , so the scaling coupling degenerates as $x \rightarrow w$ (point-level analogue of SE-3)	point-level cut-off (P2-DZZ53UF, DV-UF-1..2)
L5.3 part 1, l. 2393	TACIT	(eq-Lambda-i-not-small) stated without proof; for the L3.13 end boxes it needs u to be a good point	153_box_side_good (DV-DZZ53M-1..2)
L5.3 part 1, l. 2380–2391 (Eq.calD1)	GAP	the errors exceed the claimed $L^{\{0.95\}}$: P3.17’s own bound (l. 1514) is $L^{\{0.95\}}$, and $L^{\{0.7\}}$ ($\mathcal{E}^*$) and $L^{\{0.92\}}$ (d_i) come on top	threshold $L^{\{0.97\}}$; (eq-040418) still holds (DV-DZZ53F-a)
Thm 1.1, l. 126–139	TACIT	the a.s. statement (random $C(\tau, u, v)$) is derived from bounds in probability (P3.17, Prop 5.1); the Borel–Cantelli-along-dyadics and monotonicity step is unwritten, and convergence in probability alone does not suffice	explicit rates + Borel–Cantelli (D129, DV-D129-1)
Prop 5.1 proof, l. 2311	TYPO	the constant $2\xi/\sqrt{5}$ puts the rotated tilde boxes in $\mathbb{V}$ but not in $\mathbb{V}^\xi$ (harmless: only $y_i \in \mathbb{V}^\xi$ is used; $\xi/\sqrt{2}$ would do)	chain step with tilde boxes in $\mathbb{V}^\xi$ (DV-D129-2)
Prop 5.1 proof, l. 2340–2345	TACIT	(eq-geodesic-range) is used as an a.s.-eventually statement	one-sided first-exit bound with the wall kept (D97, DV-D129-3)
Prop 5.1 proof, l. 2336–2338	TACIT	“with high probability” at the scale $\delta e^{\sqrt{\log \delta^{-1}}}$ hides that the P3.17 tail $e^{-(\log \delta^{-1})^{0.7}}$ does not stay summable after the move to the larger scale (minor; only the polynomial bound $\delta^{c_1^2}$ is needed for the lower tail)	polynomial reference tail (P2-CHIB, DV-D129-1 addendum)

DDDF — Ding–Dubédat–Dunlap–Falconet, tightness of LFPP (arXiv:1904.08021, tightness.tex)

where	kind	problem	fix (ref)
L25, l. 1221–1258	FALSE	the printed condition drops a factor 2	$(C_1+C_2)^2 + 2(C_1+C_2)C_3 \leq C_3^2$ (P2-ELEM-b)
L9 proof, l. 689–722 (6.99), l. 1630	GAP GAP	$\alpha = s/(2\sigma^2)$ does not give the bound the case n or $n' = 0$ is not covered ((6.100) uses (5.76), which needs $n, k \geq 1$)	$\alpha = s/\sigma^2$ (D-DDDF-L9-1) constant $ \log \lambda_0 $ (DDF6-1)

DFGPS — Dubédat–Falconet–Gwynne–Pfeffer–Sun, weak LQG metrics and LFPP (arXiv:1905.00380)

where	kind	problem	fix (ref)
T:1448–1450, 2371, 2375, 697, 874, 1031, 1686, 1566, 1803, 1818	TYPO	misprints	(DEV-DFGPS-3)
Prop 4.3 proof, T:2699–2741	GAP	needs a point of ∂B_k at distance $\geq s$ from 0 for times $< s$; fails when B_k surrounds a hole of $\mathcal{B}_s$	filled-ball version DFGPSProp4_3F (D68)
L3.20, T:2321	TACIT	$D_h(u, v; B_{\{2 u-v \}}(u))$ is meaningful only for $u \neq v$	$u \neq v$ (D55)
L3.5 and Prop 3.1, T:1414–1420, 1573–1596	TACIT	Prop 3.1 says “open”, but the proof uses connectedness of U	U connected (DEV-DFGPS-L35)
P3.18 / L3.20 / L3.22, T:2247–2385	GAP	the scale $2^{\{k\}}\epsilon$ with grid mesh $2^{\{k-2\}}\epsilon r$ cannot fit $u, v \in B_{\{tr\}}(z)$ and $B_{\{2tr\}}(z) \subseteq B_{\{2 u-v \}}(u)$	scales $\epsilon(3/4)^k$, mesh $tr/40$ (DFA7-1)
Prop 4.1 Step 1, T:2464–2470	FALSE	“one block of centres for all paths” (a path around a whole circle can go either way)	block chosen per path (DEV-P41-5)
L2.14	GAP	no gap in the law of $D_h(L, R)$ is used without proof (the same gap is in GM S1.18)	proved by Cameron–Martin + Weyl + IVT, adapting Gwynne arXiv:1909.08588 L2.5 (D13)

LM — Gwynne–Miller, local metrics of the GFF (arXiv:1905.00379, local-metrics-final.tex)

where	kind	problem	fix (ref)
Thm 1.7 Step 3, l. 1075–1084	GAP	incorrect as printed: the near-geodesics P_ε vary with ε and (5.18) does not give what is claimed	repaired proof (D107, DV-D107)
L1.4 proof, tex:262	TYPO	$h _{U \setminus V}$	$h _{\{U \setminus V\}}$ (DEV-LM-a)

GMSH — Gwynne–Miller–Sheffield (arXiv:1807.07511)

where	kind	problem	fix (ref)
L2.1	TYPO	typos in the statement	read as cited (DEV-LM-h)

MQ (arXiv:1812.03913), DGo (arXiv:1610.09998)

No errors found so far.

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She — Sheffield, Gaussian free fields for mathematicians (arXiv:math/0312099)

where	kind	problem	fix (ref)
l. 277, l. 293	TYPO	the variance of $(h, f)_\nabla$ is printed as “ $(f, f)^2$ ” and “ $\ f\ _\nabla$ ”; it is $(f, f)_\nabla = \ f\ _\nabla^2$	no effect on the Lean statement (the GFF is defined by its covariance); found by P2-DEFAUDIT-A (audits/AUDIT-2026-10-03-DEFS-A.md)

Not source errors: false statements in our own earlier formalization

Kept separate for transparency; these were mistakes in our blueprint or modelling, not in the papers. Each is recorded in DECISIONS.md / DEVIATIONS.md.

- D37 (TIP_smul)
- D54 (DFGPS L2.6 transcription)
- D55
- D61
- D70
- D95
- D97
- D101–D102 (DZZ P3.2 covers)
- D106
- D108/D110 (our `IsWholePlaneGFF` allows a random additive constant, so the raw conditioning σ -algebras of CONF L3.2–L3.7 had to be taken modulo constants)
- D114 (g)
- D115 (P32StartGeom)
- D116 (s/16 tiling)
- D119 S5
- D122 (DZZCrudeMoments)
- DV-CONF33H-1 (CONFZBExt0fMarkov)

- DV-CONF210-1 (CONFZBShiftAC for general φ)
- P2-DGWIRE (L3.19 box $[1/12, 11/12]^2$)
- DEC-120 §4 (T39JArcChoice.subset : arcs $\subseteq \Gamma$ too strong for Effros-measurable arcs, DV-J3-1)
- CONFThm3_9RestC (its hsucc fails for invalid p, e.g. $p = \langle 1/2, 1/2, -1, 1 \rangle$; needs L3.5 and $0 < p.\eta$; P2-CONFJ6)
- D117 §3 (the walled tilde partition as the global $\mathcal{V}_\delta$ restricted to cells meeting the wall, whereas DZZ l. 2275 partition $\mathbb{V}_{\{u,v\}}$ itself; the L3.12 boundary case does not transfer, DV-DZZ53I-1, D131)
- DEC-130 §4 L2 (mod-constant a.s. stopping times do not give mod-constant local sets: deterministic τ , P2-T39K2, DV-D130-L2; re-decided in D130B)
- DEC-131 §3 (the P-131S coupling tail and the scale $2^\ell \approx 4C_{Mc} L$ too small with the near-pair cut-off; fixed in DEC-131-IF: $\ell = 4\kappa + \ell_0$, X4's form)
- DEC-131 §2 (the refinement with the same R on both sides, DV-D131-5/A)
- DEC-131 §3/§9 (fineChaos not local, DEC-131 §9)
- pending D130 (since decided: D130, D130B): CONFThm3_9At over raw filled-ball stopping times (our random additive constant; P2-CONFJ6)
- DEC-127 §3 (L2) / handoff P2-CONFZBM (L2): the Lipschitz route for the coarse kernel at confU (BridgeShell-Bound + reflection) does not apply — replaced by a Hölder form (P2-HEATG, DV-P127G-1..3)
- DEC-127's header calls $G_D := \pi \int_0^\infty p_D$ BP's definition, but BP Def. 1.11 is $G^D_\theta = \int_0^\infty p^D_t$ with generator Δ , while our heatKernel has generator $\frac{1}{2}\Delta$, so killedGreen = $\pi \int p_U = 2\pi G^D_\theta$ (P2-HEATA; the Lean statements are consistent)